TSMC 130nm CL013G-FSG Process
1.2V Metro™ v1.0 Standard Cell Library
Databook

October 2005, Revision 1.0

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Contents

Preface

Revision History

This document contains the release history for the TSMC 130nm CL013G-FSG Process 1.2V Metro^{TM} v1.0 Standard Cell Library Databook.

Part Number	Release Number	Date of Release	Updates
DB-Metro-TSM067-1.0/0.13@1.2-1.0	1.0	October 2005	Initial Release

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Introduction

ARM's Artisan Physical IP Metro™ standard cell library builds upon the SAGE architecture, producing the optimum combination of high-density with high-performance. The cell line-up is derived from extensive customer design, synthesis, and place-and-route benchmark analysis. Library optimization is achieved by carefully matching the library functions and drive strengths to leading synthesis and place-and-route tools, producing superior RTL-to-GDSII results.

How This Book Is Organized

This introduction is organized into three sections:

- ***Global Parameters*** provides an overview of parameters specific to your Metro library.
- ***Special Cells*** details the types of special cells included in the library.
- ***Reading the Standard Cell Datasheet*** describes the components of each datasheet.

Datasheets for each cell in this library are provided after the introduction. The datasheets are included in alphabetical order.

Global Parameters

This section specifies global parameters for the TSMC 130nm CL013G-FSG Process 1.2V Metro v1.0 Standard Cell Library. It covers physical specifications, electrical specifications, derating factors, propagation delay calculation, timing constraints, power calculation, and power-rail strapping.

Physical Specifications

Table 1 shows the physical design specifications of this library.

Table 1. Physical Specifications

Drawn Gate Length (um)	0.13
Layers of Metal	4, 5, 6, 7, or 8
Layout Grid (um)	0.005
Vertical Pin Grid (um)	0.41
Horizontal Pin Grid (um)	0.41
Cell Power and Ground Rail Width (um)	0.26
Cell Height (um)	2.87

In this library, all pins are located on the vertical and horizontal pin grids. Most place-and-route tools work more efficiently with all pins on grids, and some tools even require it.

The Metro library also supports designs with four, five, six, seven, or eight layers of metal. You may need to change the design rules in the technology file, because the top-level metal has a greater minimum width and greater minimum spacing requirement. See "TSMC 0.13UM LOGIC 1P8M SALICIDE 1.0V/2.5V,1.2V/2.5V,1.0V/3.3V,1.2V/3.3V PROCESS Design Rule" design rule manual. You must define these rules correctly for the place-and-route tool.

Table 2 describes the electrical specifications for this library.

Table 2. Electrical Specifications

Parameter	Minimum	Typical	Maximum
DC Supply Voltage (Vdd)	1.08V	1.2V	1.32V
Junction Temperature	-40°C	25°C	125°C

Table 3 shows the derating factors for this Metro Standard Cell Library.

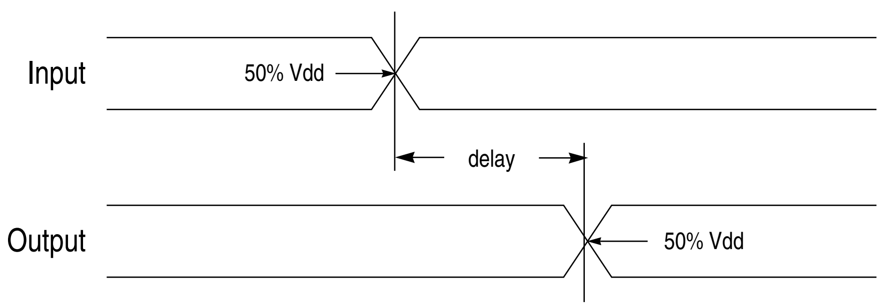
Table 3. Derating Factors

$K_{Process}$ (slow)	1.573
$K_{Process}$ (typical)	1.000 (by definition)
$K_{Process}$ (fast)	0.671
K_{Volt} (1.2V to 1.08V)	-1.422/V
K_{Volt} (1.2V to 1.32V)	-0.964/V
K_{Temp} (25°C to -40°C)	0.00115/°C
K_{Temp} (25°C to 125°C)	0.00104/°C

Propagation Delay and Transition Time

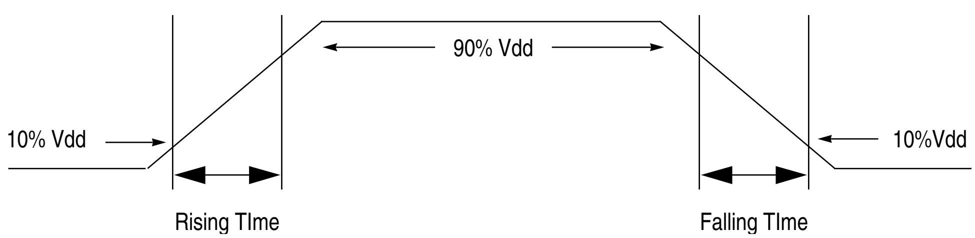
The propagation delay through a cell is the sum of the intrinsic delay, the load-dependent delay, and the input-slew dependent delay. Delays are defined as the time interval between the input stimulus crossing 50% of Vdd and the output crossing 50% of Vdd. Figure 1 illustrates the propagation delay.

Figure 1. Propagation Delay



The transition times (slews) on input and output pins are defined as the time interval between the signal crossing 10% of Vdd and 90% of Vdd. Figure 2 illustrates transition time measurements for rising and falling signals.

Figure 2. Transition Time



Factors that affect propagation delays and transition time include: temperature, supply voltage, process variations, fanout loading, interconnect loading, input-transition time, input-signal polarity, and timing constraints. The timing models provided with this library include the effects of input-transition time on propagation delays. Also, all timing models use a table lookup method to calculate accurate timing. To simplify calculations, the standard cell datasheets provide all timing numbers for an input slew of 0.023ns and a linearized load factor, K_{load}, which is not as accurate as the timing models. All cells have been characterized with a fully populated metal2 (0.41um horizontal pitch) and metal3 (0.41um vertical pitch) routing grid across the entire cell layout.

The Metro Standard Cell Library may contain negative propagation delays. Although most third-party verification tools can handle negative propagation delays, some tools will turn negative delays into a zero value.

Derating Factors

Derating factors are coefficients that the typical process characterization data is multiplied by to arrive at timing data that reflects appropriate operating conditions. Table 3 provides derating factors for variations in process case, temperature, and voltage.

Derating factors are derived by averaging the performance of many different cells in the library. A particular combination of cells may perform better or worse than indicated by these derating factors.

Delay Calculation

Using the delay data in the datasheets ($t_{intrinsic}$, K_{load}, and C_{load}) and the delay derating factors, the estimated total propagation delay is calculated as such:

$$t_{TPD} = (K_{Process}) * [1 + (K_{Volt} * \Delta Vdd)] * [1 + (K_{Temp} * \Delta T)] * t_{typical}$$

$$t_{typical} = t_{intrinsic} + (K_{load} * C_{load})$$

where:

t_{TPD} = total propagation delay (ns);

$t_{typical}$ = delay at typical corner-1.2V, 25°C, typical process (ns);

$t_{intrinsic}$ = delay through the cell when there is no output load (ns);

K_{load} = load delay multiplier (ns/pF);

C_{load} = total output load capacitance (pF);

$K_{Process}$ = process derating factor, where process is slow, typical, or fast;

K_{Volt} = voltage derating factor (/V);

ΔVdd = Vdd - 1.2V;

K_{Temp} = temperature derating factor (/°C);

ΔT = junction temperature - 25°C.

Timing Constraints

Timing constraints define minimum time intervals during which specific signals must be held steady in order to ensure the correct functioning of any given cell. Timing constraints include: setup time, hold time, recovery time, and minimum pulse width.

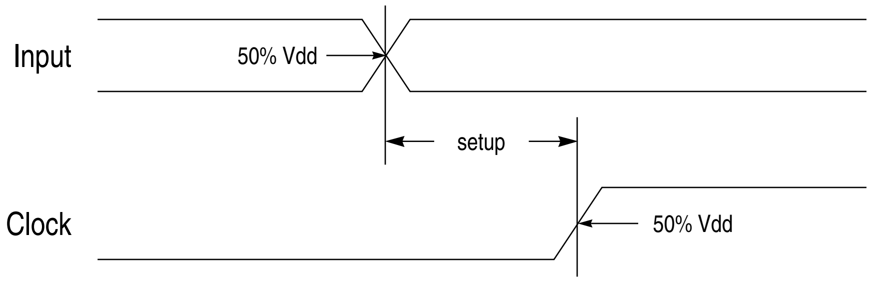
The sequential-cell timing models provided with this library include the effects of input-transition time and data-signal and clock-signal polarity on timing constraints. To simplify calculations, the datasheets specify timing constraint values for 0.023ns data slew and 0.023ns clock slew. Other factors that affect timing constraints include temperature, supply voltage, and process case variations. All cells have been characterized with a fully populated metal2 (0.41um horizontal pitch) and metal3 (0.41um vertical pitch) routing grid across the entire cell layout.

Timing constraints can affect propagation delays. The intrinsic delays given in the datasheets are measured with relaxed timing constraints (longer than necessary setup times, hold times, recovery times, and pulse widths). The use of shorter timing constraint intervals may increase delay. Each cell is considered functional as long as the actual delay does not exceed the delay given in the datasheets by more than 10%.

Setup Time

The setup time for a sequential cell is the minimum length of time the data-input signal must remain stable before the active edge of the clock (or other specified signal) to ensure correct functioning of the cell. The cell is considered functional as long as the delay for the output reaching its expected value does not exceed the reference delay (measured with a large setup time) by more than 10%. Setup-constraint values are measured as the interval between the data signal crossing 50% of Vdd for rising data (or 50% of Vdd for falling data) and the clock signal crossing 50% of Vdd for rising clocks (or 50% of Vdd for falling clocks). For the measurement of setup time, the data input signal is kept stable after the active clock edge for an infinite hold time. Figure 3 illustrates setup time for a positive-edge-triggered sequential cell.

Figure 3. Setup Time

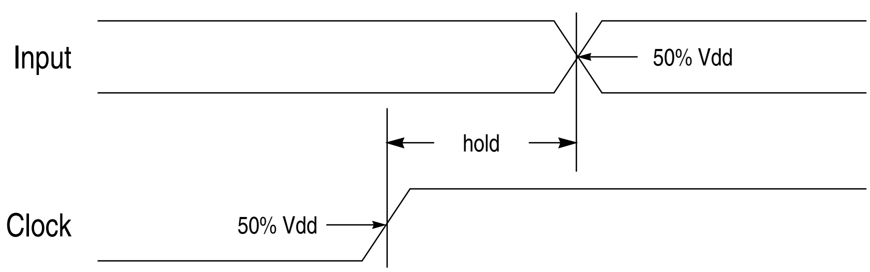


Hold Time

The hold time for a sequential cell is the minimum length of time the data-input signal must remain stable after the active edge of the clock (or other specified signal) to ensure correct functioning of the cell. The cell is considered functional as long as the delay for the output reaching its expected value does not exceed the reference delay (measured with a large hold time) by more than 10%. Hold-constraint values are measured as the interval between the data signal crossing 50% of Vdd for rising data (or 50% of Vdd for falling data) and the clock signal crossing 50% of Vdd for rising clocks (or 50% of Vdd for falling clocks). For the measurement of hold time, the data input signal is held stable before the active clock edge for an infinite setup time. Figure 4 illustrates hold time for a positive-edge-triggered sequential cell.

NOTE: ARM does not incorporate any hold time margins in the Synopsys, TLF, StarDC, or any other timing models. Chip designers should develop a timing methodology to account for chip-level timing inaccuracies inherent to extraction and timing analysis tools.

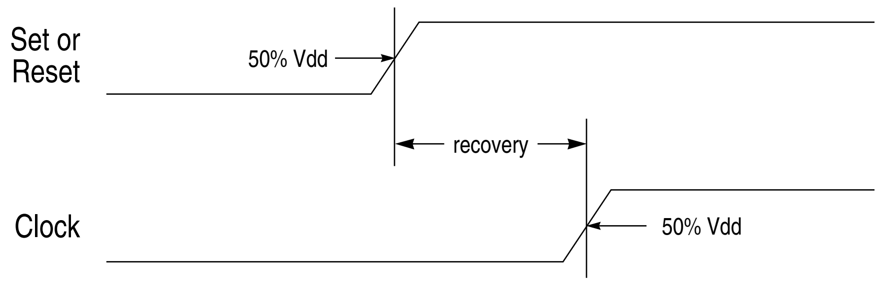
Figure 4. Hold Time



Recovery Time

Recovery time for sequential cells is the minimum length of time that the active-low set or reset signal must remain high before the active edge of the clock to ensure correct functioning of the cell. The cell is considered functional as long as the delay for the output reaching its expected value does not exceed the reference delay (measured with a large recovery time) by more than 10%. Recovery constraint values are measured as the interval between the set or reset signal crossing 50% of Vdd and the clock signal crossing 50% of Vdd for rising clocks (or 50% of Vdd for falling clocks). For the measurement of recovery time, the set or reset signal is held stable after the active clock edge for an infinite hold time. Figure 5 illustrates recovery time.

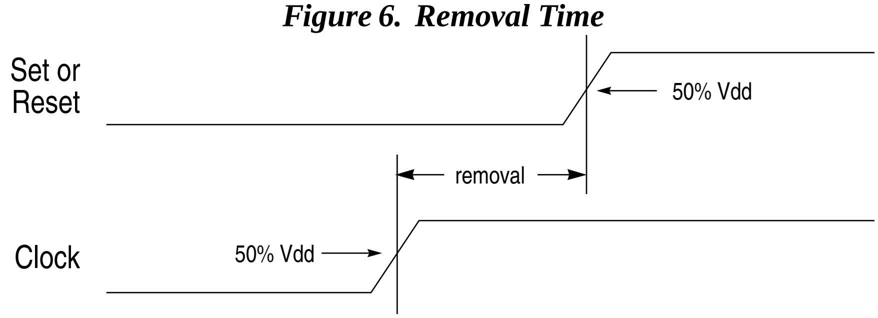
Figure 5. Recovery Time



Removal Time

Removal time for sequential cells is the minimum length of time that the active low set or reset signal must remain low after the active edge of the clock to ensure correct functioning of the cell. The cell is considered functional as long as the active clock edge does not latch in a new data value from that programmed by the asynchronous set or reset signal. Removal constraint values are measured as the interval between the set or reset signal crossing 50% of Vdd and the clock signal crossing 50% of Vdd for rising clocks (or 50% of Vdd for falling clocks). For the measurement of removal time, the set or reset signal is held stable before the active clock edge for an infinite setup time. Figure 6 illustrates removal time.

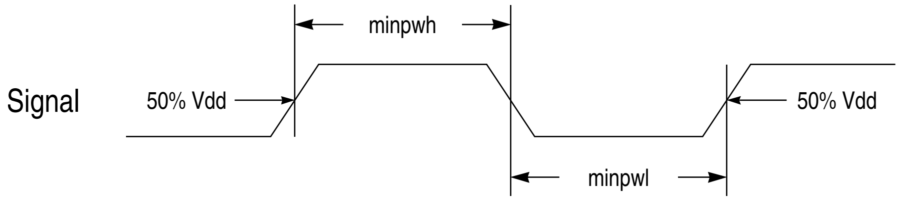
Figure 6. Removal Time



Minimum Pulse Width

Minimum pulse width is the minimum length of time between the leading and trailing edges of a pulse waveform. Minimum pulse width high (minpwh) is measured as the interval between the rising edge of the signal crossing 50% of Vdd and the falling edge of the signal crossing 50% of Vdd. Minimum pulse width low (minpwl) is measured as the interval between the falling edge of the signal crossing 50% of Vdd and the rising edge of the signal crossing 50% of Vdd. Figure 7 illustrates minimum pulse width.

Figure 7. Minimum Pulse Width



Minimum pulse width is defined as 0.997375ns for all set/reset pins (SN, RN) and 0.997375ns for all clock pins (G, GN, CK, CKN). These are the largest minimum pulse widths measured from all the cells in the library. An input pulse of shorter duration will produce unpredictable results.

Electromigration

min_period Property

All sequential cells in the .lib file have this clock pin property set:

min_period : 1.000000;

This property has the effect of limiting a design's clock frequency to 1.0GHz. Commonly used design flows may not be able to support the required accuracy for designs targeted for clock frequencies higher than this limit. Contact ARM technical support for designs targeting higher clock frequencies.

Electromigration Guideline Compliance

Artisan standard cell libraries are designed to meet foundry electromigration guidelines for normal chip design usage; however, it is the chip designer's responsibility to ensure that electromigration guidelines are met at the chip level with regard to foundry guidelines as well as ARM's guidelines for how the library will be used. The following three Electromigration guidelines must be met in order to ensure safe use of the standard cell library within the electromigration guidelines of the foundry.

1. The width of the Metal1 VDD and VSS power buses in the standard cells has been sized to provide adequate current to the cells. Vertical power straps must be placed with sufficient frequency to provide adequate current distribution to the standard cell power buses. For more details, see the section entitled Power-Rail Strapping in the standard cell user guide.
2. The output pin metal for each standard cell has been sized to accommodate multiple vias necessary (for worst case electromigration conditions) to meet via electromigration guidelines, although oversized Metal1 output pins do not necessarily require multiple vias. The number of vias required to meet electromigration guidelines is design dependent, and the chip designer must use an appropriate number of vias and wire width when routing from an output pin.
3. The internal layouts of the standard cells have been designed and verified to comply with the manufacturer's electromigration guidelines under normal usage. Normal usage is defined as follows:

- The current required by the cell does not exceed the maximum current that can be supplied by the Metal1 power buses.
- The output transition times (measured using 10% and 90% thresholds), for a cell outside the clock tree network, must be no greater than 20% of the total cycle time, or must be no greater than than 10% of the cycle time for any of the output pins of that particular cell. Limiting the output transition time has the effect of limiting the load driven by the cell which will reduce the cell's current draw, making it comply with electromigration guidelines. Ratios larger than 20% are not appropriate for commonly used design flows and are unlikely to be encountered in normal designs.
- For a cell in the clock tree network, transition times must not exceed 10% of the total cycle time for that cell.

Power Dissipation

The Metro Standard Cell Library is designed to dissipate only AC power, except for the small reverse-bias leakage currents which are normally present in all CMOS circuits.

The power dissipation internal to a cell when a given input switches is primarily dependent upon the cell design itself. The power dissipation of a complete design, or part of a design, using cells from the library is primarily a function of the switching frequency of the design's internal nets. These nets include the inputs and outputs of each cell and the capacitive load associated with the outputs of each cell.

The Metro library datasheets contain both an AC power table which documents the internal energy consumption of each cell and a pin capacitance table which gives input-pin capacitance data used to compute output loading. This information, coupled with design-specific information, can be used to estimate the total power dissipation of a cell within a design.

The AC power tables specify the amount of energy consumed within a cell (uW/MHz) when the corresponding pin changes state at 25°C, 1.2V, and typical process. The energy data in the tables were measured for an input slew of 0.023ns and no loading at the outputs.

For combinatorial cells, energy values are provided for only input pins. The energy value for each input pin is the average of energies associated with the input transitions which result in an output transition.

For sequential cells, the energy associated with each input pin is the average energy of those input transitions which *do not* result in an output transition. The energy associated with the output pin of a sequential cell is the average energy of all cases where an output transition is the result of a clock-input transition, minus the energy associated with the clock input pin. In the event that a sequential cell has multiple outputs, all output energy data will be associated with only one output pin.

Power Calculation

Power dissipation is dependent upon the power-supply voltage, frequency of operation, internal capacitance, and output load. The power dissipated by each cell is:

$$P_{avg} = \sum_{n=1}^{x}(E_{in} * f_{in}) + \sum_{n=1}^{y}(C_{on} * Vdd^2 * \frac{1}{2}f_{on}) + E_{os} * f_{01}$$

where:

P_{avg} = average power (uW);

x = number of input pins;

E_{in} = energy associated with the nth input pin (uW/MHz);

f_{in} = frequency at which the nth input pin changes state during the normal operation of the design (MHz);

y = number of output pins;

C_{on} = external capacitive loading on the nth output pin, including the capacitance of each input pin connected to the output driver, plus the route wire capacitance, actual or estimated (pF);

Vdd = operating voltage = 1.2V;

f_{on} = frequency at which the nth output pin changes state during the normal operation of the design (MHz);

E_{os} = energy associated with the output pin for sequential cells only (uW/MHz).

The switching frequency of inputs and outputs of a particular cell in a design can be obtained from a gate-level logic simulator (e.g. Verilog) by applying typical input stimuli and measuring the activity on each node of interest. The total average power for the design can be computed by adding the average power for each cell.

EXAMPLE: Calculating Power for a DFFXL Cell

For this exercise, assume that a DFFXL cell has clock switching at 133MHz (clock frequency = 66.5MHz), input and output pins switching at 20MHz, and an external capacitive loading on the output pin of 0.02pF. Using the AC Power table provided in the **sample** DFF datasheet at the end of the introduction, the power dissipated by the DFFXL can be calculated by using the following equation:

$$P_{avg} = \sum_{n=1}^{x}(E_{in} * f_{in}) + \sum_{n=1}^{y}(C_{on} * Vdd^2 * \frac{1}{2}f_{on}) + E_{os} * f_{01}$$

Given:

$x = 2;$

$E_{i1} = 0.0056$ <uW/MHz;

$E_{i2} = 0.0063$ uW/MHz;

$f_{i1} = 20$ MHz;

$f_{i2} = 133$ MHz;

$y = 2;$

$C_{o1} = 0.02$ pF;

$C_{o2} = 0.02$ pF;

$Vdd = 1.0$V;

$f_{o1} = 20$ MHz;

$f_{o2} = 20$ MHz;

$E_{os} = 0.0060$ uW/MHz,

we have:

$$P_{avg} = \sum_{n=1}^{2}(E_{in} * f_{in}) + \sum_{n=1}^{2}(C_{on} * Vdd^2 * \frac{1}{2}f_{on}) + E_{os} * f_{01}$$

$$P_{avg} = (E_{i1} * f_{i1}) + (E_{i2} * f_{i2}) + (C_{o1} * VDD^2 * \frac{1}{2}f_{o1})$$

$$+ (C_{o2} * VDD^2 * \frac{1}{2}f_{o2}) + (E_{os} * f_{01})$$

$$P_{avg} = (0.0056 * 20) + (0.0063 * 133) + \left(0.02 * 1.0 * \frac{1}{2}(20)\right)$$

$$+\left(0.02 * 1.0 * \frac{1}{2}(20)\right) + (0.0060 * 20)$$

$$P_{avg} = 1.46 \text{ uW}$$

Power-Rail Strapping

You must determine the required amount of vertical power-rail strapping to satisfy all requirements imposed by the design methodology for a given design. Power-rail strapping should be sized small enough to optimize standard cell height and maximize router efficiency, yet it must be large enough to provide sufficient power to the cells.

The guidelines below provide a rough estimate with many simplifying assumptions. For a given module design, you can estimate the amount of vertical power-rail strapping that is required to fulfill electromigration requirements.

Given:

I_{avg} = total average current for the module, calculated from previous section (mA);

w_{m1} = VSS/VDD metal1 wire width (um), see Physical Specifications;

r = number of rows in module;

d_{m1} = maximum metal1 current density allowed for the process (mA/um);

d_{m2} = maximum metal2 current density allowed for the process (mA/um);

I_{m1} = maximum current that can be supported by all horizontal metal1 wires (mA);

I_{strap} = total current that must be supported by the vertical metal2 strapping (mA);

w_{m2} = metal2 wire width required for vertical strapping (um);

c = minimum number of metal2 straps;

we have:

$$I_{m1} = w_{m1} * r * 2 * d_{m1},$$

where multiplying by 2 assumes metal1 wires are supplied from both ends;

$$I_{strap} = \frac{(I_{avg} - I_{m1})}{2},$$

where dividing by 2 assumes the metal2 vertical strap wires are supplied from both ends;

$$w_{m2} = \frac{I_{strap}}{d_{m2}},$$

It is recommended that the metal2 wire width, $w_{m2,}$ be divided into c equal portions which are spaced equidistant across the module, where

$$c = \frac{I_{avg}}{I_{m1}}$$, rounded up to the next integer.

The same consideration must be given to the number of vias used to connect the metal1 and metal2 straps.

Adding Routing Channels

In the Metro Standard Cell Library, each cell is designed with a uniform cell height of 2.87um (i.e., 7 tracks tall with 0.41um per track). The cell layouts allow neighboring rows of cells to share common power or ground rails when cells abut each other at the top and bottom edges of the cell bounding box. The sea-of-cells layout with no channels between rows will usually yield the minimum area. In case of extremely congested areas, you may want to separate some rows of cells to increase the number of routing channels within a particular layout region. Because geometries must overlap cell boundaries, a particular spacing between the rows may result in DRC violations for layer spacing. It is recommended that you do not use spacings that cause DRC violations. If these spacings must be used, the DRC violations must be fixed manually by filling the void between the rows with the appropriate layer(s).

Table 4 indicates which DRC violations to expect and how to correct them for a separation between rows of cells.

Table 4. Correcting DRC Violations

Row Separation in Number of Grids	Expected DRC Violations	Action to Correct DRC Violations
0 (Rows Abut)	None	None
1	M1 space < 0.18um	Draw Metal1 layer between rows to merge Metal1 regions above and below row separation
2	NWELL space < 0.62um	Draw NWELL layer between rows to merge NWELL regions above and below row separation
3	NWELL space < 0.62um	Draw NWELL layer between rows to merge NWELL regions above and below row separation
4	None	None
5 or more	None	None

Special Cells

This section discusses special cells in the Metro Standard Cell Library.

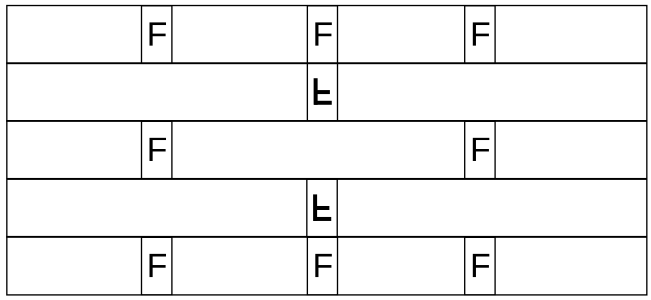
Antenna-Fix Cell

The library contains an antenna-fix cell which must be inserted manually. However, most place and route tools will indicate which nets require the antenna-fix cell. The TSMC antenna effect prevention guideline, "TSMC 0.13UM LOGIC 1P9M SALICIDE 1.0V/2.5V 1.2V/2.5V 1.0V/3.3V 1.2V/3.3V PROCESS Design Rule", specifies a maximum wire length. During place and route, the router may connect wires to the input gates of cells that are longer than the maximum length allowable by the guideline. The antenna cell can be used in this case to add an optional diode on the net close to the input gates which do not meet the guideline. Pin A on the antenna cell connects to a diode, reverse biased to ground. A diode can be added to either P or N.

NWELL and Substrate Tie Cell

The library does not have well or substrate ties inside the cells. You are required to tie the NWELLS to Vdd and the substrate to Vss before place-and-route using the FILLTIE cell. Before place-and-route, pre-place the FILLTIE cell periodically in every placement row. You must place the FILLTIE cell as frequently as the design requires. For example, if the design rules require a well or substrate connection every 20um, then the FILLTIE cell must be pre-placed every 20um. See Figure 8.

Figure 8. Sample Placement of FILLTIE Cells for 20um NWELL and Substrate Tie Design Rule



Note: The letter "F" indicates a FILLTIE cell placed in normal orientation, and the letter "F" flipped upside down indicates a FILLTIE cell placed in MY orientation.

In all rows except for the top and bottom rows, the NWELL and substrate are shared by two adjacent placement rows. This allows you to place the FILLTIE cell only half as frequently as the design rules require. But don't forget to stagger the placement in the adjacent rows by an amount equal to the design row.

Assuming that the rule is every 20um, you will need to place FILLTIE cells every 20um in the top and bottom rows. If you stagger the placement by 20um between adjacent rows, you can place FILLTIE cells every 40um for all rows between the top and bottom rows. This method will allow every row to have well and substrate ties every 20um.

FILL Cells

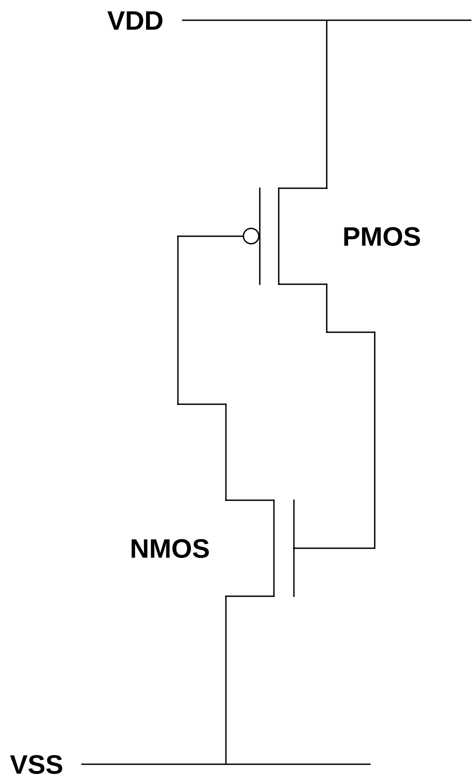
The library contains several FILL cells: FILL1, FILL2, FILL4, FILL8, FILL16, FILL32, FILL64. The number appended to FILL in the cell name denotes the width of the cell in tracks.

During place and route, the FILL cells are used to connect power and ground rails across an area containing no cells. The FILL cells are also used to ensure gaps do not occur between well or implant layers which could cause design rule violations. Using wider cells where appropriate reduces the size of the layout database.

FILLCAP Cells

FILLCAPs function as FILL cells. Inside the FILLCAP, PMOS and NMOS devices form decoupling capacitors between the VDD and VSS rails, reducing ground bounce in the power grids. Figure 9 illustrates the FILLCAP functional schematic.

Figure 9. FILLCAP Functional Schematic



Low-Power (XL) Cells

The library contains a wide variety of cells, denoted by an "XL" suffix in the cell name, that are designed specifically for low-power applications. Input capacitance for the XL cells is much lower than that for corresponding X1 (1x drive strength) cells. Because XL cells have been designed for the sole purpose of reducing power consumption, output rise and fall times for these cells may not be equal, and due to the low-drive capability of the XL cells, these cells are not intended for use in critical timing paths, or to drive heavily loaded nets.

TIEHI/LO Cells

The library contains a TIEHI cell and a TIELO cell. The outputs of the TIEHI and TIELO cells are driven through diffusion to provide isolation from the power and ground rails for better ESD protection. The standard cell abstract methodology assumes that the TIEHI and TIELO cells are used to tie off any inputs to power and ground. If these cells are not used and the router is allowed to drop vias on the power rail, DRC errors or shorts may result.

Delay Cells

The library contains delay cells that have the same width. These delay cells allow you to adjust a given delay path with a simple cell substitution after place and route.

Reading the Standard Cell Datasheet

Please refer to the **sample** datasheet for DFF at the end of the introduction for the arrangement of each of the following datasheet sections. Datasheet titles reference standard Artisan cell names. Cell names for your specific library are reflected in the cell size table on each datasheet.

NOTE: This datasheet contains **sample** characterization values.

1. Base Cell Name

The cell name field contains the cell name. The datasheets are presented alphabetically by cell name. The cell name presented here is the base cell name. The Cell Size table displays cell names for your specific library.

2. Cell Description

The cell description gives the function of the cell. When applicable, the equation(s) for the output pins are provided.

3. Functions

The function table gives all possible combinations of input and output signals for the cell. Table 5 defines the symbols used in datasheet function tables.

Table 5. Functions Key

Symbol	Description
0	Logic Low
1	Logic High
‾_	High to Low Transition
_/‾	Low to High Transition
X	Don't Care
IL	Illegal/Undefined
Z	High Impedance

4. Logic Symbol

The logic symbol is a graphical representation of the cell, similar to the view in the schematic editor when the cell is instantiated. The symbol shows the name and location of the input and output pins.

5. Cell Size

This cell size table gives the height and width (μm) for each drive strength of the cell.

6. Functional Schematic

The functional schematic provides a functional representation of the cell.

7. Drive Strength

The drive strength of each cell is indicated by an "X" followed by the unit strength.

8. AC Power

The AC power table shows the amount of energy consumed (μW/MHz) within the cell when the corresponding pin changes state. The energy data for each drive strength of the cell in the **sample** DFF datasheet are calculated at 25°C, 1.0V, typical process, input slew of 0.023ns, and no external load at the output pins.

9. Delay

The delay table shows the intrinsic delay (ns) which is the delay through the cell when there is no load on the output, and the load multiplier for load dependent delay, K_{load} (ns/pF). The delays and load multiplier for each drive strength of the cell in the sample DFF datasheet are calculated at 25°C, 1.0V, typical process, and input slew of 0.023ns.

10. Timing Constraints

The timing constraints table in the **sample** DFF datasheet shows the timing conditions (ns) required at 25°C, 1.0V, and typical process to maintain proper functionality. Setup constraint values are measured for 0.023ns data slew and 0.023ns clock slew. Hold constraint values are measured for 0.023ns data slew and 0.023ns clock slew. Minimum pulse width is defined to be 0.997375ns for all set/reset pins and 0.997375ns for all clock pins. These are the largest minimum pulse widths measured from all the cells in the library.

11. Pin Capacitance

The pin capacitance table shows the typical loading at the input pins of the cell (pF) for each drive strength of the cell.

▼ *This datasheet contains sample characterization values.* ▼

1

DFF

Process Technology:
CustomerName & Code

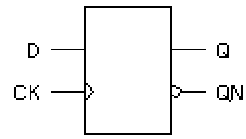
4

Cell Description

The DFF cell is a positive-edge triggered, static D-type flip-flop.

2

Logic Symbol



5

Function Table

3

D	CK	Q[n+1]	QN[n+1]
0	_/‾	0	1
1	_/‾	1	0
x	‾_	Q[n]	QN[n]

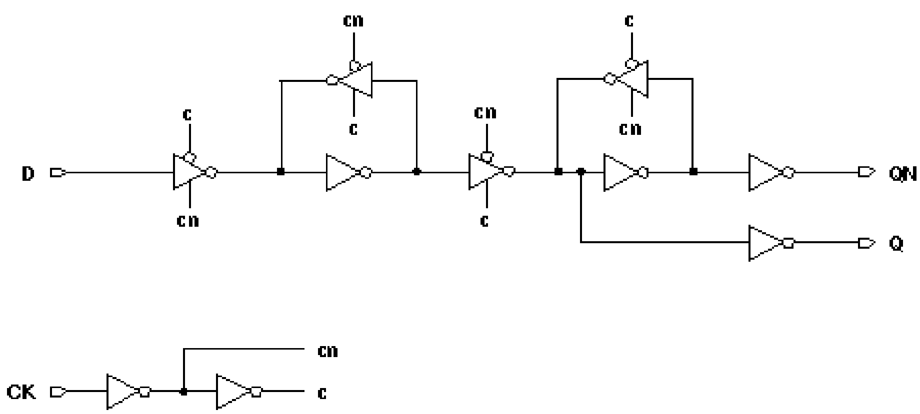
Cell Size

Drive Strength	Height (µm)	Width (µm)
DFFXL	1.000	1.000
DFFX1	2.000	2.000
DFFX2	3.000	3.000
DFFX4	4.000	4.000

6

7

Functional Schematic



ARM Sample Standard Cell Library Databook, p. **84**

▼ *This datasheet contains sample characterization values.* ▼

Process Technology:
CustomerName & Code

DFF

8 **AC Power**

7

Pin	Power (µW/MHz)			
	XL	X1	X2	X4
D	1.000	1.000	1.000	1.000
CK	2.000	2.000	2.000	2.000
Q	3.000	3.000	3.000	3.000

Pin Capacitance 11

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	1.000	1.000	1.000	1.000
CK	2.000	2.000	2.000	2.000

9 **Delays at TypTemp°C, TypVoltV, Typical Process**

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK → Q↑	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
CK → Q↓	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
CK → QN↑	3.000	3.000	3.000	3.000	3.000	3.000	3.000	3.000
CK → QN↓	4.000	4.000	4.000	4.000	4.000	4.000	4.000	4.000

10 **Timing Constraints at TypTemp°C, TypVoltV, Typical Process**

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup↑ → CK	1.000	1.000	1.000	1.000
	setup↓ → CK	2.000	2.000	2.000	2.000
	hold↑ → CK	3.000	3.000	3.000	3.000
	hold↓ → CK	4.000	4.000	4.000	4.000
CK	minpwh	5.000	5.000	5.000	5.000
	minpwh	6.000	6.000	6.000	6.000

ARM Sample Standard Cell Library Databook, p. **85**
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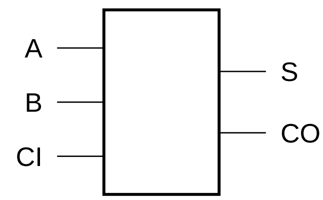
Cell Description

The ADDF cell provides the arithmetic sum (S) and carry out (CO) of two operands (A,B) with carry in (CI). The two outputs (S,CO) are represented by the logic equations:

$$S = (A \oplus B \oplus CI)$$

$$CO = (A \oplus B) \bullet CI + (A \bullet B)$$

Logic Symbol



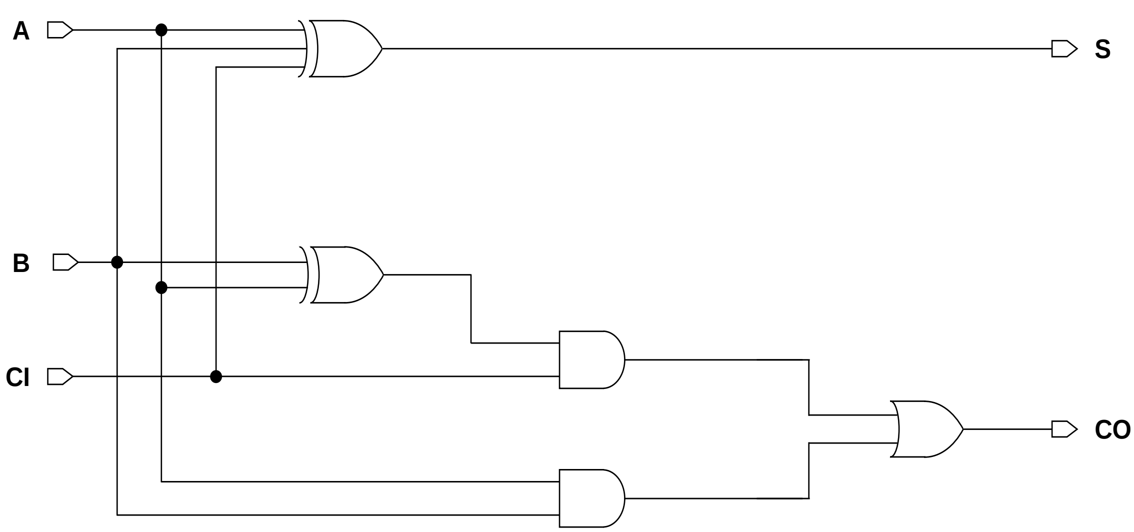
Function Table

CI	A	B	S	CO
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Cell Size

Drive Strength	Height (um)	Width (um)
ADDFXLM	2.87	10.66
ADDFX1M	2.87	10.66
ADDFX2M	2.87	10.66
ADDFX4M	2.87	11.48
ADDFX8M	2.87	13.53

Functional Schematic



AC Power

Pin	Power (uW/MHz)				
	XL	**X1**	**X2**	**X4**	**X8**
A	0.0316	0.0338	0.0375	0.0509	0.0934
B	0.0409	0.0446	0.0505	0.0673	0.1039
CI	0.0173	0.0197	0.0236	0.0367	0.0763

Pin Capacitance

Pin	Capacitance (pF)				
	XL	**X1**	**X2**	**X4**	**X8**
A	0.0038	0.0038	0.0037	0.0038	0.0038
B	0.0036	0.0036	0.0036	0.0036	0.0036
CI	0.0040	0.0040	0.0040	0.0040	0.0040

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				
	XL	**X1**	**X2**	**X4**	**X8**
A → S ↑	0.1898	0.1955	0.2064	0.2349	0.3042
A → S ↓	0.2652	0.2889	0.3073	0.3500	0.4372
B → S ↑	0.2248	0.2294	0.2391	0.2617	0.3144
B → S ↓	0.3039	0.3276	0.3459	0.3863	0.4751
CI → S ↑	0.1497	0.1580	0.1716	0.2123	0.2937
CI → S ↓	0.1070	0.1241	0.1414	0.1760	0.2549
A → CO ↑	0.2612	0.2722	0.2858	0.3325	0.4270
A → CO ↓	0.2304	0.2486	0.2648	0.2990	0.3640
B → CO ↑	0.2999	0.3108	0.3244	0.3688	0.4648
B → CO ↓	0.2617	0.2770	0.2921	0.3251	0.3855
CI → CO ↑	0.1101	0.1175	0.1309	0.1717	0.2582
CI → CO ↓	0.1534	0.1720	0.1868	0.2214	0.2927

Delays at 25°C,1.2V, Typical Process (Cont'd.)

Description	K_{load} (ns/pF)				
	XL	**X1**	**X2**	**X4**	**X8**
A → S ↑	8.9452	5.3625	3.6122	1.8708	0.9751
A → S ↓	6.6478	3.8644	2.1813	1.1368	0.6375
B → S ↑	8.9781	5.3906	3.6259	1.8772	0.9815
B → S ↓	6.6491	3.8660	2.1824	1.1374	0.6379
CI → S ↑	8.8906	5.3494	3.6080	1.8716	0.9787
CI → S ↓	6.4860	3.8076	2.1718	1.1300	0.6373
A → CO ↑	8.7207	5.2774	3.5200	1.8478	0.9632
A → CO ↓	5.8704	3.4740	1.9938	1.0495	0.5991
B → CO ↑	8.7210	5.2774	3.5200	1.8479	0.9633
B → CO ↓	5.5666	3.2830	1.8762	0.9618	0.5354
CI → CO ↑	8.8330	5.3307	3.5478	1.8641	0.9746
CI → CO ↓	6.1187	3.5854	2.0449	1.0760	0.6104

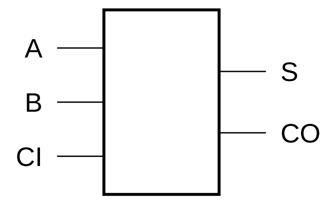
Cell Description

The ADDFH cell is a high-speed cell providing the arithmetic sum (S) and carry out (CO) of two operands (A,B) with carry in (CI). The two outputs (S,CO) are represented by the logic equations:

$S = (A \oplus B \oplus CI)$

$CO = (A \oplus B) \bullet CI + (A \bullet B)$

Logic Symbol



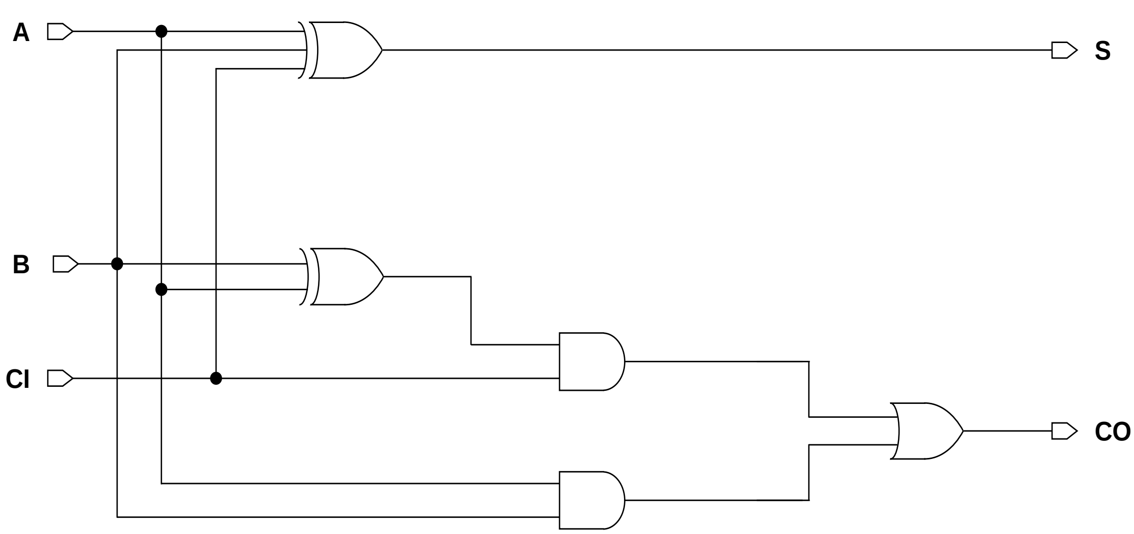
Function Table

CI	A	B	S	CO
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Cell Size

Drive Strength	Height (um)	Width (um)
ADDFHXLM	2.87	11.48
ADDFHX1M	2.87	11.48
ADDFHX2M	2.87	18.45
ADDFHX4M	2.87	22.14
ADDFHX8M	2.87	27.47

Functional Schematic



AC Power

Pin	Power (uW/MHz)				
	XL	X1	X2	X4	X8
A	0.0342	0.0448	0.0735	0.0949	0.1460
B	0.0293	0.0389	0.0606	0.0807	0.0945
CI	0.0171	0.0232	0.0357	0.0517	0.0793

Pin Capacitance

Pin	Capacitance (pF)				
	XL	X1	X2	X4	X8
A	0.0032	0.0046	0.0071	0.0128	0.0255
B	0.0065	0.0094	0.0143	0.0218	0.0172
CI	0.0019	0.0029	0.0044	0.0079	0.0163

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				
	XL	X1	X2	X4	X8
A → S ↑	0.2370	0.1994	0.2169	0.1962	0.1936
A → S ↓	0.2526	0.2253	0.2248	0.2023	0.2242
B → S ↑	0.1823	0.1580	0.1547	0.1477	0.1731
B → S ↓	0.2179	0.1949	0.1894	0.1615	0.2160
CI → S ↑	0.1968	0.1687	0.1645	0.1323	0.1592
CI → S ↓	0.1919	0.1756	0.1605	0.1278	0.2013
A → CO ↑	0.2433	0.2024	0.2210	0.2067	0.1818
A → CO ↓	0.2548	0.2274	0.2239	0.2060	0.2121
B → CO ↑	0.1609	0.1333	0.1360	0.1235	0.1303
B → CO ↓	0.2062	0.1852	0.1766	0.1646	0.1921
CI → CO ↑	0.0968	0.0856	0.0820	0.0816	0.0903
CI → CO ↓	0.1522	0.1396	0.1305	0.1269	0.1328

Delays at 25°C,1.2V, Typical Process (Cont'd.)

Description	K_{load} (ns/pF)				
	XL	**X1**	**X2**	**X4**	**X8**
A → S ↑	8.8411	5.2973	3.7672	1.9127	1.1245
A → S ↓	6.1158	3.4194	1.8754	0.9088	0.5181
B → S ↑	8.8945	5.3049	3.7779	1.9166	1.1265
B → S ↓	6.1223	3.4238	1.8776	0.9077	0.5180
CI → S ↑	8.8865	5.3097	3.7740	1.9167	1.1263
CI → S ↓	6.1954	3.4476	1.8859	0.9034	0.5187
A → CO ↑	8.8658	5.3983	3.5159	1.9254	0.9630
A → CO ↓	5.7888	3.2712	1.6426	0.9686	0.4801
B → CO ↑	8.8821	5.4023	3.5206	1.9255	0.9632
B → CO ↓	5.7981	3.2719	1.5867	0.9678	0.4827
CI → CO ↑	8.9337	5.4204	3.5217	1.9270	0.9634
CI → CO ↓	6.4560	3.5088	1.7064	0.9829	0.5010

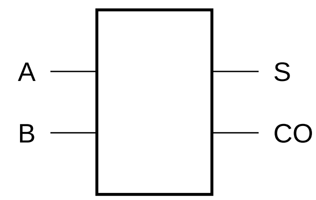
Cell Description

The ADDH cell provides the arithmetic sum (S) and carry out (CO) of two operands (A,B). The two outputs (S,CO) are represented by the logic equations:

$S = (\overline{A} \bullet B) + (A \bullet \overline{B})$

$CO = A \bullet B$

Logic Symbol



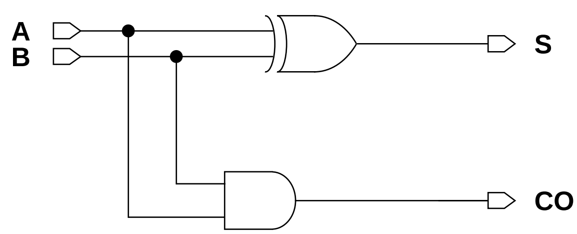
Function Table

A	B	S	CO
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Cell Size

Drive Strength	Height (um)	Width (um)
ADDHX1M	2.87	5.33
ADDHX2M	2.87	6.97
ADDHX4M	2.87	10.25
ADDHX8M	2.87	18.04

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	X1	X2	X4	X8
A	0.0194	0.0335	0.0632	0.1203
B	0.0118	0.0177	0.0304	0.0581

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
A	0.0038	0.0095	0.0188	0.0362
B	0.0036	0.0046	0.0065	0.0124

Delays at 25°C,1.2V, Typical Process

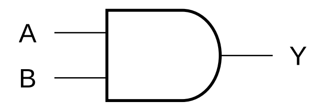
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
A → S ↑	0.0720	0.0678	0.0647	0.0629	7.1467	4.6628	2.3987	1.2555
A → S ↓	0.0936	0.0719	0.0688	0.0661	4.6868	2.5326	1.2957	0.6642
B → S ↑	0.0507	0.0525	0.0512	0.0526	7.1901	4.6678	2.3936	1.2459
B → S ↓	0.0635	0.0754	0.0685	0.0675	4.5236	2.4399	1.2530	0.6491
A → CO ↑	0.0655	0.0622	0.0625	0.0685	5.2607	3.4161	1.7823	0.9546
A → CO ↓	0.1052	0.0949	0.0893	0.0920	3.3160	1.7842	0.8534	0.4233
B → CO ↑	0.0664	0.0624	0.0603	0.0669	5.2622	3.4163	1.7816	0.9566
B → CO ↓	0.1015	0.0872	0.0785	0.0841	3.3075	1.7696	0.8462	0.4201

Cell Description

The AND2 cell provides the logical AND of two inputs (A,B). The output (Y) is represented by the logic equation:

$Y = (A \bullet B)$

Logic Symbol



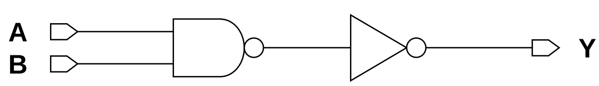
Function Table

A	B	Y
0	x	0
x	0	0
1	1	1

Cell Size

Drive Strength	Height (um)	Width (um)
AND2X1M	2.87	2.05
AND2X2M	2.87	2.05
AND2X4M	2.87	3.28
AND2X6M	2.87	4.10
AND2X8M	2.87	4.51
AND2X12M	2.87	6.56

Functional Schematic



AC Power

Pin	Power (uW/MHz)					
	X1	**X2**	**X4**	**X6**	**X8**	**X12**
A	0.0061	0.0084	0.0144	0.0206	0.0265	0.0394
B	0.0070	0.0098	0.0173	0.0238	0.0306	0.0469

Pin Capacitance

Pin	Capacitance (pF)					
	X1	**X2**	**X4**	**X6**	**X8**	**X12**
A	0.0014	0.0020	0.0036	0.0042	0.0053	0.0078
B	0.0016	0.0021	0.0042	0.0047	0.0057	0.0078

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)					
	X1	**X2**	**X4**	**X6**	**X8**	**X12**
A → Y ↑	0.0639	0.0624	0.0550	0.0611	0.0612	0.0622
A → Y ↓	0.0927	0.0855	0.0795	0.0887	0.0878	0.0880
B → Y ↑	0.0684	0.0665	0.0593	0.0651	0.0651	0.0667
B → Y ↓	0.1076	0.0997	0.0898	0.0982	0.0966	0.1027

Delays at 25°C,1.2V, Typical Process (Cont'd.)

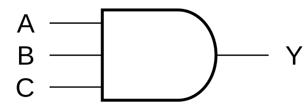
Description	K_{load} (ns/pF)					
	X1	**X2**	**X4**	**X6**	**X8**	**X12**
A → Y ↑	5.2908	3.5136	1.8157	1.2393	0.9390	0.6387
A → Y ↓	3.1632	1.7002	0.8403	0.5590	0.4192	0.2869
B → Y ↑	5.2907	3.5136	1.8155	1.2392	0.9389	0.6386
B → Y ↓	3.1903	1.7139	0.8426	0.5604	0.4200	0.2893

Cell Description

The AND3 cell provides the logical AND of three inputs (A,B,C). The output (Y) is represented by the logic equation:

$$Y = (A \bullet B \bullet C)$$

Logic Symbol



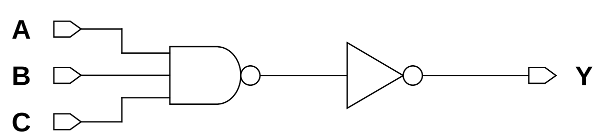
Function Table

A	B	C	Y
0	x	x	0
x	0	x	0
x	x	0	0
1	1	1	1

Cell Size

Drive Strength	Height (um)	Width (um)
AND3XLM	2.87	2.46
AND3X1M	2.87	2.46
AND3X2M	2.87	2.46
AND3X4M	2.87	2.87
AND3X6M	2.87	5.33
AND3X8M	2.87	6.97
AND3X12M	2.87	7.79

Functional Schematic



AC Power

Pin	Power (uW/MHz)						
	XL	X1	X2	X4	X6	X8	X12
A	0.0051	0.0064	0.0093	0.0152	0.0230	0.0323	0.0435
B	0.0057	0.0074	0.0110	0.0177	0.0281	0.0393	0.0518
C	0.0065	0.0086	0.0126	0.0201	0.0325	0.0447	0.0581

Pin Capacitance

Pin	Capacitance (pF)						
	XL	X1	X2	X4	X6	X8	X12
A	0.0014	0.0016	0.0021	0.0033	0.0055	0.0081	0.0092
B	0.0013	0.0015	0.0022	0.0033	0.0058	0.0081	0.0094
C	0.0013	0.0015	0.0022	0.0033	0.0063	0.0081	0.0093

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)						
	XL	X1	X2	X4	X6	X8	X12
A → Y ↑	0.0985	0.0854	0.0803	0.0808	0.0665	0.0694	0.0748
A → Y ↓	0.0962	0.1129	0.1021	0.0857	0.0888	0.0919	0.0995
B → Y ↑	0.1022	0.0902	0.0871	0.0874	0.0733	0.0759	0.0812
B → Y ↓	0.1052	0.1272	0.1185	0.0985	0.1027	0.1068	0.1143
C → Y ↑	0.1071	0.0949	0.0912	0.0912	0.0775	0.0796	0.0848
C → Y ↓	0.1144	0.1404	0.1310	0.1065	0.1137	0.1197	0.1271

Delays at 25°C,1.2V, Typical Process (Cont'd.)

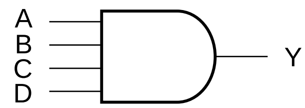
Description	K_{load} (ns/pF)						
	XL	X1	X2	X4	X6	X8	X12
A → Y ↑	8.8590	5.2108	3.5498	1.8656	1.2435	0.9399	0.6590
A → Y ↓	5.4389	3.2445	1.7352	0.8382	0.5574	0.4183	0.2961
B → Y ↑	8.8594	5.2113	3.5492	1.8655	1.2436	0.9398	0.6589
B → Y ↓	5.5030	3.2857	1.7531	0.8450	0.5608	0.4219	0.2983
C → Y ↑	8.8571	5.2101	3.5488	1.8654	1.2434	0.9398	0.6588
C → Y ↓	5.5428	3.3109	1.7658	0.8492	0.5638	0.4253	0.3005

Cell Description

The AND4 cell provides the logical AND of four inputs (A,B,C,D). The output (Y) is represented by the logic equation:

$$Y = (A \bullet B \bullet C \bullet D)$$

Logic Symbol



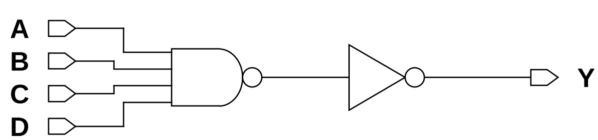
Function Table

A	B	C	D	Y
0	x	x	x	0
x	0	x	x	0
x	x	0	x	0
x	x	x	0	0
1	1	1	1	1

Cell Size

Drive Strength	Height (um)	Width (um)
AND4XLM	2.87	2.87
AND4X1M	2.87	2.87
AND4X2M	2.87	2.87
AND4X4M	2.87	5.33
AND4X6M	2.87	7.38
AND4X8M	2.87	8.61
AND4X12M	2.87	12.71

Functional Schematic



AC Power

Pin	Power (uW/MHz)						
	XL	**X1**	**X2**	**X4**	**X6**	**X8**	**X12**
A	0.0055	0.0070	0.0098	0.0174	0.0230	0.0315	0.0458
B	0.0064	0.0084	0.0118	0.0221	0.0294	0.0390	0.0588
C	0.0071	0.0096	0.0137	0.0257	0.0346	0.0462	0.0699
D	0.0079	0.0110	0.0153	0.0293	0.0402	0.0531	0.0814

Pin Capacitance

Pin	Capacitance (pF)						
	XL	**X1**	**X2**	**X4**	**X6**	**X8**	**X12**
A	0.0013	0.0016	0.0023	0.0044	0.0050	0.0073	0.0106
B	0.0015	0.0018	0.0025	0.0047	0.0048	0.0070	0.0104
C	0.0015	0.0017	0.0023	0.0050	0.0048	0.0072	0.0102
D	0.0014	0.0017	0.0024	0.0052	0.0048	0.0072	0.0104

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)						
	XL	**X1**	**X2**	**X4**	**X6**	**X8**	**X12**
A → Y ↑	0.1179	0.0937	0.0843	0.0760	0.0845	0.0850	0.0810
A → Y ↓	0.0961	0.1158	0.1006	0.0986	0.1065	0.0935	0.0985
B → Y ↑	0.1286	0.1048	0.0949	0.0870	0.0986	0.0968	0.0943
B → Y ↓	0.1054	0.1363	0.1224	0.1167	0.1290	0.1124	0.1216
C → Y ↑	0.1339	0.1103	0.1003	0.0944	0.1065	0.1050	0.1022
C → Y ↓	0.1135	0.1503	0.1361	0.1322	0.1449	0.1276	0.1385
D → Y ↑	0.1381	0.1152	0.1045	0.0982	0.1117	0.1095	0.1073
D → Y ↓	0.1242	0.1638	0.1487	0.1411	0.1566	0.1398	0.1539

Delays at 25°C,1.2V, Typical Process (Cont'd.)

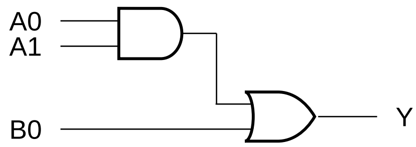
Description	K_{load} (ns/pF)						
	XL	**X1**	**X2**	**X4**	**X6**	**X8**	**X12**
A → Y ↑	8.9544	5.2727	3.5612	1.8351	1.2617	0.9522	0.6439
A → Y ↓	5.4315	3.2485	1.7166	0.8562	0.5764	0.4564	0.2917
B → Y ↑	8.9544	5.2720	3.5607	1.8348	1.2617	0.9518	0.6436
B → Y ↓	5.4556	3.2908	1.7402	0.8633	0.5827	0.4611	0.2953
C → Y ↑	8.9543	5.2711	3.5608	1.8347	1.2618	0.9515	0.6435
C → Y ↓	5.5097	3.3254	1.7561	0.8704	0.5885	0.4653	0.2985
D → Y ↑	8.9514	5.2701	3.5606	1.8344	1.2618	0.9512	0.6433
D → Y ↓	5.5882	3.3577	1.7722	0.8759	0.5938	0.4702	0.3024

Cell Description

The AO21 cell provides the logical OR of one AND group and an additional input. The output (Y) is represented by the logic equation:

$$Y = (A0 \bullet A1) + B0$$

Logic Symbol



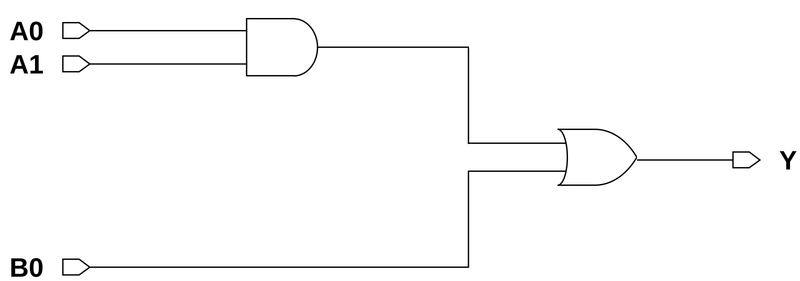
Function Table

A0	A1	B0	Y
0	x	0	0
x	0	0	0
x	x	1	1
1	1	x	1

Cell Size

Drive Strength	Height (um)	Width (um)
AO21XLM	2.87	2.46
AO21X1M	2.87	2.87
AO21X2M	2.87	2.87
AO21X4M	2.87	3.69
AO21X8M	2.87	6.15

Functional Schematic



AC Power

Pin	Power (uW/MHz)				
	XL	X1	X2	X4	X8
A0	0.0064	0.0074	0.0102	0.0182	0.0355
A1	0.0069	0.0082	0.0115	0.0206	0.0406
B0	0.0062	0.0074	0.0102	0.0186	0.0363

Pin Capacitance

Pin	Capacitance (pF)				
	XL	X1	X2	X4	X8
A0	0.0014	0.0016	0.0022	0.0035	0.0066
A1	0.0012	0.0015	0.0021	0.0033	0.0069
B0	0.0015	0.0015	0.0021	0.0034	0.0064

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				
	XL	X1	X2	X4	X8
A0 → Y ↑	0.0773	0.0718	0.0718	0.0696	0.0665
A0 → Y ↓	0.1548	0.1491	0.1362	0.1352	0.1333
A1 → Y ↑	0.0795	0.0743	0.0748	0.0728	0.0700
A1 → Y ↓	0.1675	0.1615	0.1493	0.1480	0.1477
B0 → Y ↑	0.0551	0.0532	0.0537	0.0511	0.0493
B0 → Y ↓	0.1460	0.1415	0.1293	0.1299	0.1274

Delays at 25°C,1.2V, Typical Process (Cont'd.)

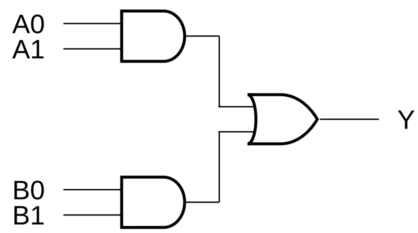
Description	K_{load} (ns/pF)				
	XL	X1	X2	X4	X8
A0 → Y ↑	9.2405	5.0588	3.4460	1.7891	0.9250
A0 → Y ↓	5.7925	3.3541	1.7820	0.8758	0.4368
A1 → Y ↑	9.2410	5.0593	3.4463	1.7892	0.9250
A1 → Y ↓	5.8528	3.3758	1.7917	0.8802	0.4392
B0 → Y ↑	9.1546	5.0186	3.4288	1.7811	0.9211
B0 → Y ↓	5.8539	3.3761	1.7918	0.8803	0.4392

Cell Description

The AO22 cell provides the logical OR of two AND groups. The output (Y) is represented by the logic equation:

$Y = (A0 \bullet A1) + (B0 \bullet B1)$

Logic Symbol



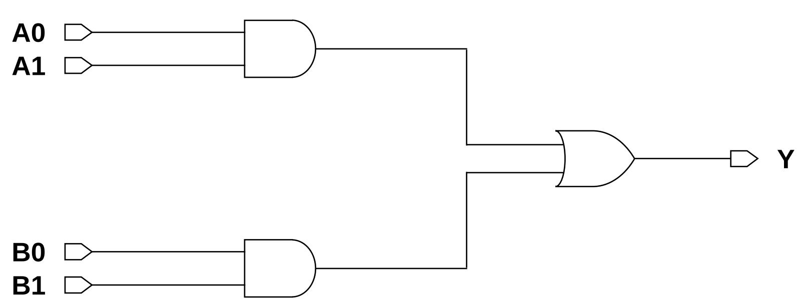
Function Table

A0	A1	B0	B1	Y
0	x	0	x	0
0	x	x	0	0
x	0	0	x	0
x	0	x	0	0
x	x	1	1	1
1	1	x	x	1

Cell Size

Drive Strength	Height (um)	Width (um)
AO22XLM	2.87	3.28
AO22X1M	2.87	3.28
AO22X2M	2.87	3.69
AO22X4M	2.87	4.10
AO22X8M	2.87	6.97

Functional Schematic



AC Power

Pin	Power (uW/MHz)				
	XL	**X1**	**X2**	**X4**	**X8**
A0	0.0063	0.0084	0.0116	0.0199	0.0387
A1	0.0069	0.0092	0.0130	0.0223	0.0438
B0	0.0073	0.0093	0.0130	0.0224	0.0435
B1	0.0080	0.0102	0.0143	0.0248	0.0486

Pin Capacitance

Pin	Capacitance (pF)				
	XL	**X1**	**X2**	**X4**	**X8**
A0	0.0014	0.0016	0.0022	0.0035	0.0068
A1	0.0014	0.0016	0.0023	0.0035	0.0071
B0	0.0014	0.0015	0.0020	0.0035	0.0066
B1	0.0013	0.0015	0.0020	0.0033	0.0069

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				
	XL	**X1**	**X2**	**X4**	**X8**
A0 → Y ↑	0.0732	0.0752	0.0709	0.0675	0.0666
A0 → Y ↓	0.1508	0.1930	0.1606	0.1396	0.1409
A1 → Y ↑	0.0756	0.0783	0.0746	0.0710	0.0703
A1 → Y ↓	0.1613	0.2080	0.1766	0.1529	0.1556
B0 → Y ↑	0.0855	0.0853	0.0824	0.0797	0.0780
B0 → Y ↓	0.1881	0.2272	0.1919	0.1671	0.1664
B1 → Y ↑	0.0874	0.0880	0.0855	0.0829	0.0815
B1 → Y ↓	0.1974	0.2415	0.2043	0.1780	0.1801

Delays at 25°C,1.2V, Typical Process (Cont'd.)

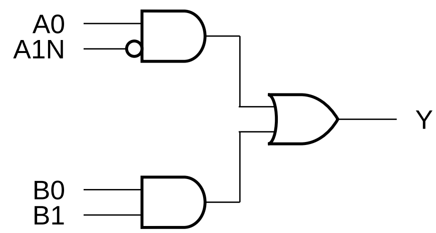
Description	K_{load} (ns/pF)				
	XL	**X1**	**X2**	**X4**	**X8**
A0 → Y ↑	8.7062	5.3186	3.4505	1.7895	0.9382
A0 → Y ↓	5.9461	3.6612	1.8555	0.8952	0.4486
A1 → Y ↑	8.7066	5.3185	3.4508	1.7893	0.9382
A1 → Y ↓	6.0040	3.6898	1.8666	0.8997	0.4510
B0 → Y ↑	8.7844	5.3557	3.4668	1.7961	0.9411
B0 → Y ↓	5.9519	3.6614	1.8582	0.8969	0.4486
B1 → Y ↑	8.7855	5.3564	3.4671	1.7963	0.9414
B1 → Y ↓	6.0085	3.6918	1.8663	0.8997	0.4510

Cell Description

The AO2B2 cell provides the logical OR of two AND groups consisting of two inputs each: (A0,A1N) and (B0,B1). The output (Y) is represented by the logic equation:

$$Y = (A0 \bullet \overline{A1N}) + (B0 \bullet B1)$$

Logic Symbol



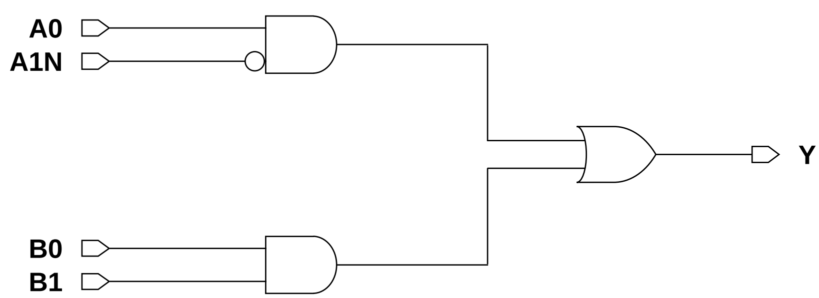
Function Table

A0	A1N	B0	B1	Y
1	0	x	x	1
x	x	1	1	1
0	x	0	x	0
x	1	0	x	0
0	x	x	0	0
x	1	x	0	0

Cell Size

Drive Strength	Height (um)	Width (um)
AO2B2XLM	2.87	4.10
AO2B2X1M	2.87	4.10
AO2B2X2M	2.87	4.10
AO2B2X4M	2.87	4.51

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	XL	X1	X2	X4
A0	0.0067	0.0085	0.0115	0.0196
A1N	0.0081	0.0102	0.0135	0.0228
B0	0.0082	0.0100	0.0135	0.0227
B1	0.0089	0.0109	0.0148	0.0251

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0010	0.0010	0.0014	0.0022
A1N	0.0015	0.0014	0.0014	0.0018
B0	0.0014	0.0015	0.0022	0.0034
B1	0.0014	0.0015	0.0021	0.0034

Delays at 25°C,1.2V, Typical Process

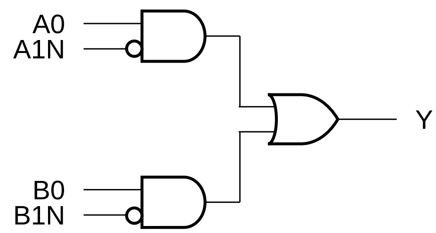
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 → Y ↑	0.0799	0.0767	0.0744	0.0704	8.6870	5.1588	3.4747	1.8016
A0 → Y ↓	0.1657	0.2008	0.1660	0.1425	6.1660	3.7020	1.9156	0.9197
A1N → Y ↑	0.1360	0.1362	0.1435	0.1386	8.7040	5.1653	3.4759	1.8021
A1N → Y ↓	0.2026	0.2431	0.2077	0.1813	6.2385	3.7373	1.9276	0.9241
B0 → Y ↑	0.0916	0.0861	0.0852	0.0814	8.7568	5.1925	3.4880	1.8075
B0 → Y ↓	0.2046	0.2397	0.1989	0.1710	6.1727	3.7067	1.9187	0.9211
B1 → Y ↑	0.0939	0.0888	0.0880	0.0848	8.7566	5.1927	3.4886	1.8072
B1 → Y ↓	0.2149	0.2540	0.2110	0.1822	6.2293	3.7369	1.9275	0.9240

Cell Description

The AO2B2B cell provides the logical OR of two AND groups consisting of two inputs each: (A0,A1N) and (B0,B1N). The output (Y) is represented by the logic equation:

$$Y = (A0 \bullet \overline{A1N}) + (B0 \bullet \overline{B1N})$$

Logic Symbol



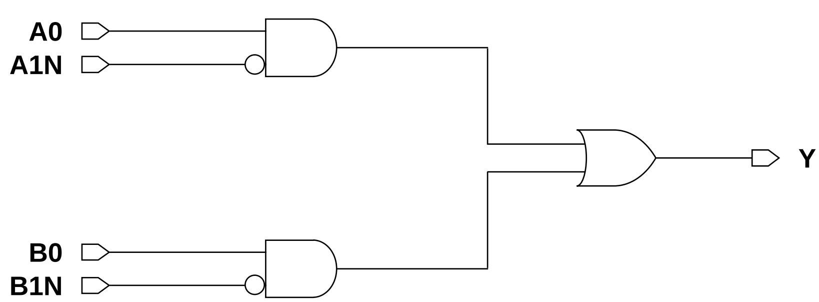
Function Table

A0	A1N	B0	B1N	Y
0	x	0	x	0
0	x	x	1	0
x	1	0	x	0
x	1	x	1	0
x	x	1	0	1
1	0	x	x	1

Cell Size

Drive Strength	Height (um)	Width (um)
AO2B2BXLM	2.87	4.92
AO2B2BX1M	2.87	4.92
AO2B2BX2M	2.87	4.92
AO2B2BX4M	2.87	5.74

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	XL	X1	X2	X4
A0	0.0066	0.0085	0.0115	0.0197
A1N	0.0080	0.0102	0.0134	0.0229
B0	0.0081	0.0101	0.0135	0.0227
B1N	0.0092	0.0109	0.0148	0.0252

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0015	0.0016	0.0022	0.0036
A1N	0.0014	0.0014	0.0014	0.0019
B0	0.0014	0.0015	0.0021	0.0034
B1N	0.0016	0.0016	0.0015	0.0019

Delays at 25°C,1.2V, Typical Process

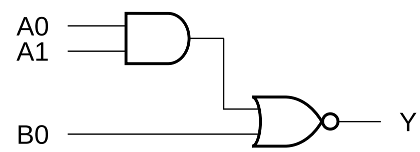
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 → Y ↑	0.0787	0.0765	0.0742	0.0707	8.6866	5.1587	3.4768	1.7955
A0 → Y ↓	0.1635	0.2009	0.1657	0.1433	6.1591	3.7037	1.9143	0.9172
A1N → Y ↑	0.1349	0.1361	0.1439	0.1371	8.7022	5.1649	3.4781	1.7962
A1N → Y ↓	0.2008	0.2429	0.2076	0.1821	6.2271	3.7396	1.9262	0.9218
B0 → Y ↑	0.0889	0.0846	0.0833	0.0805	8.7551	5.1922	3.4902	1.8011
B0 → Y ↓	0.2030	0.2404	0.1980	0.1722	6.1620	3.7078	1.9173	0.9186
B1N → Y ↑	0.1471	0.1453	0.1523	0.1445	8.7537	5.1911	3.4901	1.8014
B1N → Y ↓	0.2348	0.2766	0.2370	0.2092	6.2189	3.7380	1.9260	0.9217

Cell Description

The AOI21 cell provides the logical inverted OR of one AND group and an additional input. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 \bullet A1) + B0}$$

Logic Symbol



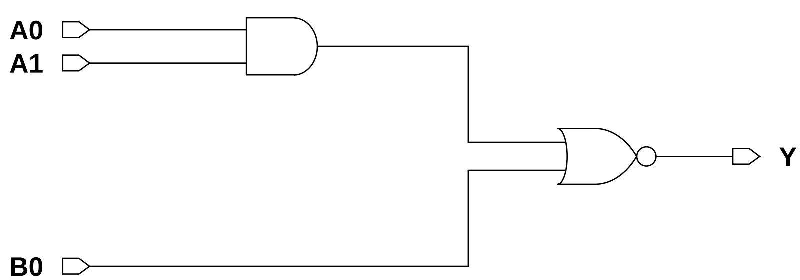
Function Table

A0	A1	B0	Y
0	x	0	1
x	0	0	1
x	x	1	0
1	1	x	0

Cell Size

Drive Strength	Height (um)	Width (um)
AOI21XLM	2.87	2.05
AOI21X1M	2.87	2.05
AOI21X2M	2.87	2.05
AOI21X3M	2.87	3.69
AOI21X4M	2.87	3.69
AOI21X6M	2.87	4.92
AOI21X8M	2.87	6.56

Functional Schematic



AC Power

Pin	Power (uW/MHz)						
	XL	X1	X2	X3	X4	X6	X8
A0	0.0039	0.0054	0.0080	0.0126	0.0156	0.0246	0.0314
A1	0.0046	0.0066	0.0103	0.0164	0.0205	0.0316	0.0409
B0	0.0036	0.0052	0.0082	0.0129	0.0161	0.0252	0.0322

Pin Capacitance

Pin	Capacitance (pF)						
	XL	X1	X2	X3	X4	X6	X8
A0	0.0016	0.0023	0.0035	0.0052	0.0066	0.0102	0.0136
A1	0.0015	0.0022	0.0033	0.0055	0.0068	0.0100	0.0135
B0	0.0016	0.0024	0.0035	0.0054	0.0067	0.0099	0.0131

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)						
	XL	X1	X2	X3	X4	X6	X8
A0 → Y ↑	0.0667	0.0585	0.0596	0.0637	0.0586	0.0618	0.0598
A0 → Y ↓	0.0339	0.0308	0.0259	0.0246	0.0242	0.0252	0.0240
A1 → Y ↑	0.0739	0.0675	0.0717	0.0780	0.0723	0.0766	0.0740
A1 → Y ↓	0.0362	0.0338	0.0291	0.0281	0.0278	0.0284	0.0274
B0 → Y ↑	0.0547	0.0491	0.0534	0.0566	0.0521	0.0578	0.0548
B0 → Y ↓	0.0187	0.0167	0.0149	0.0144	0.0140	0.0146	0.0140

Delays at 25°C,1.2V, Typical Process (Cont'd.)

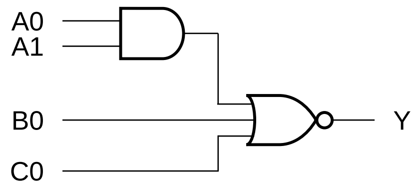
Description	K_{load} (ns/pF)						
	XL	X1	X2	X3	X4	X6	X8
A0 → Y ↑	17.8601	10.8132	7.3177	5.2110	3.7984	2.5923	1.9639
A0 → Y ↓	8.0921	4.8218	2.6187	1.6744	1.2896	0.8638	0.6432
A1 → Y ↑	17.5747	10.6509	7.1988	5.1271	3.7357	2.6039	1.9461
A1 → Y ↓	8.0930	4.8190	2.6186	1.6744	1.2898	0.8639	0.6433
B0 → Y ↑	17.6478	10.6884	7.2351	5.1501	3.7553	2.6085	1.9497
B0 → Y ↓	5.0837	2.9896	1.6156	1.0354	0.7945	0.5345	0.3976

Cell Description

The AOI211 cell provides the logical inverted OR of one AND group and two additional inputs. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 \bullet A1) + B0 + C0}$$

Logic Symbol



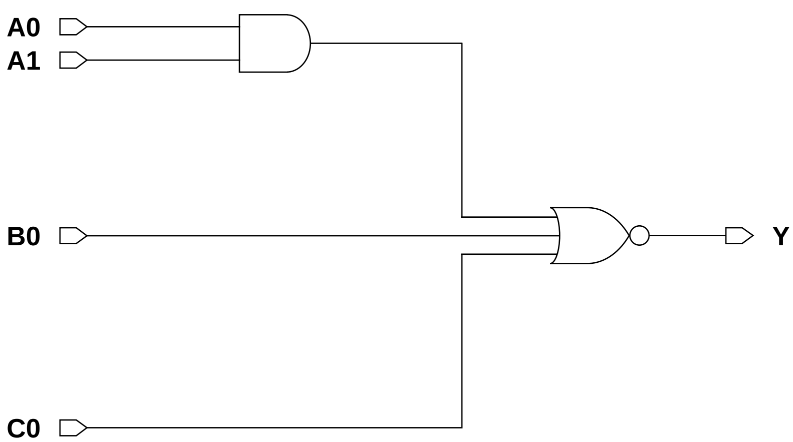
Function Table

A0	A1	B0	C0	Y
0	x	0	0	1
x	0	0	0	1
x	x	x	1	0
x	x	1	x	0
1	1	x	x	0

Cell Size

Drive Strength	Height (um)	Width (um)
AOI211XLM	2.87	2.46
AOI211X1M	2.87	2.46
AOI211X2M	2.87	2.46
AOI211X4M	2.87	4.10

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	XL	**X1**	**X2**	**X4**
A0	0.0060	0.0081	0.0116	0.0199
A1	0.0067	0.0092	0.0138	0.0245
B0	0.0050	0.0067	0.0103	0.0179
C0	0.0056	0.0077	0.0118	0.0209

Pin Capacitance

Pin	Capacitance (pF)			
	XL	**X1**	**X2**	**X4**
A0	0.0016	0.0023	0.0034	0.0062
A1	0.0017	0.0024	0.0035	0.0066
B0	0.0017	0.0025	0.0037	0.0063
C0	0.0015	0.0023	0.0035	0.0068

Delays at 25°C,1.2V, Typical Process

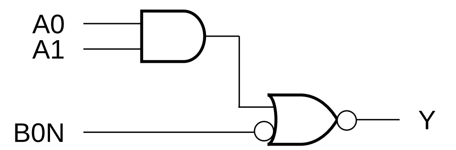
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	**X1**	**X2**	**X4**	**XL**	**X1**	**X2**	**X4**
A0 → Y ↑	0.1314	0.1113	0.1112	0.1088	26.8578	16.3132	11.0208	6.3665
A0 → Y ↓	0.0445	0.0384	0.0312	0.0262	8.3598	4.9381	2.6546	1.3211
A1 → Y ↑	0.1493	0.1309	0.1352	0.1332	27.1558	16.4927	11.1557	6.3057
A1 → Y ↓	0.0476	0.0419	0.0346	0.0300	8.3553	4.9365	2.6536	1.3206
B0 → Y ↑	0.1064	0.0889	0.0959	0.0908	27.2436	16.5336	11.1733	6.3145
B0 → Y ↓	0.0241	0.0203	0.0175	0.0156	5.1202	3.0024	1.6182	0.8579
C0 → Y ↑	0.1247	0.1101	0.1167	0.1169	27.1781	16.5090	11.1642	6.3131
C0 → Y ↓	0.0265	0.0229	0.0195	0.0191	5.1485	2.9923	1.6017	0.8993

Cell Description

The AOI21B cell provides the logical inverted OR of one AND group and an additional input. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 \bullet A1) + \overline{B0N}}$$

Logic Symbol



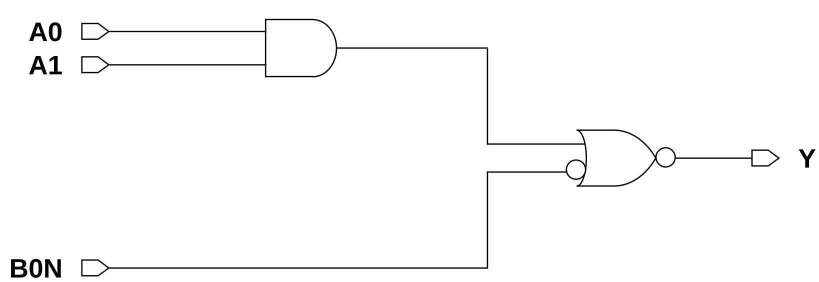
Function Table

A0	A1	B0N	Y
x	x	0	0
x	0	1	1
0	x	1	1
1	1	x	0

Cell Size

Drive Strength	Height (um)	Width (um)
AOI21BXLM	2.87	2.87
AOI21BX1M	2.87	2.87
AOI21BX2M	2.87	2.87
AOI21BX4M	2.87	3.69
AOI21BX8M	2.87	5.33

Functional Schematic



AC Power

Pin	Power (uW/MHz)				
	XL	**X1**	**X2**	**X4**	**X8**
A0	0.0092	0.0096	0.0117	0.0182	0.0345
A1	0.0087	0.0091	0.0112	0.0173	0.0324
B0N	0.0053	0.0065	0.0087	0.0154	0.0288

Pin Capacitance

Pin	Capacitance (pF)				
	XL	**X1**	**X2**	**X4**	**X8**
A0	0.0013	0.0013	0.0013	0.0018	0.0032
A1	0.0015	0.0015	0.0016	0.0020	0.0032
B0N	0.0018	0.0018	0.0021	0.0034	0.0068

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				
	XL	**X1**	**X2**	**X4**	**X8**
A0 → Y ↑	0.1346	0.1347	0.1442	0.1438	0.1370
A0 → Y ↓	0.1199	0.1290	0.1359	0.1274	0.1229
A1 → Y ↑	0.1297	0.1297	0.1400	0.1387	0.1244
A1 → Y ↓	0.1175	0.1267	0.1340	0.1251	0.1198
B0N → Y ↑	0.0632	0.0622	0.0675	0.0677	0.0635
B0N → Y ↓	0.0831	0.0921	0.0970	0.0904	0.0867

Delays at 25°C,1.2V, Typical Process (Cont'd.)

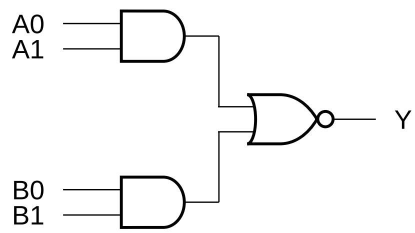
Description	K_{load} (ns/pF)				
	XL	**X1**	**X2**	**X4**	**X8**
A0 → Y ↑	8.8237	5.1825	3.4775	1.8447	0.9601
A0 → Y ↓	5.2581	3.1452	1.7218	0.8341	0.4139
A1 → Y ↑	8.8255	5.1836	3.4781	1.8445	0.9601
A1 → Y ↓	5.2584	3.1453	1.7217	0.8342	0.4138
B0N → Y ↑	8.8167	5.1773	3.4757	1.8444	0.9600
B0N → Y ↓	5.2795	3.1581	1.7292	0.8376	0.4146

Cell Description

The AOI22 cell provides the logical inverted OR of two AND groups. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 \bullet A1) + (B0 \bullet B1)}$$

Logic Symbol



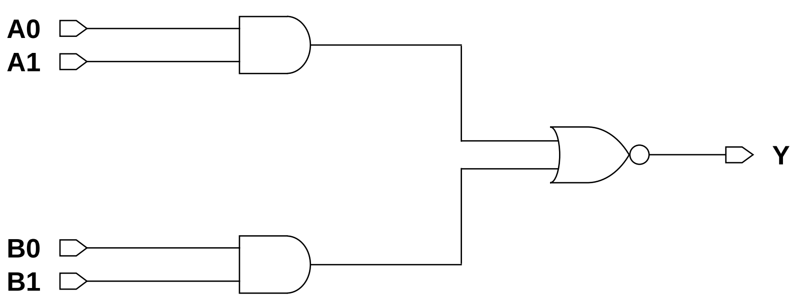
Function Table

A0	A1	B0	B1	Y
0	x	0	x	1
0	x	x	0	1
x	0	0	x	1
x	0	x	0	1
x	x	1	1	0
1	1	x	x	0

Cell Size

Drive Strength	Height (um)	Width (um)
AOI22XLM	2.87	2.46
AOI22X1M	2.87	2.46
AOI22X2M	2.87	2.87
AOI22X4M	2.87	4.92

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	XL	X1	X2	X4
A0	0.0042	0.0060	0.0093	0.0193
A1	0.0049	0.0072	0.0115	0.0233
B0	0.0054	0.0078	0.0118	0.0243
B1	0.0062	0.0091	0.0140	0.0283

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0016	0.0024	0.0035	0.0074
A1	0.0017	0.0025	0.0035	0.0066
B0	0.0016	0.0024	0.0034	0.0071
B1	0.0015	0.0023	0.0034	0.0065

Delays at 25°C,1.2V, Typical Process

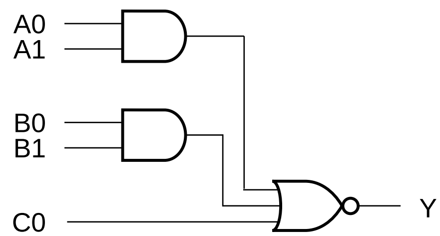
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 → Y ↑	0.0600	0.0575	0.0613	0.0680	18.0772	10.6091	7.4584	3.8510
A0 → Y ↓	0.0296	0.0272	0.0222	0.0231	7.9857	4.7810	2.5966	1.2936
A1 → Y ↑	0.0695	0.0667	0.0740	0.0818	17.6677	10.4111	7.3376	3.8968
A1 → Y ↓	0.0340	0.0312	0.0260	0.0258	7.9914	4.7821	2.5966	1.2933
B0 → Y ↑	0.1028	0.0855	0.0889	0.0920	18.0811	10.6015	7.4594	3.8357
B0 → Y ↓	0.0504	0.0439	0.0341	0.0340	8.1330	4.8068	2.6084	1.2896
B1 → Y ↑	0.1092	0.0932	0.0995	0.1074	17.7435	10.4657	7.2937	3.8953
B1 → Y ↓	0.0534	0.0473	0.0375	0.0368	8.1354	4.8060	2.6084	1.2893

Cell Description

The AOI221 cell provides the logical inverted OR of two AND groups and a third input. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 \bullet A1) + (B0 \bullet B1) + C0}$$

Logic Symbol



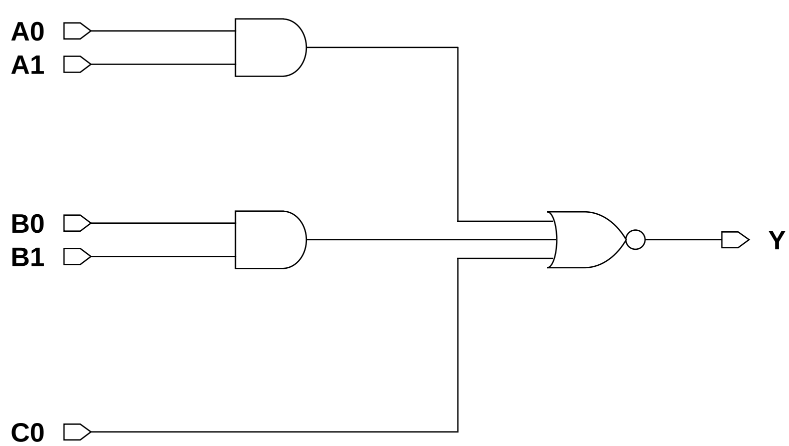
Function Table

A0	A1	B0	B1	C0	Y
0	x	0	x	0	1
0	x	x	0	0	1
x	0	0	x	0	1
x	0	x	0	0	1
x	x	x	x	1	0
x	x	1	1	x	0
1	1	x	x	x	0

Cell Size

Drive Strength	Height (um)	Width (um)
AOI221XLM	2.87	2.87
AOI221X1M	2.87	3.28
AOI221X2M	2.87	3.69
AOI221X4M	2.87	6.15

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	XL	X1	X2	X4
A0	0.0062	0.0085	0.0135	0.0241
A1	0.0068	0.0096	0.0157	0.0287
B0	0.0074	0.0103	0.0159	0.0291
B1	0.0081	0.0115	0.0181	0.0336
C0	0.0061	0.0083	0.0137	0.0247

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0016	0.0024	0.0035	0.0064
A1	0.0016	0.0024	0.0035	0.0069
B0	0.0016	0.0023	0.0034	0.0067
B1	0.0016	0.0023	0.0033	0.0069
C0	0.0018	0.0025	0.0035	0.0067

Delays at 25°C,1.2V, Typical Process

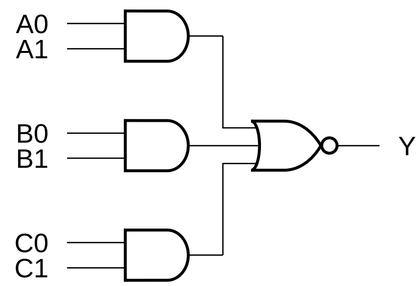
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 → Y ↑	0.1506	0.1304	0.1384	0.1249	26.3398	15.8228	10.8189	5.6742
A0 → Y ↓	0.0465	0.0394	0.0321	0.0280	7.9725	4.7478	2.5657	1.2735
A1 → Y ↑	0.1668	0.1485	0.1604	0.1461	26.6918	16.0891	10.9255	5.5968
A1 → Y ↓	0.0494	0.0425	0.0354	0.0317	7.9726	4.7474	2.5655	1.2734
B0 → Y ↑	0.1806	0.1581	0.1607	0.1501	26.4548	15.9457	10.9536	5.6755
B0 → Y ↓	0.0522	0.0451	0.0357	0.0313	8.3410	4.9650	2.6411	1.3071
B1 → Y ↑	0.1931	0.1743	0.1798	0.1682	26.6766	16.0849	10.9247	5.5652
B1 → Y ↓	0.0541	0.0480	0.0388	0.0346	8.3419	4.9641	2.6404	1.3069
C0 → Y ↑	0.1165	0.0971	0.1192	0.1047	26.7584	16.1238	10.9403	5.6084
C0 → Y ↓	0.0257	0.0205	0.0185	0.0163	5.1010	2.9415	1.6176	0.7996

Cell Description

The AOI222 cell provides the logical inverted OR of three AND groups. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 \bullet A1) + (B0 \bullet B1) + (C0 \bullet C1)}$$

Logic Symbol



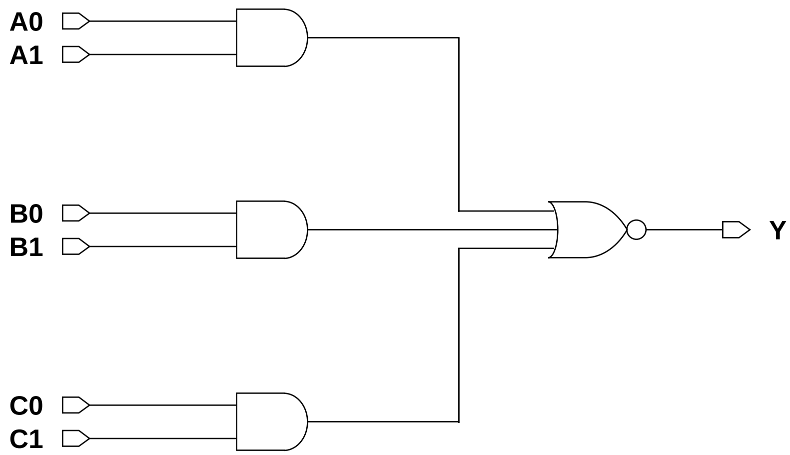
Function Table

A0	A1	B0	B1	C0	C1	Y
0	x	0	x	0	x	1
0	x	0	x	x	0	1
0	x	x	0	0	x	1
0	x	x	0	x	0	1
x	0	0	x	0	x	1
x	0	0	x	x	0	1
x	0	x	0	0	x	1
x	0	x	0	x	0	1
x	x	x	x	1	1	0
x	x	1	1	x	x	0
1	1	x	x	x	x	0

Cell Size

Drive Strength	Height (um)	Width (um)
AOI222XLM	2.87	3.69
AOI222X1M	2.87	3.69
AOI222X2M	2.87	4.10
AOI222X4M	2.87	6.97

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	XL	X1	X2	X4
A0	0.0067	0.0093	0.0145	0.0287
A1	0.0073	0.0104	0.0165	0.0326
B0	0.0077	0.0110	0.0169	0.0333
B1	0.0084	0.0121	0.0190	0.0372
C0	0.0091	0.0129	0.0192	0.0381
C1	0.0098	0.0140	0.0214	0.0421

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0017	0.0025	0.0036	0.0073
A1	0.0018	0.0026	0.0035	0.0066
B0	0.0016	0.0023	0.0034	0.0070
B1	0.0017	0.0024	0.0034	0.0064
C0	0.0016	0.0024	0.0034	0.0071
C1	0.0016	0.0023	0.0033	0.0065

Delays at 25°C,1.2V, Typical Process

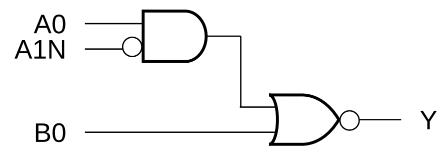
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 → Y ↑	0.1249	0.1077	0.1252	0.1315	27.0328	16.2928	11.0361	5.7777
A0 → Y ↓	0.0414	0.0333	0.0276	0.0275	7.9688	4.7761	2.5993	1.3003
A1 → Y ↑	0.1397	0.1260	0.1472	0.1527	26.7799	16.2785	11.1504	5.8205
A1 → Y ↓	0.0457	0.0377	0.0306	0.0303	7.9723	4.7769	2.5988	1.3000
B0 → Y ↑	0.1942	0.1738	0.1817	0.1851	26.2923	15.9910	11.0033	5.7667
B0 → Y ↓	0.0598	0.0513	0.0394	0.0385	7.9431	4.7365	2.5488	1.2830
B1 → Y ↑	0.2139	0.1933	0.2053	0.2078	26.7512	16.2649	11.1459	5.8195
B1 → Y ↓	0.0633	0.0547	0.0427	0.0417	7.9435	4.7364	2.5486	1.2829
C0 → Y ↑	0.2274	0.2023	0.2066	0.2069	26.4787	16.1254	11.2514	5.7576
C0 → Y ↓	0.0697	0.0617	0.0457	0.0453	8.2612	4.9193	2.6264	1.3222
C1 → Y ↑	0.2411	0.2186	0.2241	0.2303	26.7313	16.2600	11.1446	5.8187
C1 → Y ↓	0.0720	0.0646	0.0486	0.0481	8.2619	4.9188	2.6272	1.3223

Cell Description

The AOI2B1 cell provides the logical inverted OR of one AND group and an additional input. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 \bullet \overline{A1N}) + B0}$$

Logic Symbol



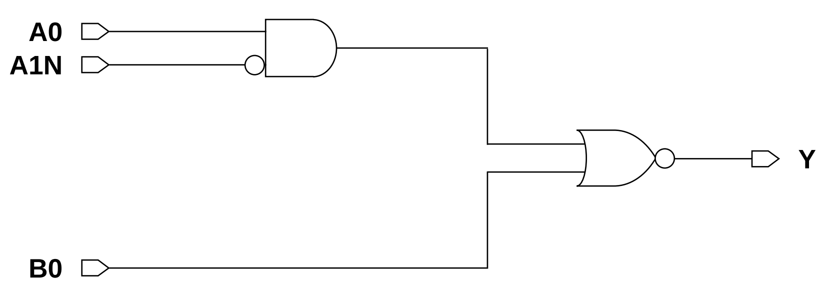
Function Table

A0	A1N	B0	Y
0	x	0	1
x	1	0	1
x	x	1	0
1	0	x	0

Cell Size

Drive Strength	Height (um)	Width (um)
AOI2B1XLM	2.87	2.46
AOI2B1X1M	2.87	2.46
AOI2B1X2M	2.87	2.87
AOI2B1X4M	2.87	4.51
AOI2B1X8M	2.87	7.38

Functional Schematic



AC Power

Pin	Power (uW/MHz)				
	XL	X1	X2	X4	X8
A0	0.0040	0.0056	0.0079	0.0152	0.0305
A1N	0.0057	0.0074	0.0104	0.0205	0.0408
B0	0.0036	0.0053	0.0083	0.0161	0.0321

Pin Capacitance

Pin	Capacitance (pF)				
	XL	X1	X2	X4	X8
A0	0.0015	0.0024	0.0035	0.0066	0.0135
A1N	0.0015	0.0016	0.0019	0.0032	0.0036
B0	0.0016	0.0024	0.0036	0.0067	0.0131

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				
	XL	X1	X2	X4	X8
A0 → Y ↑	0.0673	0.0585	0.0567	0.0586	0.0594
A0 → Y ↓	0.0334	0.0308	0.0255	0.0236	0.0232
A1N → Y ↑	0.0968	0.0947	0.0965	0.0994	0.1121
A1N → Y ↓	0.0930	0.1053	0.0921	0.0845	0.1055
B0 → Y ↑	0.0548	0.0481	0.0527	0.0519	0.0544
B0 → Y ↓	0.0188	0.0168	0.0147	0.0141	0.0140

Delays at 25°C,1.2V, Typical Process (Cont'd.)

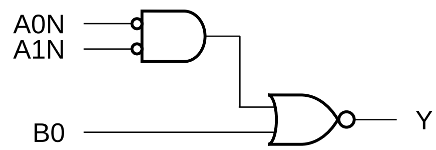
Description	K_{load} (ns/pF)				
	XL	X1	X2	X4	X8
A0 → Y ↑	17.8593	10.6511	6.9138	3.8048	1.9538
A0 → Y ↓	8.0149	4.8187	2.5902	1.2732	0.6373
A1N → Y ↑	17.5758	10.4212	7.0078	3.7346	1.9361
A1N → Y ↓	8.0406	4.8421	2.6035	1.2786	0.6431
B0 → Y ↑	17.6575	10.5247	7.0223	3.7611	1.9388
B0 → Y ↓	5.0897	2.9800	1.5839	0.7941	0.3974

Cell Description

The AOI2BB1cell provides the logical inverted OR of one AND group of two inverted inputs (A0N,A1N) and an additional non-inverted input (B0). The output (Y) is represented by the logic equation:

$$Y = \overline{(\overline{A0N} \bullet \overline{A1N}) + B0}$$

Logic Symbol



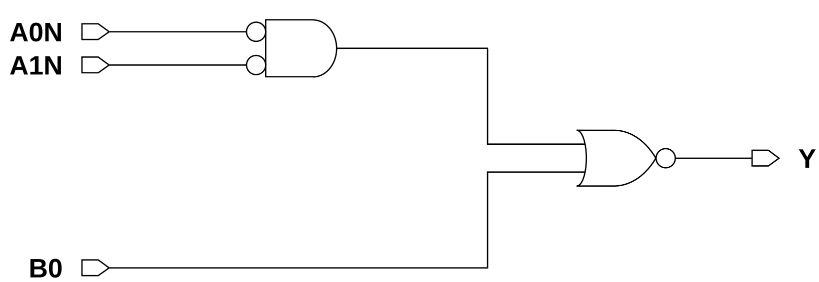
Function Table

A0N	A1N	B0	Y
1	x	0	1
x	1	0	1
x	x	1	0
0	0	x	0

Cell Size

Drive Strength	Height (um)	Width (um)
AOI2BB1XLM	2.87	2.46
AOI2BB1X1M	2.87	2.46
AOI2BB1X2M	2.87	2.46
AOI2BB1X4M	2.87	3.69
AOI2BB1X8M	2.87	6.56

Functional Schematic



AC Power

Pin	Power (uW/MHz)				
	XL	**X1**	**X2**	**X4**	**X8**
A0N	0.0057	0.0069	0.0094	0.0162	0.0319
A1N	0.0063	0.0075	0.0103	0.0177	0.0351
B0	0.0037	0.0048	0.0072	0.0144	0.0285

Pin Capacitance

Pin	Capacitance (pF)				
	XL	**X1**	**X2**	**X4**	**X8**
A0N	0.0015	0.0016	0.0020	0.0033	0.0063
A1N	0.0015	0.0016	0.0021	0.0033	0.0067
B0	0.0014	0.0023	0.0033	0.0071	0.0140

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				
	XL	**X1**	**X2**	**X4**	**X8**
A0N → Y ↑	0.0823	0.0714	0.0717	0.0682	0.0682
A0N → Y ↓	0.1234	0.1387	0.1238	0.1187	0.1187
A1N → Y ↑	0.0849	0.0736	0.0747	0.0706	0.0704
A1N → Y ↓	0.1346	0.1485	0.1351	0.1283	0.1293
B0 → Y ↑	0.0598	0.0497	0.0498	0.0516	0.0522
B0 → Y ↓	0.0212	0.0191	0.0165	0.0164	0.0162

Delays at 25°C,1.2V, Typical Process (Cont'd.)

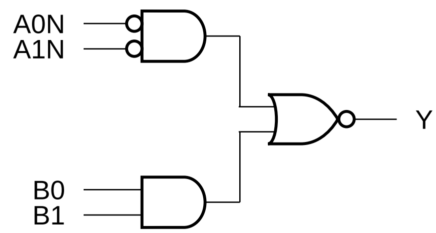
Description	K_{load} (ns/pF)				
	XL	**X1**	**X2**	**X4**	**X8**
A0N → Y ↑	18.1192	10.8163	7.2283	3.7747	1.9407
A0N → Y ↓	5.6367	3.3442	1.7756	0.8674	0.4499
A1N → Y ↑	18.1284	10.8210	7.2315	3.7762	1.9415
A1N → Y ↓	5.6360	3.3441	1.7755	0.8674	0.4499
B0 → Y ↑	18.0463	10.7962	7.2178	3.7726	1.9398
B0 → Y ↓	5.0184	2.9945	1.6111	0.7977	0.3998

Cell Description

The AOI2BB2 cell provides the logical inverted OR of one AND group of two inverted inputs (A0N,A1N) and one AND group of two non-inverted inputs (B0,B1). The output (Y) is represented by the logic equation:

$$Y = \overline{(\overline{A0N} \bullet \overline{A1N}) + (B0 \bullet B1)}$$

Logic Symbol



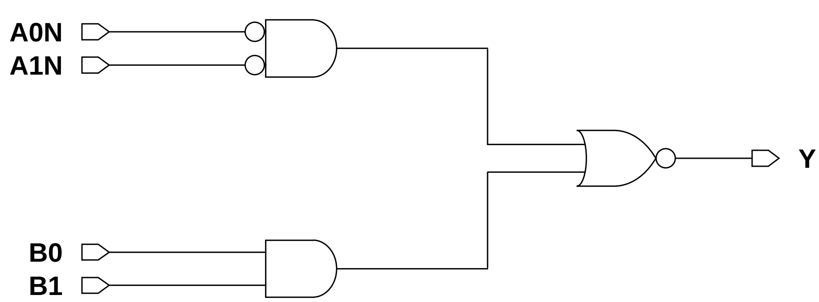
Function Table

A0N	A1N	B0	B1	Y
1	x	0	x	1
1	x	x	0	1
x	1	0	x	1
x	1	x	0	1
x	x	1	1	0
0	0	x	x	0

Cell Size

Drive Strength	Height (um)	Width (um)
AOI2BB2XLM	2.87	2.87
AOI2BB2X1M	2.87	3.28
AOI2BB2X2M	2.87	3.28
AOI2BB2X4M	2.87	4.92
AOI2BB2X8M	2.87	8.61

Functional Schematic



AC Power

Pin	Power (uW/MHz)				
	XL	**X1**	**X2**	**X4**	**X8**
A0N	0.0057	0.0070	0.0094	0.0161	0.0328
A1N	0.0063	0.0076	0.0102	0.0168	0.0345
B0	0.0039	0.0060	0.0082	0.0155	0.0317
B1	0.0044	0.0072	0.0105	0.0203	0.0408

Pin Capacitance

Pin	Capacitance (pF)				
	XL	**X1**	**X2**	**X4**	**X8**
A0N	0.0016	0.0016	0.0022	0.0035	0.0070
A1N	0.0014	0.0014	0.0020	0.0031	0.0062
B0	0.0017	0.0024	0.0034	0.0065	0.0134
B1	0.0015	0.0023	0.0033	0.0070	0.0137

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				
	XL	**X1**	**X2**	**X4**	**X8**
A0N → Y ↑	0.0811	0.0851	0.0834	0.0777	0.0817
A0N → Y ↓	0.1195	0.1393	0.1191	0.1095	0.1141
A1N → Y ↑	0.0847	0.0886	0.0878	0.0811	0.0855
A1N → Y ↓	0.1278	0.1470	0.1278	0.1170	0.1228
B0 → Y ↑	0.0668	0.0642	0.0617	0.0590	0.0613
B0 → Y ↓	0.0342	0.0321	0.0256	0.0238	0.0239
B1 → Y ↑	0.0762	0.0734	0.0737	0.0736	0.0762
B1 → Y ↓	0.0349	0.0349	0.0288	0.0277	0.0274

Delays at 25°C,1.2V, Typical Process (Cont'd.)

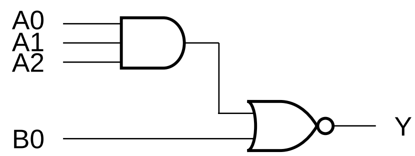
Description	K_{load} (ns/pF)				
	XL	**X1**	**X2**	**X4**	**X8**
A0N → Y ↑	18.0303	10.6966	7.2196	3.7324	1.9448
A0N → Y ↓	5.6311	3.3229	1.7603	0.8490	0.4269
A1N → Y ↑	18.0359	10.7021	7.2231	3.7344	1.9455
A1N → Y ↓	5.6305	3.3225	1.7603	0.8489	0.4268
B0 → Y ↑	17.1745	10.8124	7.3027	3.7754	1.9586
B0 → Y ↓	7.9912	4.7553	2.5768	1.2863	0.6408
B1 → Y ↑	17.9617	10.6561	7.1255	3.7151	1.9417
B1 → Y ↓	7.9896	4.7552	2.5760	1.2861	0.6408

Cell Description

The AOI31 cell provides the logical inverted OR of one AND group and an additional input. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 \bullet A1 \bullet A2) + B0}$$

Logic Symbol



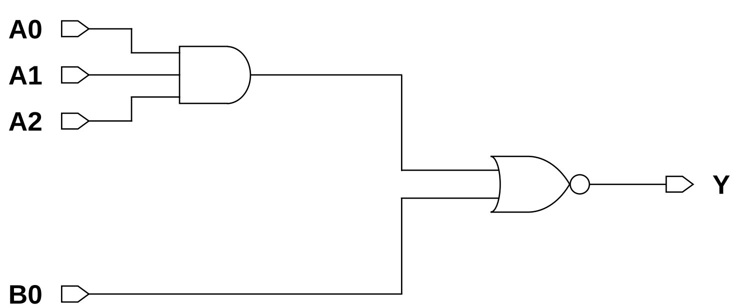
Function Table

A0	A1	A2	B0	Y
0	x	x	0	1
x	0	x	0	1
x	x	0	0	1
x	x	x	1	0
1	1	1	x	0

Cell Size

Drive Strength	Height (um)	Width (um)
AOI31XLM	2.87	2.46
AOI31X1M	2.87	2.46
AOI31X2M	2.87	2.46
AOI31X4M	2.87	4.92

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	XL	X1	X2	X4
A0	0.0041	0.0058	0.0084	0.0186
A1	0.0048	0.0071	0.0107	0.0231
A2	0.0056	0.0083	0.0130	0.0273
B0	0.0045	0.0066	0.0107	0.0227

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0016	0.0023	0.0035	0.0075
A1	0.0015	0.0023	0.0034	0.0070
A2	0.0014	0.0022	0.0033	0.0062
B0	0.0017	0.0024	0.0036	0.0068

Delays at 25°C,1.2V, Typical Process

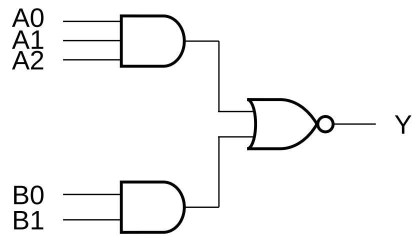
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 → Y ↑	0.0713	0.0633	0.0633	0.0683	18.0155	10.9215	7.3823	3.7561
A0 → Y ↓	0.0466	0.0422	0.0341	0.0356	10.9851	6.5890	3.5873	1.7843
A1 → Y ↑	0.0811	0.0748	0.0782	0.0853	18.0162	10.9364	7.3829	3.8185
A1 → Y ↓	0.0516	0.0483	0.0405	0.0419	10.9903	6.5880	3.5861	1.7840
A2 → Y ↑	0.0881	0.0824	0.0901	0.0983	17.4938	10.5799	7.2406	3.8421
A2 → Y ↓	0.0552	0.0521	0.0444	0.0445	10.9931	6.5882	3.5874	1.7840
B0 → Y ↑	0.0650	0.0588	0.0684	0.0740	17.8569	10.8510	7.3146	3.8496
B0 → Y ↓	0.0189	0.0167	0.0149	0.0149	5.1162	2.9833	1.6087	0.8006

Cell Description

The AOI32 cell provides the logical inverted OR of two AND groups. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 \bullet A1 \bullet A2) + (B0 \bullet B1)}$$

Logic Symbol



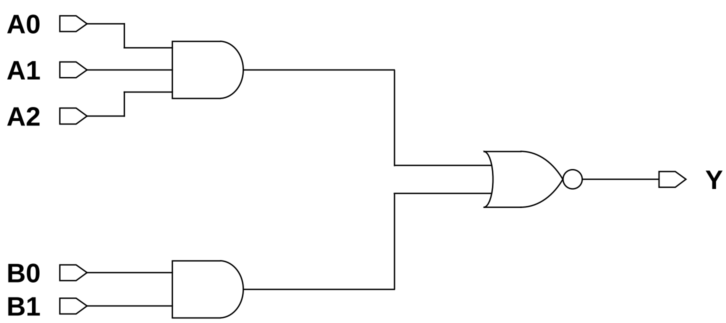
Function Table

A0	A1	A2	B0	B1	Y
0	x	x	0	x	1
0	x	x	x	0	1
x	0	x	0	x	1
x	0	x	x	0	1
x	x	0	0	x	1
x	x	0	x	0	1
x	x	x	1	1	0
1	1	1	x	x	0

Cell Size

Drive Strength	Height (um)	Width (um)
AOI32XLM	2.87	2.87
AOI32X1M	2.87	2.87
AOI32X2M	2.87	3.28
AOI32X4M	2.87	5.74

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	XL	X1	X2	X4
A0	0.0056	0.0080	0.0121	0.0261
A1	0.0063	0.0093	0.0144	0.0304
A2	0.0070	0.0105	0.0167	0.0348
B0	0.0050	0.0073	0.0117	0.0252
B1	0.0057	0.0085	0.0139	0.0292

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0016	0.0023	0.0034	0.0075
A1	0.0016	0.0023	0.0033	0.0070
A2	0.0014	0.0022	0.0033	0.0063
B0	0.0016	0.0025	0.0035	0.0073
B1	0.0016	0.0024	0.0035	0.0067

Delays at 25°C,1.2V, Typical Process

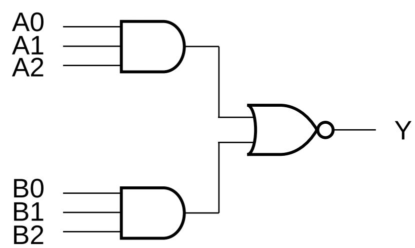
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 → Y ↑	0.1062	0.0898	0.0925	0.1001	18.2444	11.0886	7.5272	3.8463
A0 → Y ↓	0.0697	0.0578	0.0458	0.0482	11.0242	6.5810	3.5677	1.7821
A1 → Y ↑	0.1157	0.1006	0.1067	0.1175	18.2128	11.0653	7.5114	3.9013
A1 → Y ↓	0.0754	0.0643	0.0522	0.0542	11.0281	6.5810	3.5663	1.7825
A2 → Y ↑	0.1216	0.1073	0.1177	0.1339	17.7782	10.7069	7.3584	3.9713
A2 → Y ↓	0.0783	0.0679	0.0559	0.0575	11.0271	6.5810	3.5668	1.7821
B0 → Y ↑	0.0716	0.0671	0.0770	0.0875	18.2323	11.1044	7.5256	3.9281
B0 → Y ↓	0.0297	0.0259	0.0220	0.0236	8.0287	4.7700	2.5926	1.3004
B1 → Y ↑	0.0781	0.0762	0.0894	0.1025	17.7213	10.8808	7.3749	3.9741
B1 → Y ↓	0.0326	0.0291	0.0257	0.0265	8.0305	4.7698	2.5927	1.3001

Cell Description

The AOI33 cell provides the logical inverted OR of two AND groups. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 \bullet A1 \bullet A2) + (B0 \bullet B1 \bullet B2)}$$

Logic Symbol



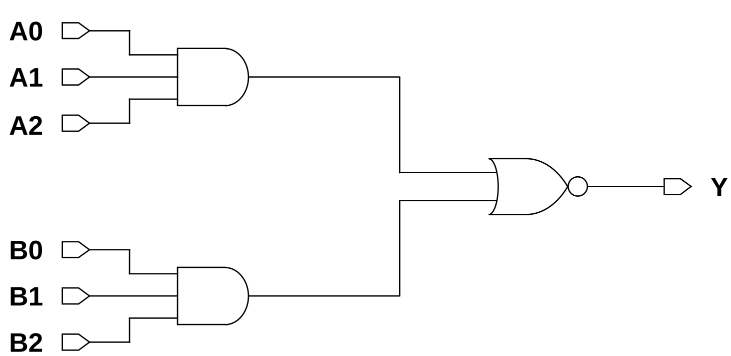
Function Table

A0	A1	A2	B0	B1	B2	Y
0	x	x	0	x	x	1
0	x	x	x	0	x	1
0	x	x	x	x	0	1
x	0	x	0	x	x	1
x	0	x	x	0	x	1
x	0	x	x	x	0	1
x	x	0	0	x	x	1
x	x	0	x	0	x	1
x	x	0	x	x	0	1
x	x	x	1	1	1	0
1	1	1	x	x	x	0

Cell Size

Drive Strength	Height (um)	Width (um)
AOI33XLM	2.87	3.69
AOI33X1M	2.87	3.69
AOI33X2M	2.87	3.69
AOI33X4M	2.87	6.97

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	XL	X1	X2	X4
A0	0.0058	0.0083	0.0130	0.0264
A1	0.0065	0.0095	0.0152	0.0306
A2	0.0073	0.0108	0.0174	0.0350
B0	0.0073	0.0106	0.0159	0.0327
B1	0.0080	0.0118	0.0182	0.0370
B2	0.0087	0.0130	0.0204	0.0412

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0017	0.0024	0.0036	0.0077
A1	0.0016	0.0024	0.0035	0.0071
A2	0.0016	0.0024	0.0035	0.0067
B0	0.0015	0.0023	0.0034	0.0075
B1	0.0015	0.0023	0.0034	0.0069
B2	0.0014	0.0022	0.0033	0.0064

Delays at 25°C,1.2V, Typical Process

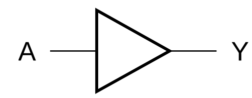
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 → Y ↑	0.0922	0.0831	0.0902	0.0944	18.4396	11.2352	7.5952	3.9480
A0 → Y ↓	0.0500	0.0410	0.0330	0.0328	10.8769	6.5310	3.5585	1.7839
A1 → Y ↑	0.1026	0.0952	0.1050	0.1103	18.3545	11.1832	7.5623	3.9845
A1 → Y ↓	0.0561	0.0480	0.0397	0.0387	10.8766	6.5308	3.5581	1.7834
A2 → Y ↑	0.1110	0.1043	0.1165	0.1262	18.0031	10.9544	7.4104	3.9992
A2 → Y ↓	0.0602	0.0521	0.0436	0.0424	10.8791	6.5316	3.5582	1.7834
B0 → Y ↑	0.1349	0.1207	0.1235	0.1293	18.4579	11.2536	7.6082	3.8798
B0 → Y ↓	0.0855	0.0758	0.0581	0.0590	10.8824	6.5228	3.5496	1.7798
B1 → Y ↑	0.1439	0.1313	0.1381	0.1475	18.3449	11.1853	7.5651	3.9290
B1 → Y ↓	0.0908	0.0817	0.0646	0.0650	10.8832	6.5228	3.5495	1.7798
B2 → Y ↑	0.1485	0.1384	0.1482	0.1642	17.8110	10.9218	7.3958	3.9985
B2 → Y ↓	0.0935	0.0851	0.0681	0.0682	10.8830	6.5225	3.5485	1.7798

Cell Description

The BUF cell provides the logical buffer of a single input (A). The output (Y) is represented by the logic equation:

$Y = A$

Logic Symbol



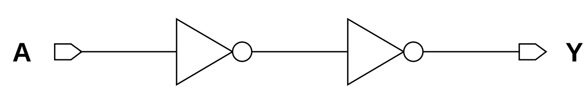
Function Table

A	Y
0	0
1	1

Cell Size

Drive Strength	Height (um)	Width (um)
BUFX2M	2.87	1.64
BUFX3M	2.87	2.05
BUFX4M	2.87	2.05
BUFX5M	2.87	2.46
BUFX6M	2.87	2.87
BUFX8M	2.87	3.28
BUFX10M	2.87	4.10
BUFX12M	2.87	4.92
BUFX14M	2.87	5.74
BUFX16M	2.87	6.15
BUFX18M	2.87	6.97
BUFX20M	2.87	7.38
BUFX24M	2.87	9.02
BUFX32M	2.87	11.89

Functional Schematic



AC Power

Pin	Power (uW/MHz)							
	X2	X3	X4	X5	X6	X8	X10	X12
A	0.0077	0.0108	0.0131	0.0166	0.0190	0.0235	0.0306	0.0362

AC Power (Cont'd.)

Pin	Power (uW/MHz)					
	X14	X16	X18	X20	X24	X32
A	0.0442	0.0476	0.0554	0.0603	0.0722	0.0972

Pin Capacitance

Pin	Capacitance (pF)							
	X2	X3	X4	X5	X6	X8	X10	X12
A	0.0019	0.0026	0.0032	0.0036	0.0046	0.0058	0.0067	0.0086

Pin Capacitance (Cont'd.)

Pin	Capacitance (pF)					
	X14	X16	X18	X20	X24	X32
A	0.0099	0.0097	0.0123	0.0132	0.0162	0.0224

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)							
	X2	X3	X4	X5	X6	X8	X10	X12
A → Y ↑	0.0477	0.0466	0.0447	0.0478	0.0434	0.0426	0.0439	0.0424
A → Y ↓	0.0778	0.0755	0.0702	0.0727	0.0681	0.0647	0.0675	0.0658

Delays at 25°C,1.2V, Typical Process (Cont'd.)

Description	Intrinsic Delay (ns)					
	X14	X16	X18	X20	X24	X32
A → Y ↑	0.0437	0.0459	0.0449	0.0447	0.0445	0.0458
A → Y ↓	0.0673	0.0724	0.0684	0.0693	0.0690	0.0695

Delays at 25°C,1.2V, Typical Process (Cont'd.)

Description	K_{load} (ns/pF)							
	X2	**X3**	**X4**	**X5**	**X6**	**X8**	**X10**	**X12**
A → Y ↑	3.5827	2.4852	1.8676	1.4839	1.2620	1.0124	0.7597	0.6761
A → Y ↓	1.7436	1.0901	0.8571	0.6633	0.5693	0.4513	0.3283	0.2932

Delays at 25°C,1.2V, Typical Process (Cont'd.)

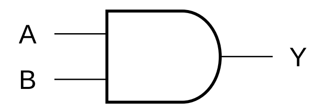
Description	K_{load} (ns/pF)					
	X14	**X16**	**X18**	**X20**	**X24**	**X32**
A → Y ↑	0.5476	0.4940	0.4301	0.3932	0.3380	0.2544
A → Y ↓	0.2322	0.2105	0.1818	0.1677	0.1433	0.1070

Cell Description

The CLKAND2 cell provides the logical AND of two inputs (A,B). The output (Y) is represented by the logic equation:

$Y = A \bullet B$

Logic Symbol



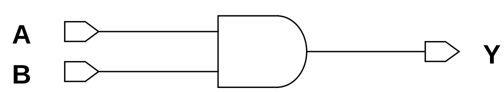
Function Table

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Cell Size

Drive Strength	Height (um)	Width (um)
CLKAND2X2M	2.87	2.05
CLKAND2X3M	2.87	2.46
CLKAND2X4M	2.87	2.46
CLKAND2X6M	2.87	3.28
CLKAND2X8M	2.87	4.51
CLKAND2X12M	2.87	5.74
CLKAND2X16M	2.87	6.97

Functional Schematic



AC Power

Pin	Power (uW/MHz)						
	X2	**X3**	**X4**	**X6**	**X8**	**X12**	**X16**
A	0.0069	0.0084	0.0105	0.0158	0.0197	0.0274	0.0373
B	0.0077	0.0093	0.0115	0.0171	0.0218	0.0300	0.0411

Pin Capacitance

Pin	Capacitance (pF)						
	X2	**X3**	**X4**	**X6**	**X8**	**X12**	**X16**
A	0.0015	0.0019	0.0025	0.0033	0.0044	0.0059	0.0085
B	0.0017	0.0020	0.0025	0.0034	0.0048	0.0065	0.0088

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)						
	X2	**X3**	**X4**	**X6**	**X8**	**X12**	**X16**
A → Y ↑	0.0747	0.0816	0.0809	0.0850	0.0842	0.0877	0.0792
A → Y ↓	0.0843	0.0800	0.0749	0.0724	0.0752	0.0699	0.0704
B → Y ↑	0.0796	0.0858	0.0844	0.0884	0.0884	0.0917	0.0832
B → Y ↓	0.0942	0.0885	0.0823	0.0769	0.0778	0.0710	0.0760

Delays at 25°C,1.2V, Typical Process (Cont'd.)

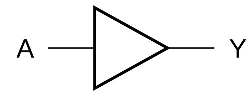
Description	K_{load} (ns/pF)						
	X2	**X3**	**X4**	**X6**	**X8**	**X12**	**X16**
A → Y ↑	3.5569	2.4988	1.8351	1.2352	0.9472	0.6833	0.4649
A → Y ↓	3.4264	2.4644	1.8761	1.2262	0.9249	0.6212	0.4779
B → Y ↑	3.5569	2.4988	1.8351	1.2351	0.9471	0.6832	0.4649
B → Y ↓	3.4316	2.4674	1.8783	1.2270	0.9247	0.6210	0.4783

Cell Description

The CLKBUF cell provides the logical buffer of a single input (A), with balanced delays for clock signals. The output (Y) is represented by the logic equation:

$Y = A$

Logic Symbol



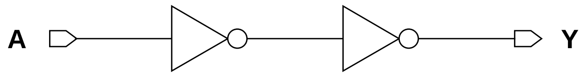
Function Table

A	Y
0	0
1	1

Cell Size

Drive Strength	Height (um)	Width (um)
CLKBUFX1M	2.87	1.64
CLKBUFX2M	2.87	1.64
CLKBUFX3M	2.87	2.05
CLKBUFX4M	2.87	2.05
CLKBUFX6M	2.87	2.46
CLKBUFX8M	2.87	3.28
CLKBUFX12M	2.87	4.51
CLKBUFX16M	2.87	5.33
CLKBUFX20M	2.87	6.56
CLKBUFX24M	2.87	6.97
CLKBUFX32M	2.87	8.20
CLKBUFX40M	2.87	9.84

Functional Schematic



AC Power

Pin	Power (uW/MHz)							
	X1	X2	X3	X4	X6	X8	X12	X16
A	0.0056	0.0068	0.0081	0.0095	0.0129	0.0168	0.0248	0.0334

AC Power (Cont'd.)

Pin	Power (uW/MHz)			
	X20	X24	X32	X40
A	0.0424	0.0497	0.0623	0.0757

Pin Capacitance

Pin	Capacitance (pF)							
	X1	X2	X3	X4	X6	X8	X12	X16
A	0.0020	0.0020	0.0020	0.0021	0.0027	0.0036	0.0049	0.0065

Pin Capacitance (Cont'd.)

Pin	Capacitance (pF)			
	X20	X24	X32	X40
A	0.0084	0.0096	0.0120	0.0147

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)							
	X1	X2	X3	X4	X6	X8	X12	X16
A → Y ↑	0.0507	0.0533	0.0603	0.0645	0.0635	0.0635	0.0605	0.0616
A → Y ↓	0.0565	0.0656	0.0771	0.0823	0.0711	0.0720	0.0701	0.0707

Delays at 25°C,1.2V, Typical Process (Cont'd.)

Description	Intrinsic Delay (ns)			
	X20	X24	X32	X40
A → Y ↑	0.0613	0.0623	0.0607	0.0619
A → Y ↓	0.0722	0.0706	0.0702	0.0696

Delays at 25°C,1.2V, Typical Process (Cont'd.)

Description	K_{load} (ns/pF)							
	X1	**X2**	**X3**	**X4**	**X6**	**X8**	**X12**	**X16**
A → Y ↑	6.4009	3.5231	2.4889	1.8097	1.3847	1.0013	0.6416	0.5044
A → Y ↓	5.0948	3.4196	2.4724	1.8665	1.2027	0.9193	0.5912	0.4416

Delays at 25°C,1.2V, Typical Process (Cont'd.)

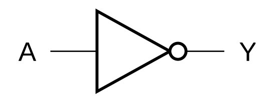
Description	K_{load} (ns/pF)			
	X20	**X24**	**X32**	**X40**
A → Y ↑	0.3795	0.3625	0.2642	0.2300
A → Y ↓	0.3482	0.2973	0.2243	0.1795

Cell Description

The CLKINV cell provides the logical inversion of a single input (A), with balanced delays for clock signals. The output (Y) is represented by the logic equation:

$Y = \overline{A}$

Logic Symbol



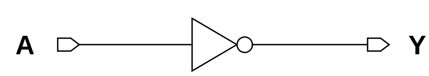
Function Table

A	Y
0	1
1	0

Cell Size

Drive Strength	Height (um)	Width (um)
CLKINVX1M	2.87	1.23
CLKINVX2M	2.87	1.23
CLKINVX3M	2.87	1.64
CLKINVX4M	2.87	1.64
CLKINVX6M	2.87	2.05
CLKINVX8M	2.87	2.46
CLKINVX12M	2.87	3.28
CLKINVX16M	2.87	4.10
CLKINVX20M	2.87	4.51
CLKINVX24M	2.87	5.74
CLKINVX32M	2.87	6.97
CLKINVX40M	2.87	8.61

Functional Schematic



AC Power

Pin	Power (uW/MHz)							
	X1	X2	X3	X4	X6	X8	X12	X16
A	0.0028	0.0040	0.0055	0.0071	0.0097	0.0129	0.0198	0.0269

AC Power (Cont'd.)

Pin	Power (uW/MHz)			
	X20	X24	X32	X40
A	0.0329	0.0381	0.0509	0.0643

Pin Capacitance

Pin	Capacitance (pF)							
	X1	X2	X3	X4	X6	X8	X12	X16
A	0.0018	0.0027	0.0039	0.0050	0.0073	0.0097	0.0148	0.0199

Pin Capacitance (Cont'd.)

Pin	Capacitance (pF)			
	X20	X24	X32	X40
A	0.0247	0.0290	0.0379	0.0476

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)							
	X1	X2	X3	X4	X6	X8	X12	X16
A → Y ↑	0.0203	0.0176	0.0166	0.0159	0.0156	0.0155	0.0169	0.0161
A → Y ↓	0.0179	0.0182	0.0177	0.0173	0.0165	0.0163	0.0163	0.0163

Delays at 25°C,1.2V, Typical Process (Cont'd.)

Description	Intrinsic Delay (ns)			
	X20	X24	X32	X40
A → Y ↑	0.0165	0.0164	0.0172	0.0180
A → Y ↓	0.0163	0.0165	0.0172	0.0177

Delays at 25°C,1.2V, Typical Process (Cont'd.)

Description	K_{load} (ns/pF)							
	X1	**X2**	**X3**	**X4**	**X6**	**X8**	**X12**	**X16**
A → Y ↑	6.2009	3.4412	2.4471	1.7795	1.2102	0.9305	0.6908	0.4633
A → Y ↓	5.1546	3.4400	2.4714	1.8519	1.2108	0.9120	0.5904	0.4380

Delays at 25°C,1.2V, Typical Process (Cont'd.)

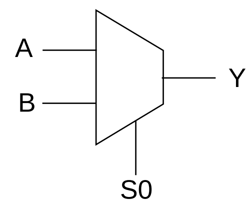
Description	K_{load} (ns/pF)			
	X20	**X24**	**X32**	**X40**
A → Y ↑	0.3928	0.3323	0.2361	0.1932
A → Y ↓	0.3522	0.3072	0.2330	0.1755

Cell Description

The CLKMX2 cell is a non-inverting 2 to 1 multiplexer with balanced delays for clock signals. The state of the select input (S0) determines which data input (A,B) is presented to the output (Y). The output (Y) is represented by the logic equation:

$Y = (S0 \bullet B) + (\overline{S0} \bullet A)$

Logic Symbol



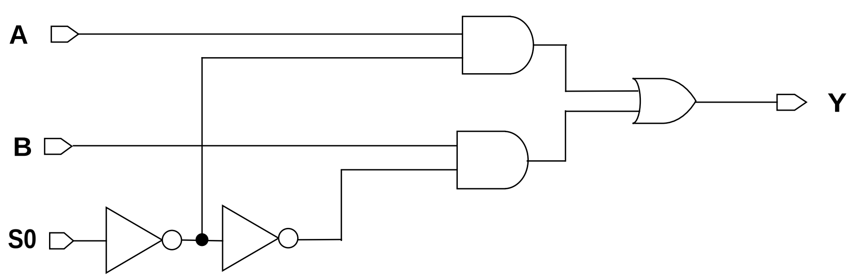
Function Table

S0	A	B	Y
0	0	x	0
0	1	x	1
1	x	0	0
1	x	1	1

Cell Size

Drive Strength	Height (um)	Width (um)
CLKMX2X2M	2.87	4.10
CLKMX2X3M	2.87	4.51
CLKMX2X4M	2.87	4.51
CLKMX2X6M	2.87	5.33
CLKMX2X8M	2.87	5.74
CLKMX2X12M	2.87	6.15
CLKMX2X16M	2.87	6.97

Functional Schematic



AC Power

Pin	Power (uW/MHz)						
	X2	**X3**	**X4**	**X6**	**X8**	**X12**	**X16**
S0	0.0135	0.0157	0.0171	0.0210	0.0246	0.0362	0.0520
B	0.0125	0.0153	0.0165	0.0211	0.0249	0.0361	0.0524
A	0.0111	0.0139	0.0152	0.0198	0.0238	0.0351	0.0514

Pin Capacitance

Pin	Capacitance (pF)						
	X2	**X3**	**X4**	**X6**	**X8**	**X12**	**X16**
S0	0.0044	0.0049	0.0048	0.0048	0.0048	0.0045	0.0048
B	0.0023	0.0027	0.0027	0.0027	0.0027	0.0028	0.0028
A	0.0025	0.0029	0.0030	0.0029	0.0029	0.0029	0.0030

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)						
	X2	**X3**	**X4**	**X6**	**X8**	**X12**	**X16**
S0 → Y ↑	0.1055	0.1043	0.1079	0.1213	0.1317	0.1607	0.1781
S0 → Y ↓	0.1086	0.1088	0.1148	0.1327	0.1520	0.1879	0.2128
B → Y ↑	0.0950	0.0970	0.1014	0.1149	0.1271	0.1474	0.1779
B → Y ↓	0.1139	0.1128	0.1198	0.1355	0.1535	0.1846	0.2150
A → Y ↑	0.0947	0.0955	0.1002	0.1136	0.1260	0.1516	0.1774
A → Y ↓	0.1107	0.1121	0.1203	0.1357	0.1555	0.1877	0.2157

Delays at 25°C,1.2V, Typical Process (Cont'd.)

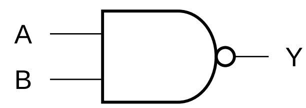
Description	K_{load} (ns/pF)						
	X2	**X3**	**X4**	**X6**	**X8**	**X12**	**X16**
S0 → Y ↑	3.6276	2.4781	1.8701	1.2612	0.9943	0.6255	0.4778
S0 → Y ↓	3.4562	2.4792	1.9078	1.2163	0.9459	0.6321	0.4727
B → Y ↑	3.6291	2.4792	1.8713	1.2620	0.9950	0.6254	0.4783
B → Y ↓	3.4567	2.4785	1.9066	1.2138	0.9431	0.6305	0.4729
A → Y ↑	3.6272	2.4772	1.8696	1.2611	0.9944	0.6255	0.4778
A → Y ↓	3.4542	2.4806	1.9090	1.2162	0.9456	0.6321	0.4727

Cell Description

The CLKNAND2 cell provides the logical NAND of two inputs (A,B), with balanced delays for clock signals. The output (Y) is represented by the logic equation:

$$Y = \overline{(A \bullet B)}$$

Logic Symbol



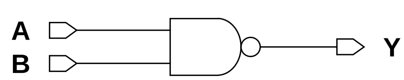
Function Table

A	B	Y
0	x	1
x	0	1
1	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
CLKNAND2X2M	2.87	1.64
CLKNAND2X4M	2.87	2.46
CLKNAND2X8M	2.87	4.51
CLKNAND2X12M	2.87	5.33
CLKNAND2X16M	2.87	8.20

Functional Schematic



AC Power

Pin	Power (uW/MHz)				
	X2	X4	X8	X12	X16
A	0.0049	0.0086	0.0170	0.0220	0.0346
B	0.0061	0.0110	0.0220	0.0283	0.0442

Pin Capacitance

Pin	Capacitance (pF)				
	X2	X4	X8	X12	X16
A	0.0028	0.0052	0.0107	0.0143	0.0220
B	0.0029	0.0056	0.0112	0.0142	0.0224

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				
	X2	X4	X8	X12	X16
A → Y ↑	0.0217	0.0214	0.0218	0.0217	0.0223
A → Y ↓	0.0271	0.0282	0.0263	0.0267	0.0265
B → Y ↑	0.0255	0.0239	0.0250	0.0254	0.0254
B → Y ↓	0.0312	0.0322	0.0301	0.0304	0.0298

Delays at 25°C,1.2V, Typical Process (Cont'd.)

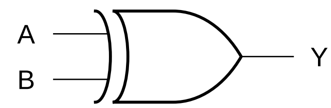
Description	K_{load} (ns/pF)				
	X2	X4	X8	X12	X16
A → Y ↑	3.5614	1.9187	0.9980	0.7493	0.5062
A → Y ↓	4.7507	2.6608	1.2424	0.9560	0.6157
B → Y ↑	3.5610	1.7992	0.9575	0.7491	0.4879
B → Y ↓	4.7540	2.6626	1.2433	0.9563	0.6160

Cell Description

The CLKXOR2 cell provides a logical EXCLUSIVE OR of two inputs (A,B) with balanced delays for clock signals. The output (Y) is represented by the logic equation:

$$Y = (A \bullet \overline{B}) + (\overline{A} \bullet B)$$

Logic Symbol



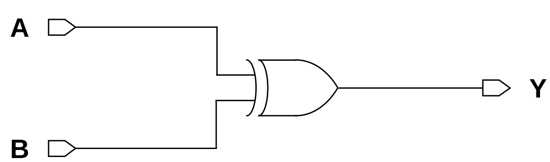
Function Table

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
CLKXOR2X2M	2.87	4.10
CLKXOR2X4M	2.87	6.56
CLKXOR2X8M	2.87	9.84
CLKXOR2X12M	2.87	12.71
CLKXOR2X16M	2.87	15.99

Functional Schematic



AC Power

Pin	Power (uW/MHz)				
	X2	X4	X8	X12	X16
A	0.0122	0.0208	0.0353	0.0541	0.0690
B	0.0154	0.0265	0.0471	0.0700	0.0892

Pin Capacitance

Pin	Capacitance (pF)				
	X2	X4	X8	X12	X16
A	0.0042	0.0066	0.0092	0.0135	0.0174
B	0.0025	0.0043	0.0086	0.0126	0.0169

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				
	X2	X4	X8	X12	X16
A → Y ↑	0.1012	0.1141	0.1152	0.1135	0.1099
A → Y ↓	0.1139	0.1243	0.1170	0.1213	0.1146
B → Y ↑	0.1467	0.1494	0.1289	0.1266	0.1328
B → Y ↓	0.1672	0.1688	0.1426	0.1442	0.1441

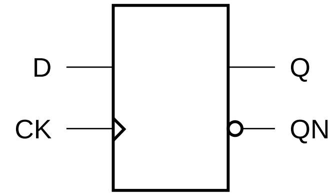
Delays at 25°C,1.2V, Typical Process (Cont'd.)

Description	K_{load} (ns/pF)				
	X2	X4	X8	X12	X16
A → Y ↑	3.5367	1.7915	0.9072	0.6259	0.4872
A → Y ↓	3.5327	1.9517	0.9255	0.6068	0.4611
B → Y ↑	3.5359	1.7953	0.9079	0.6261	0.4880
B → Y ↓	3.5329	1.9518	0.9255	0.6067	0.4610

Cell Description

The DFF cell is a positive-edge triggered, static D-type flipflop.

Logic Symbol



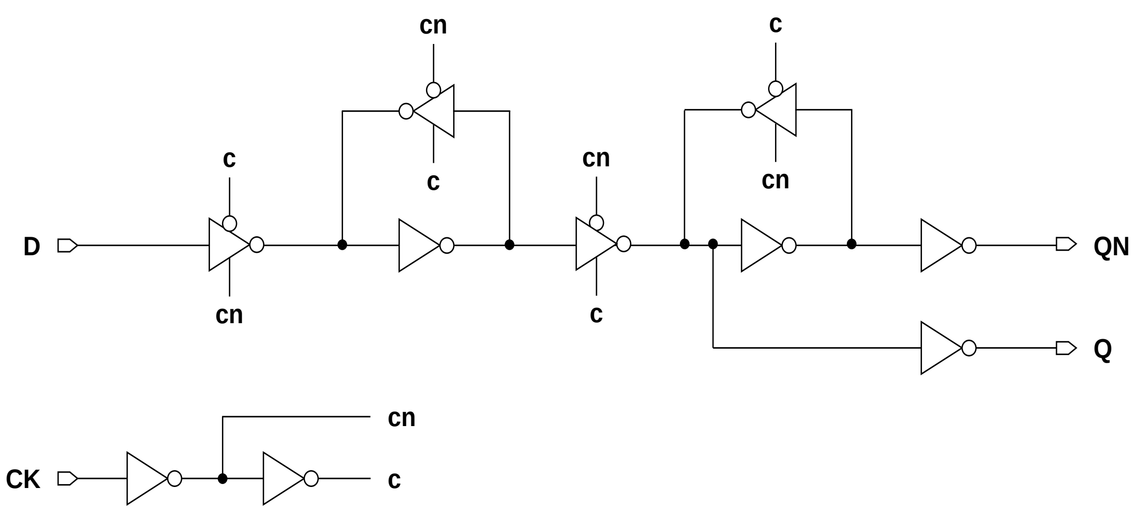
Function Table

D	CK	Q[n+1]	QN[n+1]
0	↑	0	1
1	↑	1	0
x	↓	Q[n]	QN[n]

Cell Size

Drive Strength	Height (um)	Width (um)
DFFX1M	2.87	8.20
DFFX2M	2.87	8.61
DFFX4M	2.87	9.43

Functional Schematic



AC Power

Pin	Power (uW/MHz)		
	X1	X2	X4
D	0.0082	0.0084	0.0091
CK	0.0171	0.0182	0.0201
Q	0.0123	0.0165	0.0280

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
D	0.0013	0.0013	0.0013
CK	0.0019	0.0020	0.0025

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
CK → Q ↑	0.1786	0.1880	0.1966	5.2137	3.4982	1.8991
CK → Q ↓	0.2184	0.2120	0.2218	3.5852	1.9404	0.9573
CK → QN ↑	0.2686	0.2754	0.3033	5.0997	3.4553	1.8532
CK → QN ↓	0.2661	0.2946	0.2968	3.2309	1.8242	0.8617

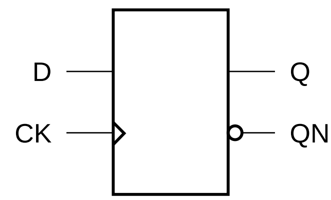
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)		
		X1	X2	X4
D	setup ↑ → CK	0.0430	0.0430	0.0469
	setup ↓ → CK	0.0977	0.0977	0.1055
	hold ↑ → CK	-0.0273	-0.0273	-0.0273
	hold ↓ → CK	-0.0430	-0.0352	-0.0430
CK	minpwh	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974

Cell Description

The DFFH cell is a positive-edge triggered, static D-type flipflop and fast clock-to-Q-path.

Logic Symbol



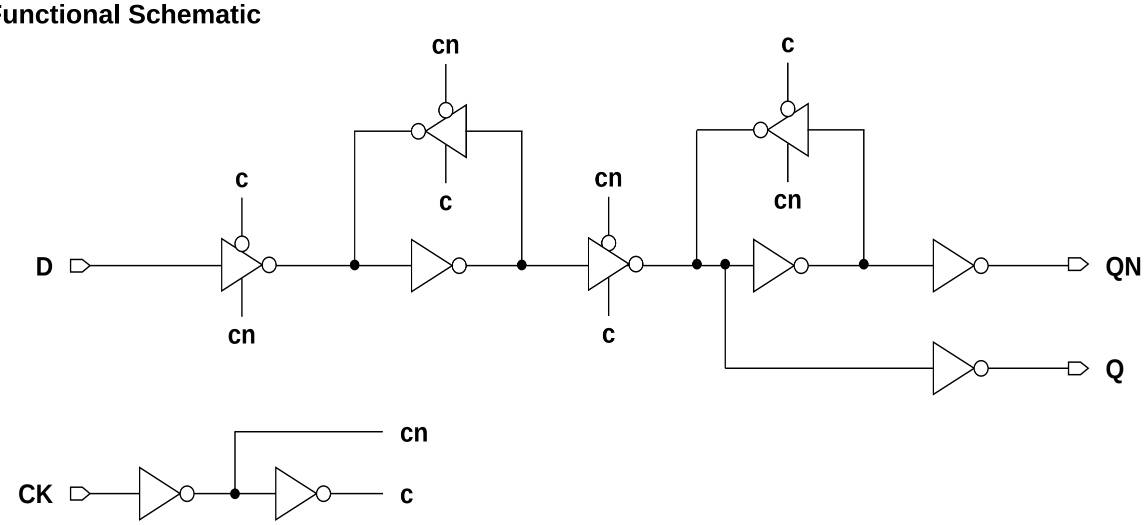
Function Table

D	CK	Q[n+1]	QN[n+1]
0	_/‾	0	1
1	_/‾	1	0
x	‾_	Q[n]	QN[n]

Cell Size

Drive Strength	Height (um)	Width (um)
DFFHX1M	2.87	10.25
DFFHX2M	2.87	10.66
DFFHX4M	2.87	11.07
DFFHX8M	2.87	11.89

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	X1	X2	X4	X8
D	0.0121	0.0155	0.0190	0.0190
CK	0.0278	0.0317	0.0368	0.0367
Q	0.0099	0.0116	0.0164	0.0273

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
D	0.0013	0.0020	0.0029	0.0029
CK	0.0029	0.0030	0.0037	0.0037

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
CK → Q ↑	0.1389	0.1378	0.1359	0.1540	5.2106	4.1158	1.7911	0.9482
CK → Q ↓	0.1343	0.1337	0.1286	0.1571	3.2727	1.7467	0.8807	0.4672
CK → QN ↑	0.1790	0.1842	0.1868	0.2204	8.8367	13.0314	13.0266	13.0828
CK → QN ↓	0.2105	0.2048	0.2200	0.2456	5.2027	5.0938	5.0707	5.0421

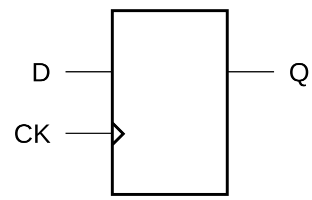
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
D	setup ↑ → CK	0.1055	0.0938	0.0703	0.0742
	setup ↓ → CK	0.1172	0.0898	0.0781	0.0742
	hold ↑ → CK	-0.0312	-0.0234	-0.0039	-0.0039
	hold ↓ → CK	-0.0625	-0.0469	-0.0352	-0.0352
CK	minpwh	0.9974	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974	0.9974

Cell Description

The DFFHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop. The cell has a single output (Q) and fast clock-to-out path.

Logic Symbol



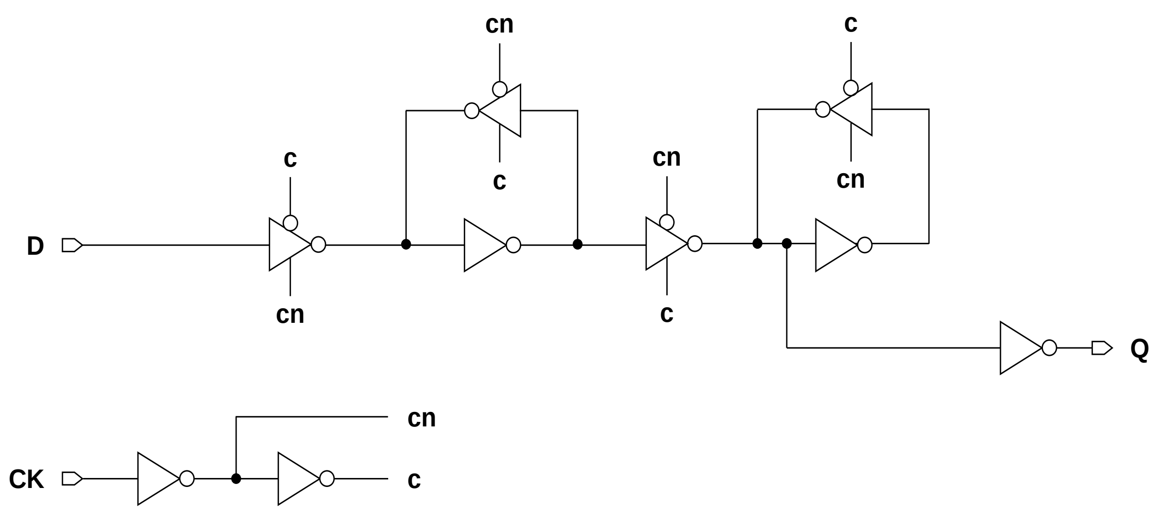
Function Table

D	CK	Q[n+1]
0	_/‾	0
1	_/‾	1
x	‾_	Q[n]

Cell Size

Drive Strength	Height (um)	Width (um)
DFFHQX1M	2.87	9.02
DFFHQX2M	2.87	9.02
DFFHQX4M	2.87	10.25
DFFHQX8M	2.87	11.89

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	X1	X2	X4	X8
D	0.0120	0.0143	0.0176	0.0184
CK	0.0266	0.0306	0.0376	0.0399
Q	0.0089	0.0103	0.0137	0.0246

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
D	0.0019	0.0015	0.0019	0.0029
CK	0.0028	0.0029	0.0037	0.0048

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
CK → Q ↑	0.1435	0.1400	0.1280	0.1307	5.2062	3.6573	1.8856	0.9763
CK → Q ↓	0.1540	0.1437	0.1278	0.1420	3.3639	1.8415	0.9076	0.4573

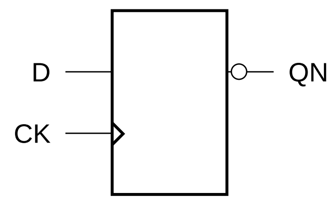
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
D	setup ↑ → CK	0.1133	0.1016	0.0938	0.1016
	setup ↓ → CK	0.0703	0.1094	0.0938	0.0781
	hold ↑ → CK	-0.0430	-0.0312	-0.0234	-0.0234
	hold ↓ → CK	-0.0312	-0.0625	-0.0508	-0.0312
CK	minpwh	0.9974	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974	0.9974

Cell Description

The DFFHQN cell is a high-speed, positive-edge triggered, static D-type flip-flop. The cell has a single output (tabpinout_) and fast clock-to-out path.

Logic Symbol



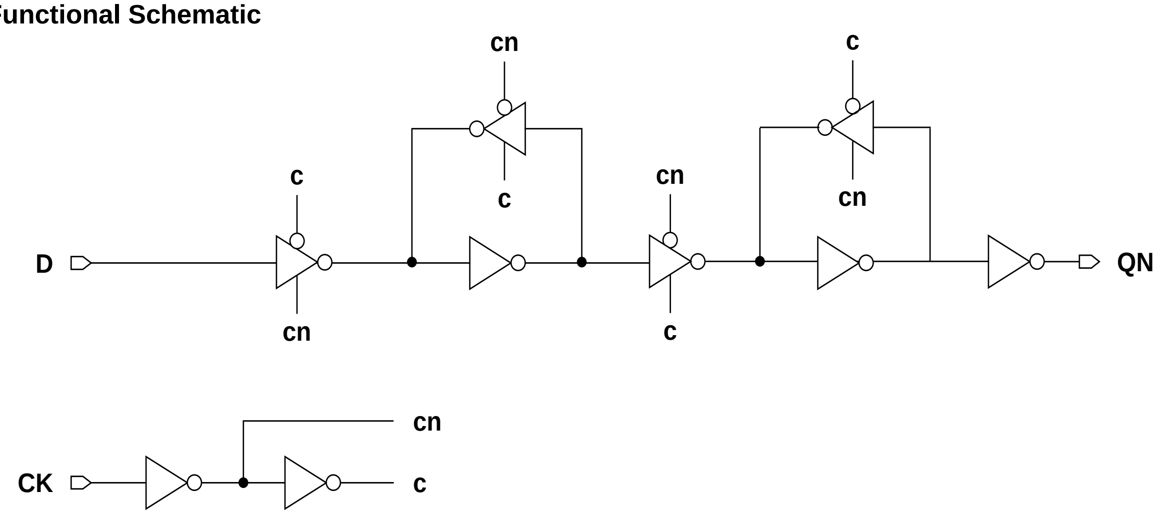
Function Table

D	CK	QN[n+1]
0	↑	1
1	↑	0
x	↓	QN[n]

Cell Size

Drive Strength	Height (um)	Width (um)
DFFHQNX1M	2.87	9.02
DFFHQNX2M	2.87	9.02
DFFHQNX4M	2.87	9.43
DFFHQNX8M	2.87	10.25

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	X1	X2	X4	X8
D	0.0099	0.0100	0.0100	0.0098
CK	0.0221	0.0221	0.0222	0.0219
QN	0.0087	0.0105	0.0157	0.0259

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
D	0.0015	0.0015	0.0015	0.0014
CK	0.0027	0.0028	0.0028	0.0029

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
CK → QN ↑	0.2007	0.2014	0.2192	0.2482	5.1217	3.5976	1.8773	1.0118
CK → QN ↓	0.2329	0.2317	0.2275	0.2501	3.2568	1.8201	0.8716	0.4497

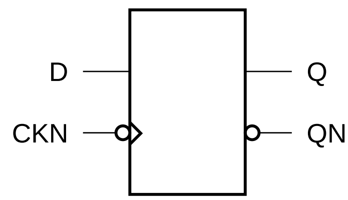
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
D	setup ↑ → CK	0.1367	0.1445	0.1445	0.1445
	setup ↓ → CK	0.1211	0.1289	0.1211	0.1172
	hold ↑ → CK	-0.0586	-0.0625	-0.0625	-0.0625
	hold ↓ → CK	-0.0703	-0.0703	-0.0703	-0.0664
CK	minpwh	0.9974	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974	0.9974

Cell Description

The DFFNH cell is a negative-edge triggered, static D-type flip-flop and fast clock-to-Q-path.

Logic Symbol



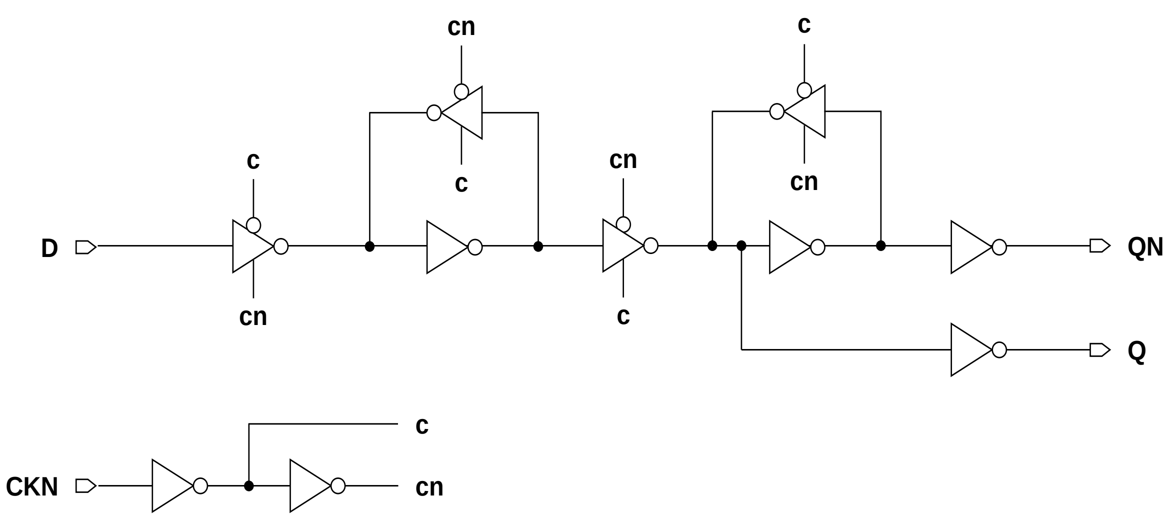
Function Table

D	CKN	Q[n+1]	QN[n+1]
0	↓	0	1
1	↓	1	0
x	↑	Q[n]	QN[n]

Cell Size

Drive Strength	Height (um)	Width (um)
DFFNHX1M	2.87	11.07
DFFNHX2M	2.87	11.48
DFFNHX4M	2.87	14.35
DFFNHX8M	2.87	15.58

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	X1	**X2**	**X4**	**X8**
D	0.0118	0.0142	0.0234	0.0281
CKN	0.0211	0.0245	0.0357	0.0397
Q	0.0133	0.0157	0.0227	0.0330

Pin Capacitance

Pin	Capacitance (pF)			
	X1	**X2**	**X4**	**X8**
D	0.0014	0.0018	0.0029	0.0050
CKN	0.0028	0.0028	0.0031	0.0045

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	**X2**	**X4**	**X8**	**X1**	**X2**	**X4**	**X8**
CKN → Q ↑	0.3264	0.3482	0.3418	0.3182	5.2607	4.1785	1.8530	0.9385
CKN → Q ↓	0.2648	0.2571	0.2445	0.2451	3.4531	1.8164	0.8880	0.4523
CKN → QN ↑	0.3151	0.3075	0.2952	0.3049	8.8441	8.9090	8.7318	8.8517
CKN → QN ↓	0.4037	0.4258	0.4263	0.4104	5.2097	5.1300	5.0534	5.0212

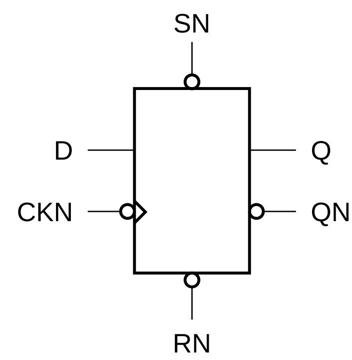
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	**X2**	**X4**	**X8**
D	setup ↑ → CKN	-0.0469	-0.0781	-0.0781	-0.0352
	setup ↓ → CKN	0.0195	-0.0078	-0.0156	-0.0039
	hold ↑ → CKN	0.1016	0.1133	0.1094	0.0898
	hold ↓ → CKN	0.0195	0.0352	0.0508	0.0391
CKN	minpwl	0.9974	0.9974	0.9974	0.9974
	minpwh	0.9974	0.9974	0.9974	0.9974

Cell Description

The DFFNSRH cell is a negative-edge triggered, static D-type flip-flop with asynchronous active-low reset (RN) and set (SN), and set dominating reset, and fast clock-to-Q-path.

Logic Symbol



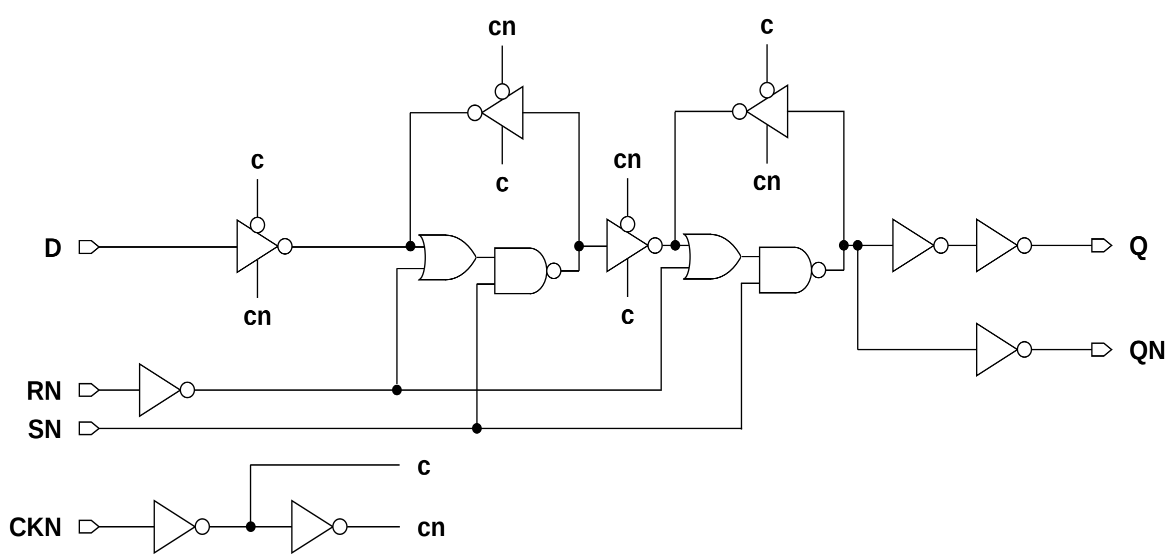
Function Table

RN	SN	D	CKN	Q[n+1]	QN[n+1]
0	1	x	x	0	1
1	0	x	x	1	0
0	0	x	x	1	0
1	1	0	↓	0	1
1	1	1	↓	1	0
1	1	x	↑	Q[n]	QN[n]

Cell Size

Drive Strength	Height (um)	Width (um)
DFFNSRHX1M	2.87	14.35
DFFNSRHX2M	2.87	14.35
DFFNSRHX4M	2.87	14.76
DFFNSRHX8M	2.87	16.40

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	X1	**X2**	**X4**	**X8**
D	0.0153	0.0176	0.0221	0.0237
CKN	0.0224	0.0254	0.0301	0.0339
SN	0.0084	0.0086	0.0091	0.0100
RN	0.0023	0.0024	0.0025	0.0026
Q	0.0153	0.0170	0.0228	0.0383

Pin Capacitance

Pin	Capacitance (pF)			
	X1	**X2**	**X4**	**X8**
D	0.0016	0.0018	0.0029	0.0035
CKN	0.0028	0.0027	0.0030	0.0047
SN	0.0027	0.0030	0.0033	0.0038
RN	0.0030	0.0032	0.0037	0.0037

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	**X2**	**X4**	**X8**	**X1**	**X2**	**X4**	**X8**
CKN → Q ↑	0.3266	0.3332	0.3313	0.2962	5.1245	3.5172	1.8082	0.9451
CKN → Q ↓	0.2554	0.2418	0.2537	0.2783	3.6775	1.8965	0.9490	0.5055
SN → Q ↑	0.2533	0.2805	0.2961	0.2635	5.0427	3.5026	1.8123	0.9311
SN → Q ↓	0.3673	0.3666	0.3870	0.4680	3.9219	2.0918	1.0712	0.5696
RN → Q ↓	0.3099	0.3123	0.3313	0.4204	3.9475	2.0914	1.0710	0.5717
CKN → QN ↑	0.3081	0.2944	0.3148	0.3480	8.8993	8.8926	8.8934	8.8734
CKN → QN ↓	0.4177	0.4240	0.4316	0.4058	5.1635	5.0641	5.0150	4.9896
SN → QN ↑	0.4167	0.4109	0.4428	0.5445	8.8961	8.8890	8.8903	8.8702
SN → QN ↓	0.3416	0.3686	0.3956	0.3676	5.1552	5.0629	5.0147	4.9894
RN → QN ↑	0.3596	0.3563	0.3867	0.4976	8.8959	8.8893	8.8912	8.8701

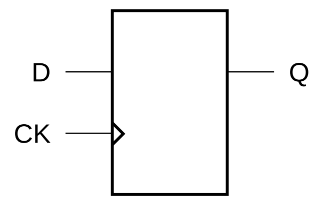
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
D	setup ↑ → CKN	0.0430	0.0000	-0.0312	0.0273
	setup ↓ → CKN	0.0938	0.0586	0.0312	0.0352
	hold ↑ → CKN	0.0586	0.0742	0.0938	0.0547
	hold ↓ → CKN	-0.0078	0.0039	0.0234	0.0195
CKN	minpwl	0.9974	0.9974	0.9974	0.9974
	minpwh	0.9974	0.9974	0.9974	0.9974
SN	minpwl	0.9974	0.9974	0.9974	0.9974
	recovery	0.0430	0.0117	-0.0039	0.0156
	removal	0.0195	0.0312	0.0391	0.0234
RN	minpwl	0.9974	0.9974	0.9974	0.9974
	recovery	-0.1602	-0.1836	-0.1875	-0.1328
	removal	0.2188	0.2500	0.2617	0.1914

Cell Description

The DFFQ cell is a positive-edge triggered, static D-type flip-flop. The cell has a single output (Q) .

Logic Symbol



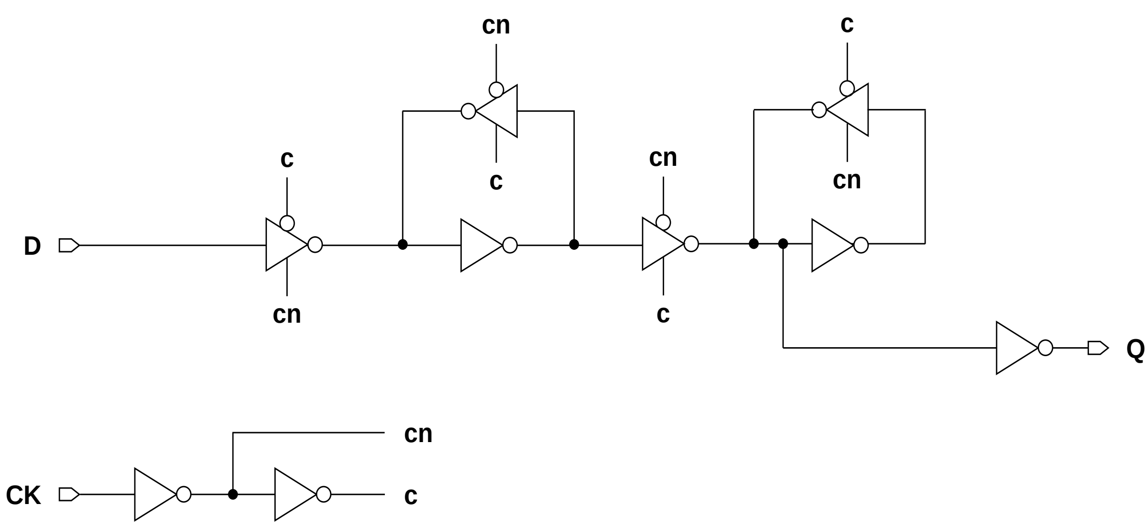
Function Table

D	CK	Q[n+1]
0	↑	0
1	↑	1
x	↓	Q[n]

Cell Size

Drive Strength	Height (um)	Width (um)
DFFQX1M	2.87	7.38
DFFQX2M	2.87	7.38
DFFQX4M	2.87	7.79

Functional Schematic



AC Power

Pin	Power (uW/MHz)		
	X1	X2	X4
D	0.0084	0.0086	0.0088
CK	0.0175	0.0180	0.0186
Q	0.0093	0.0111	0.0174

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
D	0.0014	0.0014	0.0013
CK	0.0019	0.0020	0.0022

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
CK → Q ↑	0.1811	0.1866	0.1943	5.1974	3.5261	1.8277
CK → Q ↓	0.2227	0.2084	0.2190	3.6248	1.9062	0.9830

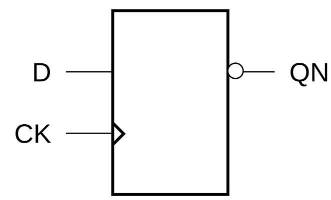
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)		
		X1	X2	X4
D	setup ↑ → CK	0.0469	0.0508	0.0508
	setup ↓ → CK	0.1055	0.1094	0.1094
	hold ↑ → CK	-0.0312	-0.0312	-0.0312
	hold ↓ → CK	-0.0469	-0.0469	-0.0508
CK	minpwh	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974

Cell Description

The DFFQN cell is a positive-edge triggered, static D-type flip-flop. The cell has a single output (QN) .

Logic Symbol



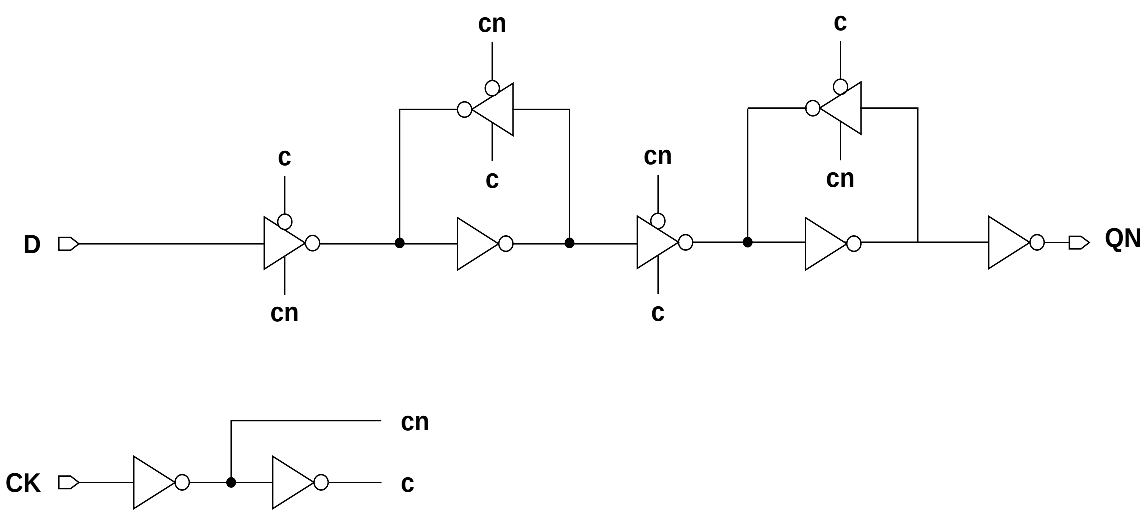
Function Table

D	CK	QN[n+1]
0	↑	1
1	↑	0
x	↓	QN[n]

Cell Size

Drive Strength	Height (um)	Width (um)
DFFQNX1M	2.87	7.38
DFFQNX2M	2.87	7.38
DFFQNX4M	2.87	7.79

Functional Schematic



AC Power

Pin	Power (uW/MHz)		
	X1	X2	X4
D	0.0077	0.0076	0.0076
CK	0.0170	0.0170	0.0169
QN	0.0089	0.0106	0.0154

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
D	0.0012	0.0012	0.0012
CK	0.0020	0.0020	0.0020

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
CK → QN ↑	0.2567	0.2677	0.2877	5.1425	3.4922	1.8096
CK → QN ↓	0.2579	0.2742	0.2686	3.2289	1.7869	0.8977

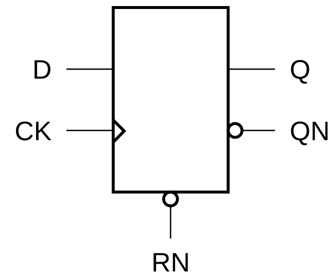
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)		
		X1	X2	X4
D	setup ↑ → CK	0.0391	0.0352	0.0352
	setup ↓ → CK	0.1211	0.1211	0.1172
	hold ↑ → CK	-0.0234	-0.0234	-0.0234
	hold ↓ → CK	-0.0781	-0.0781	-0.0781
CK	minpwh	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974

Cell Description

The DFFR cell is a positive-edge triggered, static D-type flip-flop with asynchronous active-low reset (RN).

Logic Symbol



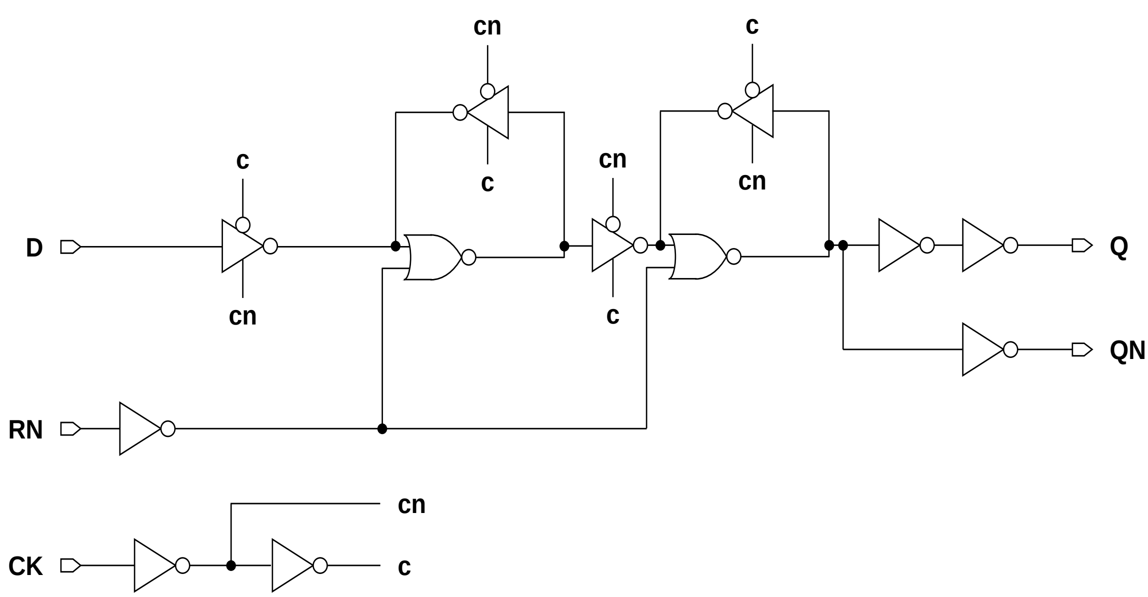
Function Table

RN	D	CK	Q[n+1]	QN[n+1]
0	x	x	0	1
1	0	↑	0	1
1	1	↑	1	0
1	x	↓	Q[n]	QN[n]

Cell Size

Drive Strength	Height (um)	Width (um)
DFFRX1M	2.87	9.43
DFFRX2M	2.87	9.84
DFFRX4M	2.87	10.66

Functional Schematic



AC Power

Pin	Power (uW/MHz)		
	X1	X2	X4
D	0.0099	0.0099	0.0099
CK	0.0200	0.0195	0.0196
RN	0.0020	0.0021	0.0023
Q	0.0130	0.0162	0.0293

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
D	0.0013	0.0013	0.0013
CK	0.0020	0.0020	0.0020
RN	0.0039	0.0041	0.0045

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
CK → Q ↑	0.2771	0.2822	0.3501	5.3480	3.5679	1.8530
CK → Q ↓	0.2901	0.3110	0.3860	3.2748	1.8011	0.9294
RN → Q ↓	0.1301	0.1417	0.1696	3.2968	1.7911	0.9236
CK → QN ↑	0.1883	0.1957	0.2263	5.3609	3.5898	1.9196
CK → QN ↓	0.1845	0.1988	0.2458	3.6701	2.0235	1.1296
RN → QN ↑	0.2529	0.2637	0.3306	5.3193	3.5704	1.9057

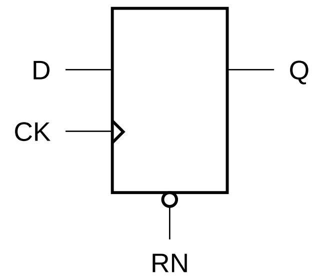
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)		
		X1	X2	X4
D	setup ↑ → CK	0.1211	0.1133	0.1133
	setup ↓ → CK	0.0938	0.0938	0.0938
	hold ↑ → CK	-0.0703	-0.0664	-0.0664
	hold ↓ → CK	-0.0156	-0.0156	-0.0156
CK	minpwh	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974
RN	minpwl	0.9974	0.9974	0.9974
	recovery	0.1172	0.1133	0.1133
	removal	-0.0938	-0.0938	-0.0977

Cell Description

The DFFRHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with asynchronous active-low reset (RN). The cell has a single output (Q) and fast clock-to-out path.

Logic Symbol



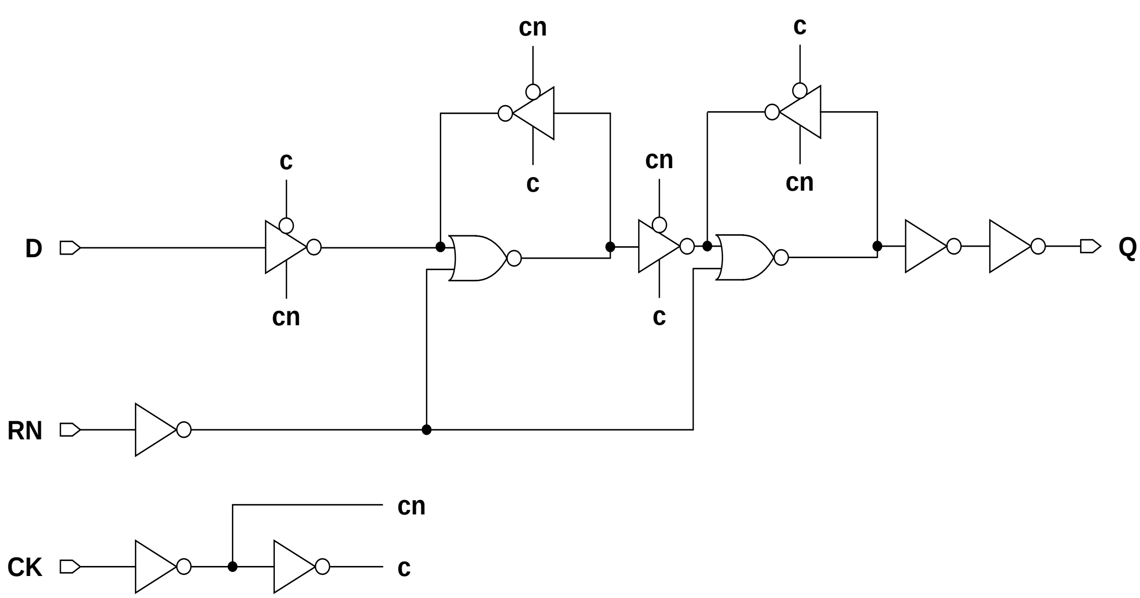
Function Table

RN	D	CK	Q[n+1]
0	x	x	0
1	0	↑	0
1	1	↑	1
1	x	↓	Q[n]

Cell Size

Drive Strength	Height (um)	Width (um)
DFFRHQX1M	2.87	10.25
DFFRHQX2M	2.87	11.07
DFFRHQX4M	2.87	11.48
DFFRHQX8M	2.87	12.30

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	X1	X2	X4	X8
D	0.0128	0.0154	0.0172	0.0172
CK	0.0276	0.0322	0.0323	0.0323
RN	0.0021	0.0029	0.0026	0.0026
Q	0.0090	0.0105	0.0146	0.0274

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
D	0.0019	0.0015	0.0029	0.0030
CK	0.0027	0.0028	0.0031	0.0031
RN	0.0028	0.0035	0.0030	0.0030

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
CK → Q ↑	0.1591	0.1458	0.1599	0.1925	5.2081	3.5645	1.8533	1.0065
CK → Q ↓	0.1494	0.1409	0.1552	0.1877	3.3385	1.7216	0.9198	0.4793
RN → Q ↓	0.1479	0.1537	0.1702	0.2349	3.3653	1.7907	0.9482	0.5001

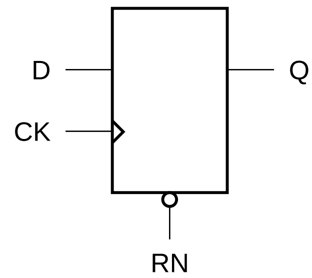
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
D	setup ↑ → CK	0.1211	0.0977	0.0977	0.0977
	setup ↓ → CK	0.0781	0.1172	0.0938	0.0898
	hold ↑ → CK	-0.0352	-0.0312	-0.0195	-0.0195
	hold ↓ → CK	-0.0273	-0.0664	-0.0391	-0.0352
CK	minpwh	0.9974	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974	0.9974
RN	minpwl	0.9974	0.9974	0.9974	0.9974
	recovery	-0.0430	-0.0391	-0.0312	-0.0312
	removal	0.0781	0.0781	0.0664	0.0625

Cell Description

The DFFRQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with asynchronous active-low reset (RN). The cell has a single output (Q).

Logic Symbol



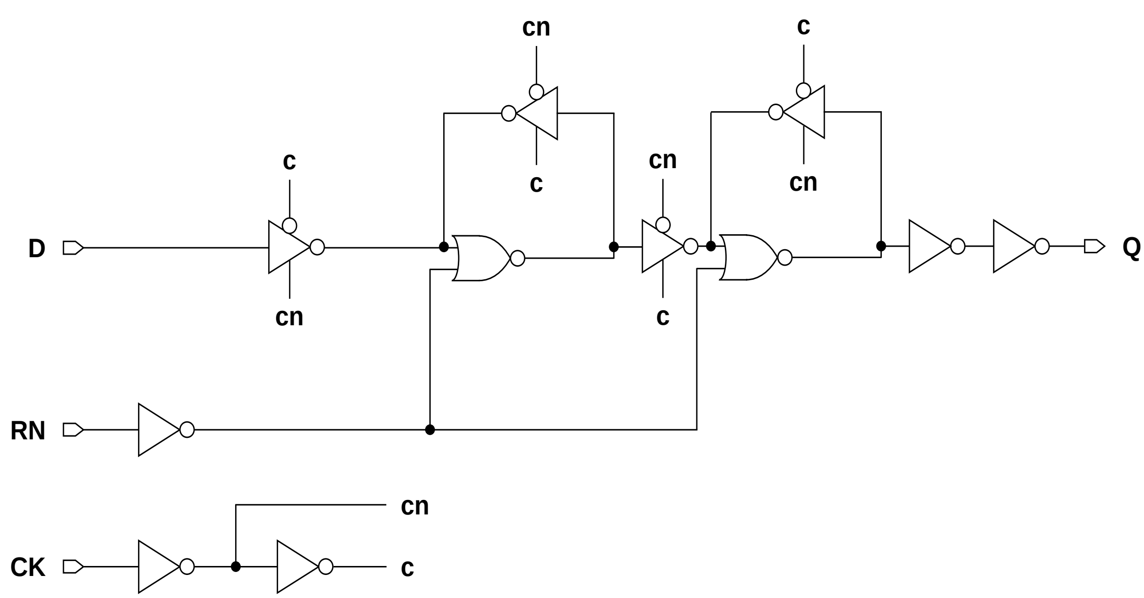
Function Table

RN	D	CK	Q[n+1]
0	x	x	0
1	0	↑	0
1	1	↑	1
1	x	↓	Q[n]

Cell Size

Drive Strength	Height (um)	Width (um)
DFFRQX1M	2.87	9.02
DFFRQX2M	2.87	9.02
DFFRQX4M	2.87	9.43

Functional Schematic



AC Power

Pin	Power (uW/MHz)		
	X1	X2	X4
D	0.0099	0.0101	0.0101
CK	0.0183	0.0183	0.0183
RN	0.0024	0.0022	0.0025
Q	0.0096	0.0109	0.0172

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
D	0.0013	0.0013	0.0013
CK	0.0015	0.0016	0.0016
RN	0.0049	0.0050	0.0055

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
CK → Q ↑	0.2386	0.2141	0.2218	5.1553	3.5632	1.8039
CK → Q ↓	0.2913	0.2834	0.3090	3.2459	1.7772	0.9091
RN → Q ↓	0.1242	0.1448	0.1704	3.2525	1.7726	0.8932

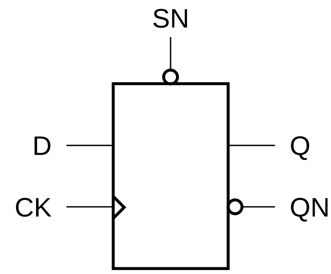
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)		
		X1	X2	X4
D	setup ↑ → CK	0.1680	0.1562	0.1562
	setup ↓ → CK	0.0742	0.0898	0.0898
	hold ↑ → CK	-0.0898	-0.0859	-0.0859
	hold ↓ → CK	-0.0078	-0.0195	-0.0156
CK	minpwh	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974
RN	minpwl	0.9974	0.9974	0.9974
	recovery	0.1719	0.1602	0.1602
	removal	-0.1172	-0.1133	-0.1133

Cell Description

The DFFS cell is a positive-edge triggered, static D-type flip-flop with asynchronous active-low set (SN).

Logic Symbol



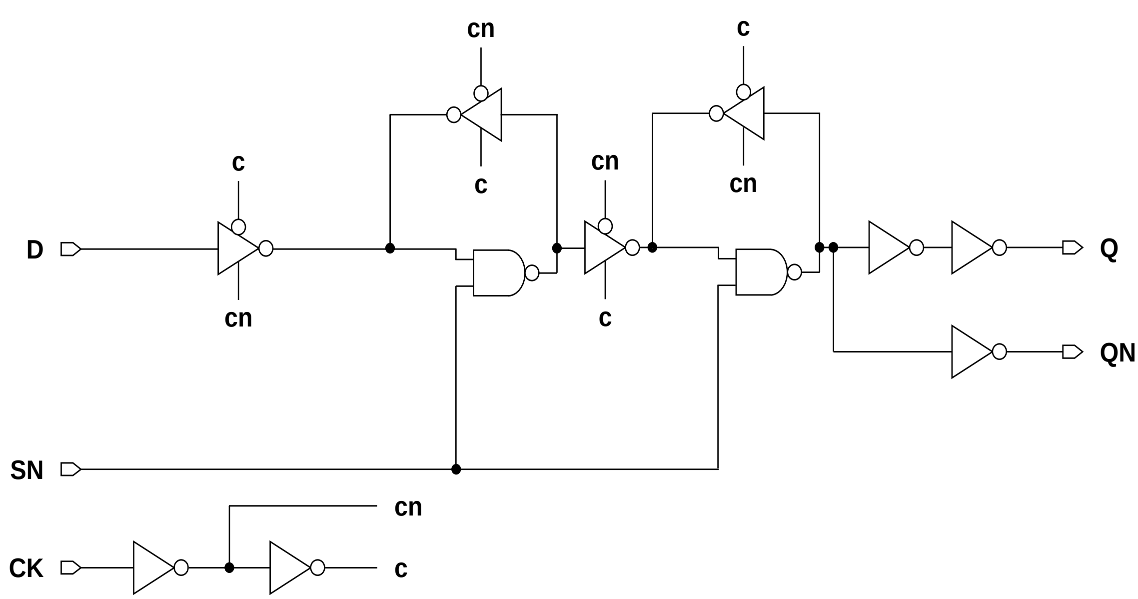
Function Table

SN	D	CK	Q[n+1]	QN[n+1]
0	x	x	1	0
1	0	↑	0	1
1	1	↑	1	0
1	x	↓	Q[n]	QN[n]

Cell Size

Drive Strength	Height (um)	Width (um)
DFFSX1M	2.87	9.43
DFFSX2M	2.87	9.84
DFFSX4M	2.87	10.66

Functional Schematic



AC Power

Pin	Power (uW/MHz)		
	X1	X2	X4
D	0.0095	0.0106	0.0121
CK	0.0200	0.0215	0.0232
SN	0.0022	0.0023	0.0023
Q	0.0130	0.0160	0.0264

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
D	0.0017	0.0017	0.0017
CK	0.0023	0.0024	0.0027
SN	0.0026	0.0030	0.0038

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
CK → Q ↑	0.2729	0.2684	0.2842	5.3566	3.5746	1.8431
CK → Q ↓	0.2894	0.2868	0.2997	3.2790	1.8035	0.9130
SN → Q ↑	0.2424	0.2655	0.3729	5.3459	3.5718	1.8420
CK → QN ↑	0.1930	0.1766	0.1653	5.4874	3.6159	1.8673
CK → QN ↓	0.2106	0.1986	0.1883	3.9839	2.0658	0.9948
SN → QN ↓	0.1848	0.1995	0.2684	3.4232	1.8782	1.0087

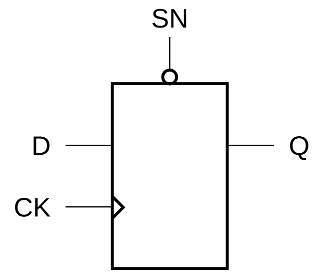
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)		
		X1	X2	X4
D	setup ↑ → CK	0.0820	0.0898	0.0742
	setup ↓ → CK	0.0547	0.0547	0.0586
	hold ↑ → CK	-0.0352	-0.0430	-0.0430
	hold ↓ → CK	-0.0039	-0.0078	-0.0156
CK	minpwh	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974
SN	minpwl	0.9974	0.9974	0.9974
	recovery	-0.0508	-0.0547	-0.0586
	removal	0.0781	0.0820	0.0820

Cell Description

The DFFSHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with asynchronous active-low set (SN). The cell has a single output (Q) and fast clock-to-out path.

Logic Symbol



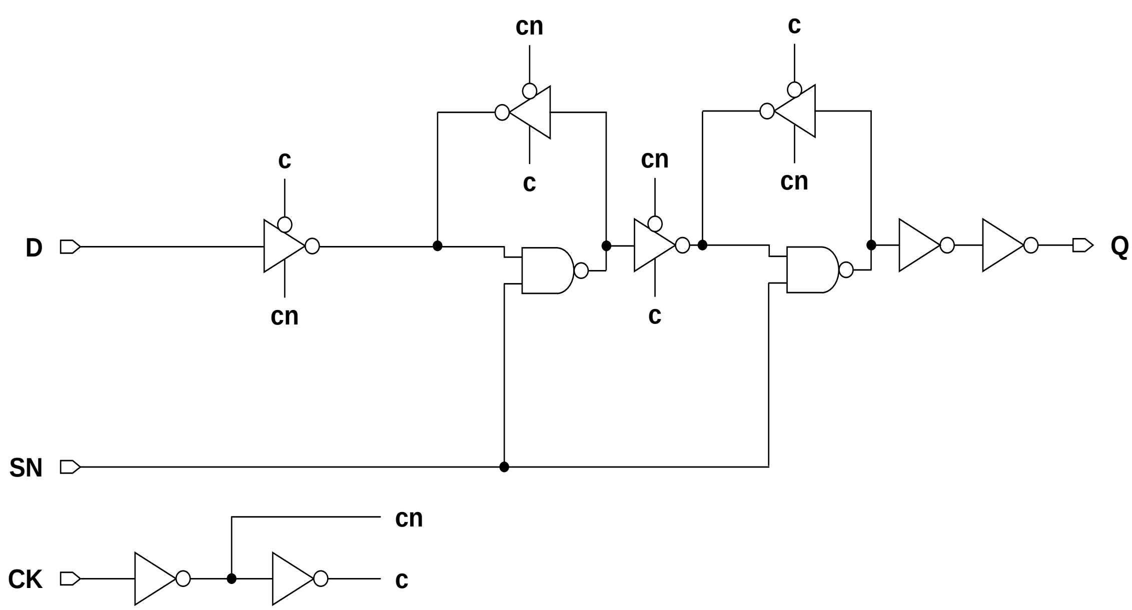
Function Table

SN	D	CK	Q[n+1]
0	x	x	1
1	0	↑	0
1	1	↑	1
1	x	↓	Q[n]

Cell Size

Drive Strength	Height (um)	Width (um)
DFFSHQX1M	2.87	11.48
DFFSHQX2M	2.87	11.48
DFFSHQX4M	2.87	12.30
DFFSHQX8M	2.87	13.53

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	X1	X2	X4	X8
D	0.0143	0.0162	0.0201	0.0202
CK	0.0262	0.0294	0.0368	0.0368
SN	0.0062	0.0062	0.0070	0.0071
Q	0.0107	0.0125	0.0180	0.0299

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
D	0.0017	0.0017	0.0019	0.0019
CK	0.0029	0.0030	0.0038	0.0038
SN	0.0030	0.0033	0.0038	0.0038

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
CK → Q ↑	0.1515	0.1421	0.1352	0.1537	5.1222	3.5837	1.8206	0.9462
CK → Q ↓	0.1622	0.1661	0.1427	0.1691	3.4130	1.8792	0.9040	0.4637
SN → Q ↑	0.2813	0.2967	0.2760	0.3065	5.4702	3.7246	1.8251	0.9414

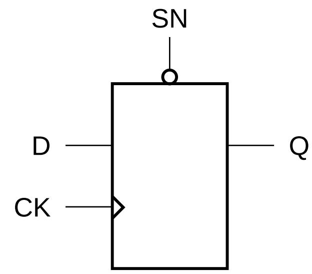
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
D	setup ↑ → CK	0.1602	0.1328	0.1328	0.1367
	setup ↓ → CK	0.1055	0.1055	0.1094	0.1055
	hold ↑ → CK	-0.0703	-0.0547	-0.0430	-0.0430
	hold ↓ → CK	-0.0430	-0.0469	-0.0469	-0.0430
CK	minpwh	0.9974	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974	0.9974
SN	minpwl	0.9974	0.9974	0.9974	0.9974
	recovery	0.0586	0.0508	0.0586	0.0547
	removal	-0.0312	-0.0312	-0.0312	-0.0312

Cell Description

The DFFSQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with asynchronous active-low set (SN). The cell has a single output (Q).

Logic Symbol



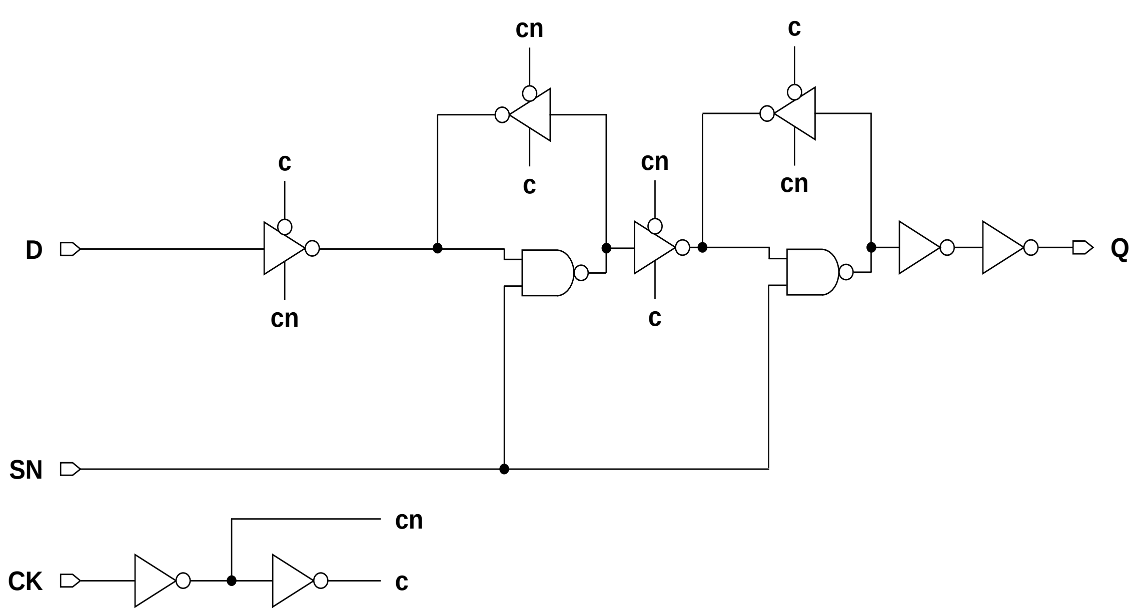
Function Table

SN	D	CK	Q[n+1]
0	x	x	1
1	0	↑	0
1	1	↑	1
1	x	↓	Q[n]

Cell Size

Drive Strength	Height (um)	Width (um)
DFFSQX1M	2.87	9.02
DFFSQX2M	2.87	9.02
DFFSQX4M	2.87	9.43

Functional Schematic



AC Power

Pin	Power (uW/MHz)		
	X1	**X2**	**X4**
D	0.0084	0.0084	0.0084
CK	0.0193	0.0190	0.0190
SN	0.0021	0.0021	0.0022
Q	0.0096	0.0111	0.0167

Pin Capacitance

Pin	Capacitance (pF)		
	X1	**X2**	**X4**
D	0.0013	0.0013	0.0012
CK	0.0024	0.0024	0.0024
SN	0.0025	0.0025	0.0025

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	**X2**	**X4**	**X1**	**X2**	**X4**
CK → Q ↑	0.2277	0.2338	0.2530	5.0617	3.4090	1.7745
CK → Q ↓	0.2617	0.2697	0.2855	3.2978	1.8198	0.9332
SN → Q ↑	0.1968	0.1990	0.2152	5.0545	3.4068	1.7733

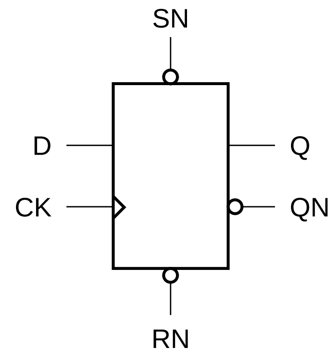
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)		
		X1	**X2**	**X4**
D	setup ↑ → CK	0.0898	0.0898	0.0859
	setup ↓ → CK	0.1406	0.1406	0.1406
	hold ↑ → CK	-0.0234	-0.0234	-0.0234
	hold ↓ → CK	-0.0586	-0.0625	-0.0625
CK	minpwh	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974
SN	minpwl	0.9974	0.9974	0.9974
	recovery	-0.0352	-0.0352	-0.0352
	removal	0.0625	0.0625	0.0625

Cell Description

The DFFSR cell is a positive-edge triggered, static D-type flip-flop with asynchronous active-low reset (RN) and set (SN), and set dominating reset.

Logic Symbol



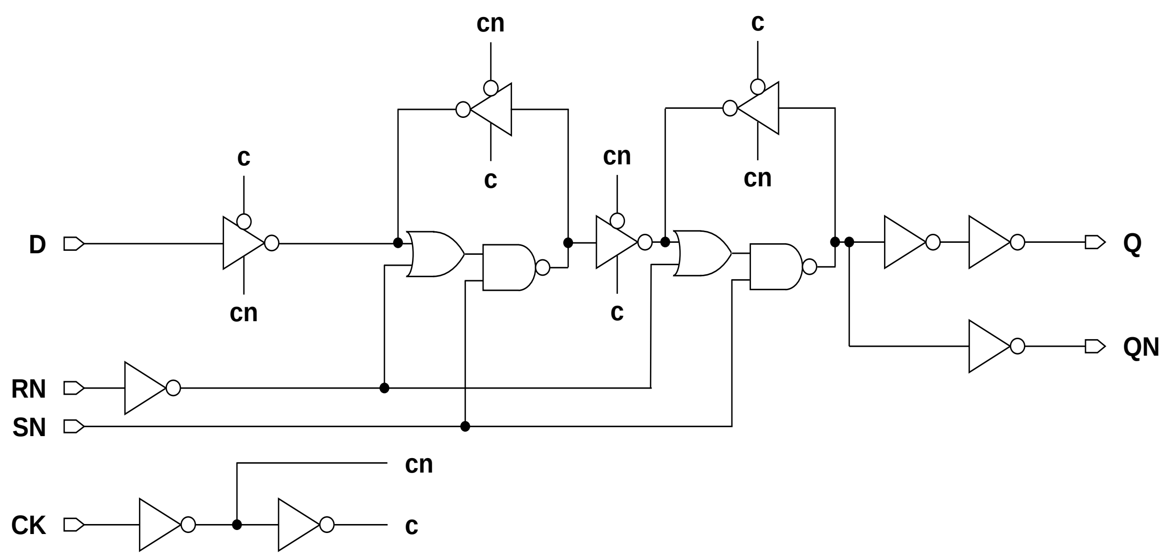
Function Table

RN	SN	D	CK	Q[n+1]	QN[n+1]
0	1	x	x	0	1
1	0	x	x	1	0
0	0	x	x	1	0
1	1	0	↑	0	1
1	1	1	↑	1	0
1	1	x	↓	Q[n]	QN[n]

Cell Size

Drive Strength	Height (um)	Width (um)
DFFSRX1M	2.87	13.53
DFFSRX2M	2.87	13.53
DFFSRX4M	2.87	14.35

Functional Schematic



AC Power

Pin	Power (uW/MHz)		
	X1	X2	X4
D	0.0111	0.0115	0.0117
CK	0.0227	0.0232	0.0244
SN	0.0026	0.0027	0.0028
RN	0.0050	0.0050	0.0051
Q	0.0154	0.0197	0.0341

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
D	0.0013	0.0013	0.0014
CK	0.0022	0.0023	0.0026
SN	0.0030	0.0034	0.0036
RN	0.0015	0.0015	0.0015

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
CK → Q ↑	0.2974	0.3243	0.3902	5.1509	3.4807	1.8589
CK → Q ↓	0.3057	0.3318	0.3671	3.3269	1.8212	0.9184
SN → Q ↑	0.2663	0.3194	0.4023	5.1177	3.4696	1.8533
SN → Q ↓	0.2592	0.2893	0.3265	3.3098	1.8134	0.9149
RN → Q ↓	0.3126	0.3506	0.3923	3.3158	1.8161	0.9155
CK → QN ↑	0.2075	0.2063	0.2141	5.3856	3.6033	1.9542
CK → QN ↓	0.2445	0.2382	0.2624	4.6571	2.3717	1.2554
SN → QN ↑	0.1625	0.1681	0.1744	5.2976	3.5630	1.9341
SN → QN ↓	0.2178	0.2408	0.2866	3.5699	1.9634	1.0340
RN → QN ↑	0.2149	0.2288	0.2413	5.3391	3.5811	1.9347

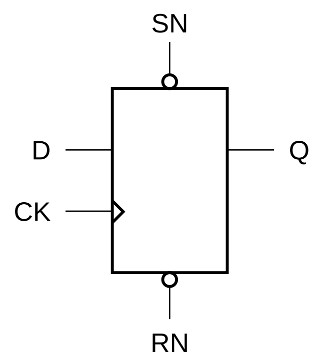
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)		
		X1	**X2**	**X4**
D	setup ↑ → CK	0.1953	0.1602	0.1523
	setup ↓ → CK	0.1797	0.1914	0.1445
	hold ↑ → CK	-0.0430	-0.0391	-0.0312
	hold ↓ → CK	-0.0664	-0.0742	-0.0391
CK	minpwh	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974
SN	minpwl	0.9974	0.9974	0.9974
	recovery	-0.0352	-0.0312	-0.0391
	removal	0.0742	0.0703	0.0781
RN	minpwl	0.9974	0.9974	0.9974
	recovery	0.1914	0.1484	0.1523
	removal	-0.0938	-0.0703	-0.0859

Cell Description

The DFFSRHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with asynchronous active-low reset (RN) and set (SN), and set dominating reset. The cell has a single output (Q) and fast clock-to-out path.

Logic Symbol



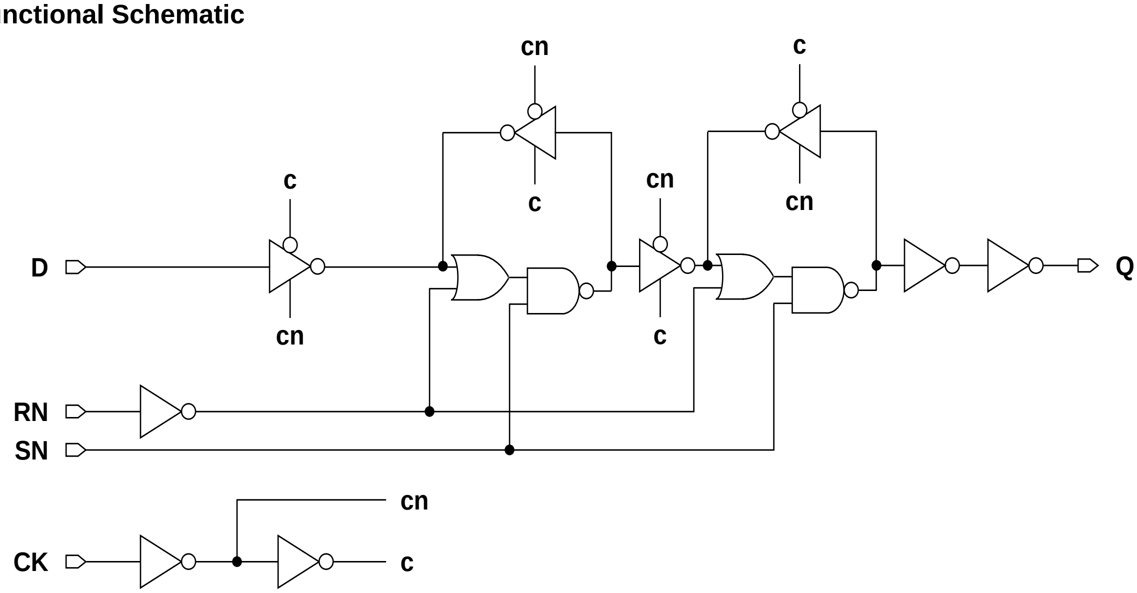
Function Table

RN	SN	D	CK	Q[n+1]
0	1	x	x	0
1	0	x	x	1
0	0	x	x	1
1	1	0	↑	0
1	1	1	↑	1
1	1	x	↓	Q[n]

Cell Size

Drive Strength	Height (um)	Width (um)
DFFSRHQX1M	2.87	13.53
DFFSRHQX2M	2.87	13.53
DFFSRHQX4M	2.87	14.35
DFFSRHQX8M	2.87	15.58

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	X1	X2	X4	X8
D	0.0145	0.0164	0.0200	0.0197
CK	0.0262	0.0291	0.0341	0.0340
SN	0.0082	0.0084	0.0088	0.0089
RN	0.0022	0.0023	0.0027	0.0027
Q	0.0118	0.0138	0.0199	0.0366

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
D	0.0014	0.0014	0.0025	0.0025
CK	0.0029	0.0028	0.0033	0.0033
SN	0.0029	0.0033	0.0035	0.0036
RN	0.0028	0.0031	0.0035	0.0035

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
CK → Q ↑	0.1792	0.1683	0.1658	0.1982	5.3389	3.5828	1.8386	0.9621
CK → Q ↓	0.1896	0.1746	0.1818	0.2202	3.7647	1.9328	1.0005	0.5320
SN → Q ↑	0.2475	0.2773	0.2709	0.3150	5.2242	3.5492	1.8189	0.9463
SN → Q ↓	0.3521	0.3688	0.3570	0.4633	3.8589	2.0788	1.0505	0.5609
RN → Q ↓	0.2992	0.3146	0.3021	0.4054	3.8801	2.0790	1.0549	0.5628

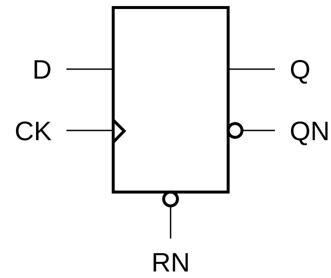
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
D	setup ↑ → CK	0.1719	0.1445	0.1133	0.1133
	setup ↓ → CK	0.1328	0.1211	0.0898	0.0859
	hold ↑ → CK	-0.0664	-0.0547	-0.0312	-0.0273
	hold ↓ → CK	-0.0508	-0.0508	-0.0234	-0.0195
CK	minpwh	0.9974	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974	0.9974
SN	minpwl	0.9974	0.9974	0.9974	0.9974
	recovery	0.0781	0.0625	0.0625	0.0586
	removal	-0.0391	-0.0312	-0.0312	-0.0352
RN	minpwl	0.9974	0.9974	0.9974	0.9974
	recovery	-0.0195	-0.0273	-0.0273	-0.0273
	removal	0.0586	0.0625	0.0703	0.0664

Cell Description

The DFFTR cell is a positive-edge triggered, static D-type flip-flop with synchronous active-low reset (RN).

Logic Symbol



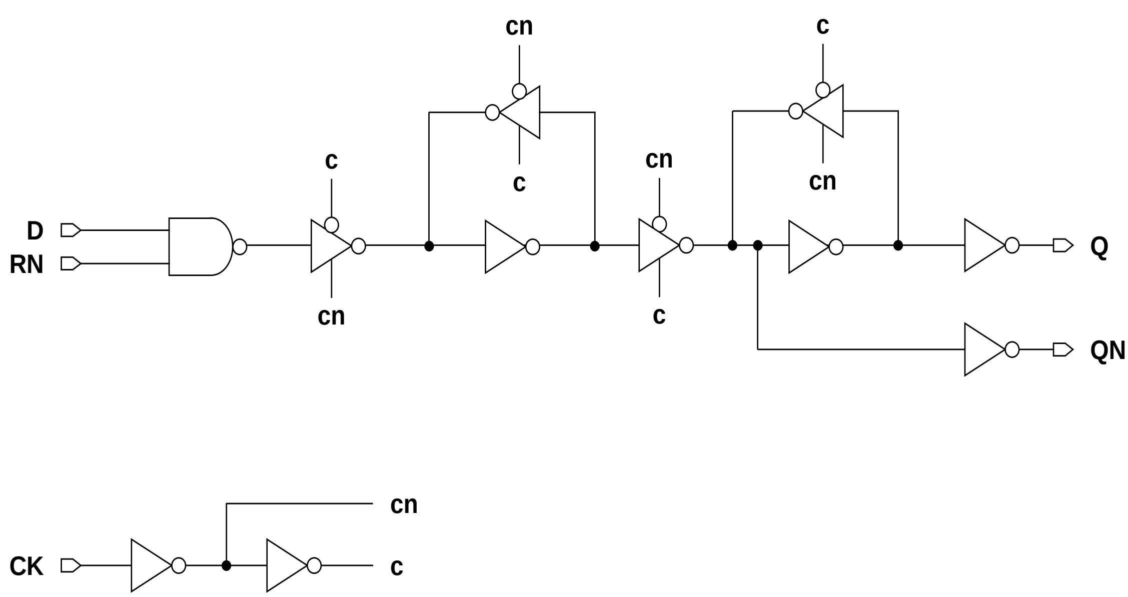
Function Table

RN	D	CK	Q[n+1]	QN[n+1]
0	x	_/¯	0	1
x	x	¯_	Q[n]	QN[n]
1	0	_/¯	0	1
1	1	_/¯	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
DFFTRX1M	2.87	9.02
DFFTRX2M	2.87	9.02
DFFTRX4M	2.87	12.30

Functional Schematic



AC Power

Pin	Power (uW/MHz)		
	X1	X2	X4
D	0.0100	0.0110	0.0193
CK	0.0201	0.0219	0.0338
RN	0.0104	0.0118	0.0223
Q	0.0125	0.0164	0.0269

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
D	0.0014	0.0016	0.0025
CK	0.0020	0.0022	0.0028
RN	0.0012	0.0014	0.0019

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
CK → Q ↑	0.2038	0.2128	0.1992	5.2875	3.5605	1.8393
CK → Q ↓	0.2572	0.2708	0.2630	3.2575	1.8009	0.8686
CK → QN ↑	0.1623	0.1603	0.1644	5.3192	3.5728	1.8460
CK → QN ↓	0.1503	0.1487	0.1337	3.4092	1.8551	0.8811

Timing Constraints at 25°C,1.2V, Typical Process

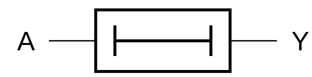
Pin	Requirement	Interval (ns)		
		X1	X2	X4
D	setup ↑ → CK	0.1211	0.1055	0.0898
	setup ↓ → CK	0.1953	0.1836	0.1484
	hold ↑ → CK	-0.0664	-0.0547	-0.0469
	hold ↓ → CK	-0.0859	-0.0742	-0.0547
CK	minpwh	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974
RN	setup ↑ → CK	0.1250	0.1094	0.0898
	setup ↓ → CK	0.1992	0.2109	0.2695
	hold ↑ → CK	-0.0664	-0.0547	-0.0469
	hold ↓ → CK	-0.0820	-0.0820	-0.1250

Cell Description

The DLY1 cell provides the logical delay of a single input (A). The output (Y) is represented by the logic equation:

$Y = A$

Logic Symbol



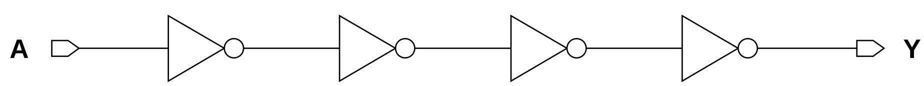
Function Table

A	Y
0	0
1	1

Cell Size

Drive Strength	Height (um)	Width (um)
DLY1X1M	2.87	3.28
DLY1X4M	2.87	4.10

Functional Schematic



AC Power

Pin	Power (uW/MHz)	
	X1	**X4**
A	0.0105	0.0179

Pin Capacitance

Pin	Capacitance (pF)	
	X1	**X4**
A	0.0015	0.0026

Delays at 25°C,1.2V, Typical Process

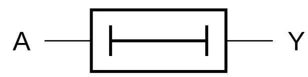
Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X1	**X4**	**X1**	**X4**
A → Y ↑	0.1398	0.1111	6.3525	1.8143
A → Y ↓	0.1788	0.1551	3.7620	1.8725

Cell Description

The DLY2 cell provides the logical delay of a single input (A). The output (Y) is represented by the logic equation:

$Y = A$

Logic Symbol



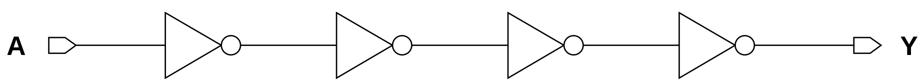
Function Table

A	Y
0	0
1	1

Cell Size

Drive Strength	Height (um)	Width (um)
DLY2X1M	2.87	3.28
DLY2X4M	2.87	4.10

Functional Schematic



AC Power

Pin	Power (uW/MHz)	
	X1	X4
A	0.0119	0.0211

Pin Capacitance

Pin	Capacitance (pF)	
	X1	X4
A	0.0015	0.0026

Delays at 25°C,1.2V, Typical Process

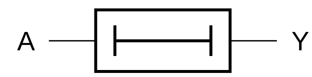
Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X1	X4	X1	X4
A → Y ↑	0.2414	0.2020	6.3986	1.8262
A → Y ↓	0.2814	0.2569	4.0504	1.9246

Cell Description

The DLY3 cell provides the logical delay of a single input (A). The output (Y) is represented by the logic equation:

$Y = A$

Logic Symbol



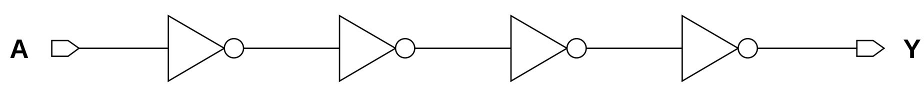
Function Table

A	Y
0	0
1	1

Cell Size

Drive Strength	Height (um)	Width (um)
DLY3X1M	2.87	3.28
DLY3X4M	2.87	4.10

Functional Schematic



AC Power

Pin	Power (uW/MHz)	
	X1	**X4**
A	0.0135	0.0246

Pin Capacitance

Pin	Capacitance (pF)	
	X1	**X4**
A	0.0015	0.0027

Delays at 25°C,1.2V, Typical Process

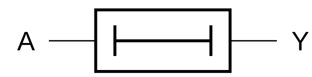
Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X1	**X4**	**X1**	**X4**
A → Y ↑	0.3650	0.3164	6.4833	1.8435
A → Y ↓	0.3897	0.3654	4.3697	1.9931

Cell Description

The DLY4 cell provides the logical delay of a single input (A). The output (Y) is represented by the logic equation:

$Y = A$

Logic Symbol



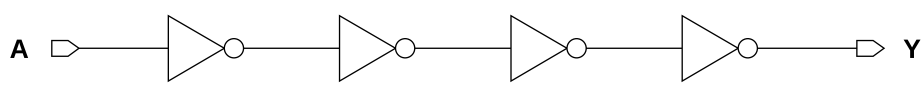
Function Table

A	Y
0	0
1	1

Cell Size

Drive Strength	Height (um)	Width (um)
DLY4X1M	2.87	3.28
DLY4X4M	2.87	4.10

Functional Schematic



AC Power

Pin	Power (uW/MHz)	
	X1	**X4**
A	0.0152	0.0284

Pin Capacitance

Pin	Capacitance (pF)	
	X1	**X4**
A	0.0015	0.0027

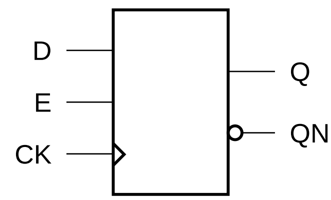
Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X1	**X4**	**X1**	**X4**
A → Y ↑	0.5057	0.4515	6.6042	1.8710
A → Y ↓	0.5042	0.4816	4.6918	2.0747

Cell Description

The EDFF cell is a positive-edge triggered, static D-type flip-flop with synchronous active-high enable (E).

Logic Symbol



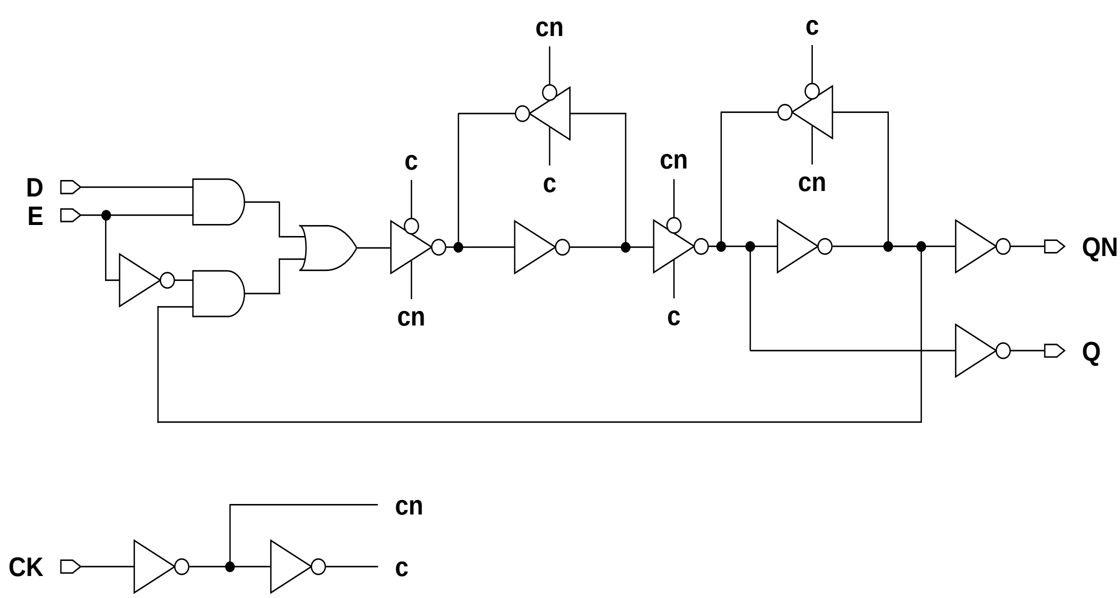
Function Table

E	D	CK	Q[n+1]	QN[n+1]
0	x	x	Q[n]	QN[n]
1	0	_/‾	0	1
1	1	_/‾	1	0
1	x	‾_	Q[n]	QN[n]

Cell Size

Drive Strength	Height (um)	Width (um)
EDFFX1M	2.87	11.48
EDFFX2M	2.87	11.48
EDFFX4M	2.87	12.30

Functional Schematic



AC Power

Pin	Power (uW/MHz)		
	X1	X2	X4
D	0.0108	0.0112	0.0118
CK	0.0203	0.0208	0.0219
E	0.0155	0.0159	0.0165
Q	0.0161	0.0202	0.0315

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
D	0.0016	0.0016	0.0016
CK	0.0017	0.0017	0.0020
E	0.0032	0.0032	0.0033

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
CK → Q ↑	0.1871	0.1886	0.1920	5.3308	3.5915	1.8374
CK → Q ↓	0.2249	0.2147	0.2196	3.6526	1.9133	0.9743
CK → QN ↑	0.3171	0.3016	0.3140	5.3525	3.5871	1.8450
CK → QN ↓	0.3198	0.3098	0.3021	3.4494	1.8267	0.8833

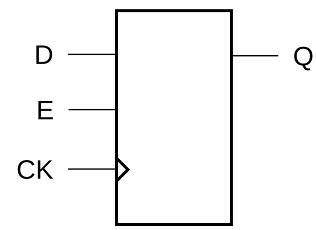
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)		
		X1	X2	X4
D	setup ↑ → CK	0.1094	0.1133	0.1172
	setup ↓ → CK	0.2148	0.2188	0.2266
	hold ↑ → CK	-0.0859	-0.0859	-0.0898
	hold ↓ → CK	-0.1484	-0.1445	-0.1523
CK	minpwh	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974
E	setup ↑ → CK	0.2578	0.2578	0.2695
	setup ↓ → CK	0.1719	0.1719	0.1719
	hold ↑ → CK	-0.0938	-0.0938	-0.0938
	hold ↓ → CK	-0.1328	-0.1289	-0.1367

Cell Description

The EDFFHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with a synchronous, active-high enable (E). The cell has a single output (Q) and fast clock-to-out path.

Logic Symbol



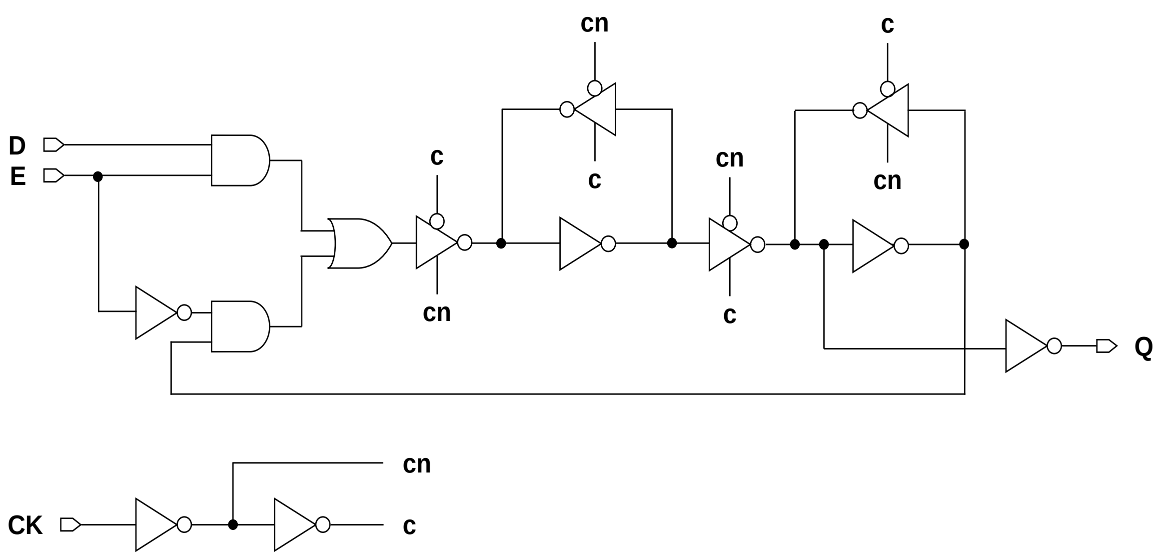
Function Table

E	D	CK	Q[n+1]
0	x	x	Q[n]
1	0	_/¯	0
1	1	_/¯	1
x	x	¯_	Q[n]

Cell Size

Drive Strength	Height (um)	Width (um)
EDFFHQX1M	2.87	11.48
EDFFHQX2M	2.87	11.48
EDFFHQX4M	2.87	12.30
EDFFHQX8M	2.87	13.53

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	X1	X2	X4	X8
D	0.0176	0.0193	0.0216	0.0219
CK	0.0281	0.0304	0.0322	0.0322
E	0.0187	0.0207	0.0221	0.0223
Q	0.0123	0.0136	0.0182	0.0316

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
D	0.0022	0.0018	0.0030	0.0030
CK	0.0029	0.0030	0.0035	0.0035
E	0.0033	0.0034	0.0034	0.0034

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
CK → Q ↑	0.1505	0.1375	0.1376	0.1580	5.0552	3.5496	1.8139	0.9438
CK → Q ↓	0.1494	0.1445	0.1537	0.1911	3.3533	1.8218	0.9372	0.4924

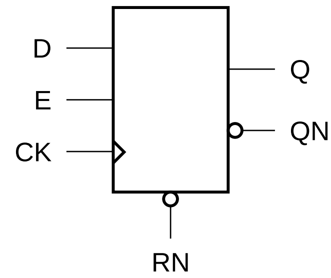
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
D	setup ↑ → CK	0.1445	0.1328	0.1289	0.1328
	setup ↓ → CK	0.1328	0.1953	0.1719	0.1680
	hold ↑ → CK	-0.0703	-0.0625	-0.0508	-0.0508
	hold ↓ → CK	-0.0781	-0.1289	-0.1055	-0.1055
CK	minpwh	0.9974	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974	0.9974
E	setup ↑ → CK	0.1289	0.1758	0.1680	0.1602
	setup ↓ → CK	0.1875	0.2344	0.2148	0.2188
	hold ↑ → CK	-0.0977	-0.0938	-0.0938	-0.0977
	hold ↓ → CK	-0.1523	-0.1562	-0.1602	-0.1602

Cell Description

The EDFFTR cell is a positive-edge triggered, static D-type flip-flop with synchronous active-high enable (E) and synchronous active-low reset (RN).

Logic Symbol



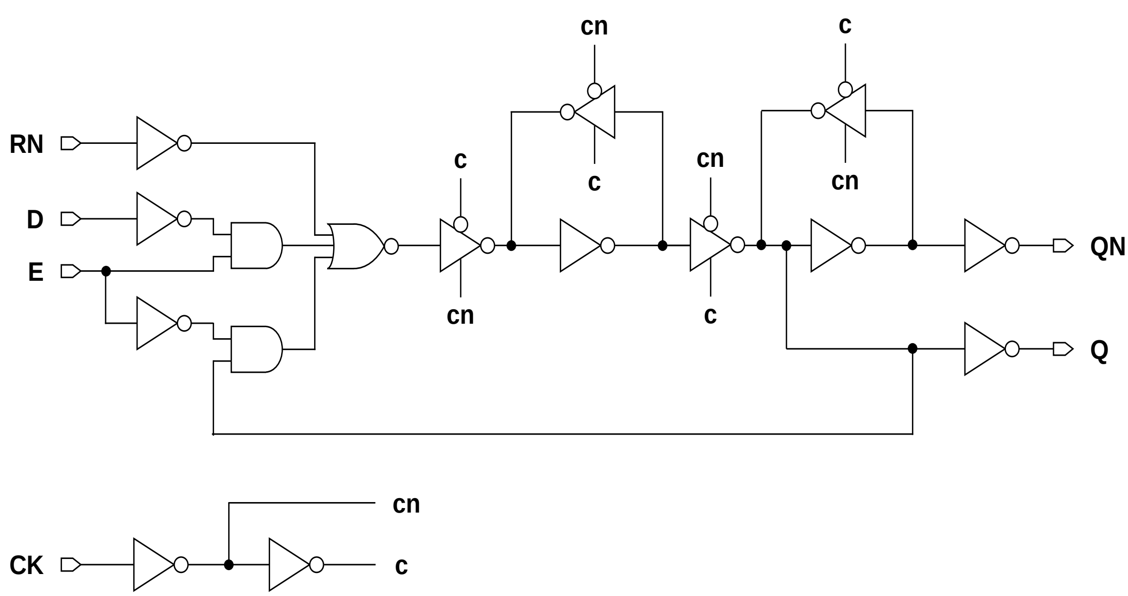
Function Table

RN	E	D	CK	Q[n+1]	QN[n+1]
0	x	x	↑	0	1
x	x	x	↓	Q[n]	QN[n]
1	0	x	↑	Q[n]	QN[n]
1	1	0	↑	0	1
1	1	1	↑	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
EDFFTRX1M	2.87	12.30
EDFFTRX2M	2.87	12.71
EDFFTRX4M	2.87	13.94

Functional Schematic



AC Power

Pin	Power (uW/MHz)		
	X1	X2	X4
D	0.0111	0.0114	0.0123
CK	0.0232	0.0235	0.0251
E	0.0157	0.0162	0.0171
RN	0.0130	0.0134	0.0143
Q	0.0161	0.0200	0.0321

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
D	0.0014	0.0014	0.0014
CK	0.0019	0.0019	0.0021
E	0.0032	0.0032	0.0033
RN	0.0014	0.0014	0.0014

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
CK → Q ↑	0.1918	0.1958	0.1987	5.1188	3.5737	1.8926
CK → Q ↓	0.2393	0.2250	0.2124	3.7004	2.0208	0.9856
CK → QN ↑	0.3275	0.3115	0.2940	5.1214	3.5569	1.8756
CK → QN ↓	0.3279	0.3272	0.2923	3.4821	1.9406	0.8800

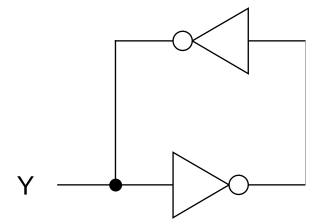
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)		
		X1	X2	X4
D	setup ↑ → CK	0.1484	0.1484	0.1562
	setup ↓ → CK	0.3594	0.3633	0.3867
	hold ↑ → CK	-0.1250	-0.1250	-0.1289
	hold ↓ → CK	-0.3047	-0.3008	-0.3242
CK	minpwh	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974
E	setup ↑ → CK	0.3867	0.3867	0.4141
	setup ↓ → CK	0.3281	0.3203	0.3438
	hold ↑ → CK	-0.1289	-0.1289	-0.1328
	hold ↓ → CK	-0.1875	-0.1875	-0.1914
RN	setup ↑ → CK	0.1641	0.1641	0.1719
	setup ↓ → CK	0.2656	0.2656	0.2891
	hold ↑ → CK	-0.1445	-0.1445	-0.1484
	hold ↓ → CK	-0.1953	-0.1836	-0.2070

Cell Description

The HOLD cell holds data at a known value. This cell is often used for holding data on a tri-state bus.

Logic Symbol



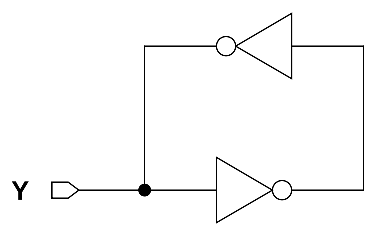
Function Table

Not Applicable

Cell Size

Drive Strength	Height (um)	Width (um)
HOLDX1M	2.87	2.05

Functional Schematic



AC Power

Pin	Power (uW/MHz)
	X1
Y	0.0099

Pin Capacitance

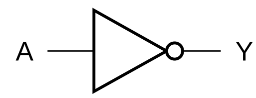
Pin	Capacitance (pF)
	X1
Y	0.1886

Cell Description

The INV cell provides the logical inversion of a single input (A). The output (Y) is represented by the logic equation:

$Y = \overline{A}$

Logic Symbol



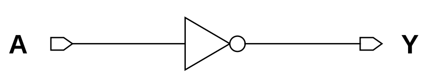
Function Table

A	Y
0	1
1	0

Cell Size

Drive Strength	Height (um)	Width (um)
INVXLM	2.87	1.23
INVX1M	2.87	1.23
INVX2M	2.87	1.23
INVX3M	2.87	1.64
INVX4M	2.87	1.64
INVX5M	2.87	2.05
INVX6M	2.87	2.05
INVX8M	2.87	2.46
INVX10M	2.87	3.28
INVX12M	2.87	3.69
INVX14M	2.87	4.10
INVX16M	2.87	4.51
INVX18M	2.87	4.92
INVX20M	2.87	5.74
INVX24M	2.87	6.56
INVX32M	2.87	8.61

Functional Schematic



AC Power

Pin	Power (uW/MHz)							
	XL	**X1**	**X2**	**X3**	**X4**	**X5**	**X6**	**X8**
A	0.0024	0.0033	0.0049	0.0067	0.0084	0.0113	0.0131	0.0165

AC Power (Cont'd.)

Pin	Power (uW/MHz)							
	X10	**X12**	**X14**	**X16**	**X18**	**X20**	**X24**	**X32**
A	0.0213	0.0249	0.0295	0.0330	0.0375	0.0410	0.0494	0.0659

Pin Capacitance

Pin	Capacitance (pF)							
	XL	**X1**	**X2**	**X3**	**X4**	**X5**	**X6**	**X8**
A	0.0015	0.0023	0.0034	0.0053	0.0066	0.0085	0.0100	0.0131

Pin Capacitance (Cont'd.)

Pin	Capacitance (pF)							
	X10	**X12**	**X14**	**X16**	**X18**	**X20**	**X24**	**X32**
A	0.0163	0.0196	0.0227	0.0261	0.0293	0.0324	0.0389	0.0517

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)							
	XL	**X1**	**X2**	**X3**	**X4**	**X5**	**X6**	**X8**
A → Y ↑	0.0225	0.0202	0.0206	0.0196	0.0183	0.0197	0.0193	0.0187
A → Y ↓	0.0159	0.0150	0.0135	0.0123	0.0121	0.0125	0.0124	0.0122

Delays at 25°C,1.2V, Typical Process (Cont'd.)

Description	Intrinsic Delay (ns)							
	X10	**X12**	**X14**	**X16**	**X18**	**X20**	**X24**	**X32**
A → Y ↑	0.0193	0.0189	0.0196	0.0193	0.0199	0.0196	0.0202	0.0216
A → Y ↓	0.0123	0.0120	0.0125	0.0124	0.0126	0.0125	0.0130	0.0143

Delays at 25°C,1.2V, Typical Process (Cont'd.)

Description	K_{load} (ns/pF)							
	XL	**X1**	**X2**	**X3**	**X4**	**X5**	**X6**	**X8**
A → Y ↑	8.5131	5.0893	3.4500	2.4556	1.7856	1.4571	1.2124	0.9286
A → Y ↓	5.1601	3.0568	1.6525	1.0509	0.8089	0.6390	0.5405	0.4157

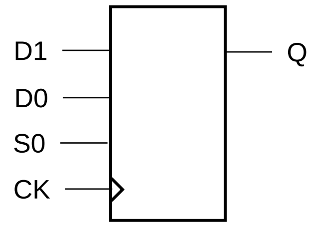
Delays at 25°C,1.2V, Typical Process (Cont'd.)

Description	K_{load} (ns/pF)							
	X10	**X12**	**X14**	**X16**	**X18**	**X20**	**X24**	**X32**
A → Y ↑	0.7440	0.6250	0.5377	0.4764	0.4346	0.3826	0.3208	0.2422
A → Y ↓	0.3193	0.2645	0.2284	0.2011	0.1858	0.1586	0.1326	0.0989

Cell Description

The MDFFHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with a 2-to-1data select control (S0) for the data inputs (D1,D0). The cell has a single output (Q) and fast clock-to-out path.

Logic Symbol



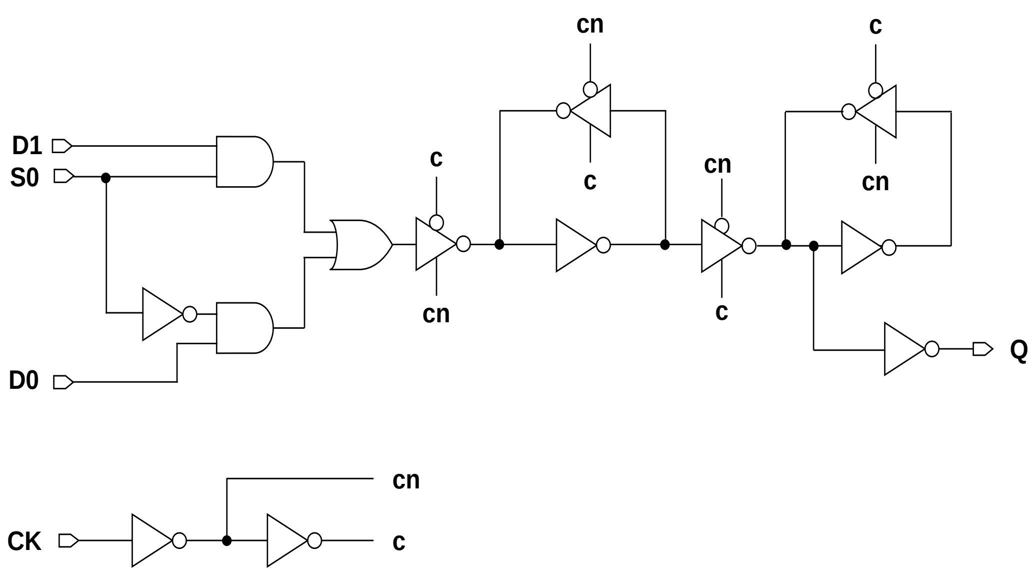
Function Table

S0	D1	D0	CK	Q[n+1]
0	x	0	↑	0
0	x	1	↑	1
1	0	x	↑	0
1	1	x	↑	1
x	x	x	↓	Q[n]

Cell Size

Drive Strength	Height (um)	Width (um)
MDFFHQX1M	2.87	12.30
MDFFHQX2M	2.87	12.71
MDFFHQX4M	2.87	13.53
MDFFHQX8M	2.87	14.35

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	X1	X2	X4	X8
D0	0.0151	0.0173	0.0215	0.0213
D1	0.0164	0.0190	0.0245	0.0244
S0	0.0196	0.0221	0.0272	0.0270
CK	0.0282	0.0325	0.0362	0.0361
Q	0.0080	0.0109	0.0156	0.0297

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
D0	0.0016	0.0019	0.0026	0.0026
D1	0.0015	0.0019	0.0029	0.0028
S0	0.0026	0.0028	0.0040	0.0041
CK	0.0029	0.0029	0.0036	0.0036

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
CK → Q ↑	0.1436	0.1436	0.1381	0.1605	5.0619	3.4806	1.7905	0.9274
CK → Q ↓	0.1433	0.1511	0.1583	0.1945	3.3447	1.8067	0.9222	0.4855

Timing Constraints at 25°C,1.2V, Typical Process

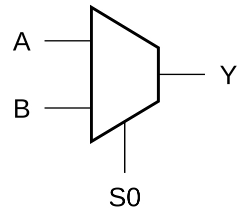
Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
D0	setup ↑ → CK	0.1719	0.1367	0.1055	0.1055
	setup ↓ → CK	0.1875	0.1562	0.1445	0.1406
	hold ↑ → CK	-0.0859	-0.0703	-0.0469	-0.0430
	hold ↓ → CK	-0.1250	-0.1055	-0.0938	-0.0898
D1	setup ↑ → CK	0.1680	0.1328	0.0820	0.0820
	setup ↓ → CK	0.2109	0.1641	0.1562	0.1523
	hold ↑ → CK	-0.0859	-0.0664	-0.0273	-0.0234
	hold ↓ → CK	-0.1484	-0.1133	-0.1094	-0.1016
S0	setup ↑ → CK	0.1914	0.1523	0.1484	0.1406
	setup ↓ → CK	0.2148	0.1875	0.1641	0.1641
	hold ↑ → CK	-0.0664	-0.0508	-0.0117	-0.0078
	hold ↓ → CK	-0.1211	-0.1055	-0.0977	-0.0938
CK	minpwh	0.9974	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974	0.9974

Cell Description

The MX2 cell is a 2-to-1 multiplexer. The state of the select input (S0) determines which data input (A,B) is presented to the output (Y).The output (Y) is represented by the logic equation:

$$Y = (\overline{S0} \bullet A) + (S0 \bullet B)$$

Logic Symbol



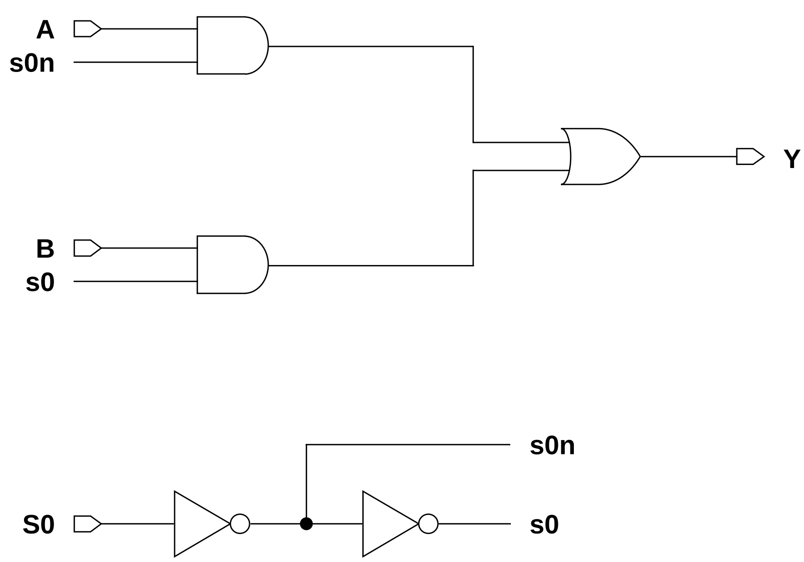
Function Table

S0	A	B	Y
0	0	x	0
0	1	x	1
1	x	0	0
1	x	1	1

Cell Size

Drive Strength	Height (um)	Width (um)
MX2XLM	2.87	4.10
MX2X1M	2.87	4.10
MX2X2M	2.87	4.10
MX2X3M	2.87	4.51
MX2X4M	2.87	4.92
MX2X6M	2.87	5.33
MX2X8M	2.87	5.74
MX2X12M	2.87	6.97

Functional Schematic



AC Power

Pin	Power (uW/MHz)							
	XL	**X1**	**X2**	**X3**	**X4**	**X6**	**X8**	**X12**
S0	0.0092	0.0119	0.0151	0.0181	0.0209	0.0280	0.0352	0.0526
A	0.0082	0.0104	0.0144	0.0182	0.0208	0.0279	0.0350	0.0529
B	0.0098	0.0115	0.0155	0.0204	0.0233	0.0305	0.0381	0.0545

Pin Capacitance

Pin	Capacitance (pF)							
	XL	**X1**	**X2**	**X3**	**X4**	**X6**	**X8**	**X12**
S0	0.0031	0.0038	0.0050	0.0056	0.0056	0.0056	0.0056	0.0061
A	0.0016	0.0023	0.0033	0.0036	0.0036	0.0037	0.0036	0.0036
B	0.0015	0.0021	0.0024	0.0034	0.0034	0.0034	0.0034	0.0035

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)							
	XL	**X1**	**X2**	**X3**	**X4**	**X6**	**X8**	**X12**
S0 → Y ↑	0.1257	0.1200	0.1094	0.1052	0.1097	0.1209	0.1296	0.1505
S0 → Y ↓	0.1375	0.1190	0.1144	0.1033	0.1160	0.1388	0.1560	0.1901
A → Y ↑	0.0911	0.0765	0.0710	0.0747	0.0783	0.0905	0.0998	0.1164
A → Y ↓	0.1466	0.1247	0.1087	0.1076	0.1172	0.1368	0.1527	0.1874
B → Y ↑	0.0923	0.0730	0.0793	0.0726	0.0767	0.0886	0.0979	0.1150
B → Y ↓	0.1527	0.1266	0.1342	0.1143	0.1241	0.1432	0.1585	0.1851

Delays at 25°C,1.2V, Typical Process (Cont'd.)

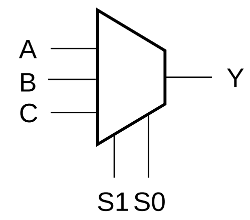
Description	K_{load} (ns/pF)							
	XL	**X1**	**X2**	**X3**	**X4**	**X6**	**X8**	**X12**
S0 → Y ↑	8.8795	5.1394	3.6090	2.5241	1.8525	1.2516	0.9476	0.6492
S0 → Y ↓	6.2472	3.3887	1.8577	1.1680	0.8950	0.6192	0.4804	0.3462
A → Y ↑	8.8846	5.1403	3.6089	2.5237	1.8525	1.2520	0.9480	0.6496
A → Y ↓	6.3092	3.4220	1.7844	1.1678	0.8994	0.6211	0.4814	0.3468
B → Y ↑	8.8968	5.1425	3.6160	2.5263	1.8542	1.2529	0.9485	0.6501
B → Y ↓	6.3675	3.4169	1.8585	1.1878	0.9165	0.6361	0.4948	0.3413

Cell Description

The MX3 cell is a 3-to-1 multiplexer. The state of the select inputs (S1,S0) determines which data input (A,B,C) is presented to the output (Y). The output (Y) is represented by the logic equation:

$$Y = (\overline{S0} \bullet \overline{S1} \bullet A) + (S0 \bullet \overline{S1} \bullet B) + (S1 \bullet C)$$

Logic Symbol



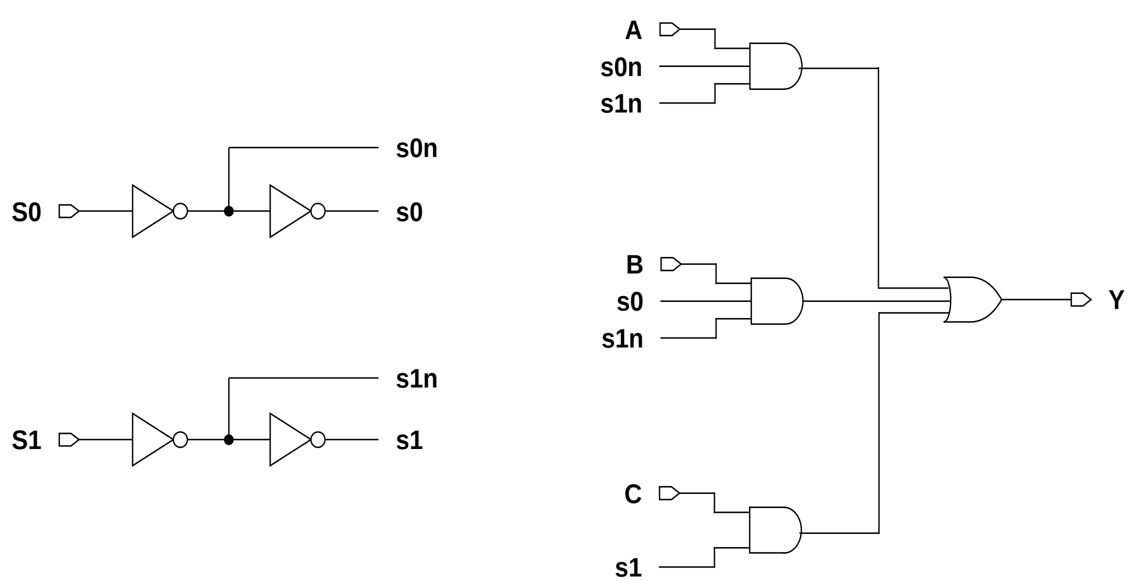
Function Table

S1	S0	A	B	C	Y
0	0	0	x	x	0
0	0	1	x	x	1
0	1	x	0	x	0
0	1	x	1	x	1
1	x	x	x	0	0
1	x	x	x	1	1

Cell Size

Drive Strength	Height (um)	Width (um)
MX3XLM	2.87	6.15
MX3X1M	2.87	6.15
MX3X2M	2.87	7.38
MX3X4M	2.87	7.79
MX3X8M	2.87	8.20

Functional Schematic



AC Power

Pin	Power (uW/MHz)				
	XL	X1	X2	X4	X8
S0	0.0136	0.0170	0.0224	0.0295	0.0470
S1	0.0092	0.0119	0.0156	0.0207	0.0388
A	0.0124	0.0165	0.0217	0.0291	0.0466
B	0.0136	0.0177	0.0246	0.0321	0.0504
C	0.0086	0.0108	0.0160	0.0224	0.0352

Pin Capacitance

Pin	Capacitance (pF)				
	XL	X1	X2	X4	X8
S0	0.0037	0.0045	0.0062	0.0062	0.0058
S1	0.0035	0.0042	0.0057	0.0059	0.0054
A	0.0020	0.0030	0.0035	0.0036	0.0034
B	0.0019	0.0024	0.0032	0.0032	0.0029
C	0.0018	0.0022	0.0032	0.0033	0.0034

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				
	XL	X1	X2	X4	X8
S0 → Y ↑	0.1614	0.1468	0.1513	0.1655	0.1922
S0 → Y ↓	0.1730	0.1638	0.1722	0.1884	0.2322
S1 → Y ↑	0.1175	0.1180	0.1131	0.1241	0.1576
S1 → Y ↓	0.0932	0.1027	0.0992	0.1237	0.1859
A → Y ↑	0.1270	0.1131	0.1169	0.1318	0.1593
A → Y ↓	0.1712	0.1589	0.1660	0.1808	0.2255
B → Y ↑	0.1229	0.1208	0.1239	0.1394	0.1743
B → Y ↓	0.1767	0.1852	0.1759	0.1915	0.2466
C → Y ↑	0.0673	0.0682	0.0676	0.0769	0.1106
C → Y ↓	0.1146	0.1176	0.1196	0.1354	0.1572

Delays at 25°C,1.2V, Typical Process (Cont'd.)

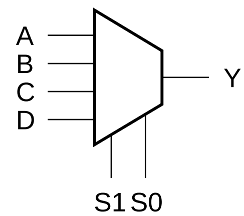
Description	K_{load} (ns/pF)				
	XL	X1	X2	X4	X8
S0 → Y ↑	8.8436	5.3088	3.5976	1.8553	0.9778
S0 → Y ↓	5.3844	3.8536	2.3121	1.0215	0.5718
S1 → Y ↑	8.8354	5.3216	3.6028	1.8589	0.9826
S1 → Y ↓	4.6482	3.4533	2.2291	0.9976	0.5760
A → Y ↑	8.8477	5.3084	3.5988	1.8558	0.9782
A → Y ↓	5.4086	3.6820	2.3147	1.0228	0.5654
B → Y ↑	8.8453	5.3297	3.6052	1.8602	0.9829
B → Y ↓	5.4405	3.8578	2.3270	1.0313	0.5824
C → Y ↑	8.6975	5.2568	3.5687	1.8385	0.9676
C → Y ↓	4.3783	3.3575	2.1448	0.9317	0.4976

Cell Description

The MX4 cell is a 4-to-1 multiplexer. The state of the select inputs (S1,S0) determines which data input (A,B,C,D) is presented to the output (Y). The output (Y) is represented by the logic equation:

$$Y = (\overline{S0} \bullet \overline{S1} \bullet A) + (S0 \bullet \overline{S1} \bullet B) + (\overline{S0} \bullet S1 \bullet C) + (S0 \bullet S1 \bullet D)$$

Logic Symbol



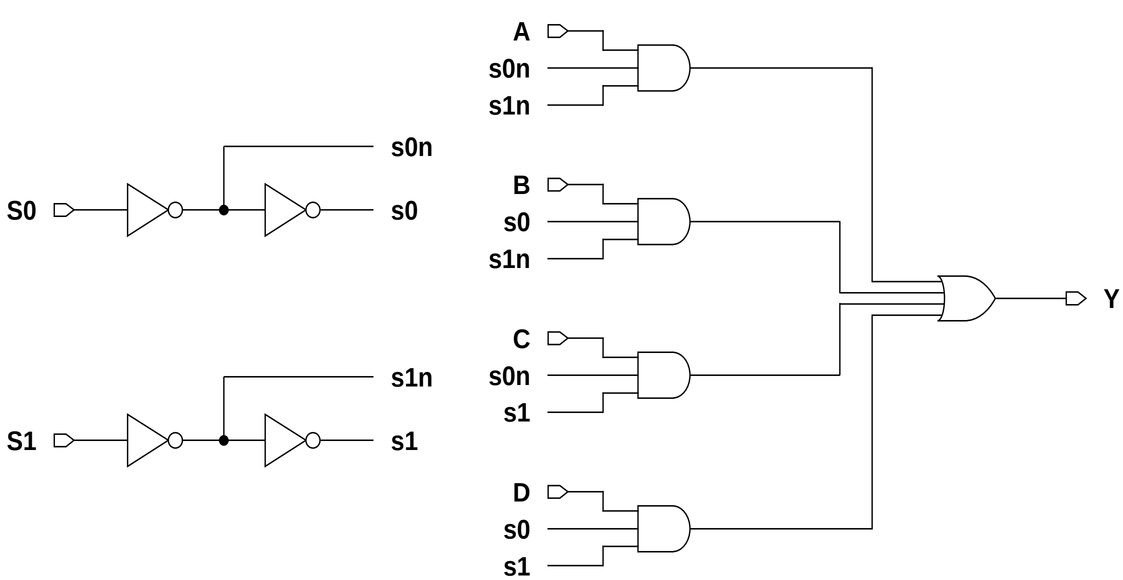
Function Table

S1	S0	A	B	C	D	Y
0	0	0	x	x	x	0
0	0	1	x	x	x	1
0	1	x	0	x	x	0
0	1	x	1	x	x	1
1	0	x	x	0	x	0
1	0	x	x	1	x	1
1	1	x	x	x	0	0
1	1	x	x	x	1	1

Cell Size

Drive Strength	Height (um)	Width (um)
MX4XLM	2.87	9.84
MX4X1M	2.87	9.84
MX4X2M	2.87	10.25
MX4X4M	2.87	10.66
MX4X8M	2.87	11.89

Functional Schematic



AC Power

Pin	Power (uW/MHz)				
	XL	X1	X2	X4	X8
S0	0.0239	0.0326	0.0408	0.0474	0.0665
S1	0.0100	0.0136	0.0173	0.0213	0.0381
A	0.0123	0.0174	0.0220	0.0287	0.0473
B	0.0137	0.0198	0.0250	0.0316	0.0504
C	0.0150	0.0210	0.0261	0.0332	0.0521
D	0.0163	0.0232	0.0286	0.0359	0.0553

Pin Capacitance

Pin	Capacitance (pF)				
	XL	X1	X2	X4	X8
S0	0.0083	0.0108	0.0132	0.0132	0.0132
S1	0.0037	0.0050	0.0059	0.0060	0.0060
A	0.0019	0.0029	0.0036	0.0036	0.0036
B	0.0018	0.0028	0.0034	0.0034	0.0034
C	0.0018	0.0028	0.0035	0.0035	0.0035
D	0.0019	0.0029	0.0035	0.0035	0.0035

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				
	XL	X1	X2	X4	X8
S0 → Y ↑	0.1778	0.1546	0.1536	0.1637	0.1920
S0 → Y ↓	0.2186	0.2059	0.1943	0.2132	0.2573
S1 → Y ↑	0.1139	0.1127	0.1143	0.1254	0.1517
S1 → Y ↓	0.0996	0.1021	0.1009	0.1247	0.1780
A → Y ↑	0.1177	0.1093	0.1131	0.1239	0.1523
A → Y ↓	0.1739	0.1692	0.1692	0.1856	0.2278
B → Y ↑	0.1181	0.1112	0.1184	0.1298	0.1593
B → Y ↓	0.1831	0.1800	0.1752	0.1935	0.2357
C → Y ↑	0.1250	0.1121	0.1139	0.1257	0.1539
C → Y ↓	0.1887	0.1821	0.1777	0.1968	0.2389
D → Y ↑	0.1226	0.1122	0.1151	0.1267	0.1555

Delays at 25°C,1.2V, Typical Process (Cont'd.)

Description	Intrinsic Delay (ns)				
	XL	X1	X2	X4	X8
D → Y ↓	0.1970	0.1901	0.1886	0.2085	0.2517

Delays at 25°C,1.2V, Typical Process (Cont'd.)

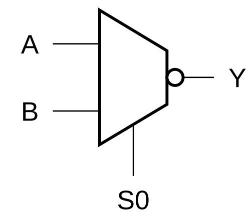
Description	K_{load} (ns/pF)				
	XL	X1	X2	X4	X8
S0 → Y ↑	8.7158	5.2056	3.4970	1.8249	0.9384
S0 → Y ↓	7.1095	3.9307	2.0442	1.0683	0.5553
S1 → Y ↑	8.6868	5.1990	3.4998	1.8259	0.9406
S1 → Y ↓	6.5202	3.8384	2.0193	1.0494	0.5465
A → Y ↑	8.6944	5.1983	3.4967	1.8238	0.9387
A → Y ↓	6.9668	3.8425	2.0428	1.0644	0.5538
B → Y ↑	8.6944	5.2029	3.5016	1.8268	0.9408
B → Y ↓	7.0270	3.9128	2.0387	1.0668	0.5545
C → Y ↑	8.7198	5.2065	3.4985	1.8253	0.9393
C → Y ↓	7.1329	3.9378	2.0476	1.0695	0.5555
D → Y ↑	8.7114	5.2080	3.5003	1.8265	0.9403
D → Y ↓	7.1735	3.8997	2.0655	1.0799	0.5612

Cell Description

The MXI2 cell is a 2-to-1multiplexer with inverted output. The state of the select input (S0) determines which data input (A,B) is presented to the output (Y). The output (Y) is represented by the logic equation:

$$Y = \overline{(\overline{S0} \bullet A) + (S0 \bullet B)}$$

Logic Symbol



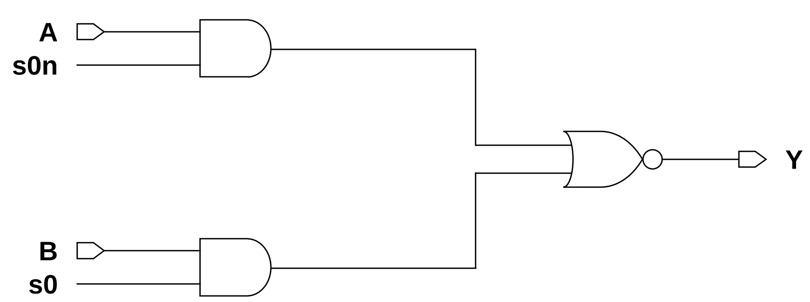
Function Table

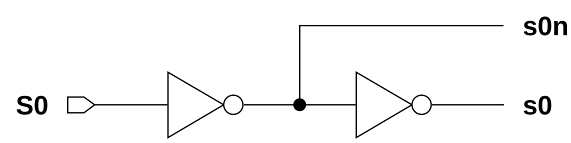
S0	A	B	Y
0	0	x	1
0	1	x	0
1	x	0	1
1	x	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
MXI2XLM	2.87	3.69
MXI2X1M	2.87	3.69
MXI2X2M	2.87	4.10
MXI2X3M	2.87	5.74
MXI2X4M	2.87	6.15
MXI2X6M	2.87	8.61
MXI2X8M	2.87	10.66
MXI2X12M	2.87	15.58

Functional Schematic





AC Power

Pin	Power (uW/MHz)							
	XL	X1	X2	X3	X4	X6	X8	X12
S0	0.0071	0.0094	0.0121	0.0196	0.0244	0.0339	0.0460	0.0675
A	0.0049	0.0066	0.0106	0.0160	0.0201	0.0296	0.0385	0.0573
B	0.0055	0.0075	0.0124	0.0197	0.0242	0.0352	0.0468	0.0690

Pin Capacitance

Pin	Capacitance (pF)							
	XL	X1	X2	X3	X4	X6	X8	X12
S0	0.0035	0.0043	0.0062	0.0079	0.0100	0.0147	0.0195	0.0288
A	0.0016	0.0024	0.0035	0.0053	0.0068	0.0100	0.0132	0.0197
B	0.0016	0.0023	0.0035	0.0054	0.0067	0.0099	0.0130	0.0194

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)							
	XL	X1	X2	X3	X4	X6	X8	X12
S0 → Y ↑	0.0510	0.0492	0.0522	0.0534	0.0500	0.0458	0.0477	0.0490
S0 → Y ↓	0.0714	0.0787	0.0692	0.0737	0.0681	0.0652	0.0655	0.0660
A → Y ↑	0.0583	0.0489	0.0509	0.0550	0.0515	0.0511	0.0508	0.0508
A → Y ↓	0.0426	0.0384	0.0338	0.0377	0.0363	0.0362	0.0354	0.0347
B → Y ↑	0.0605	0.0578	0.0584	0.0597	0.0551	0.0548	0.0552	0.0542
B → Y ↓	0.0396	0.0345	0.0320	0.0334	0.0310	0.0308	0.0311	0.0303

Delays at 25°C,1.2V, Typical Process (Cont'd.)

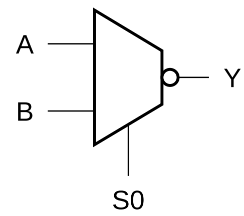
Description	K_{load} (ns/pF)							
	XL	X1	X2	X3	X4	X6	X8	X12
S0 → Y ↑	10.8221	7.1524	4.4437	3.0052	2.2372	1.5108	1.1450	0.7763
S0 → Y ↓	6.4570	3.8660	2.2025	1.4982	1.1421	0.7668	0.5687	0.3797
A → Y ↑	10.8505	6.6512	4.4629	3.0295	2.2543	1.5223	1.1549	0.7829
A → Y ↓	6.8340	4.0975	2.2869	1.5566	1.1839	0.7943	0.5877	0.3921
B → Y ↑	11.0265	7.3955	4.5400	2.9449	2.2092	1.5036	1.1466	0.7704
B → Y ↓	6.8024	4.1565	2.3459	1.5665	1.1732	0.7939	0.6054	0.3907

Cell Description

The MXI2D cell is a 2-to-1 multiplexer with inverted output. The state of the select input (S0) determines which data input (A, B) is presented to the output (Y). The output (Y) is represented by the logic equation:

$$Y = \overline{(\overline{S0} \bullet A) + (S0 \bullet B)}$$

Logic Symbol



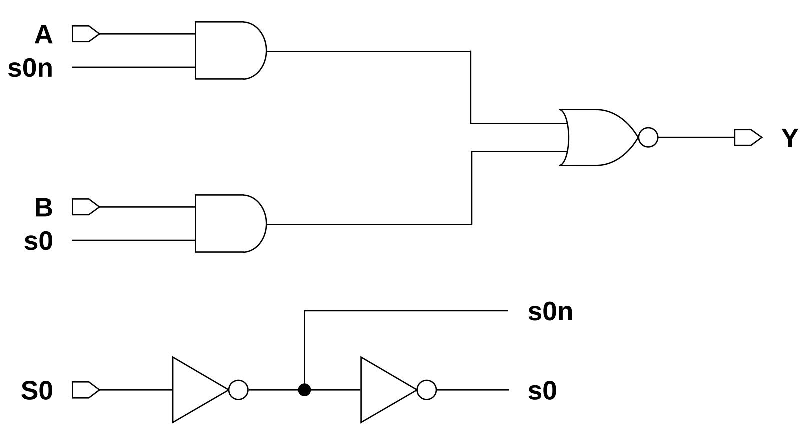
Function Table

S0	A	B	Y
0	0	x	1
0	1	x	0
1	x	0	1
1	x	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
MXI2DXLM	2.87	4.10
MXI2DX1M	2.87	4.10
MXI2DX2M	2.87	4.10
MXI2DX4M	2.87	5.74
MXI2DX8M	2.87	9.43

Functional Schematic



AC Power

Pin	Power (uW/MHz)				
	XL	X1	X2	X4	X8
S0	0.0103	0.0123	0.0163	0.0294	0.0548
A	0.0047	0.0062	0.0092	0.0168	0.0327
B	0.0050	0.0065	0.0095	0.0171	0.0329

Pin Capacitance

Pin	Capacitance (pF)				
	XL	X1	X2	X4	X8
S0	0.0015	0.0017	0.0022	0.0035	0.0067
A	0.0053	0.0067	0.0098	0.0196	0.0373
B	0.0055	0.0070	0.0100	0.0194	0.0368

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				
	XL	X1	X2	X4	X8
S0 → Y ↑	0.1311	0.1345	0.1312	0.1260	0.1169
S0 → Y ↓	0.1490	0.1452	0.1315	0.1275	0.1161
A → Y ↑	0.0422	0.0392	0.0390	0.0384	0.0383
A → Y ↓	0.0375	0.0405	0.0377	0.0383	0.0363
B → Y ↑	0.0444	0.0417	0.0422	0.0409	0.0403
B → Y ↓	0.0363	0.0388	0.0372	0.0378	0.0362

Delays at 25°C,1.2V, Typical Process (Cont'd.)

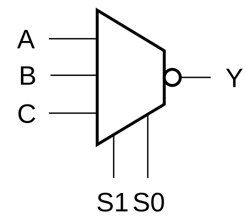
Description	K_{load} (ns/pF)				
	XL	X1	X2	X4	X8
S0 → Y ↑	8.5220	5.1086	3.4307	1.7750	0.9161
S0 → Y ↓	3.8690	3.1817	1.6972	0.8348	0.4149
A → Y ↑	8.5334	5.1123	3.4317	1.7758	0.9162
A → Y ↓	3.8650	3.1798	1.6966	0.8341	0.4147
B → Y ↑	8.5346	5.1107	3.4320	1.7759	0.9162
B → Y ↓	3.8654	3.1802	1.7057	0.8399	0.4175

Cell Description

The MXI3 cell is a 3-to-1multiplexer with inverted output. The state of the select inputs (S1,S0) determines which data input (A,B,C) is presented to the output (Y). The output (Y) is represented by the logic equation:

$$Y = (S0 \bullet S1 \bullet \overline{A}) + (\overline{S0} \bullet S1 \bullet \overline{B}) + (\overline{S1} \bullet \overline{C})$$

Logic Symbol



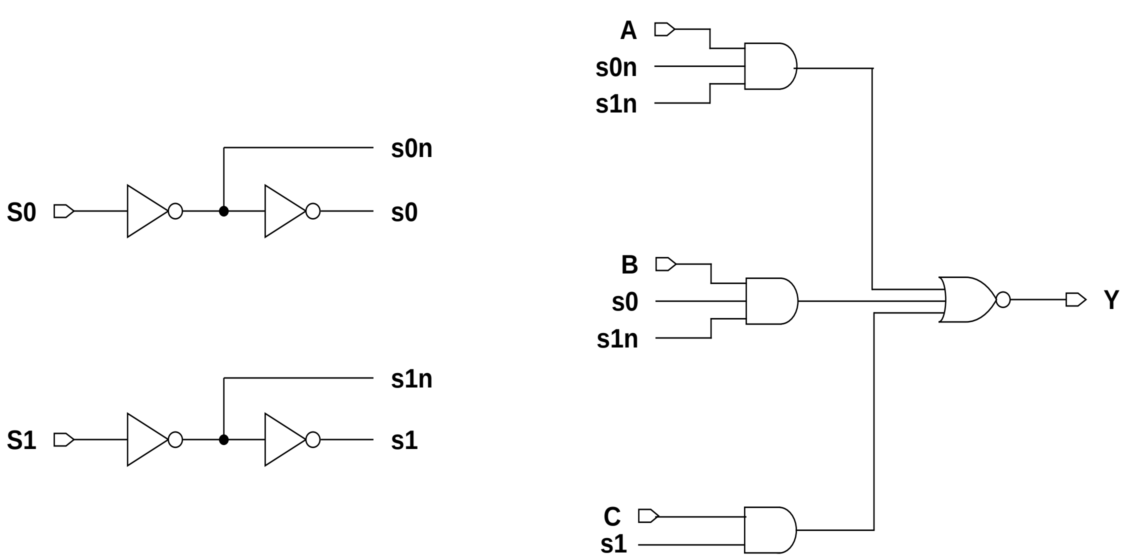
Function Table

S1	S0	A	B	C	Y
0	0	0	x	x	1
0	0	1	x	x	0
0	1	x	0	x	1
0	1	x	1	x	0
1	0	x	x	0	1
1	0	x	x	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
MXI3XLM	2.87	7.79
MXI3X1M	2.87	7.79
MXI3X2M	2.87	8.61
MXI3X4M	2.87	9.02
MXI3X8M	2.87	10.25

Functional Schematic



AC Power

Pin	Power (uW/MHz)				
	XL	**X1**	**X2**	**X4**	**X8**
S0	0.0150	0.0191	0.0257	0.0321	0.0487
S1	0.0091	0.0113	0.0151	0.0194	0.0351
A	0.0114	0.0153	0.0218	0.0271	0.0425
B	0.0122	0.0164	0.0233	0.0286	0.0406
C	0.0090	0.0119	0.0160	0.0213	0.0359

Pin Capacitance

Pin	Capacitance (pF)				
	XL	**X1**	**X2**	**X4**	**X8**
S0	0.0031	0.0036	0.0046	0.0049	0.0048
S1	0.0022	0.0025	0.0032	0.0033	0.0033
A	0.0015	0.0020	0.0025	0.0025	0.0025
B	0.0014	0.0019	0.0024	0.0024	0.0024
C	0.0015	0.0015	0.0019	0.0020	0.0020

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				
	XL	**X1**	**X2**	**X4**	**X8**
S0 → Y ↑	0.2057	0.1928	0.2021	0.2269	0.2401
S0 → Y ↓	0.2351	0.2223	0.2349	0.2466	0.2946
S1 → Y ↑	0.1132	0.1145	0.1166	0.1261	0.1424
S1 → Y ↓	0.1246	0.1122	0.1026	0.1262	0.1724
A → Y ↑	0.2179	0.1963	0.1997	0.2187	0.2322
A → Y ↓	0.2042	0.1794	0.1876	0.2078	0.2562
B → Y ↑	0.2216	0.1996	0.2164	0.2425	0.2560
B → Y ↓	0.2017	0.1805	0.1888	0.2114	0.2600
C → Y ↑	0.1445	0.1474	0.1341	0.1453	0.1685
C → Y ↓	0.1625	0.1528	0.1416	0.1480	0.1854

Delays at 25°C,1.2V, Typical Process (Cont'd.)

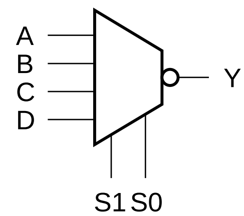
Description	K_{load} (ns/pF)				
	XL	**X1**	**X2**	**X4**	**X8**
S0 → Y ↑	8.7798	5.3047	3.4580	1.7966	0.9334
S0 → Y ↓	6.2278	3.3830	1.8677	0.9505	0.5071
S1 → Y ↑	8.7687	5.2978	3.4549	1.7945	0.9325
S1 → Y ↓	6.1484	3.3441	1.8458	0.9444	0.5058
A → Y ↑	8.7842	5.3068	3.4581	1.7962	0.9334
A → Y ↓	6.2346	3.3862	1.8675	0.9504	0.5071
B → Y ↑	8.7803	5.3039	3.4583	1.7967	0.9334
B → Y ↓	6.2258	3.3824	1.8675	0.9505	0.5072
C → Y ↑	8.7582	5.2995	3.4487	1.7885	0.9306
C → Y ↓	6.1886	3.3802	1.8395	0.9217	0.4932

Cell Description

The MXI4 cell is a 4-to-1multiplexer with inverted output. The state of the select inputs (S1,S0) determines which data input (A,B,C,D) is presented to the output (Y). The output (Y) is represented by the logic equation:

$$Y = \overline{(\overline{S0} \bullet \overline{S1} \bullet A) + (S0 \bullet \overline{S1} \bullet B) + (\overline{S0} \bullet S1 \bullet C) + (S0 \bullet S1 \bullet D)}$$

Logic Symbol



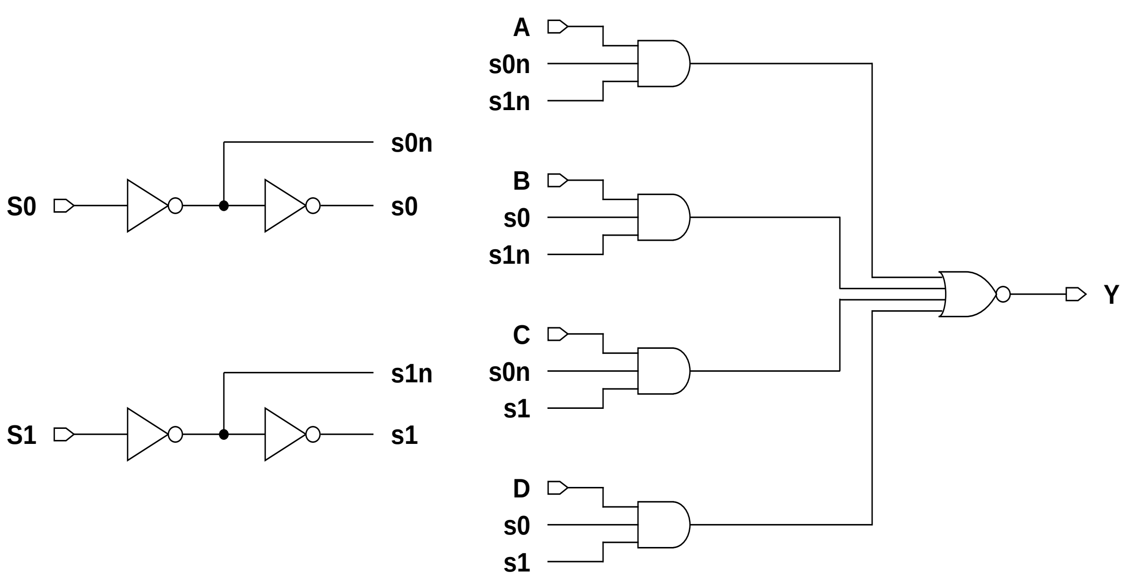
Function Table

S1	S0	A	B	C	D	Y
0	0	0	x	x	x	1
0	0	1	x	x	x	0
0	1	x	0	x	x	1
0	1	x	1	x	x	0
1	0	x	x	0	x	1
1	0	x	x	1	x	0
1	1	x	x	x	0	1
1	1	x	x	x	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
MXI4XLM	2.87	10.66
MXI4X1M	2.87	10.66
MXI4X2M	2.87	10.66
MXI4X4M	2.87	11.48
MXI4X8M	2.87	12.30

Functional Schematic



AC Power

Pin	Power (uW/MHz)				
	XL	X1	X2	X4	X8
S0	0.0259	0.0322	0.0412	0.0496	0.0637
S1	0.0087	0.0112	0.0149	0.0195	0.0324
A	0.0139	0.0175	0.0230	0.0291	0.0406
B	0.0149	0.0191	0.0241	0.0310	0.0411
C	0.0108	0.0141	0.0194	0.0253	0.0395
D	0.0116	0.0152	0.0217	0.0267	0.0384

Pin Capacitance

Pin	Capacitance (pF)				
	XL	X1	X2	X4	X8
S0	0.0052	0.0065	0.0085	0.0099	0.0099
S1	0.0022	0.0025	0.0029	0.0033	0.0032
A	0.0016	0.0018	0.0027	0.0029	0.0029
B	0.0014	0.0019	0.0023	0.0023	0.0023
C	0.0017	0.0021	0.0029	0.0033	0.0033
D	0.0014	0.0019	0.0023	0.0023	0.0023

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				
	XL	X1	X2	X4	X8
S0 → Y ↑	0.2696	0.2349	0.2023	0.2194	0.2480
S0 → Y ↓	0.2797	0.2551	0.2230	0.2254	0.2628
S1 → Y ↑	0.1113	0.1144	0.1118	0.1162	0.1394
S1 → Y ↓	0.1114	0.1062	0.0977	0.1081	0.1472
A → Y ↑	0.2410	0.2233	0.1960	0.2108	0.2393
A → Y ↓	0.2152	0.2012	0.1799	0.1871	0.2243
B → Y ↑	0.2738	0.2355	0.2144	0.2389	0.2674
B → Y ↓	0.2353	0.1983	0.1789	0.1909	0.2284
C → Y ↑	0.1864	0.1825	0.1749	0.1769	0.2040
C → Y ↓	0.1810	0.1746	0.1608	0.1709	0.2097
D → Y ↑	0.2084	0.1864	0.1891	0.2036	0.2305

Delays at 25°C,1.2V, Typical Process (Cont'd.)

Description	Intrinsic Delay (ns)				
	XL	X1	X2	X4	X8
D → Y ↓	0.1835	0.1774	0.1724	0.1863	0.2245

Delays at 25°C,1.2V, Typical Process (Cont'd.)

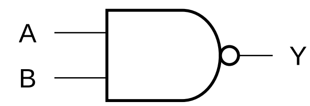
Description	K_{load} (ns/pF)				
	XL	X1	X2	X4	X8
S0 → Y ↑	8.7367	5.2318	3.5470	1.8277	0.9504
S0 → Y ↓	6.1739	3.4038	1.7928	0.9008	0.4771
S1 → Y ↑	8.7005	5.2201	3.5450	1.8258	0.9494
S1 → Y ↓	5.9760	3.3923	1.7976	0.9103	0.4786
A → Y ↑	8.7378	5.2322	3.5464	1.8276	0.9504
A → Y ↓	6.1728	3.4024	1.7919	0.9005	0.4771
B → Y ↑	8.7469	5.2336	3.5474	1.8278	0.9504
B → Y ↓	6.1822	3.4018	1.7915	0.9002	0.4770
C → Y ↑	8.7004	5.2214	3.5469	1.8258	0.9487
C → Y ↓	5.9877	3.4036	1.7992	0.9111	0.4859
D → Y ↑	8.7028	5.2210	3.5463	1.8262	0.9489
D → Y ↓	5.9893	3.4040	1.7996	0.9114	0.4859

Cell Description

The NAND2 cell provides the logical NAND of two inputs (A,B). The output (Y) is represented by the logic equation:

$Y = \overline{(A \bullet B)}$

Logic Symbol



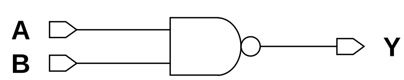
Function Table

A	B	Y
0	x	1
x	0	1
1	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
NAND2XLM	2.87	1.64
NAND2X1M	2.87	1.64
NAND2X2M	2.87	1.64
NAND2X3M	2.87	2.46
NAND2X4M	2.87	2.46
NAND2X5M	2.87	3.69
NAND2X6M	2.87	3.69
NAND2X8M	2.87	4.51
NAND2X12M	2.87	6.56

Functional Schematic



AC Power

Pin	Power (uW/MHz)								
	XL	X1	X2	X3	X4	X5	X6	X8	X12
A	0.0027	0.0039	0.0055	0.0072	0.0095	0.0127	0.0147	0.0195	0.0296
B	0.0034	0.0050	0.0077	0.0102	0.0141	0.0198	0.0229	0.0287	0.0433

Pin Capacitance

Pin	Capacitance (pF)								
	XL	X1	X2	X3	X4	X5	X6	X8	X12
A	0.0016	0.0025	0.0036	0.0049	0.0067	0.0085	0.0099	0.0136	0.0206
B	0.0015	0.0023	0.0034	0.0055	0.0073	0.0084	0.0097	0.0140	0.0210

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)								
	XL	X1	X2	X3	X4	X5	X6	X8	X12
A → Y ↑	0.0277	0.0245	0.0247	0.0237	0.0230	0.0247	0.0237	0.0242	0.0247
A → Y ↓	0.0253	0.0229	0.0197	0.0195	0.0175	0.0181	0.0179	0.0178	0.0179
B → Y ↑	0.0309	0.0289	0.0302	0.0288	0.0296	0.0334	0.0319	0.0310	0.0316
B → Y ↓	0.0267	0.0255	0.0223	0.0231	0.0212	0.0226	0.0223	0.0213	0.0214

Delays at 25°C,1.2V, Typical Process (Cont'd.)

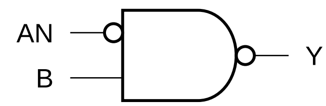
Description	K_{load} (ns/pF)								
	XL	X1	X2	X3	X4	X5	X6	X8	X12
A → Y ↑	8.8051	5.2865	3.5652	2.6124	1.8641	1.5270	1.2510	0.9731	0.6593
A → Y ↓	8.0949	4.8532	2.6506	2.0147	1.2885	1.0239	0.8693	0.6396	0.4265
B → Y ↑	8.7913	5.2796	3.5621	2.4798	1.8256	1.5269	1.2508	0.9506	0.6434
B → Y ↓	8.0952	4.8523	2.6503	2.0155	1.2889	1.0239	0.8696	0.6396	0.4265

Cell Description

The NAND2B cell provides the logical NAND of one inverted input (AN) and one non-inverted input (B). The output (Y) is represented by the logic equation:

$$Y = \overline{(\overline{AN} \bullet B)}$$

Logic Symbol



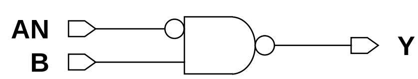
Function Table

AN	B	Y
1	x	1
x	0	1
0	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
NAND2BXLM	2.87	2.05
NAND2BX1M	2.87	2.05
NAND2BX2M	2.87	2.05
NAND2BX4M	2.87	3.28
NAND2BX8M	2.87	5.74
NAND2BX12M	2.87	8.20

Functional Schematic



AC Power

Pin	Power (uW/MHz)					
	XL	**X1**	**X2**	**X4**	**X8**	**X12**
AN	0.0051	0.0065	0.0086	0.0154	0.0293	0.0449
B	0.0027	0.0038	0.0058	0.0108	0.0221	0.0332

Pin Capacitance

Pin	Capacitance (pF)					
	XL	**X1**	**X2**	**X4**	**X8**	**X12**
AN	0.0015	0.0015	0.0019	0.0032	0.0058	0.0089
B	0.0015	0.0023	0.0033	0.0072	0.0140	0.0209

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)					
	XL	**X1**	**X2**	**X4**	**X8**	**X12**
AN → Y ↑	0.0535	0.0554	0.0531	0.0508	0.0488	0.0511
AN → Y ↓	0.0881	0.1007	0.0867	0.0777	0.0734	0.0762
B → Y ↑	0.0303	0.0280	0.0296	0.0290	0.0304	0.0322
B → Y ↓	0.0268	0.0269	0.0246	0.0239	0.0240	0.0243

Delays at 25°C,1.2V, Typical Process (Cont'd.)

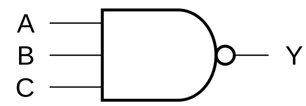
Description	K_{load} (ns/pF)					
	XL	**X1**	**X2**	**X4**	**X8**	**X12**
AN → Y ↑	8.7897	5.3065	3.5957	1.8973	0.9813	0.7014
AN → Y ↓	8.1191	4.9084	2.6653	1.3092	0.6427	0.4285
B → Y ↑	8.9299	5.3143	3.6273	1.8456	0.9660	0.6813
B → Y ↓	8.0288	4.8520	2.6449	1.3018	0.6396	0.4262

Cell Description

The NAND3 cell provides the logical NAND of three inputs (A,B,C). The output (Y) is represented by the logic equation:

$$Y = \overline{(A \bullet B \bullet C)}$$

Logic Symbol



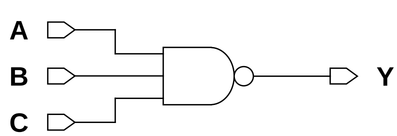
Function Table

A	B	C	Y
0	x	x	1
x	0	x	1
x	x	0	1
1	1	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
NAND3XLM	2.87	2.05
NAND3X1M	2.87	2.05
NAND3X2M	2.87	2.05
NAND3X3M	2.87	3.69
NAND3X4M	2.87	3.69
NAND3X6M	2.87	4.92
NAND3X8M	2.87	6.56
NAND3X12M	2.87	10.25

Functional Schematic



AC Power

Pin	Power (uW/MHz)							
	XL	**X1**	**X2**	**X3**	**X4**	**X6**	**X8**	**X12**
A	0.0032	0.0042	0.0061	0.0088	0.0113	0.0167	0.0184	0.0329
B	0.0040	0.0055	0.0084	0.0128	0.0164	0.0243	0.0255	0.0498
C	0.0045	0.0067	0.0107	0.0164	0.0210	0.0313	0.0329	0.0658

Pin Capacitance

Pin	Capacitance (pF)							
	XL	**X1**	**X2**	**X3**	**X4**	**X6**	**X8**	**X12**
A	0.0016	0.0024	0.0036	0.0051	0.0066	0.0106	0.0114	0.0199
B	0.0016	0.0024	0.0034	0.0058	0.0072	0.0107	0.0113	0.0194
C	0.0015	0.0023	0.0034	0.0060	0.0074	0.0104	0.0119	0.0194

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)							
	XL	**X1**	**X2**	**X3**	**X4**	**X6**	**X8**	**X12**
A → Y ↑	0.0339	0.0295	0.0297	0.0295	0.0282	0.0312	0.0317	0.0282
A → Y ↓	0.0392	0.0341	0.0278	0.0247	0.0246	0.0251	0.0261	0.0251
B → Y ↑	0.0385	0.0355	0.0376	0.0384	0.0364	0.0410	0.0393	0.0385
B → Y ↓	0.0432	0.0401	0.0341	0.0325	0.0322	0.0324	0.0321	0.0336
C → Y ↑	0.0411	0.0386	0.0417	0.0439	0.0414	0.0454	0.0464	0.0455
C → Y ↓	0.0454	0.0429	0.0375	0.0365	0.0361	0.0360	0.0365	0.0384

Delays at 25°C,1.2V, Typical Process (Cont'd.)

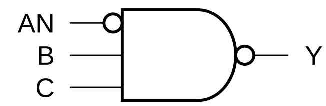
Description	K_{load} (ns/pF)							
	XL	**X1**	**X2**	**X3**	**X4**	**X6**	**X8**	**X12**
A → Y ↑	8.8353	5.3206	3.6710	2.6270	1.9224	1.4311	1.3467	0.6423
A → Y ↓	10.9674	6.6033	3.6097	2.2959	1.7682	1.1794	1.1389	0.6178
B → Y ↑	9.0558	5.4689	3.7719	2.6029	1.9038	1.4462	1.3398	0.6662
B → Y ↓	10.9650	6.6005	3.6087	2.2962	1.7679	1.1796	1.1387	0.6177
C → Y ↑	8.8617	5.3388	3.6031	2.5050	1.8222	1.3489	1.3129	0.6555
C → Y ↓	10.9588	6.6008	3.6081	2.2963	1.7682	1.1797	1.1395	0.6178

Cell Description

The NAND3B cell provides the logical NAND of one inverted input (AN) and two non-inverted inputs (B,C). The output (Y) is represented by the logic equation:

$$Y = \overline{(\overline{AN} \bullet B \bullet C)}$$

Logic Symbol



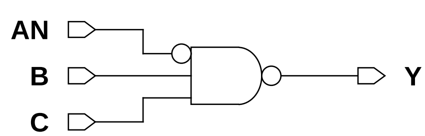
Function Table

AN	B	C	Y
1	x	x	1
x	0	x	1
x	x	0	1
0	1	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
NAND3BXLM	2.87	2.46
NAND3BX1M	2.87	2.46
NAND3BX2M	2.87	2.46
NAND3BX4M	2.87	4.10

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	XL	**X1**	**X2**	**X4**
AN	0.0056	0.0073	0.0100	0.0172
B	0.0032	0.0046	0.0067	0.0127
C	0.0038	0.0056	0.0089	0.0174

Pin Capacitance

Pin	Capacitance (pF)			
	XL	**X1**	**X2**	**X4**
AN	0.0015	0.0015	0.0019	0.0031
B	0.0016	0.0025	0.0035	0.0069
C	0.0015	0.0024	0.0034	0.0076

Delays at 25°C,1.2V, Typical Process

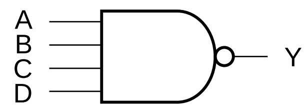
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	**X1**	**X2**	**X4**	**XL**	**X1**	**X2**	**X4**
AN → Y ↑	0.0582	0.0602	0.0594	0.0558	8.8609	5.3279	3.6828	1.9317
AN → Y ↓	0.1002	0.1121	0.0973	0.0847	10.9208	6.6417	3.6150	1.7658
B → Y ↑	0.0374	0.0350	0.0369	0.0351	9.1028	5.4841	3.7938	1.9162
B → Y ↓	0.0433	0.0420	0.0366	0.0338	10.8844	6.6130	3.6038	1.7620
C → Y ↑	0.0406	0.0383	0.0422	0.0420	8.9816	5.3543	3.6656	1.8658
C → Y ↓	0.0451	0.0447	0.0405	0.0391	10.8830	6.6121	3.6047	1.7631

Cell Description

The NAND4 cell provides a logical NAND of four inputs (A,B,C,D). The output (Y) is represented by the logic equation:

$$Y = \overline{(A \bullet B \bullet C \bullet D)}$$

Logic Symbol



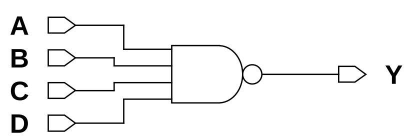
Function Table

A	B	C	D	Y
0	x	x	x	1
x	0	x	x	1
x	x	0	x	1
x	x	x	0	1
1	1	1	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
NAND4XLM	2.87	2.46
NAND4X1M	2.87	2.46
NAND4X2M	2.87	2.46
NAND4X4M	2.87	4.51
NAND4X6M	2.87	6.97
NAND4X8M	2.87	9.43
NAND4X12M	2.87	13.53

Functional Schematic



AC Power

Pin	Power (uW/MHz)						
	XL	**X1**	**X2**	**X4**	**X6**	**X8**	**X12**
A	0.0035	0.0048	0.0066	0.0122	0.0183	0.0239	0.0366
B	0.0042	0.0060	0.0089	0.0171	0.0265	0.0359	0.0541
C	0.0051	0.0075	0.0113	0.0222	0.0341	0.0460	0.0694
D	0.0055	0.0083	0.0135	0.0270	0.0417	0.0566	0.0851

Pin Capacitance

Pin	Capacitance (pF)						
	XL	**X1**	**X2**	**X4**	**X6**	**X8**	**X12**
A	0.0016	0.0024	0.0036	0.0066	0.0098	0.0132	0.0199
B	0.0016	0.0024	0.0034	0.0068	0.0097	0.0130	0.0198
C	0.0016	0.0024	0.0035	0.0076	0.0096	0.0128	0.0194
D	0.0015	0.0022	0.0034	0.0078	0.0097	0.0129	0.0195

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)						
	XL	**X1**	**X2**	**X4**	**X6**	**X8**	**X12**
A → Y ↑	0.0368	0.0325	0.0323	0.0335	0.0316	0.0314	0.0320
A → Y ↓	0.0505	0.0444	0.0347	0.0315	0.0324	0.0318	0.0323
B → Y ↑	0.0425	0.0387	0.0405	0.0416	0.0415	0.0422	0.0426
B → Y ↓	0.0578	0.0531	0.0438	0.0417	0.0439	0.0450	0.0451
C → Y ↑	0.0473	0.0437	0.0475	0.0478	0.0490	0.0497	0.0503
C → Y ↓	0.0640	0.0598	0.0509	0.0496	0.0515	0.0525	0.0527
D → Y ↑	0.0484	0.0477	0.0499	0.0514	0.0539	0.0550	0.0557
D → Y ↓	0.0662	0.0623	0.0542	0.0544	0.0559	0.0572	0.0573

Delays at 25°C,1.2V, Typical Process (Cont'd.)

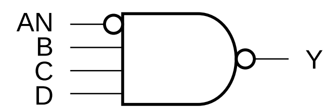
Description	K_{load} (ns/pF)						
	XL	**X1**	**X2**	**X4**	**X6**	**X8**	**X12**
A → Y ↑	8.8343	5.3476	3.6869	2.0747	1.2604	0.9540	0.6470
A → Y ↓	13.7667	8.3480	4.5700	2.2331	1.5248	1.1601	0.7767
B → Y ↑	9.1462	5.5138	3.8056	2.0321	1.2946	0.9802	0.6611
B → Y ↓	13.7627	8.3436	4.5681	2.2331	1.5246	1.1599	0.7766
C → Y ↑	9.1627	5.4982	3.8236	1.9355	1.2980	0.9830	0.6632
C → Y ↓	13.7663	8.3462	4.5684	2.2341	1.5249	1.1602	0.7765
D → Y ↑	8.8373	5.6215	3.6331	1.8372	1.2700	0.9625	0.6516
D → Y ↓	13.7630	8.3435	4.5683	2.2347	1.5249	1.1602	0.7766

Cell Description

The NAND4B cell provides a logical NAND of one inverted input (AN) and three non-inverted inputs (B,C,D). The output (Y) is represented by the logic equation:

$$Y = \overline{(\overline{AN} \bullet B \bullet C \bullet D)}$$

Logic Symbol



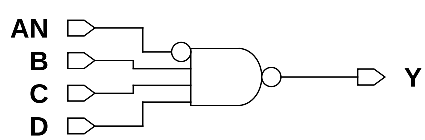
Function Table

AN	B	C	D	Y
1	x	x	x	1
x	0	x	x	1
x	x	0	x	1
x	x	x	0	1
0	1	1	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
NAND4BXLM	2.87	2.87
NAND4BX1M	2.87	2.87
NAND4BX2M	2.87	3.28
NAND4BX4M	2.87	6.15

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	XL	X1	X2	X4
AN	0.0065	0.0080	0.0109	0.0188
B	0.0035	0.0049	0.0074	0.0147
C	0.0042	0.0060	0.0095	0.0198
D	0.0048	0.0072	0.0119	0.0254

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
AN	0.0015	0.0015	0.0018	0.0032
B	0.0015	0.0023	0.0034	0.0066
C	0.0016	0.0024	0.0034	0.0065
D	0.0015	0.0023	0.0034	0.0065

Delays at 25°C,1.2V, Typical Process

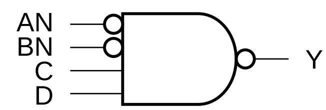
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
AN → Y ↑	0.0647	0.0641	0.0625	0.0596	8.7418	5.2528	3.6260	1.9024
AN → Y ↓	0.1198	0.1251	0.1056	0.0904	13.7287	8.3462	4.5615	2.3518
B → Y ↑	0.0411	0.0376	0.0401	0.0421	9.1671	5.5231	3.7498	1.9314
B → Y ↓	0.0567	0.0535	0.0471	0.0502	13.6851	8.3234	4.5538	2.3501
C → Y ↑	0.0468	0.0434	0.0479	0.0498	9.2225	5.5525	3.7680	1.9305
C → Y ↓	0.0647	0.0618	0.0554	0.0580	13.6860	8.3226	4.5544	2.3503
D → Y ↑	0.0487	0.0454	0.0524	0.0551	9.0359	5.3834	3.7103	1.8755
D → Y ↓	0.0662	0.0639	0.0586	0.0631	13.6899	8.3254	4.5555	2.3507

Cell Description

The NAND4BB cell provides a logical NAND of two inverted inputs (AN,BN) and two non-inverted inputs (C,D). The output (Y) is represented by the logic equation:

$$Y = \overline{(\overline{AN} \bullet \overline{BN} \bullet C \bullet D)}$$

Logic Symbol



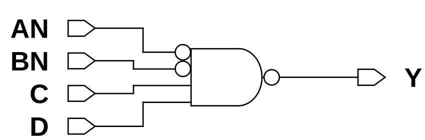
Function Table

AN	BN	C	D	Y
1	x	x	x	1
x	1	x	x	1
x	x	0	x	1
x	x	x	0	1
0	0	1	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
NAND4BBXLM	2.87	3.69
NAND4BBX1M	2.87	3.69
NAND4BBX2M	2.87	4.10
NAND4BBX4M	2.87	6.97

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	XL	**X1**	**X2**	**X4**
AN	0.0067	0.0082	0.0113	0.0199
BN	0.0068	0.0086	0.0123	0.0232
C	0.0044	0.0059	0.0085	0.0160
D	0.0051	0.0071	0.0107	0.0215

Pin Capacitance

Pin	Capacitance (pF)			
	XL	**X1**	**X2**	**X4**
AN	0.0015	0.0015	0.0018	0.0033
BN	0.0013	0.0013	0.0017	0.0031
C	0.0015	0.0023	0.0034	0.0064
D	0.0015	0.0023	0.0034	0.0065

Delays at 25°C,1.2V, Typical Process

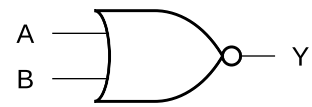
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	**X1**	**X2**	**X4**	**XL**	**X1**	**X2**	**X4**
AN → Y ↑	0.0679	0.0659	0.0642	0.0613	9.0149	5.4965	3.7199	2.0983
AN → Y ↓	0.1218	0.1275	0.1058	0.0891	14.0398	8.6990	4.5465	2.4533
BN → Y ↑	0.0782	0.0747	0.0732	0.0724	9.2340	5.6491	3.7879	1.9258
BN → Y ↓	0.1325	0.1354	0.1140	0.1112	14.0405	8.7010	4.5507	2.4588
C → Y ↑	0.0553	0.0479	0.0492	0.0486	9.2694	5.6747	3.7909	1.9425
C → Y ↓	0.0794	0.0720	0.0591	0.0613	14.0284	8.6897	4.5446	2.4538
D → Y ↑	0.0577	0.0493	0.0538	0.0554	9.0823	5.4192	3.7121	1.9246
D → Y ↓	0.0826	0.0754	0.0629	0.0674	14.0322	8.6917	4.5453	2.4544

Cell Description

The NOR2 cell provides a logical NOR of two inputs (A,B). The output (Y) is represented by the logic equation:

$Y = \overline{(A + B)}$

Logic Symbol



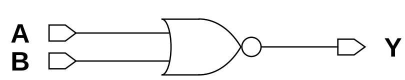
Function Table

A	B	Y
0	0	1
x	1	0
1	x	0

Cell Size

Drive Strength	Height (um)	Width (um)
NOR2XLM	2.87	1.64
NOR2X1M	2.87	1.64
NOR2X2M	2.87	1.64
NOR2X3M	2.87	2.46
NOR2X4M	2.87	2.46
NOR2X5M	2.87	3.69
NOR2X6M	2.87	3.69
NOR2X8M	2.87	4.51
NOR2X12M	2.87	6.56

Functional Schematic



AC Power

Pin	Power (uW/MHz)								
	XL	**X1**	**X2**	**X3**	**X4**	**X5**	**X6**	**X8**	**X12**
A	0.0028	0.0038	0.0058	0.0087	0.0111	0.0148	0.0170	0.0225	0.0338
B	0.0035	0.0049	0.0073	0.0113	0.0143	0.0185	0.0214	0.0286	0.0431

Pin Capacitance

Pin	Capacitance (pF)								
	XL	**X1**	**X2**	**X3**	**X4**	**X5**	**X6**	**X8**	**X12**
A	0.0016	0.0024	0.0036	0.0053	0.0067	0.0093	0.0106	0.0137	0.0208
B	0.0015	0.0023	0.0034	0.0056	0.0070	0.0088	0.0101	0.0138	0.0207

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)								
	XL	**X1**	**X2**	**X3**	**X4**	**X5**	**X6**	**X8**	**X12**
A → Y ↑	0.0458	0.0385	0.0392	0.0398	0.0375	0.0423	0.0400	0.0399	0.0406
A → Y ↓	0.0185	0.0166	0.0147	0.0139	0.0139	0.0146	0.0145	0.0146	0.0147
B → Y ↑	0.0537	0.0472	0.0477	0.0508	0.0478	0.0513	0.0490	0.0497	0.0506
B → Y ↓	0.0209	0.0193	0.0168	0.0169	0.0172	0.0164	0.0163	0.0167	0.0171

Delays at 25°C,1.2V, Typical Process (Cont'd.)

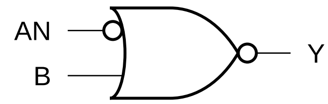
Description	K_{load} (ns/pF)								
	XL	**X1**	**X2**	**X3**	**X4**	**X5**	**X6**	**X8**	**X12**
A → Y ↑	17.3126	10.4507	7.0470	5.0381	3.6644	3.0572	2.5039	1.9166	1.2936
A → Y ↓	5.0894	2.9785	1.6133	1.0504	0.8094	0.6408	0.5445	0.4322	0.2859
B → Y ↑	17.2787	10.4375	7.0415	5.0358	3.6627	3.0547	2.5014	1.9155	1.2929
B → Y ↓	5.0971	2.9955	1.6206	1.0953	0.8676	0.6320	0.5362	0.4172	0.2859

Cell Description

The NOR2B cell provides a logical NOR of one inverted input (AN) and one non-inverted input (B). The output (Y) is represented by the logic equation:

$$Y = \overline{(\overline{AN} + B)}$$

Logic Symbol



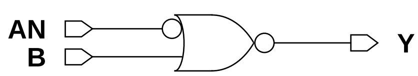
Function Table

AN	B	Y
1	0	1
x	1	0
0	x	0

Cell Size

Drive Strength	Height (um)	Width (um)
NOR2BXLM	2.87	2.05
NOR2BX1M	2.87	2.05
NOR2BX2M	2.87	2.05
NOR2BX4M	2.87	3.28
NOR2BX8M	2.87	5.74
NOR2BX12M	2.87	8.20

Functional Schematic



AC Power

Pin	Power (uW/MHz)					
	XL	**X1**	**X2**	**X4**	**X8**	**X12**
AN	0.0048	0.0060	0.0079	0.0143	0.0272	0.0416
B	0.0034	0.0049	0.0072	0.0144	0.0286	0.0431

Pin Capacitance

Pin	Capacitance (pF)					
	XL	**X1**	**X2**	**X4**	**X8**	**X12**
AN	0.0014	0.0014	0.0018	0.0032	0.0058	0.0089
B	0.0015	0.0023	0.0034	0.0070	0.0138	0.0210

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)					
	XL	**X1**	**X2**	**X4**	**X8**	**X12**
AN → Y ↑	0.0695	0.0670	0.0654	0.0644	0.0634	0.0649
AN → Y ↓	0.0782	0.0917	0.0805	0.0748	0.0711	0.0741
B → Y ↑	0.0548	0.0494	0.0496	0.0504	0.0520	0.0526
B → Y ↓	0.0202	0.0191	0.0166	0.0162	0.0164	0.0165

Delays at 25°C,1.2V, Typical Process (Cont’d.)

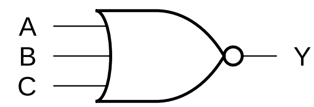
Description	K_{load} (ns/pF)					
	XL	**X1**	**X2**	**X4**	**X8**	**X12**
AN → Y ↑	17.8084	10.6314	7.1559	3.7450	1.9341	1.3066
AN → Y ↓	5.2791	3.1247	1.6818	0.8249	0.4210	0.2835
B → Y ↑	17.7699	10.6116	7.1492	3.7429	1.9330	1.3061
B → Y ↓	5.0392	2.9981	1.6262	0.8006	0.4087	0.2759

Cell Description

The NOR3 cell provides a logical NOR of three inputs (A,B,C). The output (Y) is represented by the logic equation:

$$Y = \overline{(A + B + C)}$$

Logic Symbol



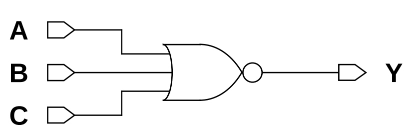
Function Table

A	B	C	Y
0	0	0	1
x	x	1	0
x	1	x	0
1	x	x	0

Cell Size

Drive Strength	Height (um)	Width (um)
NOR3XLM	2.87	2.05
NOR3X1M	2.87	2.05
NOR3X2M	2.87	2.05
NOR3X4M	2.87	3.69
NOR3X6M	2.87	4.92
NOR3X8M	2.87	6.56
NOR3X12M	2.87	9.43

Functional Schematic



AC Power

Pin	Power (uW/MHz)						
	XL	**X1**	**X2**	**X4**	**X6**	**X8**	**X12**
A	0.0039	0.0050	0.0076	0.0140	0.0215	0.0279	0.0422
B	0.0045	0.0060	0.0091	0.0172	0.0259	0.0340	0.0512
C	0.0051	0.0071	0.0105	0.0201	0.0300	0.0396	0.0593

Pin Capacitance

Pin	Capacitance (pF)						
	XL	**X1**	**X2**	**X4**	**X6**	**X8**	**X12**
A	0.0016	0.0024	0.0036	0.0066	0.0106	0.0134	0.0204
B	0.0015	0.0023	0.0034	0.0070	0.0105	0.0139	0.0212
C	0.0014	0.0022	0.0033	0.0074	0.0103	0.0144	0.0215

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)						
	XL	**X1**	**X2**	**X4**	**X6**	**X8**	**X12**
A → Y ↑	0.0799	0.0634	0.0676	0.0615	0.0686	0.0676	0.0699
A → Y ↓	0.0216	0.0188	0.0174	0.0154	0.0162	0.0156	0.0160
B → Y ↑	0.1014	0.0866	0.0894	0.0865	0.0935	0.0940	0.0976
B → Y ↓	0.0244	0.0219	0.0197	0.0182	0.0187	0.0183	0.0189
C → Y ↑	0.1093	0.0949	0.0972	0.0971	0.1035	0.1051	0.1084
C → Y ↓	0.0253	0.0226	0.0198	0.0191	0.0195	0.0192	0.0197

Delays at 25°C,1.2V, Typical Process (Cont'd.)

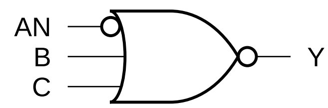
Description	K_{load} (ns/pF)						
	XL	**X1**	**X2**	**X4**	**X6**	**X8**	**X12**
A → Y ↑	26.6696	16.1588	10.8841	5.6587	4.0482	3.1127	2.1231
A → Y ↓	5.0606	2.9855	1.6838	0.7954	0.5501	0.4073	0.2796
B → Y ↑	26.6187	16.1363	10.8757	5.6566	4.0467	3.1120	2.1228
B → Y ↓	5.0377	2.9577	1.6649	0.8017	0.5488	0.4094	0.2808
C → Y ↑	26.6034	16.1326	10.8725	5.6584	4.0463	3.1126	2.1231
C → Y ↓	5.2078	3.0237	1.6355	0.8144	0.5615	0.4157	0.2854

Cell Description

The NOR3B cell provides a logical NOR of one inverted input (AN) and two non-inverted inputs (B,C). The output (Y) is represented by the logic equation:

$$Y = \overline{(\overline{AN} + B + C)}$$

Logic Symbol



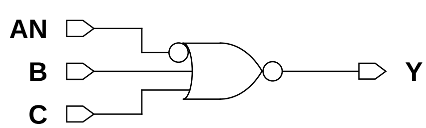
Function Table

AN	B	C	Y
1	0	0	1
x	x	1	0
x	1	x	0
0	x	x	0

Cell Size

Drive Strength	Height (um)	Width (um)
NOR3BXLM	2.87	2.46
NOR3BX1M	2.87	2.46
NOR3BX2M	2.87	2.46
NOR3BX4M	2.87	4.10

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	XL	X1	X2	X4
AN	0.0057	0.0064	0.0085	0.0141
B	0.0044	0.0059	0.0091	0.0169
C	0.0050	0.0069	0.0104	0.0198

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
AN	0.0014	0.0014	0.0019	0.0032
B	0.0015	0.0023	0.0034	0.0068
C	0.0015	0.0022	0.0034	0.0074

Delays at 25°C,1.2V, Typical Process

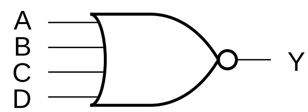
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
AN → Y ↑	0.1098	0.0927	0.0949	0.0891	27.1152	16.3191	11.0607	5.9734
AN → Y ↓	0.0942	0.0994	0.0877	0.0763	5.2806	3.1272	1.7346	0.8429
B → Y ↑	0.1029	0.0875	0.0914	0.0904	27.0415	16.2894	11.0516	5.9721
B → Y ↓	0.0240	0.0215	0.0194	0.0181	5.0261	2.9604	1.6594	0.8251
C → Y ↑	0.1123	0.0970	0.1000	0.1020	27.0371	16.2872	11.0483	5.9730
C → Y ↓	0.0249	0.0226	0.0196	0.0188	5.1089	3.0362	1.6206	0.8109

Cell Description

The NOR4 cell provides a logical NOR of four inputs (A,B,C,D). The output (Y) is represented by the logic equation:

$$Y = \overline{(A + B + C + D)}$$

Logic Symbol



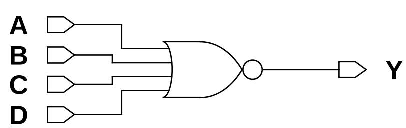
Function Table

A	B	C	D	Y
0	0	0	0	1
x	x	x	1	0
x	x	1	x	0
x	1	x	x	0
1	x	x	x	0

Cell Size

Drive Strength	Height (um)	Width (um)
NOR4XLM	2.87	2.46
NOR4X1M	2.87	2.46
NOR4X2M	2.87	4.10
NOR4X4M	2.87	7.79
NOR4X6M	2.87	11.89
NOR4X8M	2.87	15.99
NOR4X12M	2.87	24.19

Functional Schematic



AC Power

Pin	Power (uW/MHz)						
	XL	**X1**	**X2**	**X4**	**X6**	**X8**	**X12**
A	0.0047	0.0063	0.0109	0.0217	0.0330	0.0450	0.0678
B	0.0054	0.0077	0.0139	0.0275	0.0415	0.0564	0.0844
C	0.0063	0.0091	0.0170	0.0339	0.0509	0.0692	0.1033
D	0.0070	0.0105	0.0199	0.0393	0.0588	0.0801	0.1187

Pin Capacitance

Pin	Capacitance (pF)						
	XL	**X1**	**X2**	**X4**	**X6**	**X8**	**X12**
A	0.0019	0.0030	0.0052	0.0105	0.0163	0.0221	0.0329
B	0.0017	0.0027	0.0052	0.0108	0.0163	0.0217	0.0325
C	0.0018	0.0028	0.0058	0.0123	0.0182	0.0249	0.0373
D	0.0016	0.0027	0.0061	0.0118	0.0178	0.0238	0.0356

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)						
	XL	**X1**	**X2**	**X4**	**X6**	**X8**	**X12**
A → Y ↑	0.0908	0.0616	0.0509	0.0531	0.0578	0.0577	0.0600
A → Y ↓	0.0253	0.0225	0.0214	0.0229	0.0224	0.0236	0.0239
B → Y ↑	0.1267	0.0969	0.0894	0.0903	0.0949	0.0940	0.0980
B → Y ↓	0.0287	0.0268	0.0270	0.0288	0.0285	0.0306	0.0308
C → Y ↑	0.1512	0.1195	0.1145	0.1163	0.1219	0.1206	0.1264
C → Y ↓	0.0305	0.0290	0.0301	0.0308	0.0305	0.0313	0.0323
D → Y ↑	0.1609	0.1288	0.1237	0.1250	0.1310	0.1302	0.1373
D → Y ↓	0.0310	0.0292	0.0300	0.0297	0.0298	0.0304	0.0319

Delays at 25°C,1.2V, Typical Process (Cont'd.)

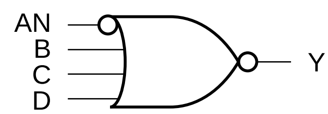
Description	K_{load} (ns/pF)						
	XL	**X1**	**X2**	**X4**	**X6**	**X8**	**X12**
A → Y ↑	28.8278	14.9163	7.6692	3.9372	2.7431	2.0051	1.4017
A → Y ↓	5.1111	3.0112	1.6372	0.9131	0.5775	0.4588	0.3096
B → Y ↑	28.7462	14.8883	7.6613	3.9333	2.7397	2.0026	1.4002
B → Y ↓	5.0227	2.9525	1.6302	0.8913	0.5878	0.4707	0.3172
C → Y ↑	28.7461	14.8877	7.6649	3.9367	2.7429	2.0051	1.4020
C → Y ↓	5.1229	3.0033	1.6575	0.8469	0.5577	0.4226	0.2913
D → Y ↑	28.7471	14.8879	7.6669	3.9356	2.7417	2.0046	1.4017
D → Y ↓	5.3596	3.1039	1.6937	0.8449	0.5632	0.4226	0.2947

Cell Description

The NOR4B cell provides a logical NOR of one inverted input (AN) and three non-inverted inputs (B,C,D). The output (Y) is represented by the logic equation:

$$Y = \overline{(\overline{AN} + B + C + D)}$$

Logic Symbol



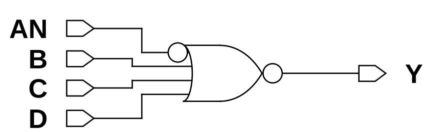
Function Table

AN	B	C	D	Y
1	0	0	0	1
x	x	x	1	0
x	x	1	x	0
x	1	x	x	0
0	x	x	x	0

Cell Size

Drive Strength	Height (um)	Width (um)
NOR4BXLM	2.87	2.87
NOR4BX1M	2.87	2.87
NOR4BX2M	2.87	4.92
NOR4BX4M	2.87	9.02

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	XL	X1	X2	X4
AN	0.0063	0.0078	0.0135	0.0250
B	0.0054	0.0078	0.0145	0.0289
C	0.0059	0.0092	0.0171	0.0336
D	0.0067	0.0106	0.0199	0.0392

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
AN	0.0015	0.0017	0.0026	0.0049
B	0.0018	0.0029	0.0059	0.0125
C	0.0017	0.0028	0.0052	0.0109
D	0.0017	0.0027	0.0050	0.0102

Delays at 25°C,1.2V, Typical Process

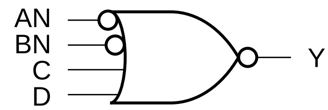
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
AN → Y ↑	0.1159	0.0938	0.0856	0.0806	28.9198	14.9391	7.6786	3.9476
AN → Y ↓	0.1051	0.1026	0.0897	0.0853	5.3725	3.1419	1.6598	0.8250
B → Y ↑	0.1229	0.0997	0.0973	0.1014	28.8359	14.9104	7.6687	3.9445
B → Y ↓	0.0277	0.0271	0.0288	0.0293	5.0196	2.9575	1.6274	0.8301
C → Y ↑	0.1465	0.1204	0.1188	0.1196	28.8331	14.9056	7.6638	3.9406
C → Y ↓	0.0296	0.0293	0.0313	0.0330	5.0993	2.9975	1.6554	0.9018
D → Y ↑	0.1550	0.1287	0.1286	0.1288	28.8249	14.9026	7.6634	3.9404
D → Y ↓	0.0291	0.0292	0.0319	0.0346	5.2320	3.0860	1.7045	0.9479

Cell Description

The NOR4BB cell provides a logical NOR of two inverted inputs (AN,BN) and two non-inverted inputs (C,D). The output (Y) is represented by the logic equation:

$$Y = \overline{(\overline{AN} + \overline{BN} + C + D)}$$

Logic Symbol



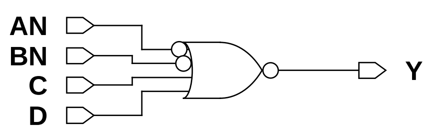
Function Table

AN	BN	C	D	Y
1	1	0	0	1
x	x	x	1	0
x	x	1	x	0
x	0	x	x	0
0	x	x	x	0

Cell Size

Drive Strength	Height (um)	Width (um)
NOR4BBXLM	2.87	4.10
NOR4BBX1M	2.87	4.10
NOR4BBX2M	2.87	5.74
NOR4BBX4M	2.87	9.84

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	XL	**X1**	**X2**	**X4**
AN	0.0049	0.0067	0.0139	0.0252
BN	0.0060	0.0083	0.0157	0.0305
C	0.0061	0.0091	0.0172	0.0337
D	0.0070	0.0105	0.0200	0.0392

Pin Capacitance

Pin	Capacitance (pF)			
	XL	**X1**	**X2**	**X4**
AN	0.0014	0.0017	0.0026	0.0049
BN	0.0015	0.0017	0.0026	0.0050
C	0.0017	0.0027	0.0052	0.0108
D	0.0017	0.0026	0.0051	0.0101

Delays at 25°C,1.2V, Typical Process

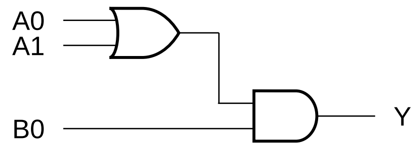
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	**X1**	**X2**	**X4**	**XL**	**X1**	**X2**	**X4**
AN → Y ↑	0.1082	0.0827	0.0858	0.0797	28.7849	14.8997	7.7072	3.9743
AN → Y ↓	0.0851	0.0882	0.0960	0.0888	5.2437	3.1061	1.6682	0.8273
BN → Y ↑	0.1563	0.1266	0.1264	0.1263	28.7729	14.8929	7.7018	3.9724
BN → Y ↓	0.1015	0.1029	0.1014	0.0973	5.1694	3.0329	1.6607	0.8440
C → Y ↑	0.1508	0.1211	0.1212	0.1214	28.7236	14.8789	7.6938	3.9681
C → Y ↓	0.0301	0.0287	0.0312	0.0325	5.1175	2.9989	1.6582	0.8959
D → Y ↑	0.1595	0.1289	0.1311	0.1310	28.7121	14.8726	7.6922	3.9677
D → Y ↓	0.0301	0.0287	0.0319	0.0341	5.3073	3.0923	1.7060	0.9411

Cell Description

The OA21 cell provides the logical AND of one OR group and an additional input. The output (Y) is represented by the logic equation:

$$Y = (A0 + A1) \bullet B0$$

Logic Symbol



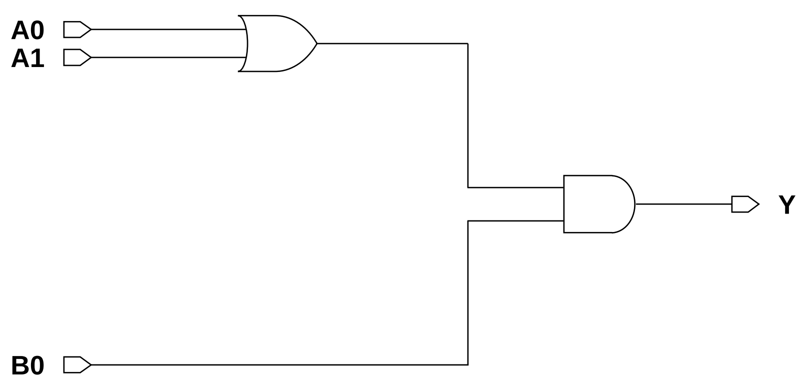
Function Table

A0	A1	B0	Y
x	x	0	0
0	0	x	0
x	1	1	1
1	x	1	1

Cell Size

Drive Strength	Height (um)	Width (um)
OA21XLM	2.87	2.87
OA21X1M	2.87	2.87
OA21X2M	2.87	2.87
OA21X4M	2.87	3.69
OA21X8M	2.87	6.56

Functional Schematic



AC Power

Pin	Power (uW/MHz)				
	XL	X1	X2	X4	X8
A0	0.0074	0.0089	0.0125	0.0205	0.0387
A1	0.0079	0.0094	0.0131	0.0219	0.0417
B0	0.0051	0.0063	0.0088	0.0158	0.0294

Pin Capacitance

Pin	Capacitance (pF)				
	XL	X1	X2	X4	X8
A0	0.0015	0.0015	0.0020	0.0032	0.0060
A1	0.0013	0.0013	0.0018	0.0031	0.0059
B0	0.0015	0.0015	0.0020	0.0034	0.0062

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				
	XL	X1	X2	X4	X8
A0 → Y ↑	0.0696	0.0739	0.0727	0.0678	0.0664
A0 → Y ↓	0.1829	0.2116	0.1985	0.1699	0.1639
A1 → Y ↑	0.0734	0.0778	0.0777	0.0732	0.0723
A1 → Y ↓	0.1903	0.2188	0.2063	0.1787	0.1780
B0 → Y ↑	0.0645	0.0689	0.0684	0.0664	0.0650
B0 → Y ↓	0.0854	0.0996	0.0925	0.0849	0.0788

Delays at 25°C,1.2V, Typical Process (Cont'd.)

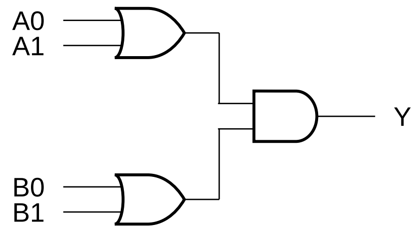
Description	K_{load} (ns/pF)				
	XL	X1	X2	X4	X8
A0 → Y ↑	8.8998	5.3209	3.5760	1.8761	0.9337
A0 → Y ↓	6.0527	3.5741	1.9300	0.9269	0.4828
A1 → Y ↑	8.9108	5.3289	3.5809	1.8783	0.9353
A1 → Y ↓	6.0532	3.5744	1.9298	0.9268	0.4828
B0 → Y ↑	8.9108	5.3293	3.5811	1.8784	0.9351
B0 → Y ↓	5.5216	3.2467	1.7833	0.8675	0.4559

Cell Description

The OA22 cell provides the logical AND of two OR groups. The output (Y) is represented by the logic equation:

$Y = (A0 + A1) \bullet (B0 + B1)$

Logic Symbol



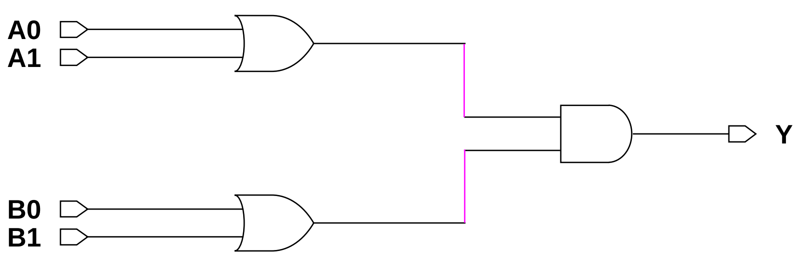
Function Table

A0	A1	B0	B1	Y
x	x	0	0	0
0	0	x	x	0
x	1	x	1	1
x	1	1	x	1
1	x	x	1	1
1	x	1	x	1

Cell Size

Drive Strength	Height (um)	Width (um)
OA22XLM	2.87	3.28
OA22X1M	2.87	3.28
OA22X2M	2.87	3.28
OA22X4M	2.87	4.10
OA22X8M	2.87	6.97

Functional Schematic



AC Power

Pin	Power (uW/MHz)				
	XL	**X1**	**X2**	**X4**	**X8**
A0	0.0061	0.0079	0.0109	0.0188	0.0376
A1	0.0064	0.0083	0.0116	0.0203	0.0404
B0	0.0100	0.0121	0.0153	0.0252	0.0487
B1	0.0104	0.0126	0.0163	0.0268	0.0514

Pin Capacitance

Pin	Capacitance (pF)				
	XL	**X1**	**X2**	**X4**	**X8**
A0	0.0015	0.0017	0.0024	0.0036	0.0073
A1	0.0013	0.0016	0.0021	0.0033	0.0065
B0	0.0015	0.0017	0.0022	0.0035	0.0074
B1	0.0014	0.0017	0.0023	0.0036	0.0065

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				
	XL	**X1**	**X2**	**X4**	**X8**
A0 → Y ↑	0.0782	0.0754	0.0755	0.0741	0.0751
A0 → Y ↓	0.1518	0.1531	0.1399	0.1369	0.1382
A1 → Y ↑	0.0842	0.0822	0.0833	0.0801	0.0807
A1 → Y ↓	0.1609	0.1626	0.1493	0.1452	0.1471
B0 → Y ↑	0.1002	0.0955	0.0911	0.0854	0.0855
B0 → Y ↓	0.2438	0.2338	0.1975	0.1841	0.1757
B1 → Y ↑	0.1061	0.1021	0.0990	0.0922	0.0906
B1 → Y ↓	0.2537	0.2442	0.2087	0.1948	0.1835

Delays at 25°C,1.2V, Typical Process (Cont'd.)

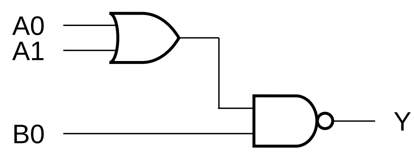
Description	K_{load} (ns/pF)				
	XL	**X1**	**X2**	**X4**	**X8**
A0 → Y ↑	8.9090	5.3409	3.5966	1.8578	0.9251
A0 → Y ↓	6.3398	3.6885	1.9494	0.9515	0.4737
A1 → Y ↑	8.9241	5.3498	3.6007	1.8596	0.9260
A1 → Y ↓	6.3396	3.6883	1.9494	0.9514	0.4737
B0 → Y ↑	8.9101	5.3421	3.5960	1.8571	0.9248
B0 → Y ↓	6.6888	3.8381	1.9978	0.9712	0.4780
B1 → Y ↑	8.9242	5.3506	3.6012	1.8589	0.9258
B1 → Y ↓	6.6877	3.8376	1.9975	0.9710	0.4778

Cell Description

The OAI21cell provides the logical inverted AND of one OR group and an additional input. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 + A1) \bullet B0}$$

Logic Symbol



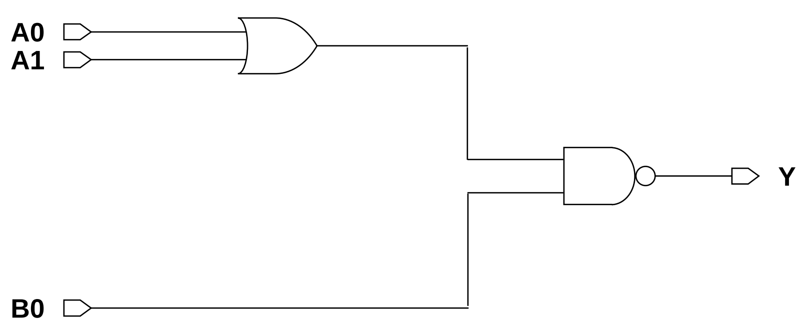
Function Table

A0	A1	B0	Y
0	0	x	1
x	x	0	1
x	1	1	0
1	x	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
OAI21XLM	2.87	2.05
OAI21X1M	2.87	2.05
OAI21X2M	2.87	2.05
OAI21X3M	2.87	3.69
OAI21X4M	2.87	3.69
OAI21X6M	2.87	5.33
OAI21X8M	2.87	6.56

Functional Schematic



AC Power

Pin	Power (uW/MHz)						
	XL	X1	X2	X3	X4	X6	X8
A0	0.0053	0.0062	0.0097	0.0142	0.0180	0.0269	0.0351
A1	0.0059	0.0074	0.0113	0.0166	0.0212	0.0314	0.0413
B0	0.0039	0.0048	0.0069	0.0096	0.0123	0.0193	0.0247

Pin Capacitance

Pin	Capacitance (pF)						
	XL	X1	X2	X3	X4	X6	X8
A0	0.0017	0.0024	0.0035	0.0052	0.0066	0.0105	0.0135
A1	0.0015	0.0022	0.0033	0.0054	0.0069	0.0101	0.0136
B0	0.0016	0.0024	0.0035	0.0053	0.0067	0.0099	0.0130

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)						
	XL	X1	X2	X3	X4	X6	X8
A0 → Y ↑	0.0840	0.0635	0.0681	0.0712	0.0662	0.0676	0.0670
A0 → Y ↓	0.0385	0.0304	0.0268	0.0257	0.0251	0.0255	0.0252
A1 → Y ↑	0.0924	0.0725	0.0767	0.0814	0.0763	0.0768	0.0770
A1 → Y ↓	0.0442	0.0363	0.0312	0.0301	0.0296	0.0298	0.0297
B0 → Y ↑	0.0324	0.0249	0.0251	0.0248	0.0234	0.0247	0.0242
B0 → Y ↓	0.0349	0.0292	0.0242	0.0225	0.0221	0.0232	0.0227

Delays at 25°C,1.2V, Typical Process (Cont'd.)

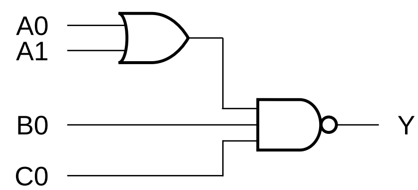
Description	K_{load} (ns/pF)						
	XL	X1	X2	X3	X4	X6	X8
A0 → Y ↑	17.8467	10.7787	7.2747	5.1823	3.7756	2.5587	1.9403
A0 → Y ↓	8.0390	4.7854	2.6141	1.7071	1.2941	0.8748	0.6598
A1 → Y ↑	17.8344	10.7717	7.2714	5.1812	3.7743	2.5575	1.9396
A1 → Y ↓	8.1347	4.8681	2.6584	1.7067	1.2971	0.8836	0.6615
B0 → Y ↑	8.8817	5.3445	3.6070	2.5519	1.8574	1.2576	0.9544
B0 → Y ↓	8.1440	4.8762	2.6612	1.7075	1.2987	0.8844	0.6619

Cell Description

The OAI211 cell provides the logical inverted OR of one OR group and two additional inputs. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 + A1) \bullet B0 \bullet C0}$$

Logic Symbol



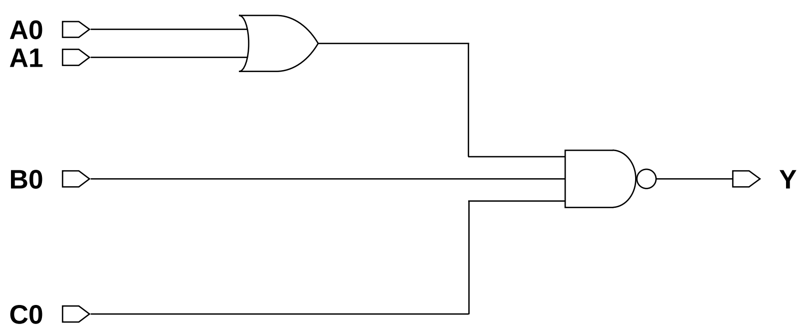
Function Table

A0	A1	B0	C0	Y
0	0	x	x	1
x	x	0	x	1
x	x	x	0	1
x	1	1	1	0
1	x	1	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
OAI211XLM	2.87	2.46
OAI211X1M	2.87	2.46
OAI211X2M	2.87	2.46
OAI211X4M	2.87	4.51
OAI211X8M	2.87	8.61

Functional Schematic



AC Power

Pin	Power (uW/MHz)				
	XL	X1	X2	X4	X8
A0	0.0062	0.0080	0.0128	0.0239	0.0473
A1	0.0067	0.0092	0.0143	0.0266	0.0530
B0	0.0041	0.0055	0.0080	0.0137	0.0276
C0	0.0047	0.0066	0.0101	0.0186	0.0370

Pin Capacitance

Pin	Capacitance (pF)				
	XL	X1	X2	X4	X8
A0	0.0016	0.0023	0.0034	0.0071	0.0140
A1	0.0017	0.0025	0.0035	0.0064	0.0132
B0	0.0016	0.0024	0.0036	0.0065	0.0133
C0	0.0016	0.0023	0.0036	0.0070	0.0137

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				
	XL	X1	X2	X4	X8
A0 → Y ↑	0.1006	0.0862	0.0935	0.0913	0.0926
A0 → Y ↓	0.0579	0.0503	0.0430	0.0412	0.0403
A1 → Y ↑	0.1126	0.0971	0.1035	0.0994	0.1016
A1 → Y ↓	0.0671	0.0592	0.0492	0.0469	0.0463
B0 → Y ↑	0.0368	0.0314	0.0307	0.0310	0.0317
B0 → Y ↓	0.0511	0.0455	0.0358	0.0332	0.0329
C0 → Y ↑	0.0421	0.0372	0.0394	0.0362	0.0393
C0 → Y ↓	0.0560	0.0507	0.0423	0.0406	0.0399

Delays at 25°C,1.2V, Typical Process (Cont'd.)

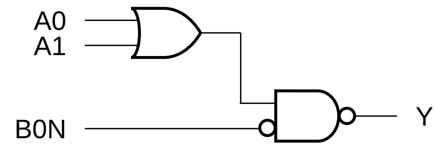
Description	K_{load} (ns/pF)				
	XL	X1	X2	X4	X8
A0 → Y ↑	18.0139	10.9075	7.4420	3.8055	1.9539
A0 → Y ↓	10.9487	6.6281	3.6258	1.8055	0.8979
A1 → Y ↑	18.0080	10.9065	7.4407	3.8042	1.9536
A1 → Y ↓	10.8987	6.5670	3.5903	1.8223	0.9025
B0 → Y ↑	8.9229	5.3652	3.6213	2.0747	1.0648
B0 → Y ↓	10.9090	6.5737	3.5925	1.8236	0.9032
C0 → Y ↑	9.1702	5.5443	3.8165	1.8605	1.0287
C0 → Y ↓	10.9074	6.5707	3.5917	1.8236	0.9033

Cell Description

The OAI21B cell provides the logical inverted AND of one OR group and an additional input. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 + A1) \bullet \overline{B0N}}$$

Logic Symbol



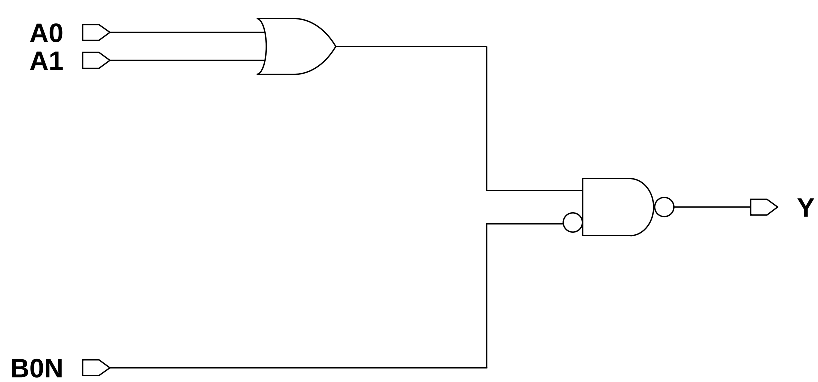
Function Table

A0	A1	B0N	Y
0	0	x	1
x	x	1	1
x	1	0	0
1	x	0	0

Cell Size

Drive Strength	Height (um)	Width (um)
OAI21BXLM	2.87	2.46
OAI21BX1M	2.87	2.46
OAI21BX2M	2.87	3.28
OAI21BX4M	2.87	4.51
OAI21BX8M	2.87	7.79

Functional Schematic



AC Power

Pin	Power (uW/MHz)				
	XL	X1	X2	X4	X8
A0	0.0044	0.0052	0.0080	0.0142	0.0276
A1	0.0048	0.0063	0.0095	0.0175	0.0340
B0N	0.0060	0.0071	0.0095	0.0173	0.0312

Pin Capacitance

Pin	Capacitance (pF)				
	XL	X1	X2	X4	X8
A0	0.0020	0.0024	0.0035	0.0066	0.0135
A1	0.0015	0.0023	0.0033	0.0070	0.0137
B0N	0.0015	0.0014	0.0020	0.0032	0.0037

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				
	XL	X1	X2	X4	X8
A0 → Y ↑	0.0853	0.0640	0.0691	0.0657	0.0666
A0 → Y ↓	0.0310	0.0343	0.0308	0.0297	0.0316
A1 → Y ↑	0.0911	0.0725	0.0772	0.0765	0.0768
A1 → Y ↓	0.0410	0.0407	0.0356	0.0349	0.0366
B0N → Y ↑	0.0586	0.0564	0.0546	0.0517	0.0638
B0N → Y ↓	0.0956	0.1081	0.0921	0.0830	0.1047

Delays at 25°C,1.2V, Typical Process (Cont'd.)

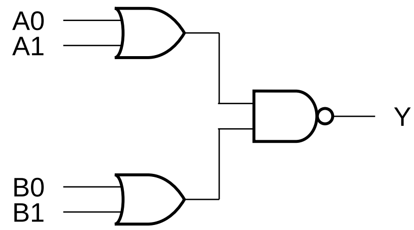
Description	K_{load} (ns/pF)				
	XL	X1	X2	X4	X8
A0 → Y ↑	17.9589	10.8173	7.3335	3.7882	1.9584
A0 → Y ↓	5.6636	4.8862	2.6251	1.3385	0.6663
A1 → Y ↑	17.9182	10.8052	7.3285	3.7873	1.9578
A1 → Y ↓	7.1058	4.9689	2.6751	1.3401	0.6646
B0N → Y ↑	9.0356	5.3956	3.7011	1.8999	0.9746
B0N → Y ↓	7.1355	4.9912	2.6855	1.3439	0.6678

Cell Description

The OAI22 cell provides the logical inverted AND of two OR groups. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 + A1) \bullet (B0 + B1)}$$

Logic Symbol



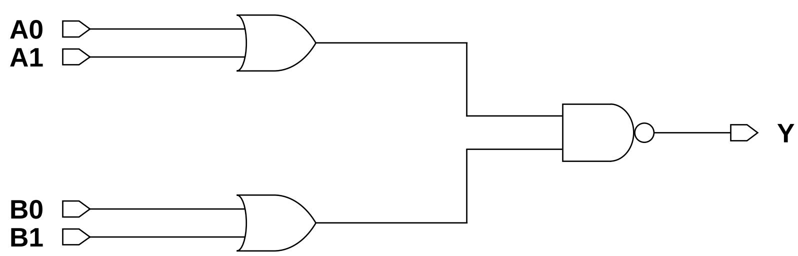
Function Table

A0	A1	B0	B1	Y
0	0	x	x	1
x	x	0	0	1
x	1	x	1	0
x	1	1	x	0
1	x	x	1	0
1	x	1	x	0

Cell Size

Drive Strength	Height (um)	Width (um)
OAI22XLM	2.87	2.46
OAI22X1M	2.87	2.46
OAI22X2M	2.87	2.87
OAI22X4M	2.87	4.92
OAI22X8M	2.87	8.61

Functional Schematic



AC Power

Pin	Power (uW/MHz)				
	XL	X1	X2	X4	X8
A0	0.0040	0.0054	0.0076	0.0159	0.0307
A1	0.0044	0.0062	0.0092	0.0187	0.0362
B0	0.0074	0.0094	0.0135	0.0263	0.0505
B1	0.0079	0.0104	0.0150	0.0290	0.0564

Pin Capacitance

Pin	Capacitance (pF)				
	XL	X1	X2	X4	X8
A0	0.0018	0.0026	0.0036	0.0074	0.0139
A1	0.0015	0.0023	0.0034	0.0065	0.0131
B0	0.0017	0.0025	0.0035	0.0073	0.0139
B1	0.0015	0.0023	0.0034	0.0064	0.0133

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				
	XL	X1	X2	X4	X8
A0 → Y ↑	0.0553	0.0486	0.0494	0.0542	0.0569
A0 → Y ↓	0.0322	0.0291	0.0257	0.0271	0.0271
A1 → Y ↑	0.0641	0.0583	0.0586	0.0625	0.0664
A1 → Y ↓	0.0387	0.0359	0.0307	0.0309	0.0313
B0 → Y ↑	0.1111	0.0937	0.0931	0.0937	0.0921
B0 → Y ↓	0.0532	0.0441	0.0365	0.0366	0.0367
B1 → Y ↑	0.1190	0.1022	0.1018	0.1016	0.1016
B1 → Y ↓	0.0590	0.0501	0.0412	0.0402	0.0413

Delays at 25°C,1.2V, Typical Process (Cont'd.)

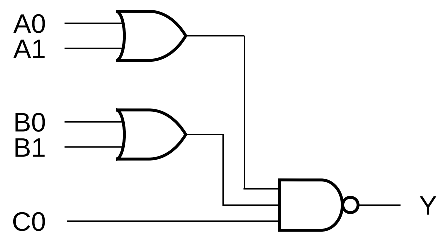
Description	K_{load} (ns/pF)				
	XL	**X1**	**X2**	**X4**	**X8**
A0 → Y ↑	18.1421	11.6917	7.4068	3.8212	2.1235
A0 → Y ↓	8.0405	4.8144	2.6095	1.2922	0.6905
A1 → Y ↑	18.1192	11.6813	7.4027	3.8181	2.1222
A1 → Y ↓	8.2253	4.9774	2.6575	1.2966	0.6923
B0 → Y ↑	18.0328	11.5553	7.5390	3.8005	1.9500
B0 → Y ↓	8.1305	4.8874	2.6224	1.2963	0.6894
B1 → Y ↑	18.0134	11.5461	7.5355	3.7982	1.9493
B1 → Y ↓	8.2227	4.9745	2.6564	1.2960	0.6921

Cell Description

The OAI221 cell provides the logical inverted AND of two OR groups and an additional input. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 + A1) \bullet (B0 + B1) \bullet C0}$$

Logic Symbol



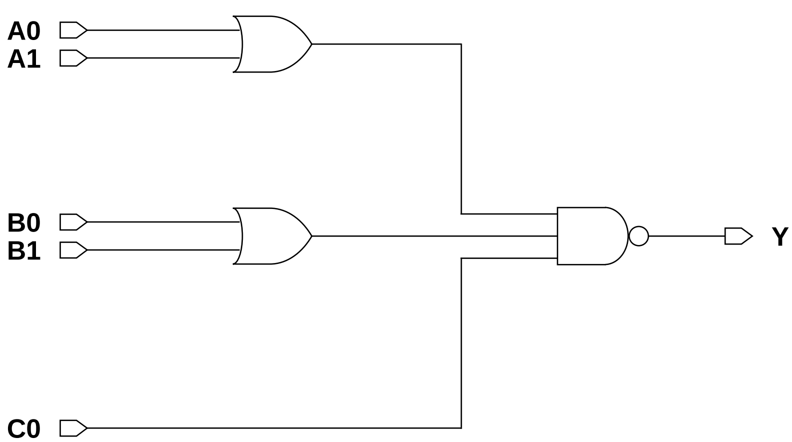
Function Table

A0	A1	B0	B1	C0	Y
0	0	x	x	x	1
x	x	0	0	x	1
x	x	x	x	0	1
x	1	x	1	1	0
x	1	1	x	1	0
1	x	x	1	1	0
1	x	1	x	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
OAI221XLM	2.87	3.28
OAI221X1M	2.87	3.28
OAI221X2M	2.87	3.69
OAI221X4M	2.87	6.15

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	XL	X1	X2	X4
A0	0.0071	0.0087	0.0121	0.0228
A1	0.0077	0.0098	0.0136	0.0256
B0	0.0092	0.0114	0.0174	0.0324
B1	0.0098	0.0125	0.0188	0.0351
C0	0.0055	0.0066	0.0094	0.0172

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0017	0.0024	0.0035	0.0073
A1	0.0016	0.0023	0.0034	0.0064
B0	0.0016	0.0023	0.0035	0.0072
B1	0.0015	0.0023	0.0034	0.0064
C0	0.0017	0.0025	0.0036	0.0069

Delays at 25°C,1.2V, Typical Process

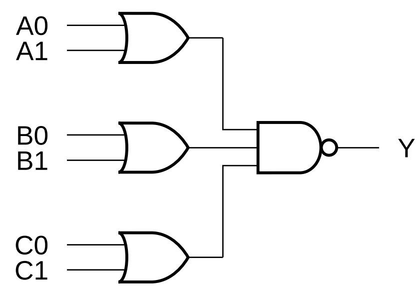
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 → Y ↑	0.1159	0.0886	0.0869	0.0854	18.6978	11.3315	7.7031	3.9379
A0 → Y ↓	0.0770	0.0628	0.0498	0.0486	11.1443	6.7135	3.7156	1.8697
A1 → Y ↑	0.1241	0.0990	0.0960	0.0936	18.6869	11.3275	7.7008	3.9361
A1 → Y ↓	0.0864	0.0734	0.0555	0.0547	11.3144	6.8405	3.6570	1.8775
B0 → Y ↑	0.1412	0.1114	0.1188	0.1169	18.2495	11.0167	7.4378	3.8411
B0 → Y ↓	0.0898	0.0736	0.0597	0.0579	11.2899	6.8304	3.6973	1.8657
B1 → Y ↑	0.1509	0.1217	0.1275	0.1249	18.2442	11.0145	7.4365	3.8400
B1 → Y ↓	0.0990	0.0833	0.0651	0.0641	11.3172	6.8430	3.6743	1.8774
C0 → Y ↑	0.0412	0.0329	0.0327	0.0308	8.9820	5.4032	3.7628	1.8742
C0 → Y ↓	0.0632	0.0562	0.0434	0.0428	11.3288	6.8487	3.6747	1.8782

Cell Description

The OAI222 cell provides the logical inverted AND of three OR groups. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 + A1) \bullet (B0 + B1) \bullet (C0 + C1)}$$

Logic Symbol



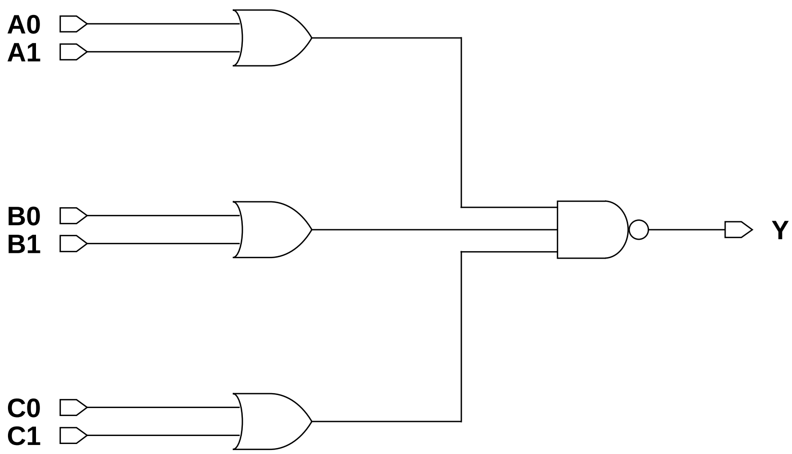
Function Table

A0	A1	B0	B1	C0	C1	Y
0	0	x	x	x	x	1
x	x	0	0	x	x	1
x	x	x	x	0	0	1
x	1	x	1	1	x	0
x	1	x	1	x	1	0
x	1	1	x	1	x	0
x	1	1	x	x	1	0
1	x	x	1	1	x	0
1	x	x	1	x	1	0
1	x	1	x	1	x	0
1	x	1	x	x	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
OAI222XLM	2.87	3.69
OAI222X1M	2.87	3.69
OAI222X2M	2.87	4.10
OAI222X4M	2.87	7.38

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	XL	**X1**	**X2**	**X4**
A0	0.0094	0.0116	0.0160	0.0306
A1	0.0098	0.0126	0.0175	0.0334
B0	0.0115	0.0143	0.0212	0.0402
B1	0.0121	0.0154	0.0226	0.0429
C0	0.0053	0.0071	0.0103	0.0203
C1	0.0060	0.0082	0.0119	0.0230

Pin Capacitance

Pin	Capacitance (pF)			
	XL	**X1**	**X2**	**X4**
A0	0.0020	0.0025	0.0036	0.0071
A1	0.0015	0.0023	0.0035	0.0064
B0	0.0016	0.0023	0.0035	0.0072
B1	0.0015	0.0024	0.0034	0.0064
C0	0.0019	0.0025	0.0036	0.0074
C1	0.0018	0.0024	0.0035	0.0065

Delays at 25°C,1.2V, Typical Process

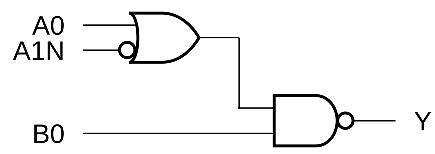
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	**X1**	**X2**	**X4**	**XL**	**X1**	**X2**	**X4**
A0 → Y ↑	0.1627	0.1153	0.1151	0.1112	18.8561	11.4316	7.7493	3.9622
A0 → Y ↓	0.0878	0.0847	0.0669	0.0632	9.2423	6.7981	3.6990	1.8229
A1 → Y ↑	0.1663	0.1242	0.1241	0.1197	18.8309	11.4260	7.7474	3.9606
A1 → Y ↓	0.1056	0.0950	0.0723	0.0694	10.5553	6.9143	3.6388	1.8278
B0 → Y ↑	0.1895	0.1375	0.1476	0.1437	18.1854	11.0094	7.4327	3.8364
B0 → Y ↓	0.1110	0.0954	0.0766	0.0723	10.5384	6.9067	3.6787	1.8211
B1 → Y ↑	0.1987	0.1479	0.1562	0.1519	18.1787	11.0078	7.4312	3.8355
B1 → Y ↓	0.1196	0.1056	0.0818	0.0785	10.5554	6.9134	3.6594	1.8281
C0 → Y ↑	0.0860	0.0647	0.0678	0.0677	18.3622	11.0810	7.4822	3.8616
C0 → Y ↓	0.0563	0.0572	0.0474	0.0475	10.4184	6.7811	3.6554	1.8343
C1 → Y ↑	0.0959	0.0746	0.0777	0.0758	18.3490	11.0743	7.4801	3.8590
C1 → Y ↓	0.0665	0.0684	0.0545	0.0531	10.5583	6.9150	3.6561	1.8280

Cell Description

The OAI2B1 cell provides the logical inverted AND of one OR group and an additional input. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 + \overline{A1N}) \bullet B0}$$

Logic Symbol



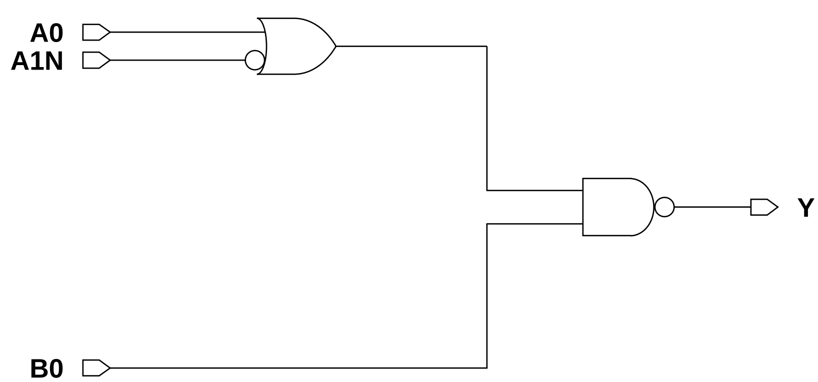
Function Table

A0	A1N	B0	Y
0	1	x	1
x	x	0	1
x	0	1	0
1	x	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
OAI2B1XLM	2.87	2.87
OAI2B1X1M	2.87	2.87
OAI2B1X2M	2.87	2.87
OAI2B1X4M	2.87	4.51
OAI2B1X8M	2.87	8.20

Functional Schematic



AC Power

Pin	Power (uW/MHz)				
	XL	X1	X2	X4	X8
A0	0.0057	0.0068	0.0097	0.0179	0.0351
A1N	0.0065	0.0080	0.0113	0.0213	0.0411
B0	0.0038	0.0046	0.0066	0.0116	0.0232

Pin Capacitance

Pin	Capacitance (pF)				
	XL	X1	X2	X4	X8
A0	0.0016	0.0024	0.0035	0.0065	0.0135
A1N	0.0015	0.0014	0.0019	0.0032	0.0059
B0	0.0016	0.0024	0.0036	0.0067	0.0131

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				
	XL	X1	X2	X4	X8
A0 → Y ↑	0.0902	0.0698	0.0699	0.0671	0.0687
A0 → Y ↓	0.0398	0.0323	0.0275	0.0254	0.0254
A1N → Y ↑	0.1228	0.1073	0.1057	0.1063	0.1046
A1N → Y ↓	0.1047	0.1105	0.0963	0.0928	0.0879
B0 → Y ↑	0.0318	0.0253	0.0253	0.0235	0.0245
B0 → Y ↓	0.0338	0.0294	0.0247	0.0222	0.0227

Delays at 25°C,1.2V, Typical Process (Cont’d.)

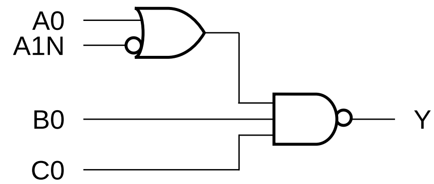
Description	K_{load} (ns/pF)				
	XL	X1	X2	X4	X8
A0 → Y ↑	18.2878	11.0491	7.4662	3.8391	1.9789
A0 → Y ↓	8.0282	4.7986	2.6873	1.3149	0.6574
A1N → Y ↑	18.2689	11.0409	7.4610	3.8383	1.9780
A1N → Y ↓	8.2394	4.9638	2.7568	1.3284	0.6646
B0 → Y ↑	8.9321	5.3613	3.6254	1.8590	0.9606
B0 → Y ↓	8.2249	4.9423	2.7469	1.3220	0.6620

Cell Description

The OAI2B11 cell provides the logical inverted OR of one OR group and two additional inputs. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 + \overline{A1N}) \bullet B0 \bullet C0}$$

Logic Symbol



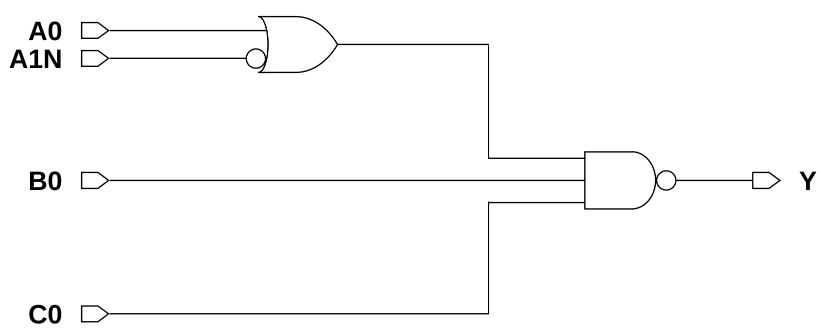
Function Table

A0	A1N	B0	C0	Y
0	1	x	x	1
x	x	0	x	1
x	x	x	0	1
x	0	1	1	0
1	x	1	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
OAI2B11XLM	2.87	3.28
OAI2B11X1M	2.87	3.28
OAI2B11X2M	2.87	3.28
OAI2B11X4M	2.87	5.33

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	XL	X1	X2	X4
A0	0.0063	0.0081	0.0124	0.0235
A1N	0.0067	0.0089	0.0137	0.0269
B0	0.0039	0.0054	0.0074	0.0129
C0	0.0045	0.0066	0.0096	0.0177

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0016	0.0024	0.0035	0.0066
A1N	0.0014	0.0014	0.0019	0.0032
B0	0.0017	0.0025	0.0036	0.0066
C0	0.0015	0.0023	0.0035	0.0071

Delays at 25°C,1.2V, Typical Process

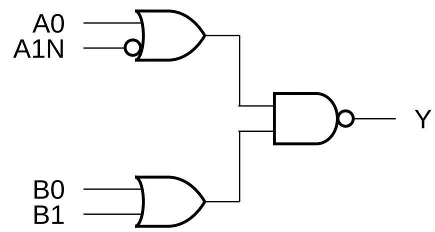
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 → Y ↑	0.1003	0.0832	0.0872	0.0897	17.6225	10.5061	7.0630	3.8749
A0 → Y ↓	0.0593	0.0513	0.0433	0.0402	11.2149	6.7791	3.7343	1.8517
A1N → Y ↑	0.1270	0.1162	0.1197	0.1270	17.6086	10.5012	7.0601	3.8736
A1N → Y ↓	0.1179	0.1242	0.1086	0.1073	11.1522	6.7749	3.7138	1.8253
B0 → Y ↑	0.0352	0.0307	0.0294	0.0287	8.6921	5.1661	3.5082	1.9616
B0 → Y ↓	0.0509	0.0470	0.0366	0.0320	11.1626	6.7733	3.7129	1.8315
C0 → Y ↑	0.0387	0.0357	0.0363	0.0357	8.6920	5.2491	3.5505	1.8810
C0 → Y ↓	0.0549	0.0525	0.0430	0.0394	11.1562	6.7713	3.7113	1.8320

Cell Description

The OAI2B2 cell provides the logical inverted AND of two OR groups. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 + \overline{A1N}) \bullet (B0 + B1)}$$

Logic Symbol



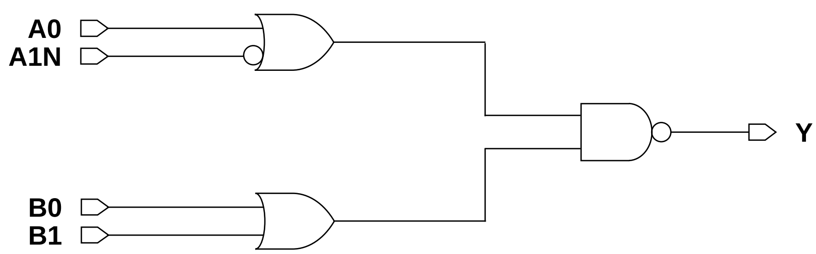
Function Table

A0	A1N	B0	B1	Y
0	1	x	x	1
x	x	0	0	1
x	0	x	1	0
x	0	1	x	0
1	x	x	1	0
1	x	1	x	0

Cell Size

Drive Strength	Height (um)	Width (um)
OAI2B2XLM	2.87	3.28
OAI2B2X1M	2.87	3.28
OAI2B2X2M	2.87	3.69
OAI2B2X4M	2.87	5.74
OAI2B2X8M	2.87	10.25

Functional Schematic



AC Power

Pin	Power (uW/MHz)				
	XL	**X1**	**X2**	**X4**	**X8**
A0	0.0038	0.0054	0.0082	0.0162	0.0319
A1N	0.0059	0.0079	0.0104	0.0209	0.0394
B0	0.0075	0.0090	0.0121	0.0232	0.0452
B1	0.0080	0.0099	0.0136	0.0265	0.0512

Pin Capacitance

Pin	Capacitance (pF)				
	XL	**X1**	**X2**	**X4**	**X8**
A0	0.0017	0.0025	0.0036	0.0068	0.0136
A1N	0.0015	0.0014	0.0019	0.0032	0.0059
B0	0.0019	0.0025	0.0036	0.0066	0.0134
B1	0.0016	0.0022	0.0034	0.0072	0.0137

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				
	XL	**X1**	**X2**	**X4**	**X8**
A0 → Y ↑	0.0591	0.0511	0.0528	0.0542	0.0568
A0 → Y ↓	0.0308	0.0319	0.0266	0.0268	0.0265
A1N → Y ↑	0.0905	0.0890	0.0884	0.0928	0.0927
A1N → Y ↓	0.0976	0.1153	0.0981	0.0964	0.0905
B0 → Y ↑	0.1248	0.0959	0.0915	0.0922	0.0963
B0 → Y ↓	0.0510	0.0518	0.0415	0.0408	0.0400
B1 → Y ↑	0.1309	0.1032	0.1008	0.1036	0.1067
B1 → Y ↓	0.0557	0.0585	0.0469	0.0459	0.0447

Delays at 25°C,1.2V, Typical Process (Cont'd.)

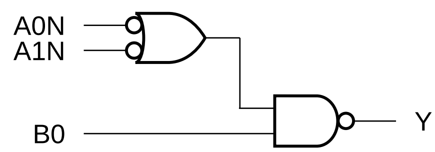
Description	K_{load} (ns/pF)				
	XL	**X1**	**X2**	**X4**	**X8**
A0 → Y ↑	18.5144	11.1309	7.6074	3.9122	2.0643
A0 → Y ↓	6.9287	4.8759	2.6388	1.3269	0.6615
A1N → Y ↑	18.4783	11.1236	7.6037	3.9121	2.0636
A1N → Y ↓	7.1705	5.0481	2.6931	1.3288	0.6634
B0 → Y ↑	17.9849	10.8453	7.3921	3.8658	2.0616
B0 → Y ↓	7.0479	4.9082	2.6573	1.3330	0.6639
B1 → Y ↑	17.9613	10.8366	7.3895	3.8657	2.0614
B1 → Y ↓	7.1504	5.0373	2.6881	1.3268	0.6629

Cell Description

The OAI2BB1cell provides the logical inverted AND of one OR group of two inverted inputs (A0N,A1N) and an additional non-inverted input (B0). The output (Y) is represented by the logic equation:

$Y = \overline{(\overline{A0N} + \overline{A1N}) \bullet B0}$

Logic Symbol



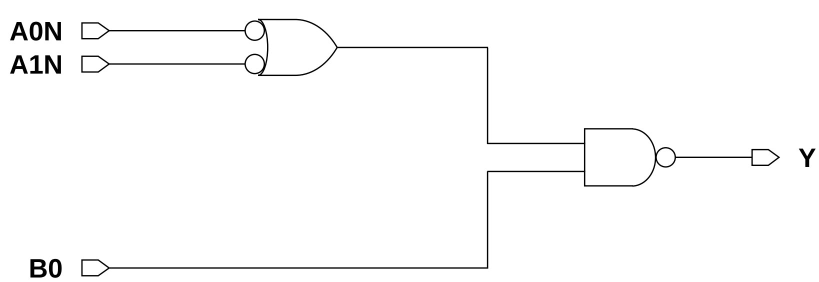
Function Table

A0N	A1N	B0	Y
1	1	x	1
x	x	0	1
x	0	1	0
0	x	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
OAI2BB1XLM	2.87	2.46
OAI2BB1X1M	2.87	2.46
OAI2BB1X2M	2.87	2.46
OAI2BB1X4M	2.87	4.10

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	XL	**X1**	**X2**	**X4**
A0N	0.0059	0.0074	0.0101	0.0192
A1N	0.0050	0.0063	0.0087	0.0170
B0	0.0027	0.0039	0.0058	0.0122

Pin Capacitance

Pin	Capacitance (pF)			
	XL	**X1**	**X2**	**X4**
A0N	0.0012	0.0012	0.0016	0.0027
A1N	0.0011	0.0011	0.0014	0.0021
B0	0.0013	0.0019	0.0029	0.0054

Delays at 25°C,1.2V, Typical Process

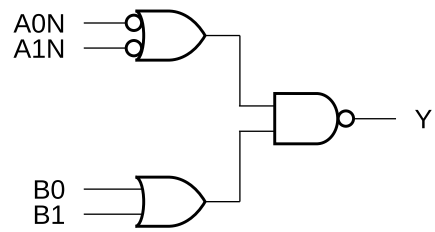
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	**X1**	**X2**	**X4**	**XL**	**X1**	**X2**	**X4**
A0N → Y ↑	0.0713	0.0696	0.0683	0.0722	8.8167	5.2932	3.6285	1.8607
A0N → Y ↓	0.0968	0.1122	0.1009	0.1002	8.1270	4.9136	2.6691	1.3631
A1N → Y ↑	0.0687	0.0670	0.0654	0.0687	8.8167	5.2926	3.6287	1.8607
A1N → Y ↓	0.0860	0.0988	0.0884	0.0878	8.0940	4.8921	2.6606	1.3590
B0 → Y ↑	0.0304	0.0278	0.0299	0.0329	8.9391	5.2840	3.6131	1.9094
B0 → Y ↓	0.0265	0.0265	0.0247	0.0267	8.0088	4.8392	2.6406	1.3502

Cell Description

The OAI2BB2 cell provides the logical inverted AND of one OR group of two inverted inputs (A0N,A1N) and one OR group of two non-inverted inputs (B0,B1). The output (Y) is represented by the logic equation:

$$Y = \overline{(\overline{A0N} + \overline{A1N}) \bullet (B0 + B1)}$$

Logic Symbol



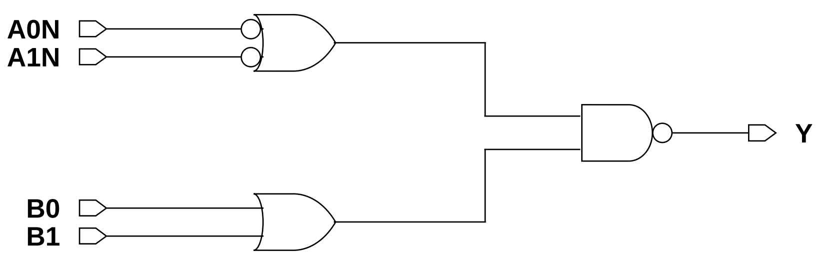
Function Table

A0N	A1N	B0	B1	Y
1	1	x	x	1
x	x	0	0	1
x	0	x	1	0
x	0	1	x	0
0	x	x	1	0
0	x	1	x	0

Cell Size

Drive Strength	Height (um)	Width (um)
OAI2BB2XLM	2.87	2.87
OAI2BB2X1M	2.87	2.87
OAI2BB2X2M	2.87	3.28
OAI2BB2X4M	2.87	4.92
OAI2BB2X8M	2.87	8.61

Functional Schematic



AC Power

Pin	Power (uW/MHz)				
	XL	**X1**	**X2**	**X4**	**X8**
A0N	0.0071	0.0089	0.0118	0.0205	0.0407
A1N	0.0065	0.0081	0.0106	0.0185	0.0367
B0	0.0044	0.0050	0.0077	0.0141	0.0278
B1	0.0047	0.0060	0.0091	0.0174	0.0341

Pin Capacitance

Pin	Capacitance (pF)				
	XL	**X1**	**X2**	**X4**	**X8**
A0N	0.0015	0.0015	0.0019	0.0032	0.0061
A1N	0.0015	0.0016	0.0020	0.0035	0.0070
B0	0.0019	0.0024	0.0035	0.0066	0.0135
B1	0.0015	0.0022	0.0033	0.0070	0.0138

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				
	XL	**X1**	**X2**	**X4**	**X8**
A0N → Y ↑	0.0843	0.0776	0.0735	0.0697	0.0696
A0N → Y ↓	0.1117	0.1295	0.1124	0.0981	0.1034
A1N → Y ↑	0.0814	0.0749	0.0705	0.0670	0.0668
A1N → Y ↓	0.1060	0.1238	0.1048	0.0922	0.0912
B0 → Y ↑	0.0847	0.0640	0.0705	0.0666	0.0677
B0 → Y ↓	0.0304	0.0339	0.0313	0.0299	0.0304
B1 → Y ↑	0.0932	0.0731	0.0794	0.0773	0.0780
B1 → Y ↓	0.0418	0.0423	0.0368	0.0350	0.0353

Delays at 25°C,1.2V, Typical Process (Cont'd.)

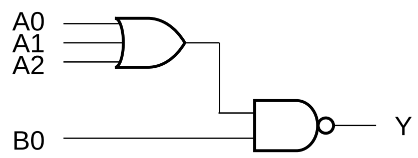
Description	K_{load} (ns/pF)				
	XL	**X1**	**X2**	**X4**	**X8**
A0N → Y ↑	8.7997	5.1499	3.6397	1.9160	0.9760
A0N → Y ↓	7.3155	5.2063	2.7802	1.3376	0.6884
A1N → Y ↑	8.7889	5.1470	3.6398	1.9161	0.9760
A1N → Y ↓	7.3105	5.2067	2.7774	1.3368	0.6866
B0 → Y ↑	18.1873	10.7883	7.5588	3.7929	1.9442
B0 → Y ↓	5.7095	4.8892	2.7008	1.3344	0.6854
B1 → Y ↑	18.1598	10.7789	7.5541	3.7917	1.9432
B1 → Y ↓	7.2195	5.1554	2.7598	1.3299	0.6842

Cell Description

The OAI31 cell provides the logical inverted AND of one OR group and an additional input. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 + A1 + A2) \bullet B0}$$

Logic Symbol



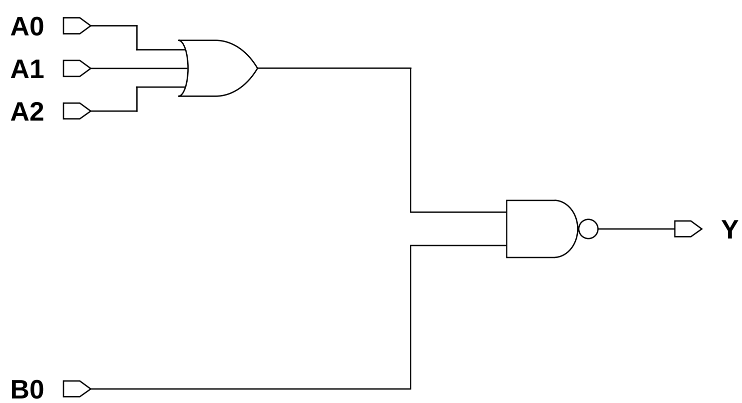
Function Table

A0	A1	A2	B0	Y
0	0	0	x	1
x	x	x	0	1
x	x	1	1	0
x	1	x	1	0
1	x	x	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
OAI31XLM	2.87	2.46
OAI31X1M	2.87	2.46
OAI31X2M	2.87	2.87
OAI31X4M	2.87	4.51

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	XL	**X1**	**X2**	**X4**
A0	0.0055	0.0065	0.0105	0.0200
A1	0.0062	0.0076	0.0120	0.0227
A2	0.0067	0.0086	0.0134	0.0258
B0	0.0045	0.0058	0.0084	0.0150

Pin Capacitance

Pin	Capacitance (pF)			
	XL	**X1**	**X2**	**X4**
A0	0.0016	0.0024	0.0035	0.0065
A1	0.0016	0.0024	0.0034	0.0066
A2	0.0014	0.0022	0.0033	0.0070
B0	0.0016	0.0024	0.0035	0.0067

Delays at 25°C,1.2V, Typical Process

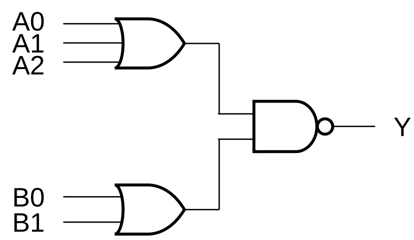
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	**X1**	**X2**	**X4**	**XL**	**X1**	**X2**	**X4**
A0 → Y ↑	0.1265	0.0937	0.1049	0.1125	27.3136	16.5469	11.1804	6.2351
A0 → Y ↓	0.0395	0.0313	0.0280	0.0283	8.0111	4.7728	2.5951	1.3866
A1 → Y ↑	0.1489	0.1168	0.1263	0.1361	27.2946	16.5326	11.1731	6.2324
A1 → Y ↓	0.0457	0.0376	0.0326	0.0320	8.0789	4.8245	2.6180	1.3440
A2 → Y ↑	0.1572	0.1252	0.1351	0.1480	27.2894	16.5311	11.1740	6.2334
A2 → Y ↓	0.0492	0.0406	0.0347	0.0344	8.2958	4.9412	2.6543	1.3550
B0 → Y ↑	0.0321	0.0251	0.0253	0.0238	8.9410	5.3659	3.6200	1.8800
B0 → Y ↓	0.0385	0.0331	0.0268	0.0242	8.3122	4.9482	2.6566	1.3694

Cell Description

The OAI32 cell provides the logical inverted AND of two OR groups. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 + A1 + A2) \bullet (B0 + B1)}$$

Logic Symbol



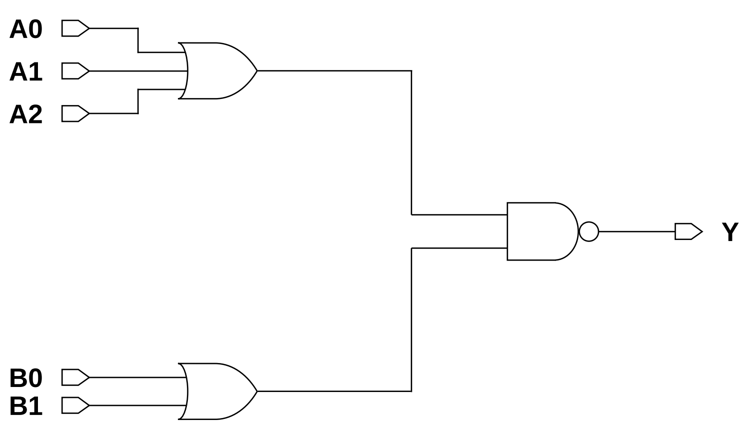
Function Table

A0	A1	A2	B0	B1	Y
0	0	0	x	x	1
x	x	x	0	0	1
x	x	1	x	1	0
x	x	1	1	x	0
x	1	x	1	x	0
x	1	x	x	1	0
1	x	x	1	x	0
1	x	x	x	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
OAI32XLM	2.87	2.87
OAI32X1M	2.87	2.87
OAI32X2M	2.87	3.28
OAI32X4M	2.87	5.74

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	XL	**X1**	**X2**	**X4**
A0	0.0077	0.0091	0.0140	0.0277
A1	0.0082	0.0101	0.0155	0.0310
A2	0.0088	0.0111	0.0168	0.0340
B0	0.0044	0.0064	0.0091	0.0181
B1	0.0051	0.0075	0.0106	0.0214

Pin Capacitance

Pin	Capacitance (pF)			
	XL	**X1**	**X2**	**X4**
A0	0.0017	0.0024	0.0036	0.0064
A1	0.0015	0.0022	0.0034	0.0069
A2	0.0014	0.0022	0.0032	0.0073
B0	0.0017	0.0025	0.0035	0.0066
B1	0.0015	0.0024	0.0035	0.0073

Delays at 25°C,1.2V, Typical Process

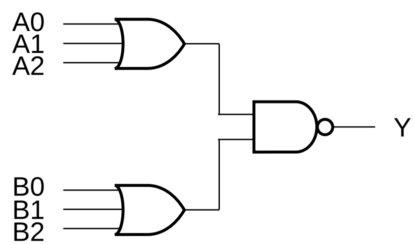
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	**X1**	**X2**	**X4**	**XL**	**X1**	**X2**	**X4**
A0 → Y ↑	0.1699	0.1271	0.1388	0.1386	27.5408	16.6865	11.3048	5.8154
A0 → Y ↓	0.0545	0.0448	0.0375	0.0366	8.1130	4.8624	2.6055	1.3070
A1 → Y ↑	0.1915	0.1485	0.1599	0.1643	27.5166	16.6742	11.2995	5.8137
A1 → Y ↓	0.0610	0.0513	0.0423	0.0422	8.1833	4.8864	2.6239	1.3156
A2 → Y ↑	0.2011	0.1595	0.1685	0.1761	27.5168	16.6748	11.2989	5.8143
A2 → Y ↓	0.0657	0.0560	0.0451	0.0450	8.3875	5.0024	2.6630	1.3253
B0 → Y ↑	0.0582	0.0505	0.0497	0.0494	18.1799	11.0026	7.4331	3.8266
B0 → Y ↓	0.0376	0.0363	0.0285	0.0279	8.2059	4.9545	2.6289	1.3296
B1 → Y ↑	0.0673	0.0601	0.0594	0.0608	18.1590	10.9934	7.4290	3.8262
B1 → Y ↓	0.0454	0.0436	0.0341	0.0335	8.3919	5.0054	2.6637	1.3254

Cell Description

The OAI33 cell provides the logical inverted AND of two OR groups. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 + A1 + A2) \bullet (B0 + B1 + B2)}$$

Logic Symbol



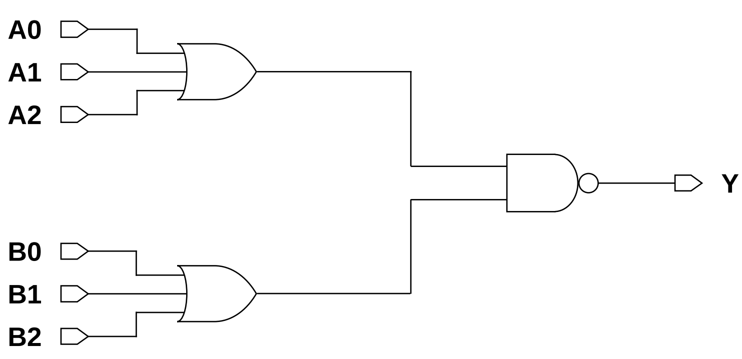
Function Table

A0	A1	A2	B0	B1	B2	Y
0	0	0	x	x	x	1
x	x	x	0	0	0	1
x	x	1	x	x	1	0
x	x	1	x	1	x	0
x	x	1	1	x	x	0
x	1	x	x	x	1	0
x	1	x	x	1	x	0
x	1	x	1	x	x	0
1	x	x	x	x	1	0
1	x	x	x	1	x	0
1	x	x	1	x	x	0

Cell Size

Drive Strength	Height (um)	Width (um)
OAI33XLM	2.87	3.69
OAI33X1M	2.87	3.69
OAI33X2M	2.87	3.69
OAI33X4M	2.87	6.97

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	XL	X1	X2	X4
A0	0.0056	0.0077	0.0113	0.0218
A1	0.0064	0.0089	0.0129	0.0250
A2	0.0070	0.0100	0.0143	0.0282
B0	0.0090	0.0120	0.0180	0.0360
B1	0.0097	0.0131	0.0196	0.0392
B2	0.0102	0.0140	0.0208	0.0422

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0016	0.0024	0.0035	0.0067
A1	0.0018	0.0026	0.0036	0.0070
A2	0.0016	0.0024	0.0035	0.0078
B0	0.0016	0.0023	0.0035	0.0065
B1	0.0016	0.0023	0.0034	0.0069
B2	0.0014	0.0021	0.0032	0.0072

Delays at 25°C,1.2V, Typical Process

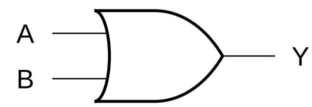
Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 → Y ↑	0.1061	0.0838	0.0875	0.0871	28.3024	17.1580	11.4127	5.9931
A0 → Y ↓	0.0444	0.0390	0.0321	0.0319	8.1254	4.8131	2.6428	1.3482
A1 → Y ↑	0.1388	0.1141	0.1131	0.1141	28.3152	17.1571	11.4097	5.9920
A1 → Y ↓	0.0544	0.0481	0.0383	0.0381	8.1643	4.8361	2.6533	1.3512
A2 → Y ↑	0.1485	0.1236	0.1228	0.1266	28.3111	17.1552	11.4103	5.9936
A2 → Y ↓	0.0594	0.0529	0.0420	0.0418	8.3183	4.9376	2.7084	1.3614
B0 → Y ↑	0.1955	0.1585	0.1740	0.1769	27.6355	16.7218	11.3553	5.8376
B0 → Y ↓	0.0642	0.0542	0.0468	0.0449	8.0930	4.8446	2.7122	1.3226
B1 → Y ↑	0.2196	0.1828	0.1956	0.2028	27.6141	16.7122	11.3500	5.8362
B1 → Y ↓	0.0717	0.0617	0.0524	0.0520	8.1242	4.8629	2.7225	1.3537
B2 → Y ↑	0.2282	0.1910	0.2038	0.2146	27.6143	16.7109	11.3496	5.8364
B2 → Y ↓	0.0771	0.0661	0.0543	0.0554	8.3180	4.9365	2.6880	1.3614

Cell Description

The OR2 cell provides the logical OR of two inputs (A,B). The output (Y) is represented by the logic equation:

$$Y = (A + B)$$

Logic Symbol



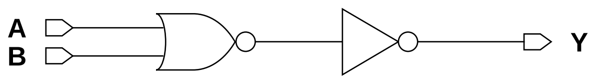
Function Table

A	B	Y
0	0	0
x	1	1
1	x	1

Cell Size

Drive Strength	Height (um)	Width (um)
OR2X1M	2.87	2.05
OR2X2M	2.87	2.05
OR2X4M	2.87	2.46
OR2X6M	2.87	4.10
OR2X8M	2.87	4.51
OR2X12M	2.87	6.97

Functional Schematic



AC Power

Pin	Power (uW/MHz)					
	X1	**X2**	**X4**	**X6**	**X8**	**X12**
A	0.0073	0.0091	0.0154	0.0243	0.0303	0.0446
B	0.0081	0.0103	0.0170	0.0278	0.0337	0.0511

Pin Capacitance

Pin	Capacitance (pF)					
	X1	**X2**	**X4**	**X6**	**X8**	**X12**
A	0.0017	0.0023	0.0033	0.0058	0.0063	0.0107
B	0.0018	0.0024	0.0034	0.0065	0.0068	0.0110

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)					
	X1	**X2**	**X4**	**X6**	**X8**	**X12**
A → Y ↑	0.0568	0.0532	0.0492	0.0484	0.0474	0.0552
A → Y ↓	0.1273	0.1023	0.1129	0.0952	0.1108	0.0911
B → Y ↑	0.0615	0.0581	0.0517	0.0537	0.0502	0.0619
B → Y ↓	0.1399	0.1135	0.1233	0.1064	0.1214	0.1014

Delays at 25°C,1.2V, Typical Process (Cont'd.)

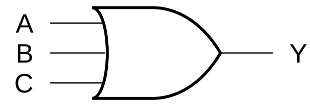
Description	K_{load} (ns/pF)					
	X1	**X2**	**X4**	**X6**	**X8**	**X12**
A → Y ↑	5.2656	3.5580	1.8343	1.2552	0.9459	0.6514
A → Y ↓	3.3497	1.7540	0.9073	0.5748	0.4400	0.2886
B → Y ↑	5.2704	3.5602	1.8346	1.2562	0.9463	0.6522
B → Y ↓	3.3497	1.7540	0.9073	0.5747	0.4399	0.2885

Cell Description

The OR3 cell provides the logical OR of three inputs (A,B,C). The output (Y) is represented by the logic equation:

$Y = (A + B + C)$

Logic Symbol



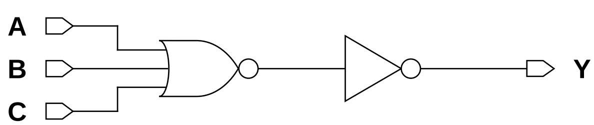
Function Table

A	B	C	Y
0	0	0	0
x	x	1	1
x	1	x	1
1	x	x	1

Cell Size

Drive Strength	Height (um)	Width (um)
OR3X1M	2.87	2.46
OR3X2M	2.87	2.46
OR3X4M	2.87	4.51
OR3X6M	2.87	4.92
OR3X8M	2.87	6.97
OR3X12M	2.87	11.07

Functional Schematic



AC Power

Pin	Power (uW/MHz)					
	X1	**X2**	**X4**	**X6**	**X8**	**X12**
A	0.0090	0.0110	0.0192	0.0266	0.0370	0.0566
B	0.0098	0.0123	0.0221	0.0295	0.0417	0.0646
C	0.0107	0.0136	0.0248	0.0322	0.0464	0.0725

Pin Capacitance

Pin	Capacitance (pF)					
	X1	**X2**	**X4**	**X6**	**X8**	**X12**
A	0.0018	0.0025	0.0046	0.0047	0.0080	0.0129
B	0.0018	0.0026	0.0050	0.0052	0.0077	0.0124
C	0.0018	0.0025	0.0053	0.0054	0.0078	0.0126

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)					
	X1	**X2**	**X4**	**X6**	**X8**	**X12**
A → Y ↑	0.0636	0.0597	0.0547	0.0588	0.0531	0.0549
A → Y ↓	0.1673	0.1304	0.1148	0.1417	0.1275	0.1224
B → Y ↑	0.0694	0.0663	0.0621	0.0649	0.0588	0.0626
B → Y ↓	0.1932	0.1553	0.1413	0.1683	0.1566	0.1525
C → Y ↑	0.0733	0.0705	0.0666	0.0685	0.0624	0.0673
C → Y ↓	0.2053	0.1655	0.1525	0.1791	0.1690	0.1650

Delays at 25°C,1.2V, Typical Process (Cont'd.)

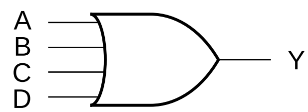
Description	K_{load} (ns/pF)					
	X1	**X2**	**X4**	**X6**	**X8**	**X12**
A → Y ↑	5.2948	3.5965	1.8525	1.2529	0.9967	0.6836
A → Y ↓	3.6611	1.8641	0.9046	0.6202	0.4594	0.3127
B → Y ↑	5.3030	3.6004	1.8550	1.2541	0.9978	0.6845
B → Y ↓	3.6608	1.8640	0.9046	0.6202	0.4593	0.3125
C → Y ↑	5.3149	3.6076	1.8585	1.2560	1.0001	0.6866
C → Y ↓	3.6609	1.8640	0.9046	0.6202	0.4593	0.3123

Cell Description

The OR4 cell provides the logical OR of four inputs (A,B,C,D). The output (Y) is represented by the logic equation:

$Y = (A + B + C + D)$

Logic Symbol



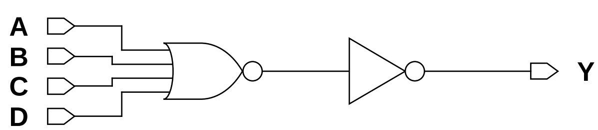
Function Table

A	B	C	D	Y
0	0	0	0	0
x	x	x	1	1
x	x	1	x	1
x	1	x	x	1
1	x	x	x	1

Cell Size

Drive Strength	Height (um)	Width (um)
OR4X1M	2.87	2.87
OR4X2M	2.87	3.28
OR4X4M	2.87	6.56
OR4X6M	2.87	8.20
OR4X8M	2.87	11.48
OR4X12M	2.87	14.35

Functional Schematic



AC Power

Pin	Power (uW/MHz)					
	X1	**X2**	**X4**	**X6**	**X8**	**X12**
A	0.0099	0.0122	0.0208	0.0346	0.0430	0.0653
B	0.0107	0.0135	0.0239	0.0391	0.0491	0.0738
C	0.0116	0.0148	0.0271	0.0432	0.0548	0.0820
D	0.0125	0.0162	0.0300	0.0476	0.0604	0.0900

Pin Capacitance

Pin	Capacitance (pF)					
	X1	**X2**	**X4**	**X6**	**X8**	**X12**
A	0.0019	0.0026	0.0047	0.0070	0.0094	0.0137
B	0.0019	0.0026	0.0045	0.0067	0.0090	0.0128
C	0.0019	0.0025	0.0045	0.0067	0.0090	0.0134
D	0.0019	0.0026	0.0045	0.0069	0.0089	0.0129

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)					
	X1	**X2**	**X4**	**X6**	**X8**	**X12**
A → Y ↑	0.0657	0.0620	0.0558	0.0586	0.0555	0.0597
A → Y ↓	0.2069	0.1624	0.1297	0.1524	0.1359	0.1415
B → Y ↑	0.0709	0.0687	0.0656	0.0663	0.0643	0.0677
B → Y ↓	0.2473	0.2026	0.1796	0.1996	0.1853	0.1863
C → Y ↑	0.0750	0.0730	0.0708	0.0711	0.0695	0.0677
C → Y ↓	0.2723	0.2257	0.2061	0.2276	0.2128	0.2129
D → Y ↑	0.0771	0.0754	0.0724	0.0729	0.0720	0.0722
D → Y ↓	0.2862	0.2380	0.2177	0.2401	0.2245	0.2233

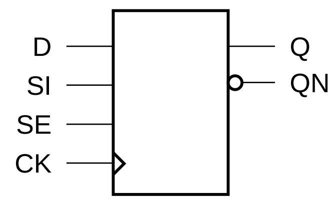
Delays at 25°C,1.2V, Typical Process (Cont'd.)

Description	K_{load} (ns/pF)					
	X1	**X2**	**X4**	**X6**	**X8**	**X12**
A → Y ↑	5.2559	3.6092	1.7784	1.2704	0.9343	0.6258
A → Y ↓	3.9940	2.0157	0.9786	0.6951	0.4917	0.3327
B → Y ↑	5.2620	3.6131	1.7823	1.2721	0.9361	0.6269
B → Y ↓	3.9929	2.0156	0.9791	0.6951	0.4919	0.3327
C → Y ↑	5.2785	3.6213	1.7885	1.2754	0.9390	0.6288
C → Y ↓	3.9934	2.0154	0.9791	0.6950	0.4918	0.3328
D → Y ↑	5.3011	3.6345	1.7989	1.2812	0.9442	0.6321
D → Y ↓	3.9933	2.0158	0.9791	0.6949	0.4918	0.3327

Cell Description

The SDFF cell is a positive-edge triggered, static D-type flip-flop with scan input (SI) and active-high scan enable (SE).

Logic Symbol



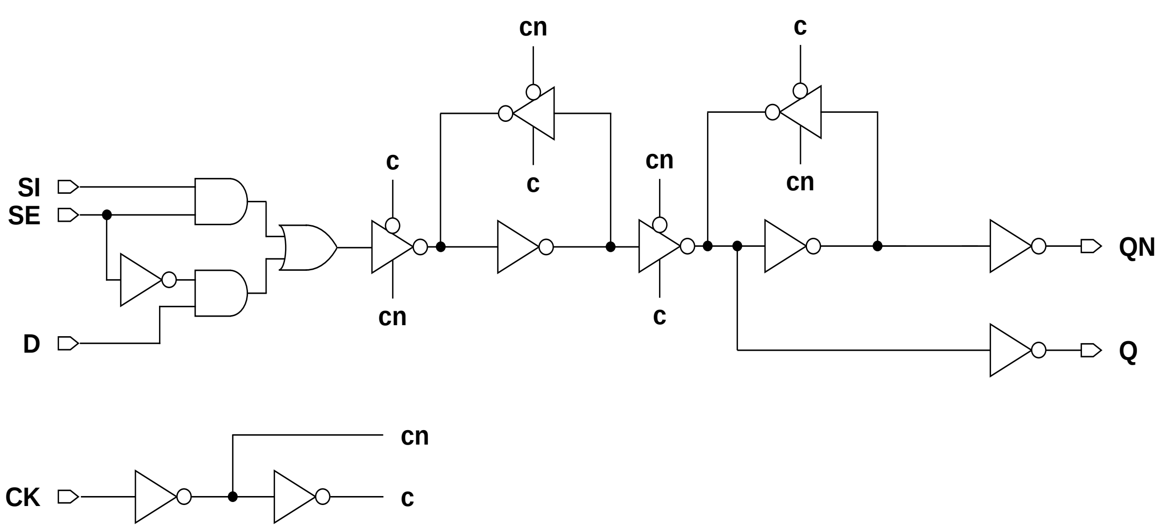
Function Table

D	SI	SE	CK	Q[n+1]	QN[n+1]
1	x	0	↑	1	0
0	x	0	↑	0	1
x	x	x	↓	Q[n]	QN[n]
x	1	1	↑	1	0
x	0	1	↑	0	1

Cell Size

Drive Strength	Height (um)	Width (um)
SDFFX1M	2.87	10.25
SDFFX2M	2.87	10.66
SDFFX4M	2.87	11.48

Functional Schematic



AC Power

Pin	Power (uW/MHz)		
	X1	X2	X4
SI	0.0128	0.0129	0.0135
SE	0.0145	0.0146	0.0151
D	0.0108	0.0112	0.0117
CK	0.0204	0.0214	0.0228
Q	0.0125	0.0164	0.0271

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
SI	0.0014	0.0015	0.0015
SE	0.0032	0.0030	0.0029
D	0.0016	0.0014	0.0014
CK	0.0019	0.0019	0.0021

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
CK → Q ↑	0.1771	0.1906	0.1927	5.1782	3.5851	1.9462
CK → Q ↓	0.2194	0.2118	0.2193	3.9355	2.0049	0.9898
CK → QN ↑	0.2679	0.2744	0.2996	5.1157	3.5799	1.9028
CK → QN ↓	0.2643	0.2976	0.2923	3.2034	1.8701	0.8956

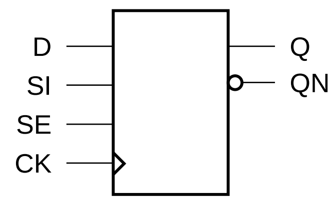
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)		
		X1	X2	X4
SI	setup ↑ → CK	0.1250	0.1172	0.1211
	setup ↓ → CK	0.3438	0.3633	0.3828
	hold ↑ → CK	-0.1016	-0.0938	-0.0977
	hold ↓ → CK	-0.2852	-0.3008	-0.3125
SE	setup ↑ → CK	0.3594	0.3828	0.4062
	setup ↓ → CK	0.2031	0.3047	0.3203
	hold ↑ → CK	-0.0898	-0.0781	-0.0820
	hold ↓ → CK	-0.1445	-0.1406	-0.1445
D	setup ↑ → CK	0.1094	0.1016	0.1055
	setup ↓ → CK	0.2109	0.3164	0.3359
	hold ↑ → CK	-0.0859	-0.0781	-0.0820
	hold ↓ → CK	-0.1523	-0.2539	-0.2695
CK	minpwh	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974

Cell Description

The SDFFH cell is a positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and fast clock-to-Q-path.

Logic Symbol



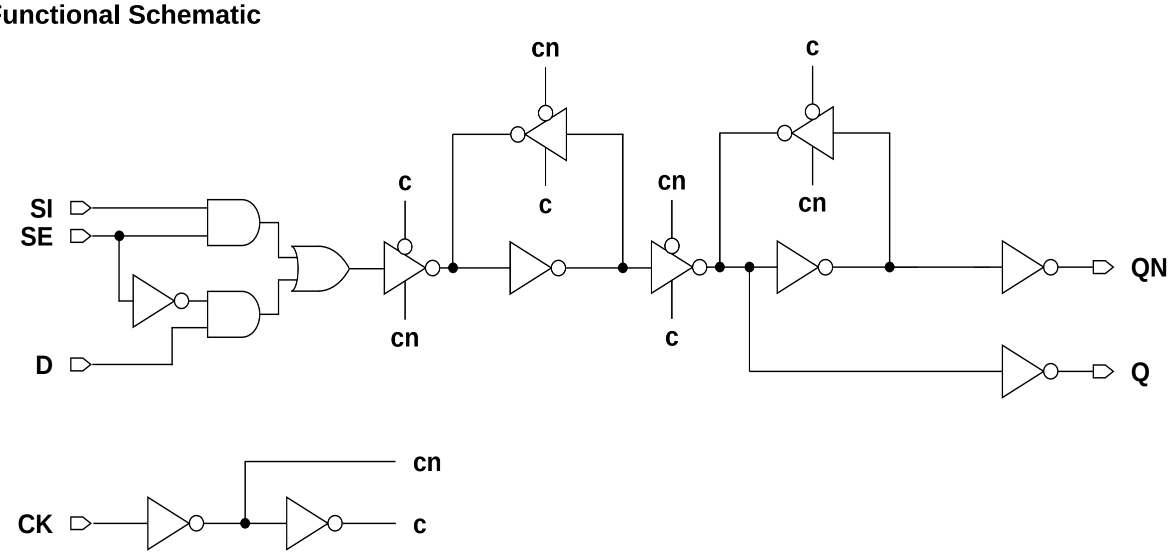
Function Table

D	SI	SE	CK	Q[n+1]	QN[n+1]
1	x	0	_/‾	1	0
0	x	0	_/‾	0	1
x	x	x	‾_	Q[n]	QN[n]
x	1	1	_/‾	1	0
x	0	1	_/‾	0	1

Cell Size

Drive Strength	Height (um)	Width (um)
SDFFHX1M	2.87	13.12
SDFFHX2M	2.87	13.53
SDFFHX4M	2.87	13.94
SDFFHX8M	2.87	14.76

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	X1	X2	X4	X8
SI	0.0141	0.0174	0.0206	0.0207
SE	0.0195	0.0222	0.0260	0.0260
D	0.0167	0.0203	0.0253	0.0253
CK	0.0306	0.0350	0.0408	0.0410
Q	0.0104	0.0118	0.0166	0.0274

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
SI	0.0016	0.0016	0.0020	0.0020
SE	0.0037	0.0039	0.0044	0.0044
D	0.0016	0.0019	0.0029	0.0029
CK	0.0028	0.0029	0.0037	0.0037

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
CK → Q ↑	0.1508	0.1480	0.1466	0.1657	5.2061	4.1069	1.7915	0.9486
CK → Q ↓	0.1403	0.1401	0.1344	0.1643	3.2725	1.7471	0.8809	0.4678
CK → QN ↑	0.1850	0.1906	0.1927	0.2277	8.8405	13.0249	13.0259	13.0823
CK → QN ↓	0.2226	0.2149	0.2307	0.2571	5.2024	5.0938	5.0708	5.0426

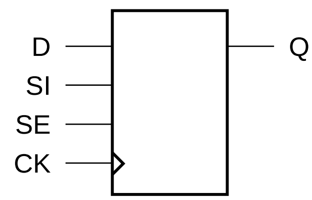
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
SI	setup ↑ → CK	0.1406	0.1406	0.1094	0.1133
	setup ↓ → CK	0.2500	0.2617	0.2227	0.2188
	hold ↑ → CK	-0.0625	-0.0703	-0.0430	-0.0391
	hold ↓ → CK	-0.1875	-0.2031	-0.1680	-0.1641
SE	setup ↑ → CK	0.2812	0.2930	0.2656	0.2617
	setup ↓ → CK	0.2148	0.1836	0.1523	0.1562
	hold ↑ → CK	-0.0703	-0.0742	-0.0469	-0.0469
	hold ↓ → CK	-0.1055	-0.1016	-0.0859	-0.0820
D	setup ↑ → CK	0.1484	0.1328	0.1016	0.1055
	setup ↓ → CK	0.2109	0.1797	0.1484	0.1445
	hold ↑ → CK	-0.0703	-0.0625	-0.0352	-0.0312
	hold ↓ → CK	-0.1445	-0.1250	-0.0977	-0.0938
CK	minpwh	0.9974	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974	0.9974

Cell Description

The SDFFHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with scan input (SI) and active-high scan enable (SE). The cell has a single output (Q) and fast clock-to-out path.

Logic Symbol



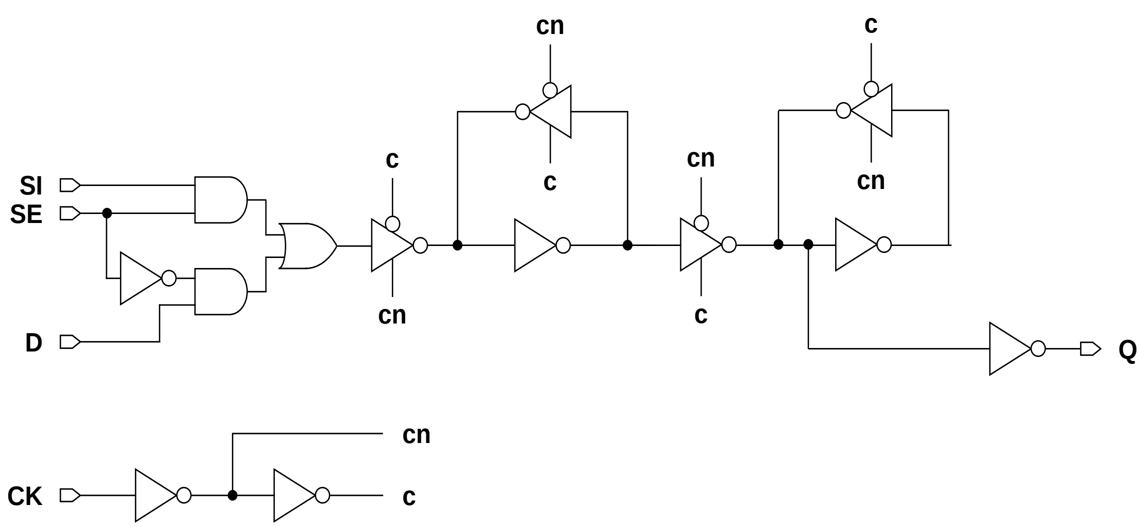
Function Table

D	SI	SE	CK	Q[n+1]
1	x	0	↑	1
0	x	0	↑	0
x	x	x	↓	Q[n]
x	1	1	↑	1
x	0	1	↑	0

Cell Size

Drive Strength	Height (um)	Width (um)
SDFFHQX1M	2.87	12.71
SDFFHQX2M	2.87	13.12
SDFFHQX4M	2.87	13.53
SDFFHQX8M	2.87	14.76

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	X1	X2	X4	X8
SI	0.0142	0.0179	0.0219	0.0235
SE	0.0175	0.0200	0.0249	0.0261
D	0.0157	0.0187	0.0240	0.0244
CK	0.0296	0.0339	0.0401	0.0430
Q	0.0074	0.0094	0.0138	0.0241

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
SI	0.0016	0.0017	0.0020	0.0026
SE	0.0033	0.0036	0.0042	0.0047
D	0.0020	0.0019	0.0026	0.0026
CK	0.0028	0.0028	0.0037	0.0052

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
CK → Q ↑	0.1510	0.1469	0.1383	0.1346	5.2062	4.0686	1.8705	0.9697
CK → Q ↓	0.1389	0.1424	0.1285	0.1335	3.2567	1.8184	0.8829	0.4516

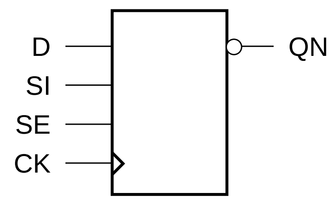
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
SI	setup ↑ → CK	0.1328	0.1328	0.1328	0.1328
	setup ↓ → CK	0.2578	0.2617	0.2617	0.2344
	hold ↑ → CK	-0.0586	-0.0664	-0.0586	-0.0469
	hold ↓ → CK	-0.1953	-0.2031	-0.1992	-0.1758
SE	setup ↑ → CK	0.2617	0.2695	0.2773	0.2500
	setup ↓ → CK	0.1719	0.1602	0.1562	0.1875
	hold ↑ → CK	-0.0508	-0.0586	-0.0508	-0.0391
	hold ↓ → CK	-0.0938	-0.0898	-0.0820	-0.0938
D	setup ↑ → CK	0.1406	0.1172	0.1133	0.1406
	setup ↓ → CK	0.1406	0.1562	0.1602	0.1758
	hold ↑ → CK	-0.0625	-0.0508	-0.0430	-0.0547
	hold ↓ → CK	-0.0859	-0.1055	-0.1055	-0.1172
CK	minpwh	0.9974	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974	0.9974

Cell Description

The SDFFHQN cell is a high-speed, positive-edge triggered, static D-type flip-flop with scan input (SI) and active-high scan enable (SE). The cell has a single output (QN) and fast clock-to-out path.

Logic Symbol



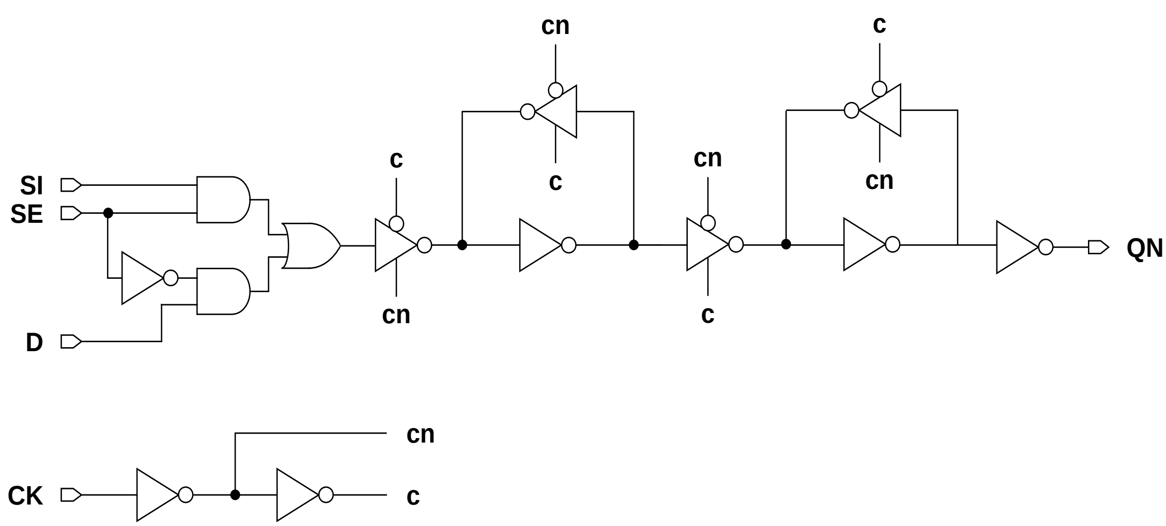
Function Table

D	SI	SE	CK	Q[n+1]
1	x	0	↑	0
0	x	0	↑	1
x	x	x	↓	QN[n]
x	1	1	↑	0
x	0	1	↑	1

Cell Size

Drive Strength	Height (um)	Width (um)
SDFFHQNX1M	2.87	11.48
SDFFHQNX2M	2.87	11.48
SDFFHQNX4M	2.87	11.89
SDFFHQNX8M	2.87	13.53

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	X1	X2	X4	X8
SI	0.0119	0.0121	0.0130	0.0183
SE	0.0147	0.0147	0.0160	0.0203
D	0.0124	0.0126	0.0137	0.0196
CK	0.0224	0.0228	0.0249	0.0328
QN	0.0087	0.0110	0.0160	0.0281

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
SI	0.0012	0.0012	0.0012	0.0012
SE	0.0034	0.0034	0.0033	0.0037
D	0.0015	0.0015	0.0015	0.0021
CK	0.0030	0.0030	0.0029	0.0031

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
CK → QN ↑	0.1994	0.2066	0.1901	0.1728	5.1215	3.6821	1.8150	0.9490
CK → QN ↓	0.2326	0.2258	0.2144	0.1974	3.2616	1.8103	0.8485	0.4275

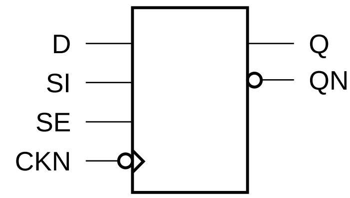
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
SI	setup ↑ → CK	0.1680	0.1406	0.1484	0.1328
	setup ↓ → CK	0.2422	0.2539	0.2227	0.2422
	hold ↑ → CK	-0.0898	-0.0742	-0.0781	-0.0781
	hold ↓ → CK	-0.1797	-0.1875	-0.1719	-0.1953
SE	setup ↑ → CK	0.2578	0.2734	0.2383	0.2656
	setup ↓ → CK	0.2148	0.2031	0.1914	0.1562
	hold ↑ → CK	-0.0820	-0.0664	-0.0703	-0.0703
	hold ↓ → CK	-0.1289	-0.1211	-0.1172	-0.0859
D	setup ↑ → CK	0.1719	0.1445	0.1523	0.1133
	setup ↓ → CK	0.1914	0.2031	0.1719	0.1289
	hold ↑ → CK	-0.0938	-0.0781	-0.0820	-0.0625
	hold ↓ → CK	-0.1289	-0.1367	-0.1211	-0.0859
CK	minpwh	0.9974	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974	0.9974

Cell Description

The SDFFNH cell is a negative-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and fast clock-to-Q-path.

Logic Symbol



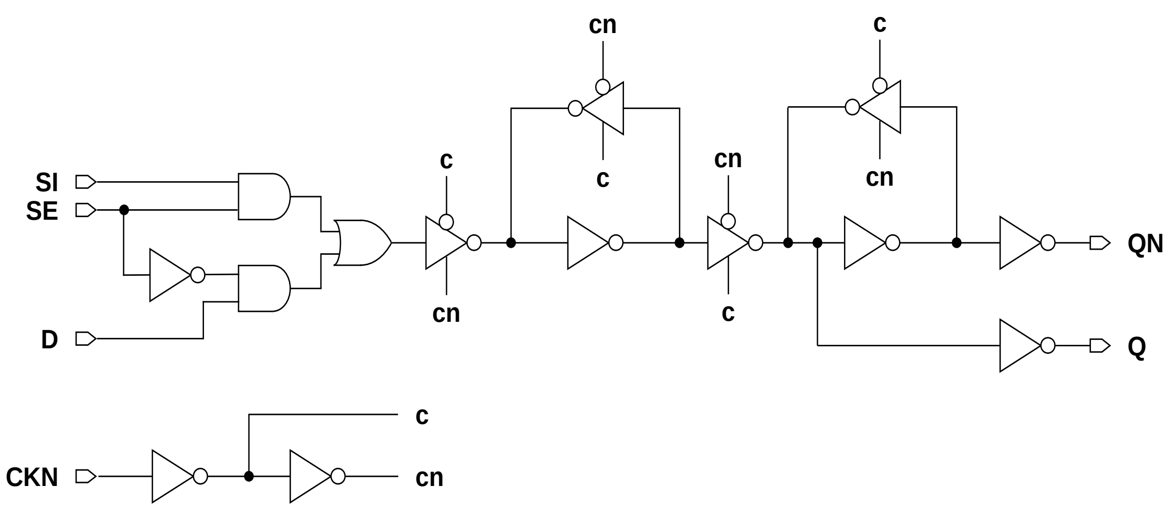
Function Table

D	SI	SE	CKN	Q[n+1]	QN[n+1]
1	x	0	⎺╲⎽	1	0
0	x	0	⎺╲⎽	0	1
x	x	x	⎽╱⎺	Q[n]	QN[n]
x	1	1	⎺╲⎽	1	0
x	0	1	⎺╲⎽	0	1

Cell Size

Drive Strength	Height (um)	Width (um)
SDFFNHX1M	2.87	13.53
SDFFNHX2M	2.87	13.94
SDFFNHX4M	2.87	16.81
SDFFNHX8M	2.87	17.63

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	X1	**X2**	**X4**	**X8**
SI	0.0134	0.0156	0.0245	0.0290
SE	0.0184	0.0205	0.0294	0.0346
D	0.0158	0.0184	0.0283	0.0330
CKN	0.0231	0.0262	0.0374	0.0418
Q	0.0129	0.0155	0.0225	0.0335

Pin Capacitance

Pin	Capacitance (pF)			
	X1	**X2**	**X4**	**X8**
SI	0.0016	0.0016	0.0020	0.0024
SE	0.0036	0.0037	0.0039	0.0044
D	0.0014	0.0018	0.0026	0.0028
CKN	0.0028	0.0028	0.0032	0.0045

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	**X2**	**X4**	**X8**	**X1**	**X2**	**X4**	**X8**
CKN $\rightarrow$ Q $\uparrow$	0.3655	0.3839	0.3689	0.3429	5.2231	4.1676	1.8521	1.0147
CKN $\rightarrow$ Q $\downarrow$	0.2813	0.2738	0.2559	0.2551	3.4481	1.8108	0.8804	0.5010
CKN $\rightarrow$ QN $\uparrow$	0.3308	0.3244	0.3066	0.3202	8.8206	8.8942	8.7195	8.5588
CKN $\rightarrow$ QN $\downarrow$	0.4436	0.4621	0.4540	0.4446	5.2246	5.1266	5.0558	5.0663

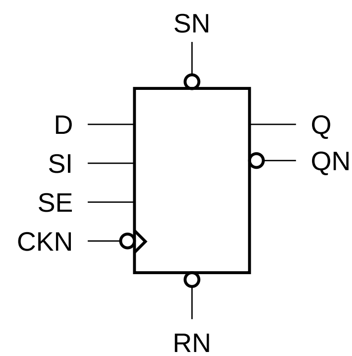
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
SI	setup ↑ → CKN	-0.0469	-0.0742	-0.0508	-0.0156
	setup ↓ → CKN	0.1172	0.1172	0.1094	0.1641
	hold ↑ → CKN	0.0898	0.0977	0.0898	0.0703
	hold ↓ → CKN	-0.0703	-0.0781	-0.0625	-0.1172
SE	setup ↑ → CKN	0.1445	0.1484	0.1445	0.2031
	setup ↓ → CKN	0.0977	0.0781	0.0742	0.1211
	hold ↑ → CKN	0.0977	0.1094	0.0977	0.0742
	hold ↓ → CKN	0.0312	0.0430	0.0312	-0.0039
D	setup ↑ → CKN	-0.0352	-0.0742	-0.0547	-0.0078
	setup ↓ → CKN	0.0898	0.0586	0.0430	0.0781
	hold ↑ → CKN	0.0781	0.0977	0.0898	0.0625
	hold ↓ → CKN	-0.0430	-0.0195	0.0000	-0.0312
CKN	minpwl	0.9974	0.9974	0.9974	0.9974
	minpwh	0.9974	0.9974	0.9974	0.9974

Cell Description

The SDFFNSRH cell is a negative-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and asynchronous active-low reset (RN) and set (SN). Set (SN) dominates reset (RN). This cell has a fast clock-to-Q path.

Logic Symbol



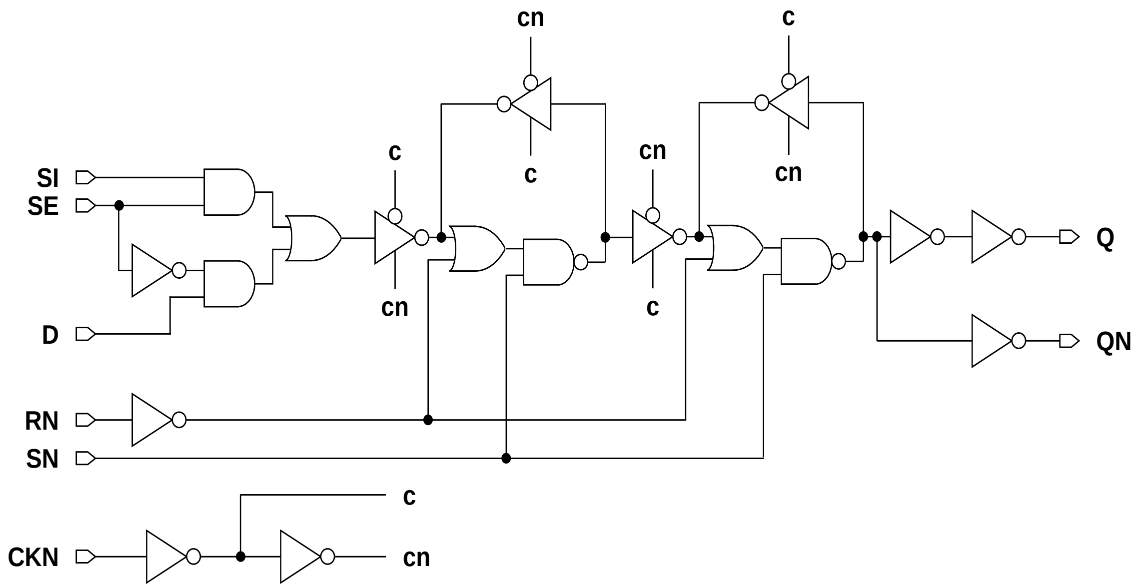
Function Table

RN	SN	D	SI	SE	CKN	Q[n+1]	QN[n+1]
1	1	1	x	0	⎺\⎽	1	0
1	1	0	x	0	⎺\⎽	0	1
1	1	x	x	x	⎽/⎺	Q[n]	QN[n]
1	1	x	1	1	⎺\⎽	1	0
1	1	x	0	1	⎺\⎽	0	1
0	1	x	x	x	x	0	1
1	0	x	x	x	x	1	0
0	0	x	x	x	x	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
SDFFNSRHX1M	2.87	17.22
SDFFNSRHX2M	2.87	17.22
SDFFNSRHX4M	2.87	17.63
SDFFNSRHX8M	2.87	19.27

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	X1	**X2**	**X4**	**X8**
SI	0.0174	0.0196	0.0230	0.0249
SE	0.0203	0.0225	0.0261	0.0287
D	0.0196	0.0220	0.0268	0.0295
CKN	0.0234	0.0265	0.0311	0.0346
SN	0.0087	0.0088	0.0091	0.0099
RN	0.0023	0.0024	0.0026	0.0026
Q	0.0150	0.0176	0.0225	0.0381

Pin Capacitance

Pin	Capacitance (pF)			
	X1	**X2**	**X4**	**X8**
SI	0.0016	0.0016	0.0019	0.0028
SE	0.0033	0.0033	0.0037	0.0045
D	0.0015	0.0019	0.0029	0.0035
CKN	0.0027	0.0028	0.0030	0.0042
SN	0.0027	0.0031	0.0034	0.0039
RN	0.0030	0.0033	0.0038	0.0038

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	**X2**	**X4**	**X8**	**X1**	**X2**	**X4**	**X8**
CKN → Q ↑	0.3472	0.3499	0.3454	0.3026	5.1281	3.5204	1.8122	0.9473
CKN → Q ↓	0.2697	0.2472	0.2577	0.2801	3.7002	1.8892	0.9506	0.5065
SN → Q ↑	0.2528	0.2806	0.2946	0.2623	5.0498	3.5058	1.8144	0.9322
SN → Q ↓	0.3696	0.3682	0.3879	0.4710	3.9610	2.0876	1.0742	0.5740
RN → Q ↓	0.3126	0.3134	0.3332	0.4220	3.9872	2.0874	1.0741	0.5772
CKN → QN ↑	0.3217	0.3004	0.3179	0.3509	8.8834	8.6700	8.6751	9.0297
CKN → QN ↓	0.4394	0.4420	0.4470	0.4133	5.1719	5.0463	5.0304	5.0093
SN → QN ↑	0.4190	0.4135	0.4429	0.5514	8.8909	8.6664	8.6714	9.0276
SN → QN ↓	0.3415	0.3693	0.3944	0.3665	5.1655	5.0453	5.0301	5.0093
RN → QN ↑	0.3623	0.3583	0.3877	0.5037	8.8909	8.6673	8.6730	9.0293

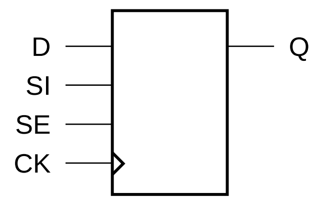
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
SI	setup ↑ → CKN	0.0703	0.0391	0.0078	0.0469
	setup ↓ → CKN	0.2305	0.2188	0.1758	0.1484
	hold ↑ → CKN	0.0312	0.0352	0.0586	0.0391
	hold ↓ → CKN	-0.1406	-0.1445	-0.1133	-0.0859
SE	setup ↑ → CKN	0.2695	0.2578	0.2188	0.1953
	setup ↓ → CKN	0.1836	0.1523	0.1250	0.1250
	hold ↑ → CKN	0.0312	0.0391	0.0586	0.0391
	hold ↓ → CKN	-0.0195	-0.0078	0.0078	-0.0352
D	setup ↑ → CKN	0.0859	0.0430	0.0078	0.0664
	setup ↓ → CKN	0.1953	0.1484	0.1055	0.0977
	hold ↑ → CKN	0.0156	0.0312	0.0547	0.0195
	hold ↓ → CKN	-0.1055	-0.0742	-0.0469	-0.0352
CKN	minpwl	0.9974	0.9974	0.9974	0.9974
	minpwh	0.9974	0.9974	0.9974	0.9974
SN	minpwl	0.9974	0.9974	0.9974	0.9974
	recovery	0.0352	0.0078	0.0000	0.0195
	removal	0.0234	0.0352	0.0391	0.0195
RN	minpwl	0.9974	0.9974	0.9974	0.9974
	recovery	-0.1797	-0.1953	-0.1953	-0.1367
	removal	0.2383	0.2656	0.2734	0.1953

Cell Description

The SDFFQ cell is a positive-edge triggered, static D-type flip-flop with scan input (SI) and active-high scan enable (SE). The cell has a single output (Q).

Logic Symbol



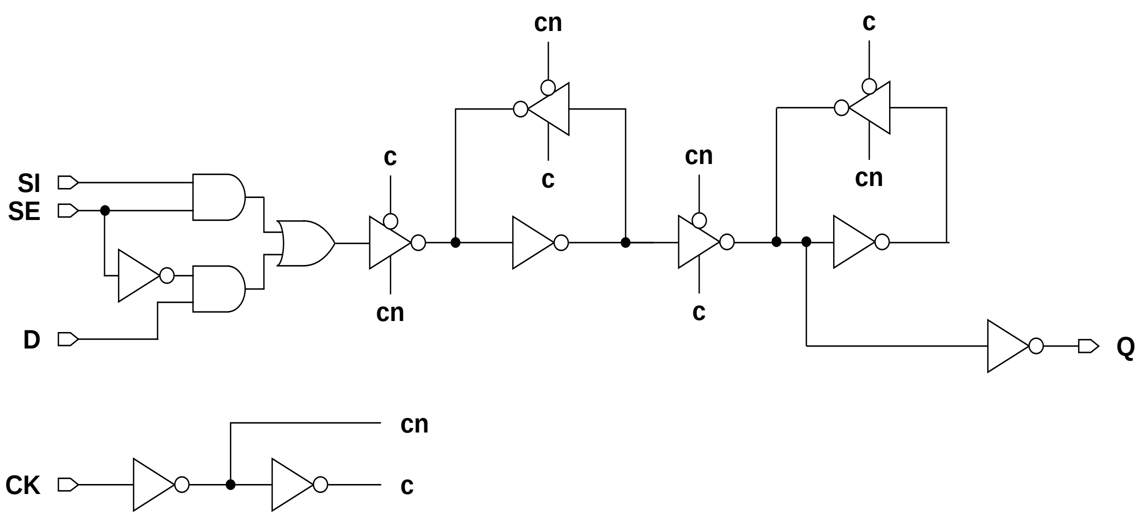
Function Table

D	SI	SE	CK	Q[n+1]
1	x	0	↑	1
0	x	0	↑	0
x	x	x	↓	Q[n]
x	1	1	↑	1
x	0	1	↑	0

Cell Size

Drive Strength	Height (um)	Width (um)
SDFFQX1M	2.87	9.43
SDFFQX2M	2.87	9.43
SDFFQX4M	2.87	10.25

Functional Schematic



AC Power

Pin	Power (uW/MHz)		
	X1	**X2**	**X4**
SI	0.0127	0.0129	0.0134
SE	0.0146	0.0148	0.0153
D	0.0109	0.0111	0.0116
CK	0.0209	0.0210	0.0225
Q	0.0092	0.0111	0.0177

Pin Capacitance

Pin	Capacitance (pF)		
	X1	**X2**	**X4**
SI	0.0014	0.0014	0.0014
SE	0.0032	0.0032	0.0032
D	0.0016	0.0017	0.0017
CK	0.0020	0.0020	0.0021

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	**X2**	**X4**	**X1**	**X2**	**X4**
CK → Q ↑	0.1800	0.1816	0.1998	5.2017	3.5265	1.8016
CK → Q ↓	0.2188	0.2081	0.2152	3.6186	1.9124	0.9808

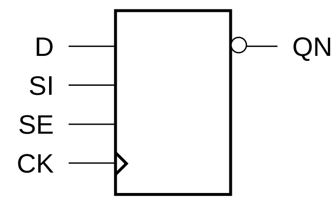
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)		
		X1	**X2**	**X4**
SI	setup ↑ → CK	0.1211	0.1250	0.1328
	setup ↓ → CK	0.3438	0.3438	0.3516
	hold ↑ → CK	-0.1016	-0.1016	-0.1055
	hold ↓ → CK	-0.2812	-0.2812	-0.2812
SE	setup ↑ → CK	0.3555	0.3594	0.3633
	setup ↓ → CK	0.2070	0.2109	0.2148
	hold ↑ → CK	-0.0898	-0.0898	-0.0938
	hold ↓ → CK	-0.1445	-0.1445	-0.1445
D	setup ↑ → CK	0.1055	0.1094	0.1133
	setup ↓ → CK	0.2109	0.2148	0.2188
	hold ↑ → CK	-0.0859	-0.0859	-0.0898
	hold ↓ → CK	-0.1523	-0.1484	-0.1484
CK	minpwh	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974

Cell Description

The SDFFQN cell is a positive-edge triggered, static D-type flip-flop with scan input (SI) and active-high scan enable (SE). The cell has a single output (QN).

Logic Symbol



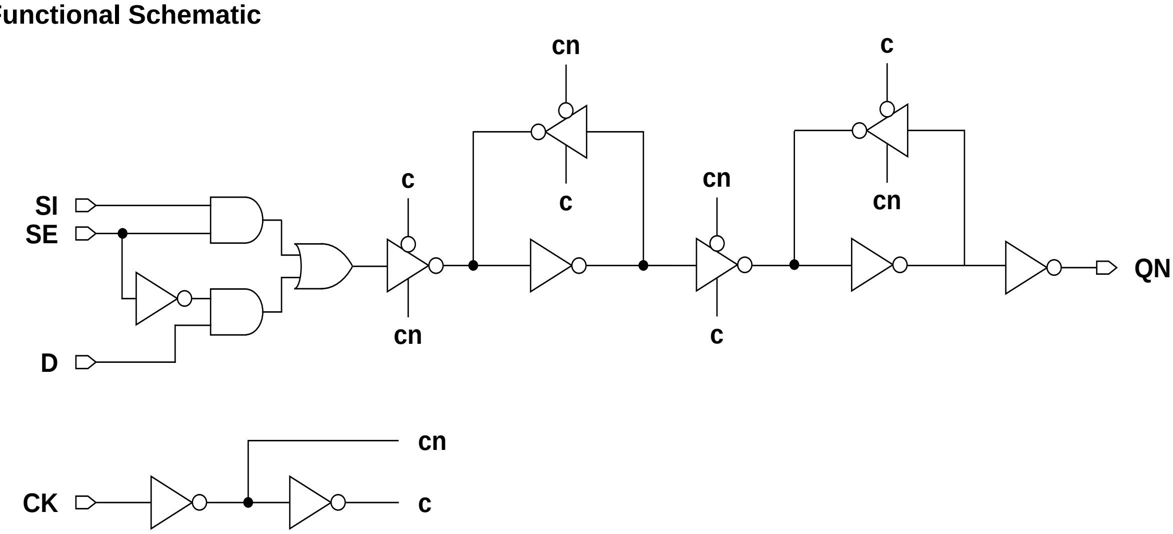
Function Table

D	SI	SE	CK	QN[n+1]
1	x	0	↑	0
0	x	0	↑	1
x	x	x	↓	QN[n]
x	1	1	↑	0
x	0	1	↑	1

Cell Size

Drive Strength	Height (um)	Width (um)
SDFFQNX1M	2.87	9.43
SDFFQNX2M	2.87	9.43
SDFFQNX4M	2.87	9.84

Functional Schematic



AC Power

Pin	Power (uW/MHz)		
	X1	X2	X4
SI	0.0126	0.0122	0.0122
SE	0.0145	0.0141	0.0141
D	0.0108	0.0106	0.0106
CK	0.0206	0.0204	0.0205
QN	0.0089	0.0106	0.0149

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
SI	0.0014	0.0014	0.0014
SE	0.0033	0.0030	0.0030
D	0.0017	0.0014	0.0014
CK	0.0020	0.0020	0.0020

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
CK → QN ↑	0.2560	0.2675	0.2839	5.1456	3.4832	1.7976
CK → QN ↓	0.2584	0.2738	0.2648	3.2331	1.7853	0.8729

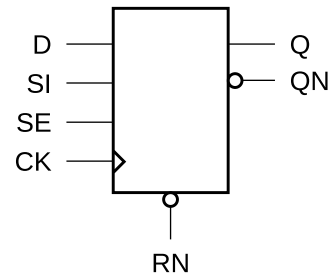
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)		
		X1	X2	X4
SI	setup ↑ → CK	0.1211	0.1094	0.1094
	setup ↓ → CK	0.3477	0.3594	0.3594
	hold ↑ → CK	-0.1055	-0.0898	-0.0898
	hold ↓ → CK	-0.3008	-0.3164	-0.3164
SE	setup ↑ → CK	0.3594	0.3828	0.3828
	setup ↓ → CK	0.2109	0.3086	0.3047
	hold ↑ → CK	-0.0938	-0.0781	-0.0781
	hold ↓ → CK	-0.1484	-0.1406	-0.1406
D	setup ↑ → CK	0.1055	0.0977	0.0977
	setup ↓ → CK	0.2148	0.3203	0.3164
	hold ↑ → CK	-0.0898	-0.0781	-0.0781
	hold ↓ → CK	-0.1680	-0.2773	-0.2773
CK	minpwh	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974

Cell Description

The SDFFR cell is a positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and asynchronous active-low reset (RN).

Logic Symbol



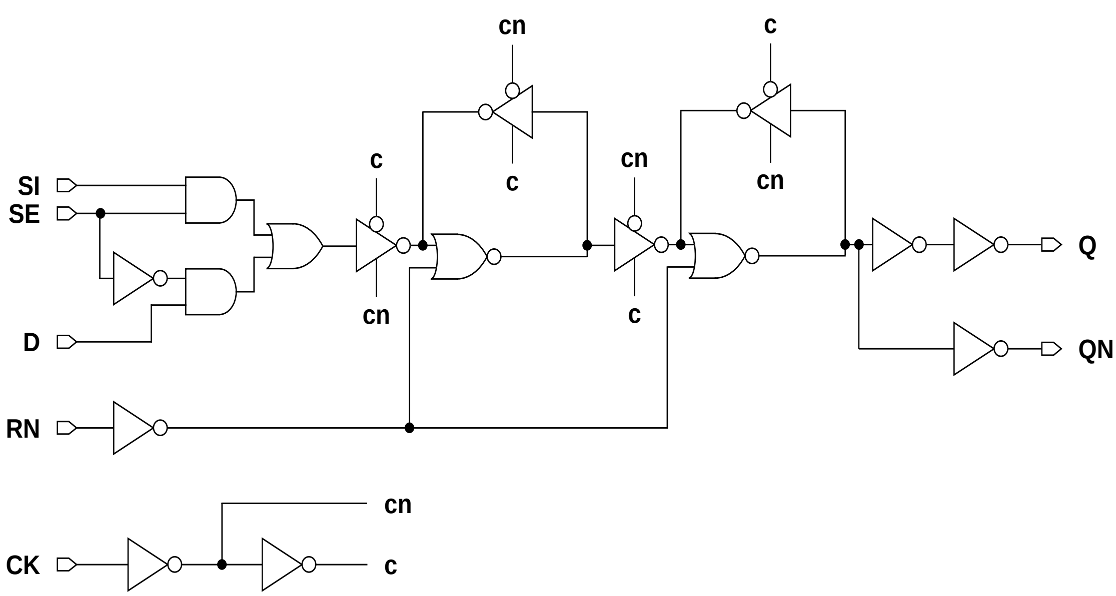
Function Table

RN	D	SI	SE	CK	Q[n+1]	QN[n+1]
1	1	x	0	↑	1	0
1	0	x	0	↑	0	1
1	x	x	x	↓	Q[n]	QN[n]
1	x	1	1	↑	1	0
1	x	0	1	↑	0	1
0	x	x	x	x	0	1

Cell Size

Drive Strength	Height (um)	Width (um)
SDFFRX1M	2.87	11.89
SDFFRX2M	2.87	12.30
SDFFRX4M	2.87	13.12

Functional Schematic



AC Power

Pin	Power (uW/MHz)		
	X1	X2	X4
SI	0.0142	0.0151	0.0164
SE	0.0158	0.0166	0.0186
D	0.0134	0.0142	0.0159
CK	0.0189	0.0200	0.0214
RN	0.0021	0.0021	0.0025
Q	0.0122	0.0159	0.0274

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
SI	0.0010	0.0011	0.0012
SE	0.0027	0.0027	0.0032
D	0.0012	0.0012	0.0019
CK	0.0020	0.0021	0.0028
RN	0.0050	0.0052	0.0060

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
CK → Q ↑	0.2340	0.2242	0.2677	5.3552	3.5768	1.8526
CK → Q ↓	0.2567	0.2725	0.3148	3.2748	1.8059	0.9247
RN → Q ↓	0.1257	0.1468	0.1685	3.2630	1.8022	0.9194
CK → QN ↑	0.1585	0.1602	0.1671	5.3032	3.5815	1.8931
CK → QN ↓	0.1501	0.1470	0.1711	3.3879	1.8431	0.9678
RN → QN ↑	0.2211	0.2623	0.3401	5.2770	3.6060	1.9246

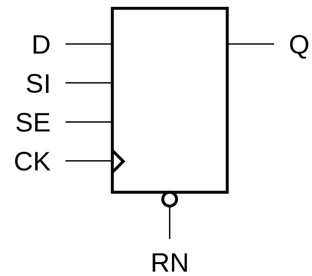
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)		
		X1	X2	X4
SI	setup ↑ → CK	0.1602	0.1680	0.1523
	setup ↓ → CK	0.3125	0.3281	0.3398
	hold ↑ → CK	-0.1055	-0.0977	-0.0820
	hold ↓ → CK	-0.1875	-0.1758	-0.1797
SE	setup ↑ → CK	0.3242	0.3438	0.3672
	setup ↓ → CK	0.2109	0.2266	0.1953
	hold ↑ → CK	-0.0898	-0.0820	-0.0664
	hold ↓ → CK	-0.1094	-0.1016	-0.0625
D	setup ↑ → CK	0.1445	0.1484	0.1172
	setup ↓ → CK	0.2461	0.2617	0.1875
	hold ↑ → CK	-0.0938	-0.0859	-0.0625
	hold ↓ → CK	-0.1445	-0.1367	-0.0938
CK	minpwh	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974
RN	minpwl	0.9974	0.9974	0.9974
	recovery	0.1562	0.1602	0.1289
	removal	-0.1250	-0.1211	-0.0977

Cell Description

The SDFFRHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and asynchronous active-low reset (RN). The cell has a single output (Q) and fast clock-to-out path.

Logic Symbol



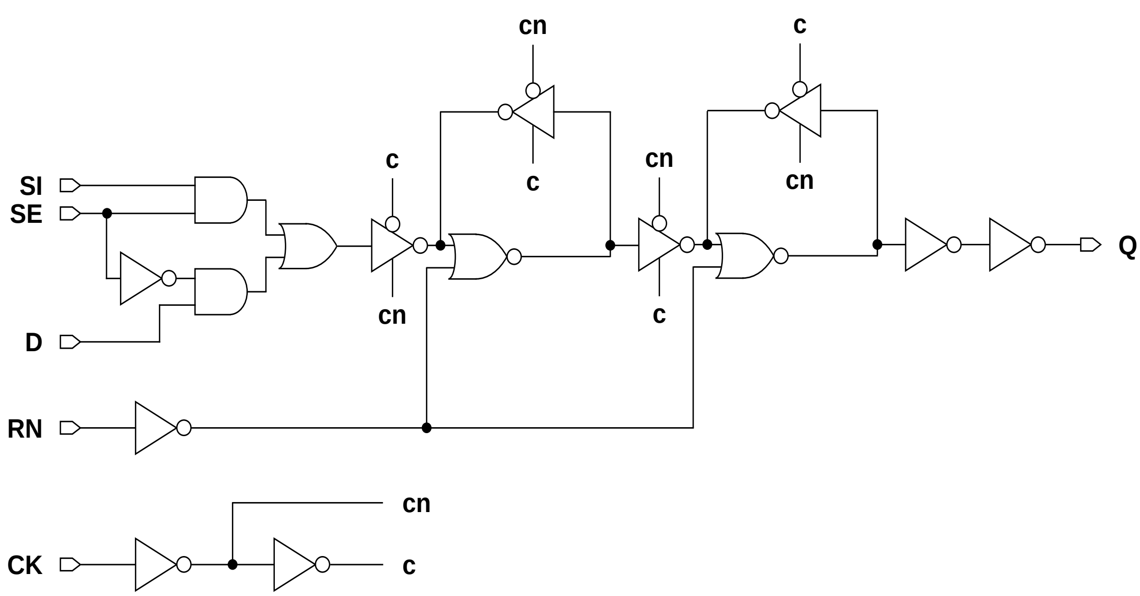
Function Table

RN	D	SI	SE	CK	Q[n+1]
1	1	x	0	↑	1
1	0	x	0	↑	0
1	x	x	x	↓	Q[n]
1	x	1	1	↑	1
1	x	0	1	↑	0
0	x	x	x	x	0

Cell Size

Drive Strength	Height (um)	Width (um)
SDFFRHQX1M	2.87	12.71
SDFFRHQX2M	2.87	13.53
SDFFRHQX4M	2.87	13.94
SDFFRHQX8M	2.87	14.76

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	X1	X2	X4	X8
SI	0.0151	0.0187	0.0183	0.0185
SE	0.0171	0.0207	0.0203	0.0205
D	0.0167	0.0199	0.0207	0.0209
CK	0.0285	0.0338	0.0340	0.0342
RN	0.0024	0.0027	0.0027	0.0027
Q	0.0087	0.0097	0.0143	0.0272

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
SI	0.0013	0.0013	0.0014	0.0014
SE	0.0032	0.0034	0.0035	0.0035
D	0.0022	0.0020	0.0030	0.0030
CK	0.0028	0.0028	0.0031	0.0031
RN	0.0030	0.0034	0.0034	0.0034

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
CK → Q ↑	0.1630	0.1548	0.1666	0.1992	5.2262	3.5280	1.8433	1.0085
CK → Q ↓	0.1665	0.1457	0.1614	0.1944	3.4018	1.7656	0.8981	0.4796
RN → Q ↓	0.1489	0.1516	0.1703	0.2357	3.3770	1.8491	0.9276	0.5004

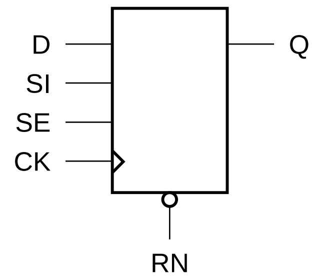
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	**X2**	**X4**	**X8**
SI	setup ↑ → CK	0.1523	0.1484	0.1484	0.1484
	setup ↓ → CK	0.2344	0.2852	0.2734	0.2695
	hold ↑ → CK	-0.0742	-0.0703	-0.0703	-0.0703
	hold ↓ → CK	-0.1758	-0.2109	-0.2031	-0.1992
SE	setup ↑ → CK	0.2500	0.3047	0.2969	0.2930
	setup ↓ → CK	0.1914	0.1758	0.1680	0.1719
	hold ↑ → CK	-0.0703	-0.0625	-0.0664	-0.0664
	hold ↓ → CK	-0.0859	-0.0938	-0.0859	-0.0820
D	setup ↑ → CK	0.1562	0.1328	0.1250	0.1250
	setup ↓ → CK	0.1172	0.1797	0.1367	0.1367
	hold ↑ → CK	-0.0781	-0.0547	-0.0469	-0.0469
	hold ↓ → CK	-0.0664	-0.1133	-0.0781	-0.0742
CK	minpwh	0.9974	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974	0.9974
RN	minpwl	0.9974	0.9974	0.9974	0.9974
	recovery	-0.0469	-0.0391	-0.0391	-0.0391
	removal	0.0820	0.0781	0.0742	0.0664

Cell Description

The SDFFRQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and asynchronous active-low reset (RN). The cell has a single output (Q).

Logic Symbol



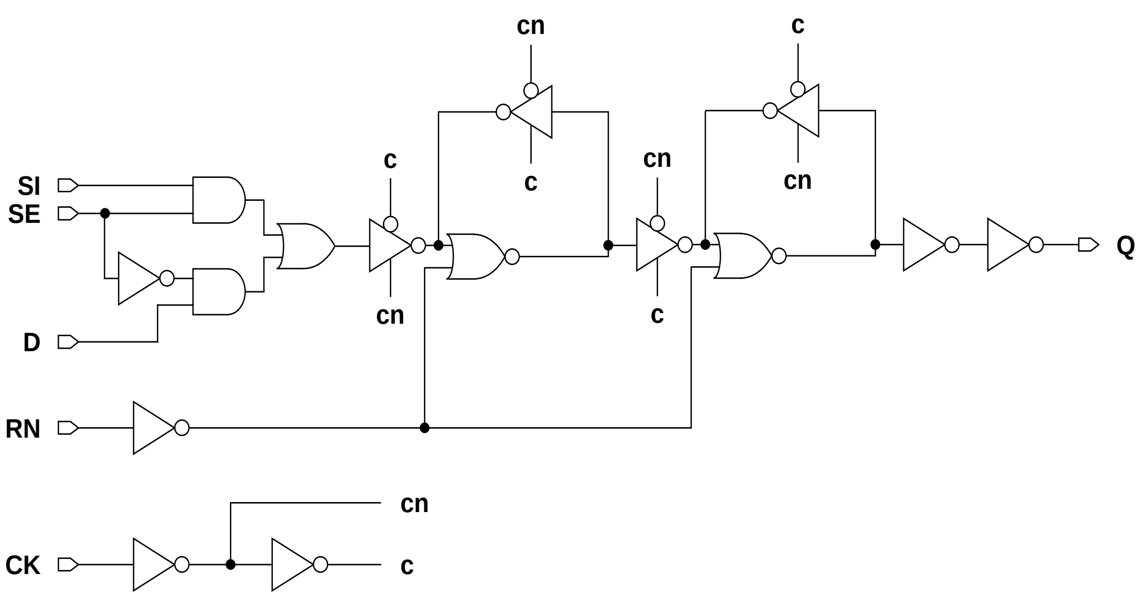
Function Table

RN	D	SI	SE	CK	Q[n+1]
1	1	x	0	↑	1
1	0	x	0	↑	0
1	x	x	x	↓	Q[n]
1	x	1	1	↑	1
1	x	0	1	↑	0
0	x	x	x	x	0

Cell Size

Drive Strength	Height (um)	Width (um)
SDFFRQX1M	2.87	11.48
SDFFRQX2M	2.87	11.48
SDFFRQX4M	2.87	11.89

Functional Schematic



AC Power

Pin	Power (uW/MHz)		
	X1	X2	X4
SI	0.0127	0.0124	0.0125
SE	0.0144	0.0143	0.0143
D	0.0120	0.0117	0.0118
CK	0.0175	0.0168	0.0168
RN	0.0021	0.0022	0.0024
Q	0.0085	0.0104	0.0169

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
SI	0.0010	0.0010	0.0010
SE	0.0026	0.0026	0.0026
D	0.0011	0.0011	0.0011
CK	0.0019	0.0019	0.0019
RN	0.0051	0.0052	0.0057

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
CK → Q ↑	0.2118	0.2072	0.2152	5.3193	3.5622	1.8196
CK → Q ↓	0.2501	0.2627	0.2911	3.3574	1.8305	0.9601
RN → Q ↓	0.1252	0.1421	0.1689	3.3470	1.7996	0.9315

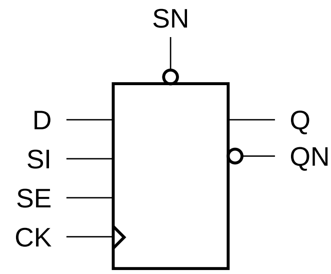
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)		
		X1	**X2**	**X4**
SI	setup ↑ → CK	0.1602	0.1562	0.1562
	setup ↓ → CK	0.2852	0.2891	0.2852
	hold ↑ → CK	-0.1016	-0.0938	-0.0977
	hold ↓ → CK	-0.1719	-0.1719	-0.1680
SE	setup ↑ → CK	0.3008	0.3047	0.3008
	setup ↓ → CK	0.2109	0.2070	0.2031
	hold ↑ → CK	-0.0859	-0.0781	-0.0781
	hold ↓ → CK	-0.0977	-0.0977	-0.0977
D	setup ↑ → CK	0.1484	0.1445	0.1445
	setup ↓ → CK	0.2305	0.2305	0.2305
	hold ↑ → CK	-0.0938	-0.0859	-0.0898
	hold ↓ → CK	-0.1328	-0.1328	-0.1328
CK	minpwh	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974
RN	minpwl	0.9974	0.9974	0.9974
	recovery	0.1602	0.1562	0.1523
	removal	-0.1250	-0.1172	-0.1211

Cell Description

The SDFFS cell is a positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and asynchronous active-low set (SN).

Logic Symbol



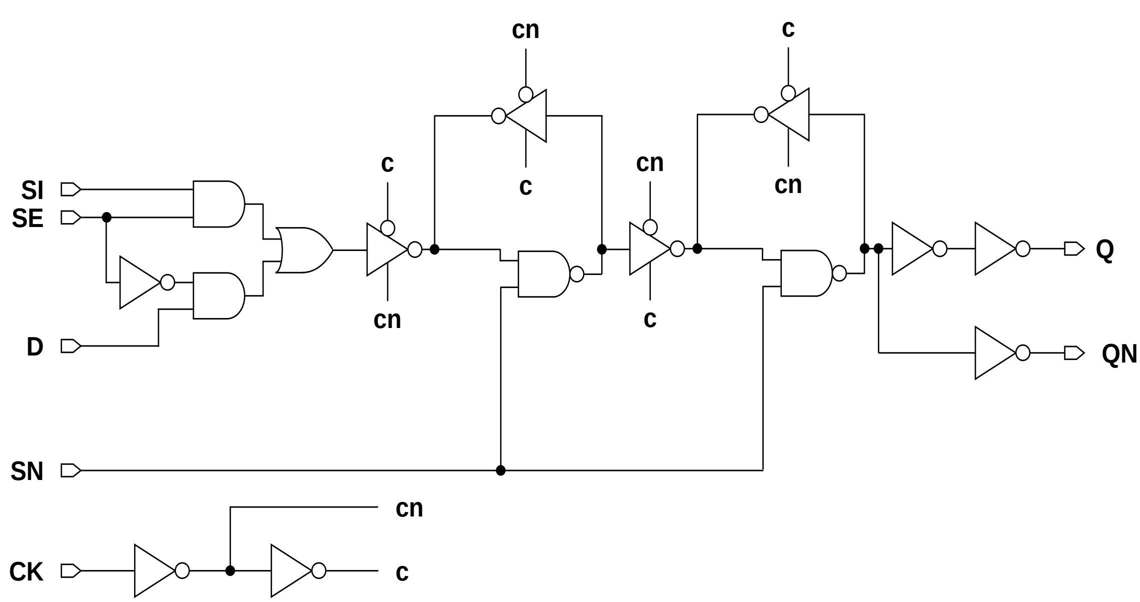
Function Table

SN	D	SI	SE	CK	Q[n+1]	QN[n+1]
1	1	x	0	↑	1	0
1	0	x	0	↑	0	1
1	x	x	x	↓	Q[n]	QN[n]
1	x	1	1	↑	1	0
1	x	0	1	↑	0	1
0	x	x	x	x	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
SDFFSX1M	2.87	11.48
SDFFSX2M	2.87	11.89
SDFFSX4M	2.87	12.71

Functional Schematic



AC Power

Pin	Power (uW/MHz)		
	X1	**X2**	**X4**
SI	0.0131	0.0150	0.0180
SE	0.0152	0.0172	0.0191
D	0.0125	0.0142	0.0169
CK	0.0205	0.0222	0.0256
SN	0.0022	0.0023	0.0023
Q	0.0130	0.0161	0.0267

Pin Capacitance

Pin	Capacitance (pF)		
	X1	**X2**	**X4**
SI	0.0014	0.0014	0.0014
SE	0.0028	0.0028	0.0028
D	0.0018	0.0018	0.0018
CK	0.0024	0.0025	0.0026
SN	0.0026	0.0030	0.0037

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	**X2**	**X4**	**X1**	**X2**	**X4**
CK → Q ↑	0.2707	0.2676	0.2862	5.3171	3.5786	1.8450
CK → Q ↓	0.2877	0.2867	0.3048	3.2834	1.8025	0.9089
SN → Q ↑	0.2422	0.2683	0.3763	5.3057	3.5760	1.8442
CK → QN ↑	0.1909	0.1759	0.1700	5.4506	3.6234	1.8702
CK → QN ↓	0.2092	0.1972	0.1901	3.9913	2.0585	0.9896
SN → QN ↓	0.1853	0.2013	0.2706	3.4255	1.8826	1.0125

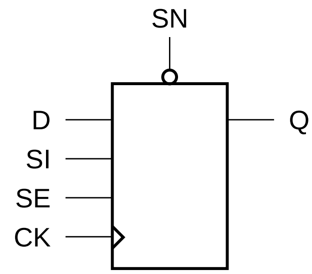
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)		
		X1	X2	X4
SI	setup ↑ → CK	0.1406	0.1562	0.1406
	setup ↓ → CK	0.3008	0.2969	0.3242
	hold ↑ → CK	-0.0781	-0.0898	-0.0938
	hold ↓ → CK	-0.2188	-0.2148	-0.2344
SE	setup ↑ → CK	0.3281	0.3242	0.3516
	setup ↓ → CK	0.2070	0.2266	0.2344
	hold ↑ → CK	-0.0664	-0.0742	-0.0781
	hold ↓ → CK	-0.1289	-0.1250	-0.1523
D	setup ↑ → CK	0.1484	0.1602	0.1367
	setup ↓ → CK	0.2422	0.2383	0.2812
	hold ↑ → CK	-0.0820	-0.0938	-0.0938
	hold ↓ → CK	-0.1680	-0.1641	-0.1992
CK	minpwh	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974
SN	minpwl	0.9974	0.9974	0.9974
	recovery	-0.0391	-0.0469	-0.0547
	removal	0.0625	0.0703	0.0781

Cell Description

The SDFFSHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and asynchronous active-low set (SN). The cell has a single output (Q) and fast clock-to-out path.

Logic Symbol



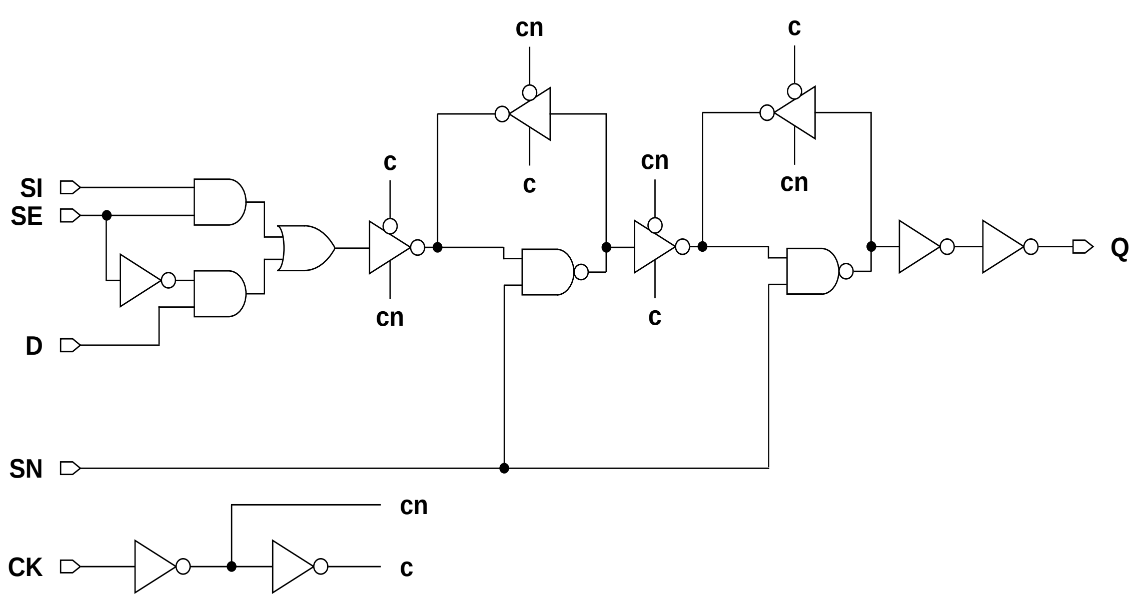
Function Table

SN	D	SI	SE	CK	Q[n+1]
1	1	x	0	↑	1
1	0	x	0	↑	0
1	x	x	x	↓	Q[n]
1	x	1	1	↑	1
1	x	0	1	↑	0
0	x	x	x	x	1

Cell Size

Drive Strength	Height (um)	Width (um)
SDFFSHQX1M	2.87	14.35
SDFFSHQX2M	2.87	14.76
SDFFSHQX4M	2.87	15.17
SDFFSHQX8M	2.87	16.40

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	X1	X2	X4	X8
SI	0.0169	0.0210	0.0236	0.0237
SE	0.0192	0.0234	0.0262	0.0263
D	0.0179	0.0226	0.0256	0.0257
CK	0.0275	0.0331	0.0386	0.0386
SN	0.0062	0.0063	0.0070	0.0071
Q	0.0107	0.0123	0.0176	0.0295

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
SI	0.0016	0.0017	0.0017	0.0017
SE	0.0044	0.0046	0.0053	0.0053
D	0.0017	0.0021	0.0028	0.0027
CK	0.0028	0.0029	0.0033	0.0033
SN	0.0030	0.0033	0.0038	0.0038

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
CK $\rightarrow$ Q $\uparrow$	0.1590	0.1490	0.1350	0.1541	5.1059	3.5441	1.8115	0.9445
CK $\rightarrow$ Q $\downarrow$	0.1702	0.1511	0.1454	0.1707	3.4131	1.7889	0.8970	0.4636
SN $\rightarrow$ Q $\uparrow$	0.2789	0.3856	0.2707	0.3004	5.4518	3.8308	1.8161	0.9398

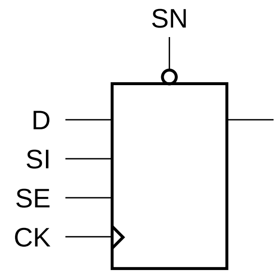
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
SI	setup ↑ → CK	0.1953	0.1914	0.1875	0.1953
	setup ↓ → CK	0.2461	0.2773	0.2891	0.2852
	hold ↑ → CK	-0.1055	-0.1055	-0.0977	-0.0977
	hold ↓ → CK	-0.1758	-0.1953	-0.2031	-0.1992
SE	setup ↑ → CK	0.2578	0.2930	0.3125	0.3086
	setup ↓ → CK	0.2266	0.2148	0.2109	0.2148
	hold ↑ → CK	-0.0977	-0.0977	-0.0938	-0.0938
	hold ↓ → CK	-0.0977	-0.0859	-0.0508	-0.0469
D	setup ↑ → CK	0.1914	0.1719	0.1719	0.1758
	setup ↓ → CK	0.1758	0.1680	0.1289	0.1289
	hold ↑ → CK	-0.1016	-0.0859	-0.0859	-0.0820
	hold ↓ → CK	-0.1094	-0.0938	-0.0625	-0.0586
CK	minpwh	0.9974	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974	0.9974
SN	minpwl	0.9974	0.9974	0.9974	0.9974
	recovery	0.0547	0.0508	0.0547	0.0547
	removal	-0.0312	-0.0234	-0.0312	-0.0312

Cell Description

The SDFFSQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and asynchronous active-low set (SN). The cell has a single output (Q).

Logic Symbol



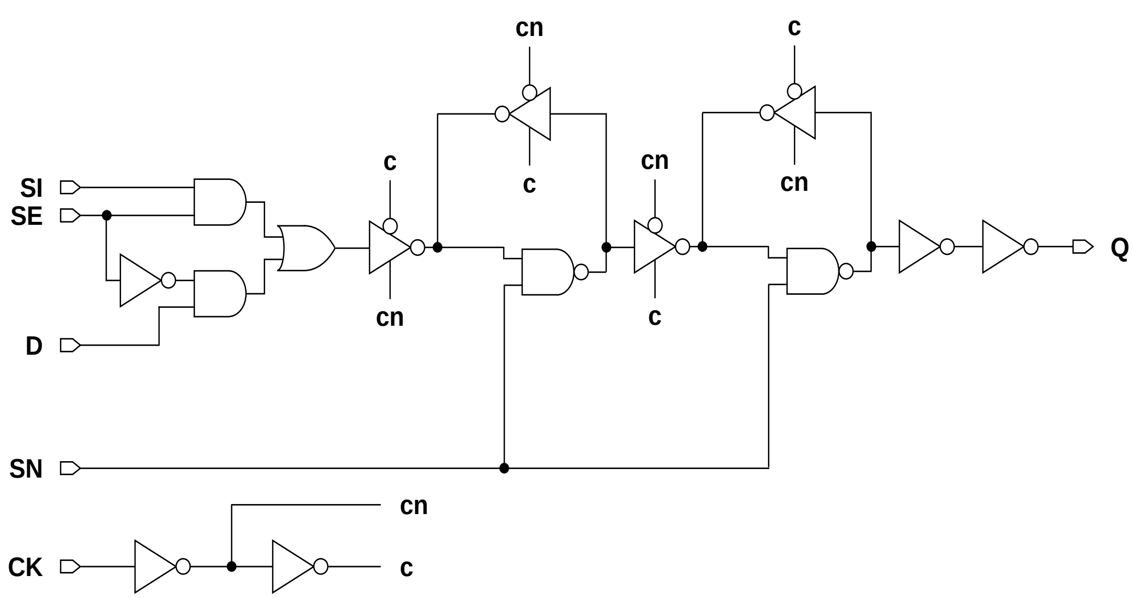
Function Table

SN	D	SI	SE	CK	Q[n+1]
1	1	x	0	↑	1
1	0	x	0	↑	0
1	x	x	x	↓	Q[n]
1	x	1	1	↑	1
1	x	0	1	↑	0
0	x	x	x	x	1

Cell Size

Drive Strength	Height (um)	Width (um)
SDFFSQX1M	2.87	11.07
SDFFSQX2M	2.87	11.07
SDFFSQX4M	2.87	11.48

Functional Schematic



AC Power

Pin	Power (uW/MHz)		
	X1	X2	X4
SI	0.0130	0.0130	0.0130
SE	0.0153	0.0153	0.0152
D	0.0124	0.0124	0.0124
CK	0.0206	0.0204	0.0204
SN	0.0022	0.0021	0.0022
Q	0.0094	0.0110	0.0166

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
SI	0.0015	0.0015	0.0015
SE	0.0028	0.0028	0.0028
D	0.0018	0.0018	0.0018
CK	0.0024	0.0024	0.0024
SN	0.0025	0.0025	0.0025

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
CK → Q ↑	0.2249	0.2325	0.2518	5.0452	3.4096	1.7941
CK → Q ↓	0.2603	0.2694	0.2855	3.3087	1.8199	0.9333
SN → Q ↑	0.1950	0.1985	0.2148	5.0374	3.4074	1.7931

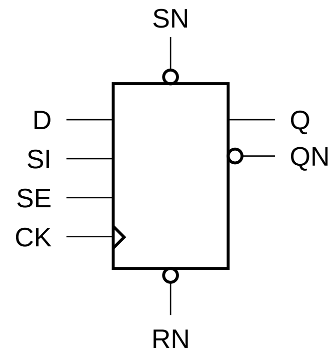
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)		
		X1	X2	X4
SI	setup ↑ → CK	0.1641	0.1602	0.1562
	setup ↓ → CK	0.3008	0.3008	0.2969
	hold ↑ → CK	-0.0781	-0.0781	-0.0781
	hold ↓ → CK	-0.2070	-0.2070	-0.2070
SE	setup ↑ → CK	0.3281	0.3242	0.3242
	setup ↓ → CK	0.2305	0.2266	0.2227
	hold ↑ → CK	-0.0664	-0.0664	-0.0664
	hold ↓ → CK	-0.1211	-0.1211	-0.1211
D	setup ↑ → CK	0.1680	0.1641	0.1602
	setup ↓ → CK	0.2461	0.2422	0.2422
	hold ↑ → CK	-0.0820	-0.0820	-0.0820
	hold ↓ → CK	-0.1641	-0.1641	-0.1641
CK	minpwh	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974
SN	minpwl	0.9974	0.9974	0.9974
	recovery	-0.0352	-0.0352	-0.0391
	removal	0.0625	0.0625	0.0625

Cell Description

The SDFFSR cell is a positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and asynchronous active-low reset (RN) and set (SN). Set (SN) dominates reset (RN).

Logic Symbol



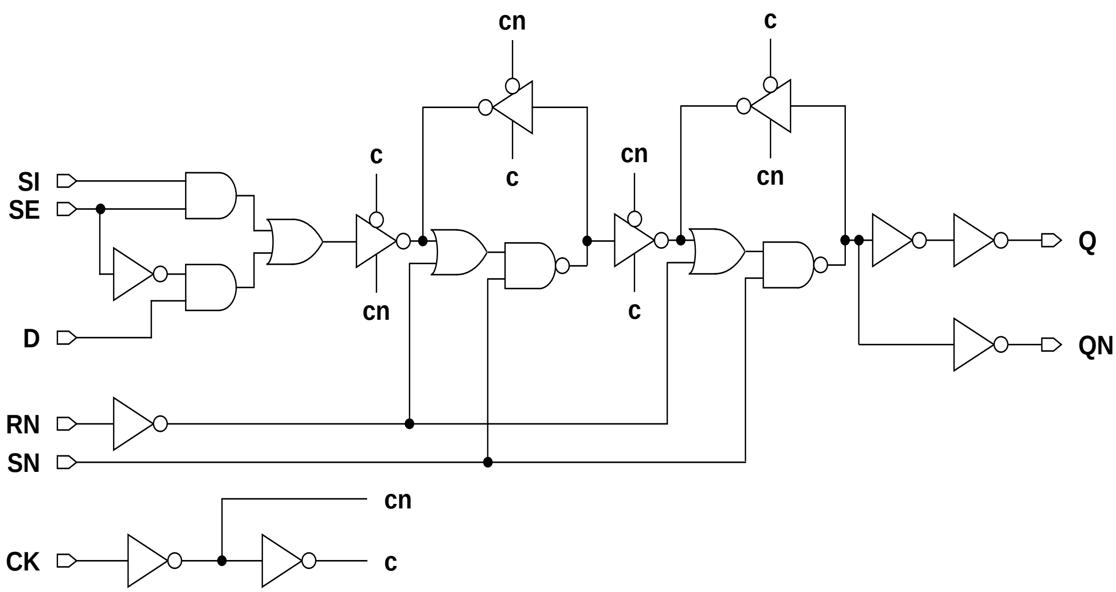
Function Table

RN	SN	D	SI	SE	CK	Q[n+1]	QN[n+1]
1	1	1	x	0	_/‾	1	0
1	1	0	x	0	_/‾	0	1
1	1	x	x	x	‾_	Q[n]	QN[n]
1	1	x	1	1	_/‾	1	0
1	1	x	0	1	_/‾	0	1
0	1	x	x	x	x	0	1
1	0	x	x	x	x	1	0
0	0	x	x	x	x	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
SDFFSRX1M	2.87	15.17
SDFFSRX2M	2.87	15.17
SDFFSRX4M	2.87	16.40

Functional Schematic



AC Power

Pin	Power (uW/MHz)		
	X1	X2	X4
SI	0.0158	0.0163	0.0169
SE	0.0173	0.0178	0.0185
D	0.0149	0.0154	0.0161
CK	0.0246	0.0253	0.0268
SN	0.0026	0.0028	0.0028
RN	0.0051	0.0051	0.0051
Q	0.0153	0.0197	0.0342

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
SI	0.0015	0.0015	0.0015
SE	0.0028	0.0028	0.0028
D	0.0018	0.0018	0.0020
CK	0.0021	0.0023	0.0026
SN	0.0030	0.0034	0.0037
RN	0.0015	0.0015	0.0015

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
CK → Q ↑	0.2988	0.3247	0.3895	5.2035	3.4884	1.8631
CK → Q ↓	0.3077	0.3315	0.3663	3.3451	1.8180	0.9187
SN → Q ↑	0.2682	0.3179	0.4037	5.1704	3.4775	1.8578
SN → Q ↓	0.2609	0.2874	0.3271	3.3275	1.8103	0.9153
RN → Q ↓	0.3142	0.3487	0.3931	3.3333	1.8127	0.9156
CK → QN ↑	0.2089	0.2088	0.2138	5.4310	3.6302	1.9693
CK → QN ↓	0.2459	0.2401	0.2617	4.6732	2.3723	1.2638
SN → QN ↑	0.1634	0.1683	0.1756	5.3407	3.5895	1.9484
SN → QN ↓	0.2195	0.2412	0.2886	3.5880	1.9593	1.0372
RN → QN ↑	0.2157	0.2291	0.2425	5.3821	3.6073	1.9493

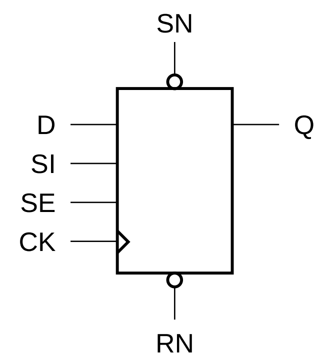
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)		
		X1	X2	X4
SI	setup ↑ → CK	0.2734	0.2383	0.2422
	setup ↓ → CK	0.3477	0.3633	0.3789
	hold ↑ → CK	-0.1094	-0.1055	-0.1016
	hold ↓ → CK	-0.2070	-0.2148	-0.2188
SE	setup ↑ → CK	0.3594	0.3750	0.3945
	setup ↓ → CK	0.3320	0.2969	0.2812
	hold ↑ → CK	-0.0977	-0.0938	-0.0898
	hold ↓ → CK	-0.1211	-0.1250	-0.1289
D	setup ↑ → CK	0.2773	0.2422	0.2227
	setup ↓ → CK	0.2773	0.2930	0.3086
	hold ↑ → CK	-0.1133	-0.1094	-0.0859
	hold ↓ → CK	-0.1602	-0.1641	-0.1719
CK	minpwh	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974
SN	minpwl	0.9974	0.9974	0.9974
	recovery	-0.0391	-0.0352	-0.0391
	removal	0.0820	0.0742	0.0781
RN	minpwl	0.9974	0.9974	0.9974
	recovery	0.1953	0.1523	0.1562
	removal	-0.0938	-0.0703	-0.0820

Cell Description

The SDFFSRHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and asynchronous active-low reset (RN) and set (SN), and set dominating reset. The cell has a single output (Q) and fast clock-to-out path.

Logic Symbol



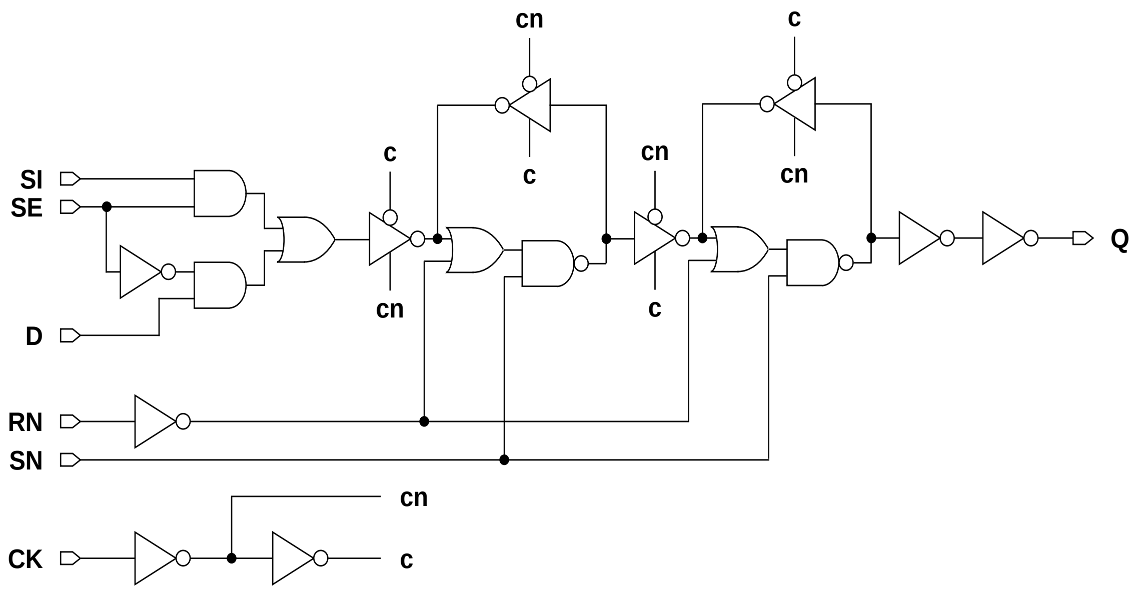
Function Table

RN	SN	D	SI	SE	CK	Q[n+1]
1	1	1	x	0	_/‾	1
1	1	0	x	0	_/‾	0
1	1	x	x	x	‾_	Q[n]
1	1	x	1	1	_/‾	1
1	1	x	0	1	_/‾	0
0	1	x	x	x	x	0
1	0	x	x	x	x	1
0	0	x	x	x	x	1

Cell Size

Drive Strength	Height (um)	Width (um)
SDFFSRHQX1M	2.87	16.40
SDFFSRHQX2M	2.87	16.81
SDFFSRHQX4M	2.87	17.22
SDFFSRHQX8M	2.87	18.45

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	X1	**X2**	**X4**	**X8**
SI	0.0172	0.0196	0.0219	0.0220
SE	0.0195	0.0218	0.0238	0.0239
D	0.0179	0.0210	0.0232	0.0233
CK	0.0277	0.0312	0.0356	0.0356
SN	0.0083	0.0084	0.0089	0.0089
RN	0.0022	0.0024	0.0027	0.0028
Q	0.0117	0.0137	0.0200	0.0363

Pin Capacitance

Pin	Capacitance (pF)			
	X1	**X2**	**X4**	**X8**
SI	0.0016	0.0016	0.0018	0.0018
SE	0.0042	0.0042	0.0046	0.0046
D	0.0016	0.0019	0.0026	0.0026
CK	0.0028	0.0029	0.0031	0.0032
SN	0.0030	0.0032	0.0035	0.0035
RN	0.0028	0.0032	0.0036	0.0036

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	**X2**	**X4**	**X8**	**X1**	**X2**	**X4**	**X8**
CK → Q ↑	0.1890	0.1766	0.1723	0.2027	5.3406	3.6002	1.8471	0.9631
CK → Q ↓	0.2000	0.1839	0.1884	0.2253	3.7859	1.9291	0.9990	0.5318
SN → Q ↑	0.2460	0.2759	0.2776	0.3103	5.2250	3.5679	1.8266	0.9470
SN → Q ↓	0.3568	0.3709	0.3606	0.4603	3.8843	2.0870	1.0501	0.5607
RN → Q ↓	0.3020	0.3165	0.3034	0.4043	3.9066	2.0873	1.0554	0.5626

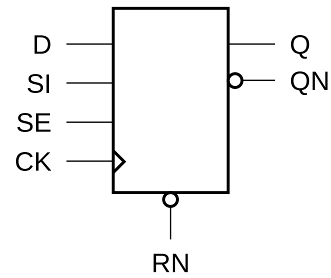
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
SI	setup ↑ → CK	0.2070	0.1875	0.1484	0.1523
	setup ↓ → CK	0.2617	0.2539	0.2539	0.2500
	hold ↑ → CK	-0.1055	-0.0938	-0.0664	-0.0625
	hold ↓ → CK	-0.1719	-0.1719	-0.1680	-0.1641
SE	setup ↑ → CK	0.2695	0.2617	0.2656	0.2617
	setup ↓ → CK	0.2461	0.2227	0.1914	0.1953
	hold ↑ → CK	-0.0977	-0.0898	-0.0586	-0.0586
	hold ↓ → CK	-0.1133	-0.0977	-0.0820	-0.0781
D	setup ↑ → CK	0.2070	0.1797	0.1484	0.1523
	setup ↓ → CK	0.2109	0.1758	0.1484	0.1406
	hold ↑ → CK	-0.1055	-0.0898	-0.0664	-0.0625
	hold ↓ → CK	-0.1211	-0.0938	-0.0664	-0.0625
CK	minpwh	0.9974	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974	0.9974
SN	minpwl	0.9974	0.9974	0.9974	0.9974
	recovery	0.0742	0.0547	0.0586	0.0586
	removal	-0.0352	-0.0273	-0.0312	-0.0312
RN	minpwl	0.9974	0.9974	0.9974	0.9974
	recovery	-0.0312	-0.0352	-0.0312	-0.0312
	removal	0.0664	0.0742	0.0742	0.0703

Cell Description

The SDFFTR cell is a positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and synchronous active-low reset (RN). Scan enable (SE) dominates reset (RN).

Logic Symbol



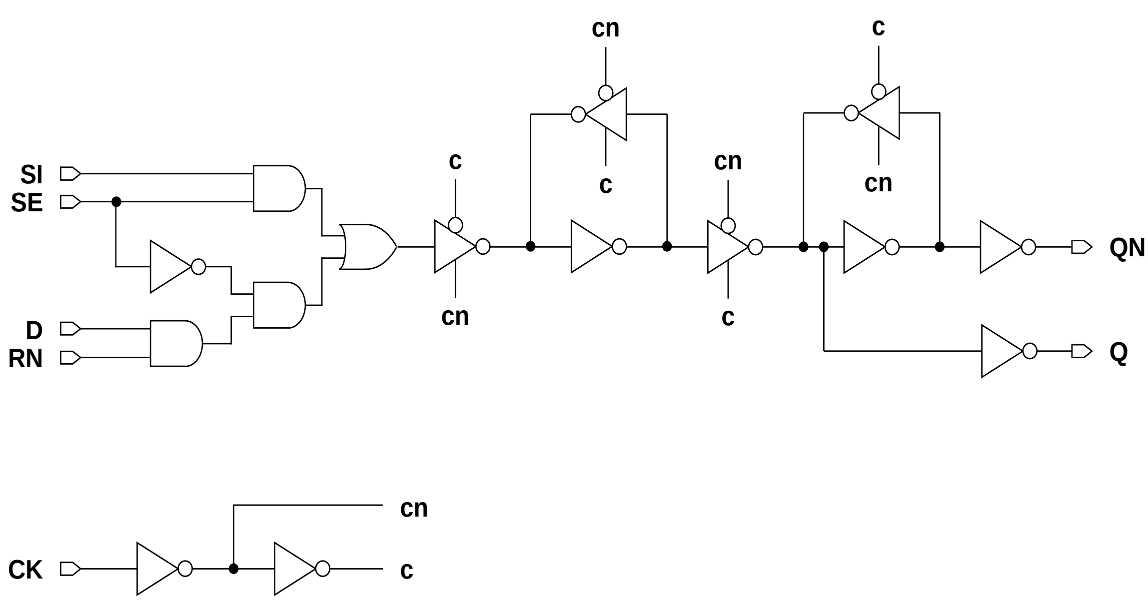
Function Table

RN	D	SI	SE	CK	Q[n+1]	QN[n+1]
x	x	0	1	↑	0	1
x	x	1	1	↑	1	0
0	x	x	0	↑	0	1
1	0	x	0	↑	0	1
1	1	x	0	↑	1	0
x	x	x	x	↓	Q[n]	QN[n]

Cell Size

Drive Strength	Height (um)	Width (um)
SDFFTRX1M	2.87	11.07
SDFFTRX2M	2.87	11.48
SDFFTRX4M	2.87	14.35

Functional Schematic



AC Power

Pin	Power (uW/MHz)		
	X1	**X2**	**X4**
SI	0.0142	0.0169	0.0284
SE	0.0171	0.0195	0.0276
D	0.0138	0.0162	0.0246
CK	0.0192	0.0216	0.0328
RN	0.0156	0.0186	0.0322
Q	0.0121	0.0169	0.0271

Pin Capacitance

Pin	Capacitance (pF)		
	X1	**X2**	**X4**
SI	0.0014	0.0014	0.0016
SE	0.0045	0.0045	0.0050
D	0.0013	0.0013	0.0018
CK	0.0020	0.0021	0.0023
RN	0.0025	0.0025	0.0030

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	**X2**	**X4**	**X1**	**X2**	**X4**
CK $\rightarrow$ Q $\uparrow$	0.2097	0.2030	0.2041	5.2919	3.6562	1.8393
CK $\rightarrow$ Q $\downarrow$	0.2540	0.2743	0.2657	3.2661	1.8724	0.8683
CK $\rightarrow$ QN $\uparrow$	0.1598	0.1621	0.1679	5.3161	3.6633	1.8470
CK $\rightarrow$ QN $\downarrow$	0.1538	0.1372	0.1394	3.4124	1.8730	0.8841

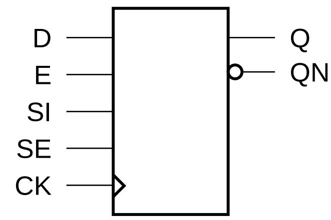
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)		
		X1	X2	X4
SI	setup ↑ → CK	0.1328	0.1406	0.1602
	setup ↓ → CK	0.3047	0.3477	0.4180
	hold ↑ → CK	-0.0859	-0.0898	-0.1016
	hold ↓ → CK	-0.1875	-0.2109	-0.2461
SE	setup ↑ → CK	0.3047	0.3438	0.4180
	setup ↓ → CK	0.2305	0.2656	0.2109
	hold ↑ → CK	-0.0820	-0.0859	-0.0977
	hold ↓ → CK	-0.1328	-0.1523	-0.1094
D	setup ↑ → CK	0.1562	0.1641	0.1562
	setup ↓ → CK	0.2500	0.2852	0.2344
	hold ↑ → CK	-0.1055	-0.1055	-0.1016
	hold ↓ → CK	-0.1523	-0.1680	-0.1328
CK	minpwh	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974
RN	setup ↑ → CK	0.1641	0.1719	0.1602
	setup ↓ → CK	0.3438	0.3867	0.4727
	hold ↑ → CK	-0.1133	-0.1133	-0.1055
	hold ↓ → CK	-0.2109	-0.2344	-0.2773

Cell Description

The SEDFF cell is a positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and synchronous active-high enable (E).

Logic Symbol



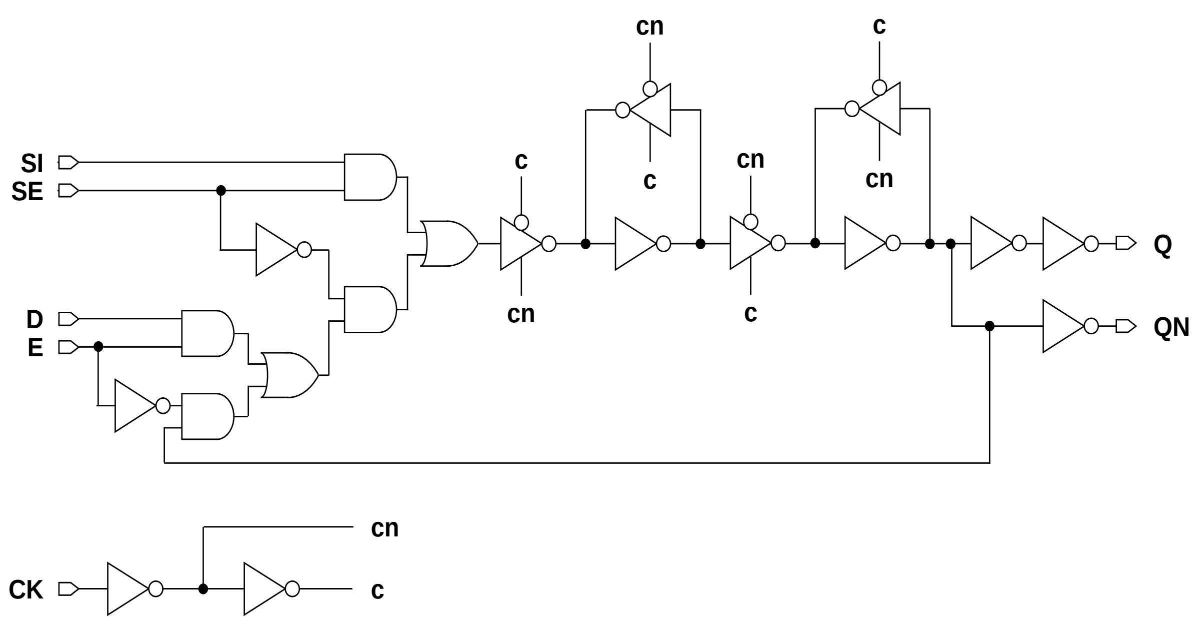
Function Table

D	E	SI	SE	CK	Q[n+1]	QN[n+1]
x	x	1	1	_/‾	1	0
x	x	0	1	_/‾	0	1
x	0	x	0	_/‾	Q[n]	QN[n]
0	1	x	0	_/‾	0	1
1	1	x	0	_/‾	1	0
x	x	x	x	‾_	Q[n]	QN[n]

Cell Size

Drive Strength	Height (um)	Width (um)
SEDFFX1M	2.87	14.35
SEDFFX2M	2.87	14.35
SEDFFX4M	2.87	15.58

Functional Schematic



AC Power

Pin	Power (uW/MHz)		
	X1	X2	X4
SI	0.0164	0.0163	0.0172
SE	0.0187	0.0187	0.0194
D	0.0132	0.0132	0.0139
CK	0.0228	0.0231	0.0245
E	0.0177	0.0179	0.0185
Q	0.0173	0.0206	0.0318

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
SI	0.0013	0.0014	0.0014
SE	0.0029	0.0030	0.0029
D	0.0018	0.0018	0.0018
CK	0.0017	0.0017	0.0020
E	0.0032	0.0034	0.0033

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.1905	0.1904	0.1930	5.2011	3.4899	1.8320
CK $\rightarrow$ Q $\downarrow$	0.2342	0.2107	0.2128	3.6822	1.9560	0.9633
CK $\rightarrow$ QN $\uparrow$	0.3242	0.2925	0.3082	5.1811	3.4973	1.8441
CK $\rightarrow$ QN $\downarrow$	0.3277	0.3128	0.3061	3.4633	1.8774	0.8866

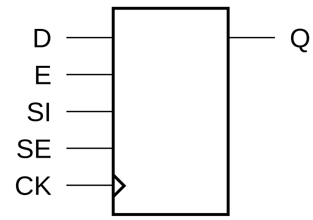
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)		
		X1	X2	X4
SI	setup ↑ → CK	0.1875	0.1797	0.1875
	setup ↓ → CK	0.4648	0.4609	0.4805
	hold ↑ → CK	-0.1562	-0.1484	-0.1484
	hold ↓ → CK	-0.3867	-0.3711	-0.3867
SE	setup ↑ → CK	0.4648	0.4609	0.4805
	setup ↓ → CK	0.3242	0.3203	0.3281
	hold ↑ → CK	-0.1406	-0.1367	-0.1406
	hold ↓ → CK	-0.2344	-0.2266	-0.2305
D	setup ↑ → CK	0.1953	0.1914	0.1992
	setup ↓ → CK	0.2891	0.2891	0.2969
	hold ↑ → CK	-0.1602	-0.1523	-0.1562
	hold ↓ → CK	-0.2188	-0.2109	-0.2148
CK	minpwh	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974
E	setup ↑ → CK	0.3477	0.3477	0.3594
	setup ↓ → CK	0.2656	0.2578	0.2617
	hold ↑ → CK	-0.1680	-0.1602	-0.1641
	hold ↓ → CK	-0.1875	-0.1797	-0.1875

Cell Description

The SEDFFHQ cell is a positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and synchronous active-high enable (E). The cell has a single output (Q) and fast clock-to-output path.

Logic Symbol



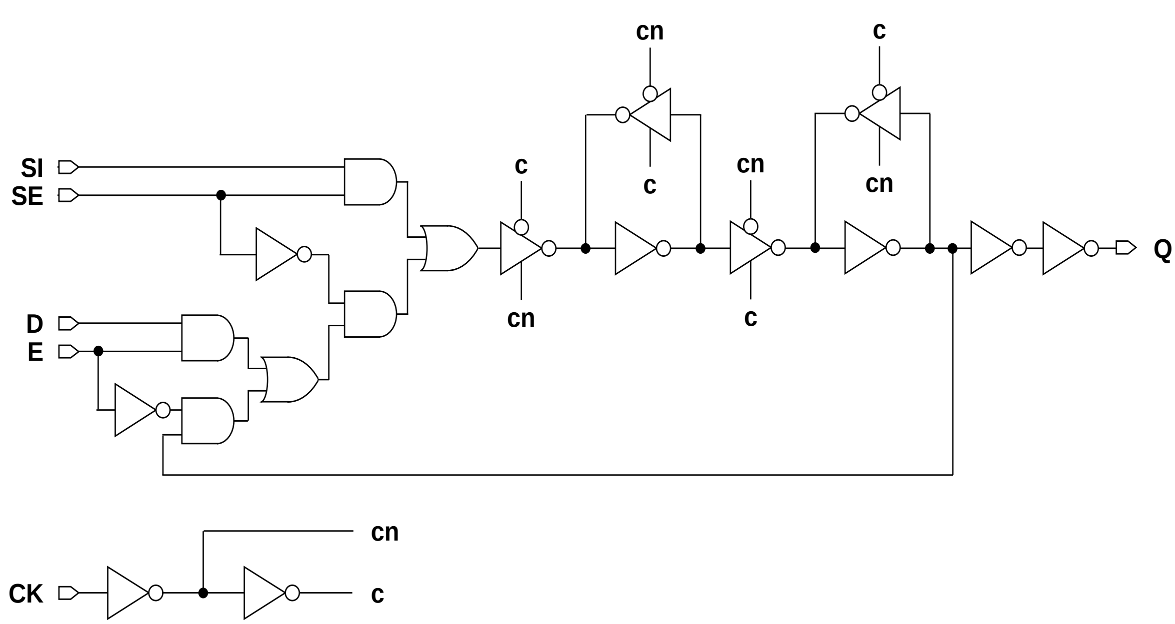
Function Table

D	E	SI	SE	CK	Q[n+1]
x	x	1	1	_/‾	1
x	x	0	1	_/‾	0
x	0	x	0	_/‾	Q[n]
0	1	x	0	_/‾	0
1	1	x	0	_/‾	1
x	x	x	x	‾_	Q[n]

Cell Size

Drive Strength	Height (um)	Width (um)
SEDFFHQX1M	2.87	15.99
SEDFFHQX2M	2.87	15.99
SEDFFHQX4M	2.87	16.81
SEDFFHQX8M	2.87	18.04

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	X1	**X2**	**X4**	**X8**
SI	0.0155	0.0191	0.0195	0.0196
SE	0.0271	0.0282	0.0292	0.0293
D	0.0167	0.0199	0.0217	0.0217
CK	0.0291	0.0306	0.0322	0.0322
E	0.0277	0.0294	0.0304	0.0305
Q	0.0116	0.0135	0.0182	0.0317

Pin Capacitance

Pin	Capacitance (pF)			
	X1	**X2**	**X4**	**X8**
SI	0.0013	0.0013	0.0013	0.0013
SE	0.0040	0.0040	0.0043	0.0043
D	0.0023	0.0022	0.0030	0.0030
CK	0.0029	0.0029	0.0035	0.0035
E	0.0032	0.0031	0.0031	0.0031

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	**X2**	**X4**	**X8**	**X1**	**X2**	**X4**	**X8**
CK $\rightarrow$ Q $\uparrow$	0.1460	0.1376	0.1377	0.1582	5.2416	3.5523	1.8150	0.9452
CK $\rightarrow$ Q $\downarrow$	0.1396	0.1447	0.1535	0.1910	3.3341	1.8299	0.9380	0.4923

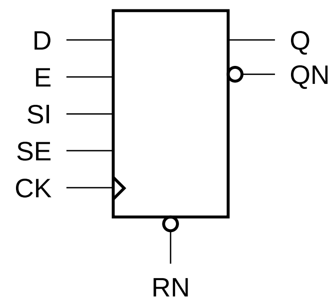
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
SI	setup ↑ → CK	0.1680	0.1758	0.1914	0.1953
	setup ↓ → CK	0.3164	0.3750	0.4023	0.3945
	hold ↑ → CK	-0.0938	-0.1055	-0.1055	-0.1055
	hold ↓ → CK	-0.2500	-0.3008	-0.3203	-0.3164
SE	setup ↑ → CK	0.3906	0.4688	0.5039	0.4961
	setup ↓ → CK	0.3008	0.3320	0.3047	0.3008
	hold ↑ → CK	-0.1133	-0.1250	-0.1445	-0.1484
	hold ↓ → CK	-0.2266	-0.2227	-0.2188	-0.2188
D	setup ↑ → CK	0.1602	0.1484	0.1406	0.1445
	setup ↓ → CK	0.1484	0.2227	0.1875	0.1797
	hold ↑ → CK	-0.0898	-0.0781	-0.0625	-0.0625
	hold ↓ → CK	-0.0898	-0.1562	-0.1172	-0.1172
CK	minpwh	0.9974	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974	0.9974
E	setup ↑ → CK	0.2266	0.2578	0.2344	0.2266
	setup ↓ → CK	0.3594	0.4180	0.3906	0.3906
	hold ↑ → CK	-0.1758	-0.1953	-0.1953	-0.1992
	hold ↓ → CK	-0.1914	-0.2031	-0.1914	-0.1953

Cell Description

The SEDFFTR cell is a positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), synchronous active-high enable (E) and synchronous active low reset (RN). Scan enable (SE) dominates reset (RN) and enable (E).

Logic Symbol



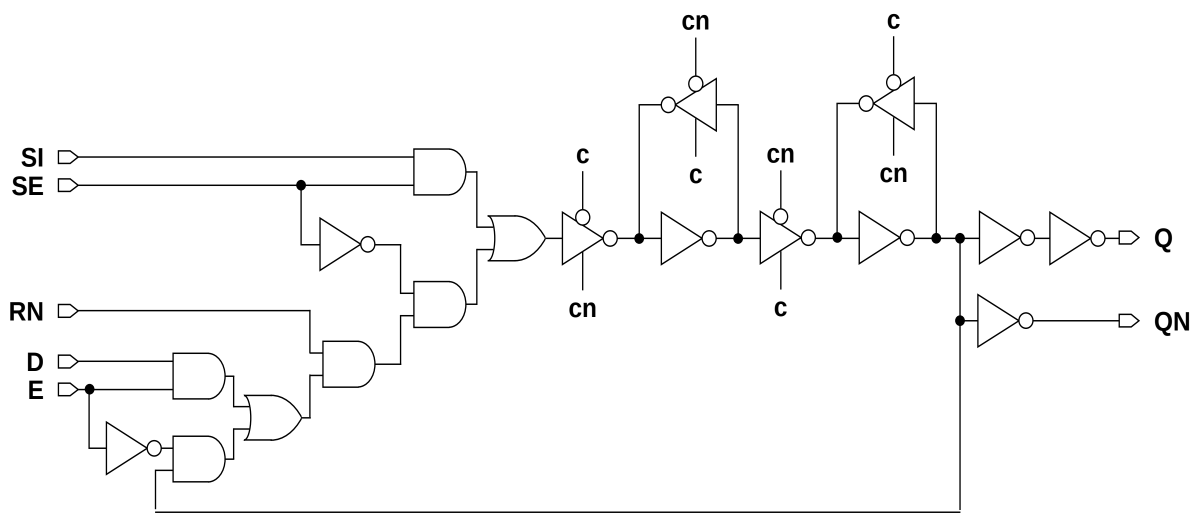
Function Table

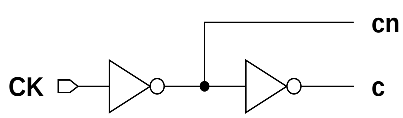
RN	D	E	SI	SE	CK	Q[n+1]	QN[n+1]
x	x	x	0	1	↑	0	1
x	x	x	1	1	↑	1	0
1	x	0	x	0	↑	Q[n]	QN[n]
0	x	x	x	0	↑	0	1
1	1	1	x	0	↑	1	0
1	0	1	x	0	↑	0	1
x	x	x	x	x	↓	Q[n]	QN[n]

Cell Size

Drive Strength	Height (um)	Width (um)
SEDFFTRX1M	2.87	18.86
SEDFFTRX2M	2.87	18.86
SEDFFTRX4M	2.87	20.09

Functional Schematic





AC Power

Pin	Power (uW/MHz)		
	X1	X2	X4
SI	0.0172	0.0177	0.0190
SE	0.0244	0.0250	0.0262
D	0.0196	0.0203	0.0213
CK	0.0226	0.0232	0.0249
E	0.0257	0.0263	0.0273
RN	0.0166	0.0171	0.0183
Q	0.0173	0.0211	0.0310

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
SI	0.0015	0.0015	0.0015
SE	0.0026	0.0026	0.0026
D	0.0014	0.0014	0.0014
CK	0.0020	0.0020	0.0022
E	0.0015	0.0015	0.0015
RN	0.0016	0.0016	0.0016

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
CK → Q ↑	0.2837	0.2755	0.2878	5.3095	3.6345	1.8514
CK → Q ↓	0.2799	0.2757	0.2730	3.4032	1.8798	0.8786
CK → QN ↑	0.1632	0.1641	0.1669	5.3096	3.6450	1.8717
CK → QN ↓	0.1945	0.1897	0.1844	3.6549	1.9965	0.9551

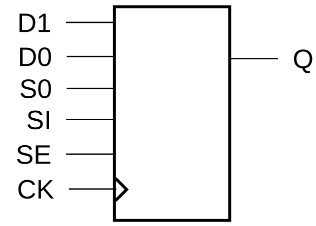
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)		
		X1	**X2**	**X4**
SI	setup ↑ → CK	0.2734	0.2617	0.2695
	setup ↓ → CK	0.2266	0.2305	0.2383
	hold ↑ → CK	-0.2227	-0.2031	-0.2109
	hold ↓ → CK	-0.2031	-0.1992	-0.2031
SE	setup ↑ → CK	0.3359	0.3242	0.3281
	setup ↓ → CK	0.3672	0.3555	0.3633
	hold ↑ → CK	-0.2227	-0.2148	-0.2227
	hold ↓ → CK	-0.2852	-0.2812	-0.2852
D	setup ↑ → CK	0.3359	0.3242	0.3320
	setup ↓ → CK	0.3125	0.3125	0.3203
	hold ↑ → CK	-0.2773	-0.2617	-0.2695
	hold ↓ → CK	-0.2812	-0.2773	-0.2812
CK	minpwh	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974
E	setup ↑ → CK	0.4141	0.4023	0.4102
	setup ↓ → CK	0.3398	0.3203	0.3359
	hold ↑ → CK	-0.3008	-0.2969	-0.3008
	hold ↓ → CK	-0.1016	-0.1094	-0.1562
RN	setup ↑ → CK	0.2539	0.2383	0.2422
	setup ↓ → CK	0.2305	0.2305	0.2383
	hold ↑ → CK	-0.1992	-0.1758	-0.1836
	hold ↓ → CK	-0.2070	-0.2031	-0.2031

Cell Description

The SMDFFHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with a 2to1data select control (S0) for the data inputs (D1,D0), scan input (SI), and activehigh scan enable (SE). The cell has a single output (Q) and fast clocktoout path.

Logic Symbol



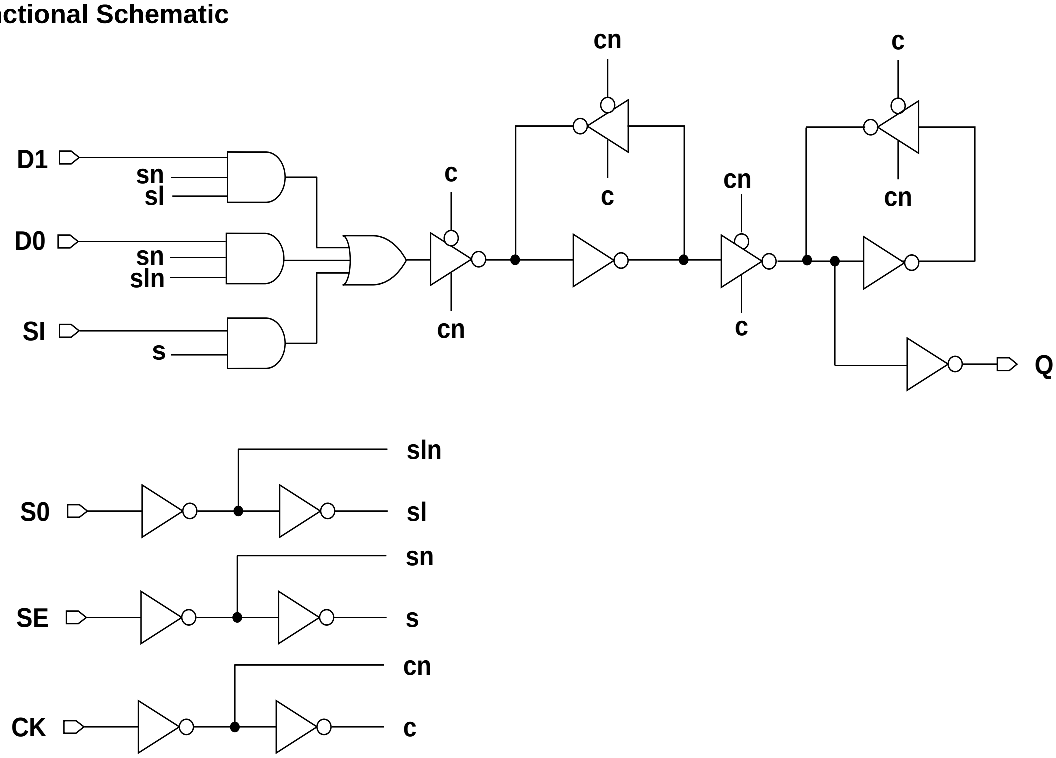
Function Table

SE	SI	S0	D1	D0	CK	Q[n+1]
0	x	0	x	0	↑	0
0	x	0	x	1	↑	1
0	x	1	0	x	↑	0
0	x	1	1	x	↑	1
1	0	x	x	x	↑	0
1	1	x	x	x	↑	1
x	x	x	x	x	↓	Q[n]

Cell Size

Drive Strength	Height (um)	Width (um)
SMDFFHQX1M	2.87	17.22
SMDFFHQX2M	2.87	17.63
SMDFFHQX4M	2.87	18.04
SMDFFHQX8M	2.87	18.86

Functional Schematic



AC Power

Pin	Power (uW/MHz)			
	X1	**X2**	**X4**	**X8**
SI	0.0163	0.0185	0.0214	0.0217
SE	0.0266	0.0285	0.0322	0.0322
D0	0.0166	0.0188	0.0224	0.0227
D1	0.0168	0.0193	0.0230	0.0233
S0	0.0271	0.0293	0.0325	0.0326
CK	0.0283	0.0326	0.0362	0.0362
Q	0.0080	0.0109	0.0153	0.0294

Pin Capacitance

Pin	Capacitance (pF)			
	X1	**X2**	**X4**	**X8**
SI	0.0013	0.0013	0.0017	0.0017
SE	0.0041	0.0042	0.0043	0.0043
D0	0.0016	0.0019	0.0026	0.0026
D1	0.0016	0.0020	0.0030	0.0030
S0	0.0031	0.0031	0.0031	0.0031
CK	0.0029	0.0029	0.0036	0.0036

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	**X2**	**X4**	**X8**	**X1**	**X2**	**X4**	**X8**
CK → Q ↑	0.1442	0.1437	0.1376	0.1602	5.0625	3.4820	1.7911	0.9283
CK → Q ↓	0.1436	0.1513	0.1572	0.1940	3.3473	1.8087	0.9199	0.4858

Timing Constraints at 25°C,1.2V, Typical Process

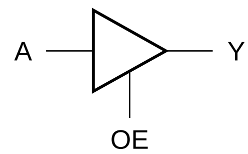
Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
SI	setup ↑ → CK	0.1953	0.1797	0.1211	0.1289
	setup ↓ → CK	0.3281	0.3242	0.3164	0.3125
	hold ↑ → CK	-0.1094	-0.1055	-0.0625	-0.0625
	hold ↓ → CK	-0.2578	-0.2656	-0.2578	-0.2539
SE	setup ↑ → CK	0.3438	0.3438	0.3359	0.3320
	setup ↓ → CK	0.3398	0.3047	0.2891	0.2852
	hold ↑ → CK	-0.1133	-0.1133	-0.0781	-0.0781
	hold ↓ → CK	-0.2031	-0.1836	-0.1562	-0.1523
D0	setup ↑ → CK	0.1914	0.1523	0.1094	0.1133
	setup ↓ → CK	0.2109	0.1797	0.1562	0.1523
	hold ↑ → CK	-0.1055	-0.0820	-0.0508	-0.0508
	hold ↓ → CK	-0.1445	-0.1250	-0.1055	-0.1016
D1	setup ↑ → CK	0.1914	0.1484	0.0898	0.0938
	setup ↓ → CK	0.2344	0.1836	0.1602	0.1562
	hold ↑ → CK	-0.1055	-0.0820	-0.0352	-0.0352
	hold ↓ → CK	-0.1641	-0.1289	-0.1094	-0.1055
S0	setup ↑ → CK	0.2617	0.2266	0.2109	0.2109
	setup ↓ → CK	0.2891	0.2617	0.2539	0.2539
	hold ↑ → CK	-0.1758	-0.1562	-0.1211	-0.1211
	hold ↓ → CK	-0.1914	-0.1719	-0.1445	-0.1445
CK	minpwh	0.9974	0.9974	0.9974	0.9974
	minpwl	0.9974	0.9974	0.9974	0.9974

Cell Description

The TBUF cell provides the logical buffer of a single input (A) with an active-high output enable (OE). When the enable is high, the output (Y) is represented by the logic equation:

$Y = A$

Logic Symbol



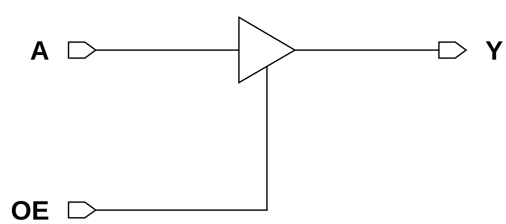
Function Table

OE	A	Y
0	x	Z
1	0	0
1	1	1

Cell Size

Drive Strength	Height (um)	Width (um)
TBUFX1M	2.87	3.69
TBUFX2M	2.87	3.69
TBUFX3M	2.87	4.10
TBUFX4M	2.87	4.10
TBUFX6M	2.87	4.51
TBUFX8M	2.87	6.15
TBUFX12M	2.87	6.97
TBUFX16M	2.87	7.79
TBUFX20M	2.87	10.66
TBUFX24M	2.87	12.71

Functional Schematic



AC Power

Pin	Power (uW/MHz)							
	X1	**X2**	**X3**	**X4**	**X6**	**X8**	**X12**	**X16**
A	0.0097	0.0110	0.0146	0.0169	0.0220	0.0310	0.0434	0.0551
OE	0.0072	0.0082	0.0111	0.0130	0.0175	0.0238	0.0336	0.0430

AC Power (Cont'd.)

Pin	Power (uW/MHz)	
	X20	**X24**
A	0.0692	0.0857
OE	0.0543	0.0644

Pin Capacitance

Pin	Capacitance (pF)							
	X1	**X2**	**X3**	**X4**	**X6**	**X8**	**X12**	**X16**
A	0.0017	0.0021	0.0027	0.0033	0.0036	0.0060	0.0087	0.0100
OE	0.0026	0.0026	0.0027	0.0029	0.0033	0.0043	0.0056	0.0069
Y	0.0019	0.0024	0.0032	0.0038	0.0060	0.0075	0.0114	0.0155

Pin Capacitance (Cont'd.)

Pin	Capacitance (pF)	
	X20	**X24**
A	0.0130	0.0165
OE	0.0083	0.0106
Y	0.0191	0.0227

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)							
	X1	**X2**	**X3**	**X4**	**X6**	**X8**	**X12**	**X16**
A → Y ↑	0.0946	0.0887	0.0853	0.0816	0.0874	0.0753	0.0713	0.0725
A → Y ↓	0.1683	0.1432	0.1376	0.1261	0.1290	0.1165	0.1136	0.1197
OE → Y ↑	0.0618	0.0637	0.0632	0.0612	0.0679	0.0577	0.0547	0.0555
OE → Y ↓	0.1097	0.1073	0.1032	0.1011	0.1031	0.0936	0.0905	0.0937

Delays at 25°C,1.2V, Typical Process (Cont'd.)

Description	Intrinsic Delay (ns)	
	X20	X24
A → Y ↑	0.0763	0.0727
A → Y ↓	0.1160	0.1136
OE → Y ↑	0.0613	0.0592
OE → Y ↓	0.0969	0.0915

Delays at 25°C,1.2V, Typical Process (Cont'd.)

Description	K_{load} (ns/pF)							
	X1	X2	X3	X4	X6	X8	X12	X16
A → Y ↑	5.1691	4.4587	2.4948	2.0404	1.4321	0.9875	0.6679	0.5364
A → Y ↓	3.6886	2.1429	1.2555	0.9878	0.6855	0.4786	0.3112	0.2457
OE → Y ↑	5.1600	4.4592	2.4952	2.0415	1.4330	0.9880	0.6688	0.5366
OE → Y ↓	3.6543	2.1289	1.2460	0.9834	0.6822	0.4768	0.3098	0.2445

Delays at 25°C,1.2V, Typical Process (Cont'd.)

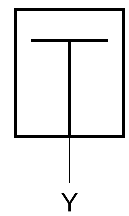
Description	K_{load} (ns/pF)	
	X20	X24
A → Y ↑	0.4173	0.3525
A → Y ↓	0.2000	0.1586
OE → Y ↑	0.4177	0.3528
OE → Y ↓	0.1993	0.1578

Cell Description

The TIEHI cell drives the output (Y) to a logic high. The output is driven through diffusion and not tied directly to the power rail to provide some ESD protection. The output (Y) is represented by the logic equation:

$Y = 1$

Logic Symbol



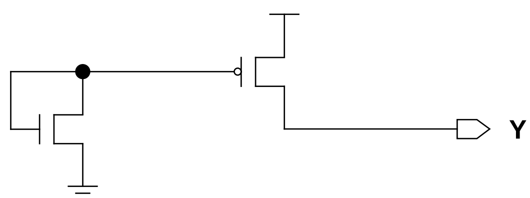
Function Table

Y
1

Cell Size

Drive Strength	Height (um)	Width (um)
TIEHIM	2.87	1.23

Functional Schematic

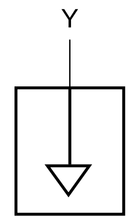


Cell Description

The TIELO cell drives the output (Y) to a logic low. The output is driven through diffusion and not tied directly to the power rail to provide some ESD protection. The output (Y) is represented by the logic equation:

$Y = 0$

Logic Symbol



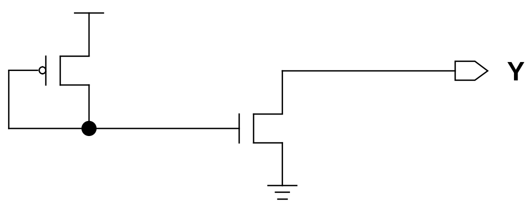
Function Table

Y
0

Cell Size

Drive Strength	Height (um)	Width (um)
TIELOM	2.87	1.23

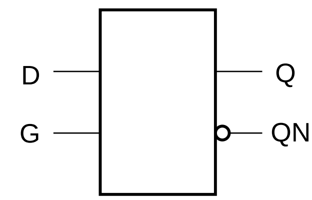
Functional Schematic



Cell Description

The TLAT cell is an active-high D-type transparent latch. When the enable (G) is high, data is transferred to the outputs (Q,QN).

Logic Symbol



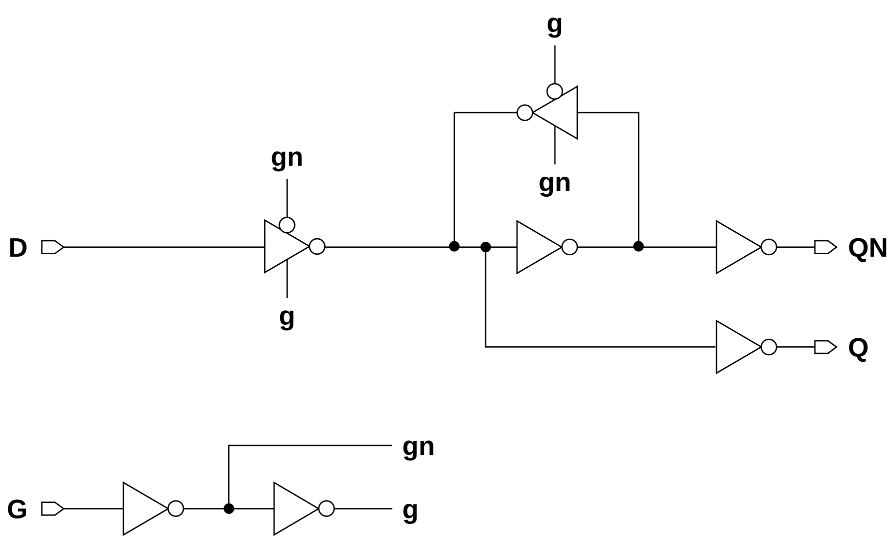
Function Table

G	D	Q[n+1]	QN[n+1]
1	0	0	1
1	1	1	0
0	x	Q[n]	QN[n]

Cell Size

Drive Strength	Height (um)	Width (um)
TLATX1M	2.87	5.74
TLATX2M	2.87	5.74
TLATX4M	2.87	6.56

Functional Schematic



AC Power

Pin	Power (uW/MHz)		
	X1	X2	X4
D	0.0019	0.0022	0.0020
G	0.0075	0.0077	0.0107
Q	0.0123	0.0159	0.0276

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
D	0.0026	0.0031	0.0030
G	0.0015	0.0018	0.0029

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
D → Q ↑	0.0683	0.0716	0.0769	5.2883	3.5599	1.8560
D → Q ↓	0.1340	0.1308	0.1866	3.3526	1.8018	0.9821
G → Q ↑	0.1660	0.1645	0.1383	5.2872	3.5582	1.8556
G → Q ↓	0.1359	0.1311	0.1753	3.3501	1.8018	0.9819
D → QN ↑	0.1845	0.1827	0.2566	5.2827	3.5499	1.8459
D → QN ↓	0.1572	0.1499	0.1527	3.2457	1.7349	0.8537
G → QN ↑	0.1871	0.1834	0.2455	5.2827	3.5500	1.8460
G → QN ↓	0.2558	0.2435	0.2145	3.2455	1.7350	0.8537

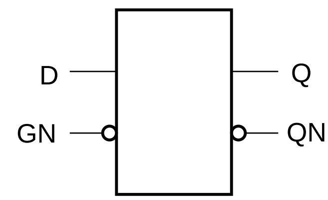
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)		
		X1	X2	X4
D	setup ↑ → G	-0.0234	-0.0117	0.0117
	setup ↓ → G	0.0859	0.0898	0.1641
	hold ↑ → G	0.0312	0.0234	0.0039
	hold ↓ → G	-0.0820	-0.0859	-0.1562
G	minpwh	0.9974	0.9974	0.9974

Cell Description

The TLATN cell is an active-low D-type transparent latch. When the enable (GN) is low, data is transferred to the outputs (Q,QN).

Logic Symbol



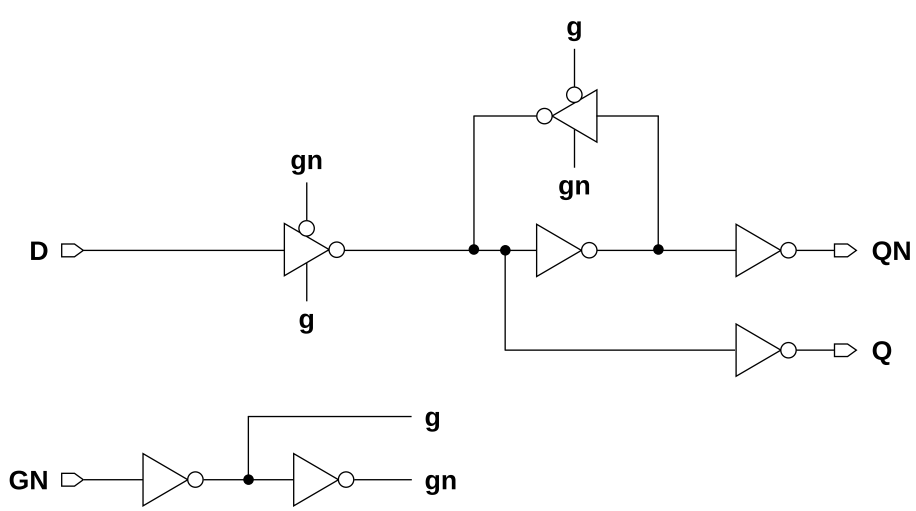
Function Table

GN	D	Q[n+1]	QN[n+1]
0	0	0	1
0	1	1	0
1	x	Q[n]	QN[n]

Cell Size

Drive Strength	Height (um)	Width (um)
TLATNX1M	2.87	5.74
TLATNX2M	2.87	5.74
TLATNX4M	2.87	6.56

Functional Schematic



AC Power

Pin	Power (uW/MHz)		
	X1	X2	X4
D	0.0020	0.0022	0.0022
GN	0.0080	0.0089	0.0115
Q	0.0127	0.0170	0.0282

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
D	0.0026	0.0030	0.0031
GN	0.0016	0.0018	0.0029

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
D → Q ↑	0.0673	0.0654	0.0759	5.2866	3.5570	1.8580
D → Q ↓	0.1335	0.1507	0.1875	3.3466	1.8402	0.9817
GN → Q ↑	0.1256	0.1194	0.1154	5.2937	3.5586	1.8580
GN → Q ↓	0.2082	0.2160	0.2261	3.3455	1.8395	0.9817
D → QN ↑	0.1825	0.2038	0.2571	5.2814	3.5506	1.8464
D → QN ↓	0.1548	0.1439	0.1514	3.2450	1.7334	0.8535
GN → QN ↑	0.2578	0.2692	0.2960	5.2810	3.5505	1.8464
GN → QN ↓	0.2143	0.1990	0.1915	3.2450	1.7336	0.8536

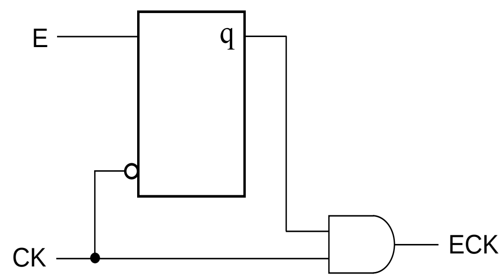
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)		
		X1	X2	X4
D	setup ↑ → GN	0.0430	0.0430	0.0547
	setup ↓ → GN	0.0625	0.0820	0.1406
	hold ↑ → GN	-0.0391	-0.0312	-0.0391
	hold ↓ → GN	-0.0469	-0.0664	-0.1328
GN	minpwl	0.9974	0.9974	0.9974

Cell Description

The TLATNCA cell is a positive-edge triggered clock-gating latch. The positive-edge clock (CK) is qualified by the latched enable signal (E) to create the gated positive-edge clock (ECK).

Logic Symbol



Function Table

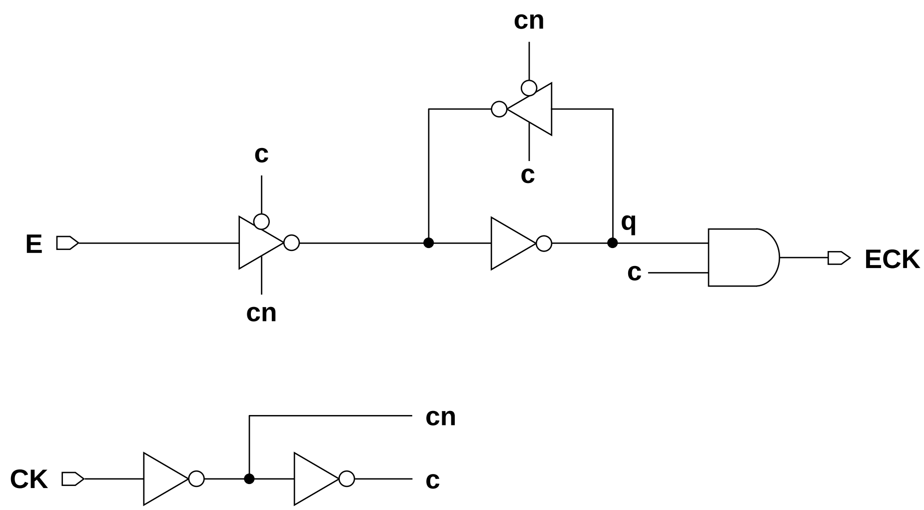
CK	E	q[n+1]	ECK[n+1]
1	x	q[n]	q[n]
0	0	0	0
0	1	1	0

- *Note: q is an internal node, and is not accessible.*

Cell Size

Drive Strength	Height (um)	Width (um)
TLATNCAX2M	2.87	5.33
TLATNCAX3M	2.87	5.74
TLATNCAX4M	2.87	7.38
TLATNCAX6M	2.87	8.61
TLATNCAX8M	2.87	9.43
TLATNCAX12M	2.87	13.94
TLATNCAX16M	2.87	15.58
TLATNCAX20M	2.87	20.09

Functional Schematic



AC Power

Pin	Power (uW/MHz)							
	X2	**X3**	**X4**	**X6**	**X8**	**X12**	**X16**	**X20**
E	0.0119	0.0126	0.0169	0.0205	0.0218	0.0416	0.0449	0.0569
CK	0.0114	0.0137	0.0178	0.0205	0.0235	0.0392	0.0418	0.0608
ECK	0.0123	0.0141	0.0186	0.0255	0.0307	0.0460	0.0584	0.0725

Pin Capacitance

Pin	Capacitance (pF)							
	X2	**X3**	**X4**	**X6**	**X8**	**X12**	**X16**	**X20**
E	0.0029	0.0031	0.0048	0.0054	0.0054	0.0133	0.0133	0.0175
CK	0.0026	0.0033	0.0036	0.0056	0.0067	0.0084	0.0084	0.0165

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)							
	X2	**X3**	**X4**	**X6**	**X8**	**X12**	**X16**	**X20**
CK → ECK ↑	0.0754	0.0769	0.0793	0.0765	0.0813	0.0951	0.1061	0.0891
CK → ECK ↓	0.0999	0.0971	0.1021	0.0925	0.0858	0.0979	0.1057	0.0904

Delays at 25°C,1.2V, Typical Process (Cont'd.)

Description	K_{load} (ns/pF)							
	X2	**X3**	**X4**	**X6**	**X8**	**X12**	**X16**	**X20**
CK → ECK ↑	7.2295	5.0688	3.7006	2.5279	1.9204	1.3224	1.0043	0.8283
CK → ECK ↓	3.4293	2.4378	1.9149	1.2132	0.8934	0.6919	0.4822	0.3563

Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)							
		X2	**X3**	**X4**	**X6**	**X8**	**X12**	**X16**	**X20**
E	setup ↑ → CK	0.0625	0.0664	0.0547	0.0586	0.0625	0.0391	0.0391	0.0430
	setup ↓ → CK	0.0273	0.0234	0.0273	0.0352	0.0391	0.0156	0.0156	0.0273
	hold ↑ → CK	-0.0312	-0.0352	-0.0312	-0.0312	-0.0352	-0.0195	-0.0234	-0.0273
	hold ↓ → CK	0.0547	0.0547	0.0469	0.0508	0.0508	0.0508	0.0547	0.0430
CK	minpwl	0.9974	0.9974	0.9974	0.9974	0.9974	0.9974	0.9974	0.9974

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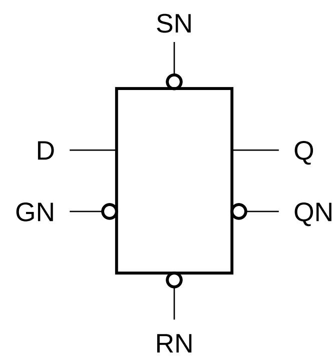
Cell Description

The TLATNSR cell is an active-low D-type transparent latch with asynchronous active-low set (SN) and reset (RN), and set dominating reset. When the enable (GN) is low, data is transferred to the outputs (Q,QN).

Logic Symbol



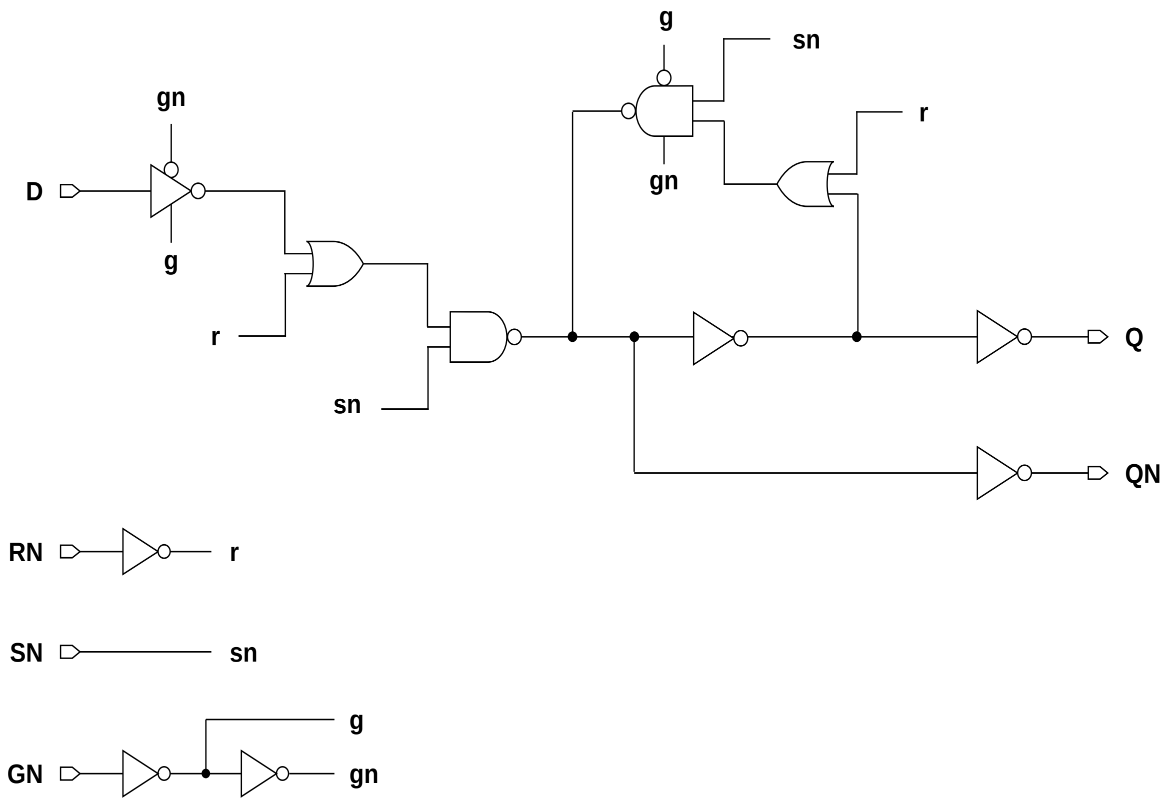
Function Table

RN	SN	GN	D	Q[n+1]	QN[n+1]
1	1	0	0	0	1
1	1	0	1	1	0
1	1	1	x	Q[n]	QN[n]
0	1	x	x	0	1
1	0	x	x	1	0
0	0	x	x	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
TLATNSRX1M	2.87	9.02
TLATNSRX2M	2.87	11.89
TLATNSRX4M	2.87	13.53

Functional Schematic



AC Power

Pin	Power (uW/MHz)		
	X1	X2	X4
D	0.0017	0.0041	0.0041
GN	0.0104	0.0155	0.0184
SN	0.0064	0.0094	0.0137
RN	0.0031	0.0048	0.0048
Q	0.0238	0.0325	0.0530

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
D	0.0026	0.0056	0.0056
GN	0.0017	0.0022	0.0035
SN	0.0022	0.0031	0.0057
RN	0.0044	0.0066	0.0065

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
D → Q ↑	0.1404	0.1458	0.1661	5.4166	3.6234	2.0184
D → Q ↓	0.3420	0.2946	0.3378	4.4926	2.3054	1.1733
GN → Q ↑	0.1934	0.1898	0.1921	5.4156	3.6244	2.0177
GN → Q ↓	0.3898	0.3367	0.3558	4.4911	2.3051	1.1729
SN → Q ↑	0.1633	0.1826	0.1165	5.3276	3.5969	1.9746
SN → Q ↓	0.3606	0.3034	0.3445	4.4455	2.2895	1.1651
RN → Q ↑	0.1397	0.1422	0.1631	5.4167	3.6234	2.0184
RN → Q ↓	0.2187	0.2373	0.2737	4.1062	2.4107	1.2053
D → QN ↑	0.4010	0.3305	0.3692	5.2838	3.5410	1.8738
D → QN ↓	0.1891	0.1905	0.2008	3.1110	1.7653	0.8380
GN → QN ↑	0.4493	0.3729	0.3874	5.2853	3.5410	1.8739
GN → QN ↓	0.2431	0.2353	0.2271	3.1129	1.7657	0.8380
SN → QN ↑	0.4186	0.3391	0.3760	5.2831	3.5409	1.8737
SN → QN ↓	0.2115	0.2273	0.1517	3.1010	1.7630	0.8323
RN → QN ↑	0.2695	0.2720	0.3036	5.2776	3.5423	1.8742
RN → QN ↓	0.1888	0.1872	0.1979	3.1114	1.7652	0.8380

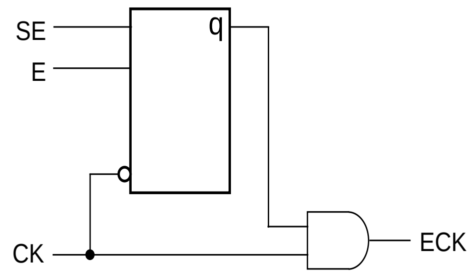
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)		
		X1	X2	X4
D	setup ↑ → GN	0.1094	0.1172	0.1406
	setup ↓ → GN	0.2656	0.1953	0.2578
	hold ↑ → GN	-0.0977	-0.0977	-0.1133
	hold ↓ → GN	-0.2344	-0.1680	-0.2266
GN	minpwl	0.9974	0.9974	0.9974
SN	minpwl	0.9974	0.9974	0.9974
	recovery	0.2812	0.1953	0.2539
	removal	-0.2773	-0.1914	-0.2500
RN	minpwl	0.9974	0.9974	0.9974
	recovery	0.1016	0.1016	0.1172
	removal	-0.0977	-0.0977	-0.1133

Cell Description

The TLATNTSCA cell is a positive-edge triggered clock-gating latch. The positive-edge clock (CK) is qualified by the latched enable signals (SE) and (E) to create the gated positive-edge clock (ECK).

Logic Symbol



Function Table

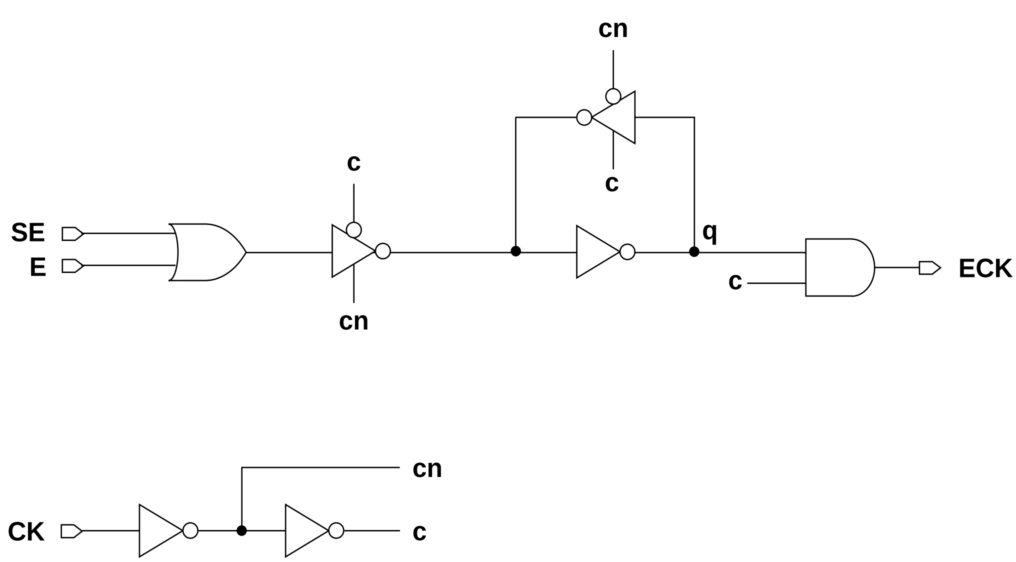
CK	SE	E	q[n+1]	ECK[n+1]
1	x	x	q[n]	q[n]
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	1	0

- *Note: q is an internal node and is not accessible.*

Cell Size

Drive Strength	Height (um)	Width (um)
TLATNTSCAX2M	2.87	6.97
TLATNTSCAX3M	2.87	8.20
TLATNTSCAX4M	2.87	9.84
TLATNTSCAX6M	2.87	11.48
TLATNTSCAX8M	2.87	13.12
TLATNTSCAX12M	2.87	16.81
TLATNTSCAX16M	2.87	22.55
TLATNTSCAX20M	2.87	25.01

Functional Schematic



AC Power

Pin	Power (uW/MHz)							
	X2	X3	X4	X6	X8	X12	X16	X20
E	0.0176	0.0194	0.0235	0.0322	0.0388	0.0533	0.0704	0.0823
SE	0.0179	0.0197	0.0241	0.0331	0.0399	0.0549	0.0727	0.0847
CK	0.0117	0.0134	0.0192	0.0252	0.0310	0.0449	0.0594	0.0685
ECK	0.0123	0.0145	0.0185	0.0271	0.0324	0.0421	0.0580	0.0688

Pin Capacitance

Pin	Capacitance (pF)							
	X2	X3	X4	X6	X8	X12	X16	X20
E	0.0016	0.0017	0.0018	0.0020	0.0024	0.0031	0.0043	0.0045
SE	0.0014	0.0015	0.0019	0.0021	0.0024	0.0032	0.0043	0.0047
CK	0.0027	0.0032	0.0044	0.0059	0.0068	0.0113	0.0143	0.0166

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)							
	X2	X3	X4	X6	X8	X12	X16	X20
CK $\rightarrow$ ECK $\uparrow$	0.0761	0.0764	0.0729	0.0743	0.0798	0.0802	0.0900	0.0864
CK $\rightarrow$ ECK $\downarrow$	0.1017	0.0947	0.0977	0.0914	0.0939	0.0868	0.0906	0.0906

Delays at 25°C,1.2V, Typical Process (Cont'd.)

Description	K_{load} (ns/pF)							
	X2	X3	X4	X6	X8	X12	X16	X20
CK $\rightarrow$ ECK $\uparrow$	7.2399	5.1047	3.7008	2.4978	1.9008	1.5725	1.1417	0.9041
CK $\rightarrow$ ECK $\downarrow$	3.4189	2.4402	1.9137	1.2226	0.9289	0.6142	0.4571	0.3661

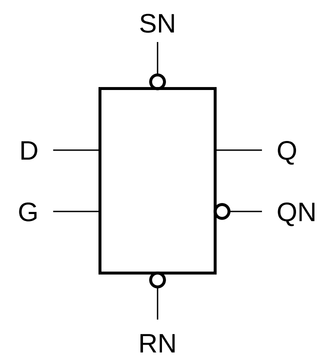
Timing Constraints at 25°C,1.2V, Typical Process

Pin	Requirement	Interval (ns)							
		X2	X3	X4	X6	X8	X12	X16	X20
E	setup ↑ → CK	0.1680	0.1602	0.1445	0.1406	0.1367	0.1328	0.1289	0.1250
	setup ↓ → CK	0.1562	0.1562	0.1719	0.1641	0.1523	0.1445	0.1484	0.1641
	hold ↑ → CK	-0.1289	-0.1289	-0.1133	-0.1172	-0.1172	-0.1133	-0.1094	-0.1055
	hold ↓ → CK	-0.0742	-0.0820	-0.1016	-0.0977	-0.0898	-0.0898	-0.0938	-0.1133
SE	setup ↑ → CK	0.1719	0.1641	0.1445	0.1445	0.1406	0.1367	0.1328	0.1250
	setup ↓ → CK	0.1641	0.1641	0.1875	0.1797	0.1641	0.1562	0.1562	0.1758
	hold ↑ → CK	-0.1367	-0.1328	-0.1172	-0.1211	-0.1211	-0.1172	-0.1133	-0.1094
	hold ↓ → CK	-0.0820	-0.0898	-0.1133	-0.1133	-0.1055	-0.1016	-0.1055	-0.1250
CK	minpwl	0.9974	0.9974	0.9974	0.9974	0.9974	0.9974	0.9974	0.9974

Cell Description

The TLATSR cell is an active-high D-type transparent latch with asynchronous active-low set (SN) and reset (RN), and set dominating reset. When the enable (G) is high, data is transferred to the outputs (Q,QN).

Logic Symbol



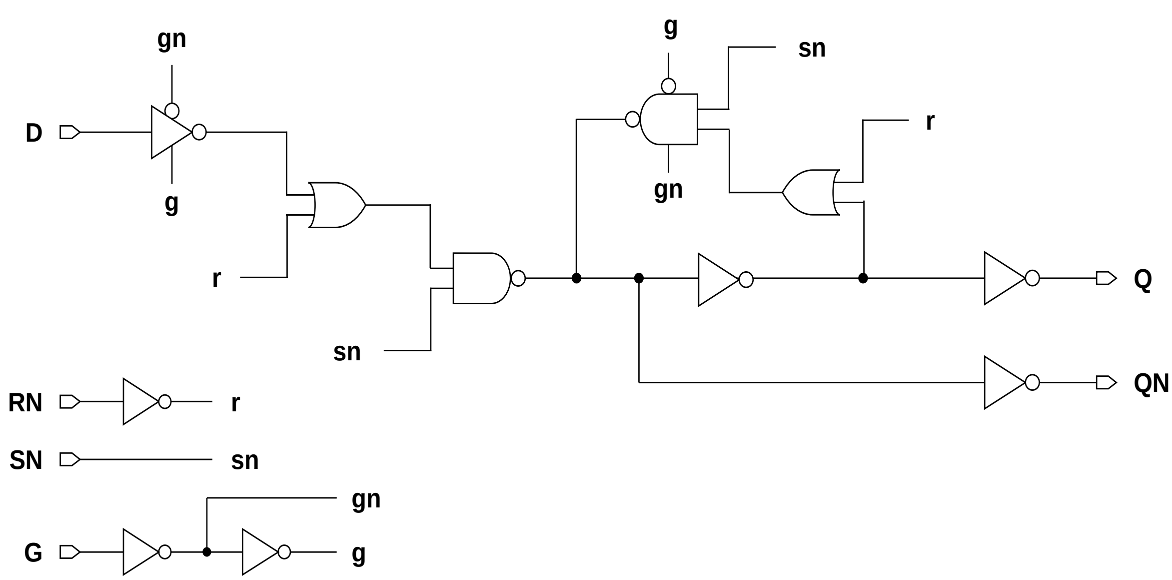
Function Table

RN	SN	G	D	Q[n+1]	QN[n+1]
1	1	1	0	0	1
1	1	1	1	1	0
1	1	0	x	Q[n]	QN[n]
0	1	x	x	0	1
1	0	x	x	1	0
0	0	x	x	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
TLATSRX1M	2.87	8.61
TLATSRX2M	2.87	8.61
TLATSRX4M	2.87	9.84

Functional Schematic



AC Power

Pin	Power (uW/MHz)		
	X1	X2	X4
D	0.0016	0.0023	0.0023
G	0.0096	0.0099	0.0115
SN	0.0059	0.0072	0.0079
RN	0.0022	0.0026	0.0027
Q	0.0222	0.0262	0.0421

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
D	0.0022	0.0027	0.0027
G	0.0017	0.0018	0.0026
SN	0.0018	0.0024	0.0028
RN	0.0030	0.0034	0.0034

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
D → Q ↑	0.1525	0.1765	0.2018	5.4712	3.6328	1.9119
D → Q ↓	0.3492	0.2930	0.3363	4.6075	2.3524	1.2152
G → Q ↑	0.2443	0.2558	0.2558	5.4675	3.6306	1.9118
G → Q ↓	0.3308	0.2831	0.3219	4.6128	2.3542	1.2163
SN → Q ↑	0.1811	0.1647	0.1710	5.3450	3.5439	1.8526
SN → Q ↓	0.3769	0.3149	0.3538	4.5676	2.3313	1.2045
RN → Q ↑	0.1496	0.1726	0.1982	5.4711	3.6324	1.9120
RN → Q ↓	0.2403	0.1929	0.2239	4.3197	2.2324	1.1293
D → QN ↑	0.4117	0.3634	0.4294	5.3053	3.5618	1.8391
D → QN ↓	0.2503	0.2726	0.2940	3.2702	1.8118	0.8579
G → QN ↑	0.3942	0.3542	0.4154	5.3074	3.5623	1.8393
G → QN ↓	0.3436	0.3532	0.3488	3.2730	1.8124	0.8581
SN → QN ↑	0.4391	0.3846	0.4455	5.3039	3.5619	1.8392
SN → QN ↓	0.2742	0.2537	0.2541	3.2549	1.8014	0.8503
RN → QN ↑	0.2993	0.2577	0.3031	5.2972	3.5622	1.8391
RN → QN ↓	0.2480	0.2693	0.2907	3.2705	1.8116	0.8578

Timing Constraints at 25°C,1.2V, Typical Process

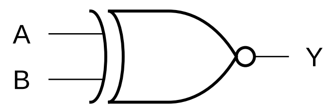
Pin	Requirement	Interval (ns)		
		X1	X2	X4
D	setup ↑ → G	0.0430	0.0742	0.1328
	setup ↓ → G	0.2852	0.2344	0.3047
	hold ↑ → G	-0.0273	-0.0508	-0.1016
	hold ↓ → G	-0.2773	-0.2266	-0.2930
G	minpwh	0.9974	0.9974	0.9974
SN	minpwl	0.9974	0.9974	0.9974
	recovery	0.3164	0.2578	0.3203
	removal	-0.3125	-0.2539	-0.3164
RN	minpwl	0.9974	0.9974	0.9974
	recovery	0.0391	0.0664	0.1133
	removal	-0.0352	-0.0625	-0.1094

Cell Description

The XNOR2 cell provides a logical EXCLUSIVE NOR of two inputs (A,B). The output (Y) is represented by the logic equation:

$Y = (A \bullet B) + (\overline{A} \bullet \overline{B})$

Logic Symbol



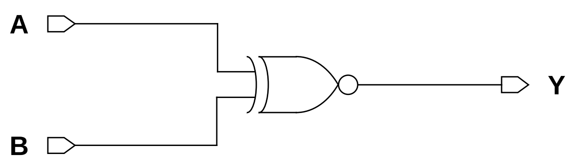
Function Table

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Cell Size

Drive Strength	Height (um)	Width (um)
XNOR2XLM	2.87	4.10
XNOR2X1M	2.87	4.10
XNOR2X2M	2.87	4.10
XNOR2X4M	2.87	5.74
XNOR2X8M	2.87	10.66

Functional Schematic



AC Power

Pin	Power (uW/MHz)				
	XL	X1	X2	X4	X8
A	0.0055	0.0068	0.0098	0.0189	0.0364
B	0.0096	0.0115	0.0168	0.0296	0.0582

Pin Capacitance

Pin	Capacitance (pF)				
	XL	X1	X2	X4	X8
A	0.0022	0.0024	0.0032	0.0048	0.0095
B	0.0017	0.0025	0.0036	0.0068	0.0133

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				
	XL	X1	X2	X4	X8
A → Y ↑	0.0455	0.0473	0.0430	0.0461	0.0464
A → Y ↓	0.0574	0.0628	0.0602	0.0633	0.0616
B → Y ↑	0.1152	0.0876	0.0826	0.0750	0.0737
B → Y ↓	0.1239	0.0935	0.0874	0.0758	0.0769

Delays at 25°C,1.2V, Typical Process (Cont'd.)

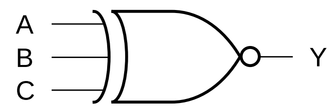
Description	K_{load} (ns/pF)				
	XL	X1	X2	X4	X8
A → Y ↑	11.0224	7.3358	4.7539	2.4377	1.2161
A → Y ↓	6.9703	4.5080	2.4824	1.2239	0.6145
B → Y ↑	11.1452	7.2981	4.8718	2.4942	1.2459
B → Y ↓	7.3174	4.6490	2.6239	1.2945	0.6459

Cell Description

The XNOR3 cell provides a logical EXCLUSIVE NOR of three inputs (A,B,C). The output (Y) is represented by the following equation:

$Y = \overline{A \oplus B \oplus C}$

Logic Symbol



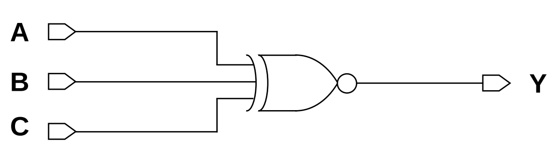
Function Table

A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
XNOR3XLM	2.87	8.20
XNOR3X1M	2.87	8.20
XNOR3X2M	2.87	8.20
XNOR3X4M	2.87	9.43
XNOR3X8M	2.87	11.48

Functional Schematic



AC Power

Pin	Power (uW/MHz)				
	XL	**X1**	**X2**	**X4**	**X8**
A	0.0268	0.0334	0.0385	0.0543	0.0979
B	0.0200	0.0244	0.0285	0.0438	0.0883
C	0.0101	0.0126	0.0154	0.0197	0.0380

Pin Capacitance

Pin	Capacitance (pF)				
	XL	**X1**	**X2**	**X4**	**X8**
A	0.0025	0.0034	0.0034	0.0035	0.0035
B	0.0030	0.0034	0.0034	0.0036	0.0037
C	0.0026	0.0029	0.0031	0.0031	0.0031

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				
	XL	**X1**	**X2**	**X4**	**X8**
A → Y ↑	0.2366	0.2101	0.2097	0.2224	0.2835
A → Y ↓	0.2749	0.2430	0.2372	0.2768	0.3722
B → Y ↑	0.2048	0.1799	0.1830	0.2059	0.2675
B → Y ↓	0.2645	0.2360	0.2253	0.2570	0.3525
C → Y ↑	0.1089	0.1141	0.1221	0.1361	0.1671
C → Y ↓	0.1225	0.1165	0.1006	0.1246	0.1738

Delays at 25°C,1.2V, Typical Process (Cont'd.)

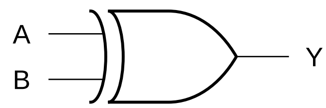
Description	K_{load} (ns/pF)				
	XL	**X1**	**X2**	**X4**	**X8**
A → Y ↑	8.6790	5.1691	3.4932	1.8255	0.9610
A → Y ↓	6.1991	3.4576	2.1030	1.1167	0.6239
B → Y ↑	8.6780	5.1687	3.5118	1.8372	0.9607
B → Y ↓	6.1999	3.4575	2.0453	1.1160	0.6239
C → Y ↑	8.7689	5.2110	3.5234	1.8316	0.9597
C → Y ↓	6.1979	3.4576	1.9902	1.0378	0.5818

Cell Description

The XOR2 cell provides a logical EXCLUSIVE OR of two inputs (A,B). The output (Y) is represented by the logic equation:

$Y = (A \bullet \overline{B}) + (\overline{A} \bullet B)$

Logic Symbol



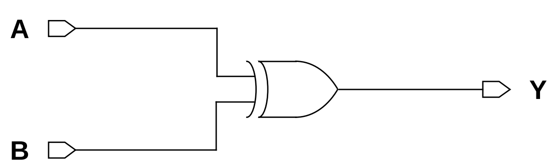
Function Table

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Cell Size

Drive Strength	Height (um)	Width (um)
XOR2XLM	2.87	4.10
XOR2X1M	2.87	4.10
XOR2X2M	2.87	4.10
XOR2X3M	2.87	5.74
XOR2X4M	2.87	5.74
XOR2X8M	2.87	11.07

Functional Schematic



AC Power

Pin	Power (uW/MHz)					
	XL	**X1**	**X2**	**X3**	**X4**	**X8**
A	0.0054	0.0067	0.0098	0.0165	0.0190	0.0367
B	0.0104	0.0128	0.0188	0.0272	0.0333	0.0660

Pin Capacitance

Pin	Capacitance (pF)					
	XL	**X1**	**X2**	**X3**	**X4**	**X8**
A	0.0022	0.0024	0.0032	0.0041	0.0048	0.0095
B	0.0017	0.0025	0.0036	0.0053	0.0068	0.0134

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)					
	XL	**X1**	**X2**	**X3**	**X4**	**X8**
A → Y ↑	0.0470	0.0481	0.0448	0.0536	0.0501	0.0469
A → Y ↓	0.0569	0.0626	0.0615	0.0719	0.0639	0.0618
B → Y ↑	0.1114	0.0826	0.0766	0.0778	0.0720	0.0719
B → Y ↓	0.1198	0.0945	0.0890	0.0841	0.0782	0.0781

Delays at 25°C,1.2V, Typical Process (Cont'd.)

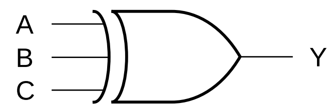
Description	K_{load} (ns/pF)					
	XL	**X1**	**X2**	**X3**	**X4**	**X8**
A → Y ↑	11.0210	7.3226	4.7255	3.2057	2.3944	1.2179
A → Y ↓	6.9653	4.5045	2.4939	1.5975	1.2340	0.6187
B → Y ↑	11.1712	7.2938	4.8623	3.2125	2.4606	1.2539
B → Y ↓	7.3073	4.6558	2.5746	1.6521	1.2730	0.6367

Cell Description

The XOR3 cell provides a logical EXCLUSIVE OR of three inputs (A,B,C). The output (Y) is represented by the following equation:

$$Y = A \oplus B \oplus C$$

Logic Symbol



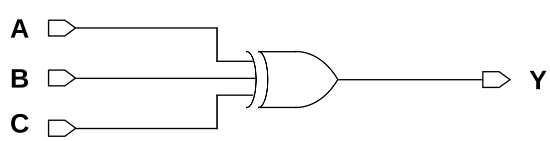
Function Table

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Cell Size

Drive Strength	Height (um)	Width (um)
XOR3XLM	2.87	7.38
XOR3X1M	2.87	8.20
XOR3X2M	2.87	8.20
XOR3X4M	2.87	9.43
XOR3X8M	2.87	11.07

Functional Schematic



AC Power

Pin	Power (uW/MHz)				
	XL	**X1**	**X2**	**X4**	**X8**
A	0.0253	0.0349	0.0403	0.0560	0.1015
B	0.0196	0.0247	0.0299	0.0455	0.0901
C	0.0100	0.0118	0.0150	0.0195	0.0403

Pin Capacitance

Pin	Capacitance (pF)				
	XL	**X1**	**X2**	**X4**	**X8**
A	0.0025	0.0035	0.0035	0.0035	0.0034
B	0.0028	0.0034	0.0034	0.0036	0.0036
C	0.0038	0.0043	0.0050	0.0051	0.0051

Delays at 25°C,1.2V, Typical Process

Description	Intrinsic Delay (ns)				
	XL	**X1**	**X2**	**X4**	**X8**
A → Y ↑	0.2352	0.2094	0.2132	0.2236	0.2817
A → Y ↓	0.2636	0.2370	0.2394	0.2789	0.3777
B → Y ↑	0.2136	0.1790	0.1848	0.2028	0.2608
B → Y ↓	0.2589	0.2301	0.2235	0.2588	0.3514
C → Y ↑	0.1163	0.1150	0.1057	0.1069	0.1208
C → Y ↓	0.1197	0.1160	0.1131	0.1310	0.1702

Delays at 25°C,1.2V, Typical Process (Cont'd.)

Description	K_{load} (ns/pF)				
	XL	**X1**	**X2**	**X4**	**X8**
A → Y ↑	8.8489	5.1643	3.4929	1.8018	0.9573
A → Y ↓	6.2003	3.4661	2.1403	1.1338	0.6376
B → Y ↑	8.8486	5.1641	3.4933	1.8301	0.9572
B → Y ↓	6.2013	3.4454	2.1025	1.1328	0.6374
C → Y ↑	8.8246	5.1557	3.4896	1.7994	0.9316
C → Y ↓	6.1420	3.4141	2.0693	1.0861	0.6075